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Research

Private capital monitor

Media narratives v the hard reality

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Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Executive summary

- Private capital managers' confident Q1 performance is a stark contrast to the media and investor narratives that have focussed on concerns about credit quality, AI disruption and redemptions. We take a closer look at what this divide means for PC in 2026.
- AI disruption: Our risk scoring and indexes show that PC managers were more worried about the 2022-23 rate rises, but today, the media is far more worried about AI.
- Overall in Q1, PC managers became more attuned to various external risks including AI disruption. Still, the large managers remain confident about their portfolios and fundraising.
- Managers emphasised that retail fundraising woes have been more concentrated and less severe than the media presents.
- We show our proprietary survey of individuals in the UK and the US that highlights the virality of negative headlines in the financial press.
- We track decompression in BDCs' non-accrual rates to show that the deterioration comes after quarters of strong growth. Thus, further defaults will require significant stress in the equity layer.
- We continue to expect PC dealmaking to recover as the most intense adverse headlines subside.
- Last, we take a look at the impact of market volatility amidst geopolitical tensions on realisations and dealmaking. We expect both to be resilient, including IPOs, as both corporates and dealmakers have become accustomed to market turbulence this decade. We also note that softness in sponsored deals is less pronounced outside tech and managers are eager to invest in key themes such as infrastructure, energy and defence.

01.
Q1 2026 results

What do earnings calls from large, listed managers say about the private capital environment?
Key themes

Q1 was characterised by a complex macro environment, with managers navigating geopolitical tensions, fears of AI-driven disruption, and a bifurcation in private credit markets. Despite this, management teams are confident amidst strong fundraising, accelerating demand for AI-related infrastructure, and strategic advantage of diversified, scaled platforms. Firms are differentiating themselves through operational expertise and a disciplined investment processes, while being ready for opportunities from market dislocations.

1. AI as a generational investment catalyst, especially in infrastructure

- The AI revolution is the most dominant theme, viewed as a massive, multi-decade tailwind. The focus is less on direct software investment and more on providing the essential physical infrastructure required to power AI. Management consistently highlighted the enormous capital required for data centres, power generation, transmission, cooling systems, and the broader energy transition.
    - Blackstone noted its portfolio of data centres exceeds $150 billion, with a $160 billion prospective pipeline, and stated AI infrastructure is the "single most important thing for the performance of our clients". Ares sized the third-party data centre market opportunity at around $900 billion.
- Firms are positioning themselves as beneficiaries of the AI build-out, regardless of which software models win.
- Internally, firms are adopting AI to enhance productivity. TPG noted that nearly 80% of its staff uses its customized AI tools daily

2. A bifurcation in private credit markets with strong institutional backing

- Despite negative headlines and "noise" surrounding retail-focused BDCs, managers reported robust institutional demand for private credit, emphasizing a "flight to quality.“ Several firms noted a clear divergence. While retail channels experienced redemption requests, institutional demand remained strong.
    - KKR noted "meaningful inbound interest from institutions around our direct lending business".
    - Blackstone stated its institutional and insurance clients, representing 75% of its credit AUM, "continued to commit large-scale capital".
- Managers stressed their disciplined underwriting, low LTVs, and focus on senior secured debt as key risk mitigants. Diversification away from sponsor-backed direct lending into areas like asset-based finance (ABF) and investment-grade private credit is a major trend.

3. Fundraising strength driven by scale and client consolidation

- Despite the environment, fundraising was a significant bright spot. Across the board, large, diversified managers are benefiting from a trend of investors consolidating their relationships with fewer, strategic partners who can offer a wide range of products – clients are increasingly seeking multi-asset, customized solutions rather than single-fund products. The GP-led secondaries market is seen as a major growth area, driven by the need for liquidity solutions for fund sponsors.

4. Cautious optimism on monetisations amid geopolitical uncertainty

- The M&A and IPO environment is viewed with caution due to geopolitical instability, but firms with high-quality, well-performing assets see a path to monetization. Firms are actively exiting assets at strong multiples.

Source: Bloomberg Finance LP, Deutsche Bank.

What do earnings calls from large, listed managers say about the private capital environment?
Key risks

The primary risks centre on the disruption of software and other "white-collar" industries by AI, the potential for a broader credit cycle turn, and redemption pressures in the retail wealth channel for private credit.

1. AI disruption to software and “white-collar" business models

- While AI is seen as an opportunity, its potential to disrupt existing business models, particularly in software, is a frequently cited risk. While there is a general acknowledgment that the full impact of AI is unknown and will play out over several years, managers are actively reviewing their software portfolios for AI risk:
  - Ares had a review that concluded that 86% of its portfolio had low AI disruption risk, 13% had medium risk, and only 1% had high risk.
  - TPG noted one of its most exposed funds has 7% exposure to companies in a "mitigate category" where material AI risk is perceived.
  - Blackstone mentioned that software "has come into focus as an at-risk area".
- The consensus is to be defensively positioned, focusing on software with proprietary data and high barriers to entry, while avoiding areas like content generation or simple productivity tools.

2. Redemption pressures in retail private credit vehicles

- The slowdown and net outflows in non-traded BDCs and other retail credit funds are a headwind, though managers frame it as manageable and cyclical.
- While the overall impact on AUM and fees is presented as minimal to firm-wide earnings, it is causing a near-term deceleration in fee growth from these specific products.
- Managers also defended the redemption gates (typically 5% of NAV per quarter) as an essential feature to protect investors from forced asset sales and align the fund's liquidity with its illiquid underlying assets.

3. Geopolitical Instability and Macro Headwinds

- The conflict in Iran and broader geopolitical tensions were cited as creating uncertainty, volatility, and a more challenging operating environment – the conflict has caused a near-term "pause" in transaction activity as sponsors and corporates re-evaluate the landscape.
- Despite the risks, managers believe periods of uncertainty favour scaled, patient investors.
  - Ares noted it experienced its two fastest growth periods during the GFC and COVID crises.

Source: Bloomberg Finance LP, Deutsche Bank.

What do earnings calls from large, listed managers say about the private capital environment?

Scoring key concerns using earnings transcripts and our AI tool, dbLumina – Q4 vs Q1

Managers became on average more concerned about risks last quarter. They are also relatively more concerned about AI disruption and it is now the top 3 risk in our list, though it still trails macroeconomic environment risk by half. Importantly, managers continue to stress they're comfortable with own credit portfolios and fundraising.

AI & digital infrastructure

- Earlier this year, AI was discussed more as an opportunity. Software exposure was noted, but the risk of disruption was less of a focus. Now. the risk of AI-driven disruption to software business models has become a central topic of questioning and a key area of risk management. Firms are proactively quantifying and managing this risk.

Private credit:

- The narrative has become more nuanced. While underlying credit quality remains strong, the sector is experiencing a turbulence in the US retail wealth channel, while institutional demand and opportunities in non-corporate or non-US credit are accelerating.

Realisations & IPOs

- The strong monetisation momentum from late 2025 was tempered by early 2026 volatility. However, the underlying pipelines remain robust, suggesting that a stabilisation in markets could lead to an acceleration in H2.

Retail channel volatility

- The wealth channel was a key growth engine, with record inflows reported by many firms for 2025. But the "social media and press reporting" on private credit led to a tangible slowdown and increased redemptions in US-focused retail credit funds – the risk of pro-cyclical flows in the retail channel, previously a more theoretical concern, materialized in Q1 2026.
- Managers also noted that redemption requests are often concentrated among a small number of larger, often institutional-like, investors within these funds, rather than a broad-based retail exodus.
- Firms are emphasizing the structural features of these semi-liquid vehicles and the fact that they have met all redemption requests, which they believe will ultimately build long-term confidence.

Average concern score across 8 large listed managers (from 0, least concerned, to 100, most concerned)

Source: Bloomberg Finance LP, Deutsche Bank.

Q1 earnings: reported performance by asset class – performance has slowed outside infrastructure

KKR	Q1	LTM
Private equity		
Traditional PE portfolio	1%	10%
Real assets		
Opportunistic real estate	-1%	2%
Infrastructure	2%	10%
Credit		
Leveraged credit composite	-1%	5%
Alternative credit composite	-1%	4%

Apollo	Q1	LTM
Direct origination	0.5%	8.5%
Opportunistic credit	2.5%	11.4%
Multi-credit	1.0%	7.9%
Asset-backed finance	-1.0%	8.2%
Flagship PE	-0.3%	4.7%

Carlyle	Q1	LTM
Global private equity		
Corporate PE	-2%	3%
Real estate	1%	3%
Infrastructure & natural resources	9%	25%
Global credit	4%	15%

Ares	Q1	LTM
Credit		
Alternative credit	3.9%	15.3%
Opportunistic credit	-0.2%	11.7%
US senior direct lending	2.1%	11.8%
US junior direct lending	1.5%	14.8%
European direct lending	2.6%	9.2%
APAC credit	2.5%	20.3%
Corporate private equity	-1.1%	3.5%
Real assets		
Americas real estate equity	0.6%	5.9%
European RE equity	0.1%	3.1%
Infrastructure debt	1.9%	9.0%
Secondaries		
Private equity secondaries	0.9%	7.2%
Real estate secondaries	2.5%	7.9%

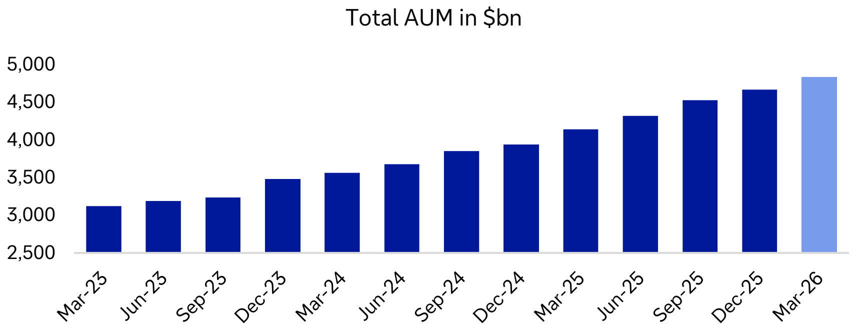
Blue Owl	Q1	LTM
Direct lending	-0.4%	8.5%
Alternative credit	1.7%	11.0%
Net lease	2.6%	14.7%
Real estate credit	1.9%	8.5%
Digital infrastructure	3.9%	2.6%

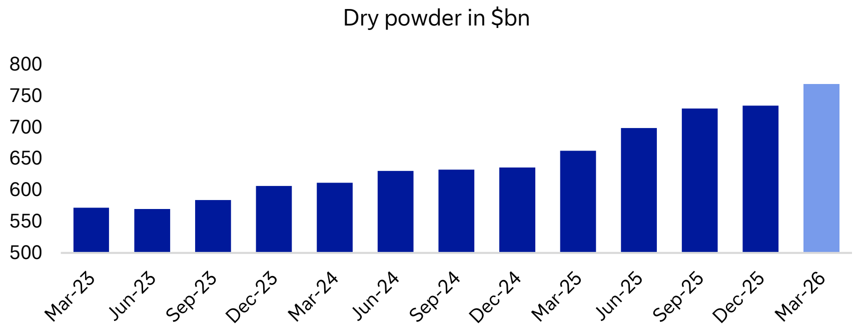
Blackstone	Q1	LTM
Real estate		
Opportunistic	-0.9%	-1.6%
Core+	0.8%	2.6%
Private equity		
Corporate private equity	3.2%	15.7%
Tactical opportunities	3.6%	10.9%
Secondaries	0.7%	12.7%
Infrastructure	7.8%	24.8%
Credit & insurance		
Private credit	0.6%	8.9%
Liquid credit	-1.3%	4.1%

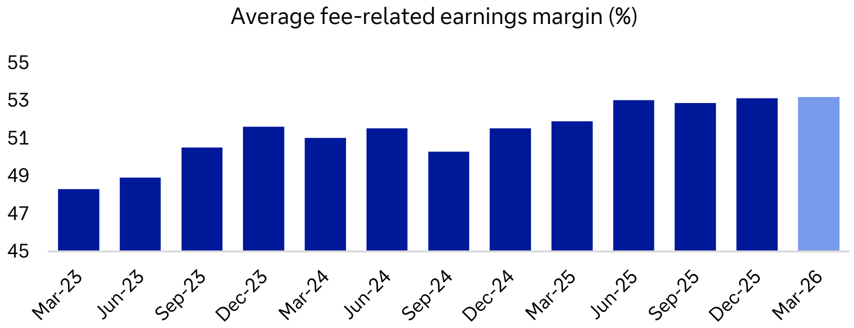
TPG	Q1	LTM
Capital	-2.3%	6.7%
Growth	-1.5%	2.0%
Impact	0.3%	12.5%
Credit	2.2%	10.8%
Real Estate	1.9%	8.3%

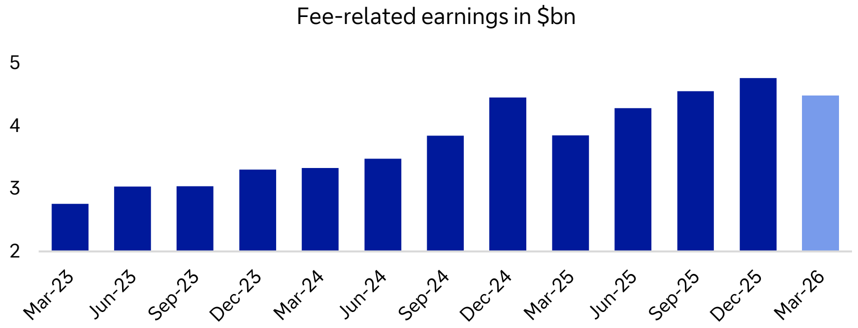
Source: Company reports, Bloomberg Finance LP, Deutsche Bank. Note: Performance reflects gross returns or appreciation depending on the manager, strategy and time period.

Growth in key metrics for selected large private capital managers – strong AUM growth but softer earnings QoQ









Source: Bloomberg Finance LP, Deutsche Bank.

Management comments – focus on resilience amidst a more volatile backdrop

- Stephen Schwarzman of Blackstone: “Blackstone delivered outstanding first-quarter results despite the turbulent environment, highlighted by almost $70 billion of inflows and positive appreciation across nearly all of our flagship strategies. Our all-weather model protects us in these times of disruption while also allowing us to invest where we see the greatest opportunity”
- Joseph Bae and Scott Nuttall of KKR: “Against a volatile backdrop, monetization activity accelerated, and over the past 12 months we’ve invested more capital on behalf of our clients than at any point in our history. We are highly confident in the strength of our platform and our long-term positioning.”
- Marc Rowan of Apollo: “Our first quarter results set a strong tone for the year, with record fee-related earnings and assets under management surpassing $1 trillion – a testament to the trust our clients place in us and a reminder of the value we create at scale. Whether we’re originating investment grade credit, delivering retirement solutions, or developing new products and capabilities, innovation and discipline continue to drive us forward.”
- Harvey Schwartz of Carlyle: “Our first quarter results reflect continued momentum executing against our strategic plan. Carlyle AlpInvest delivered another quarter of exceptional growth, fundraising for both institutional and wealth clients had a strong start to the year, and we had a record quarter for U.S. Buyout realizations – each reinforcing our confidence in the path to achieving the 2028 targets…”
- Michael Arougheti of Ares: “We reported strong first quarter results highlighted by continued growth across our key financial metrics, including record first quarter fundraising of $30 billion, up more than 45% year over year. We are on track for another record year of fundraising as we continue to see broad-based investor demand across our platform. We also continue to see strong fundamental performance across our investment portfolios despite the volatile market environment”.”
- Jon Winkelried of TPG: ““TPG’s strong first quarter performance reflects the significant momentum across our global platform, as the scale of our franchise and consistent execution continue to translate into powerful results. Over the past year, despite an uncertain macro environment, we delivered step function growth across capital formation, deployment, and realizations. Our resilient business model is intentionally built to navigate complexity and capitalize on opportunity. As our clients deepen their engagement with TPG, we are confident in our positioning and ability to deliver long-term growth and differentiated value for our stakeholders..“
- Doug Ostrover and Marc Lipschultz of Blue Owl: “Blue Owl's results for the first quarter of 2026 demonstrate the power of our three differentiated and scaled platforms, each of which has contributed to our continued expansion to $315 billion of AUM. Our financial results reflect stability, stemming from our durable capital base, and growth, driven by fundraising and ongoing capital deployment."

Source: Company reports, Deutsche Bank.

02.

Spotlight section: the key narratives weighing on private capital

1. Private capital outflows fears

Private capital managers argue that the "headline-driven" anxiety, particularly in the press and social media, is disconnected from the fundamental reality they are observing. While acknowledging a cyclical slowdown and elevated redemptions in some US retail vehicles, managers assert that the impact is contained and far less severe than portrayed. They point to strong portfolio performance, structural resilience of perpetual funds, and robust demand from institutional clients as counterweights. They highlight investor education, product diversification, and frame the current dislocation as a market share opportunity for scaled, trusted platforms.

Redemptions are concentrated and not systemic:

- Ares: for its non-traded BDC, the firm noted that the "majority of repurchase requests... came from a limited number of family offices and smaller institutions in select regions… Over 95% of our investors did not request redemptions"
- Blue Owl: for its OCIC fund, redemption requests were "concentrated with 1% of investors representing a majority of tenders" and approximately 90% of the investor base did not tender at all.
- Blackstone: on redemptions: "contrary to this popular idea that it's small investors that are leading the charge here, it is actually a smaller number of large investors who are double the size basically on average of the typical investor in these vehicles."

In sharp contrast to the retail wealth channel, managers report strong demand from institutional and insurance clients, who view the dislocation as an opportunity.

- Blackstone: mentioned the "striking the difference in terms of what we've seen from the institutional and insurance clients relative to the wealth channel," and noted that Q1 was one of their best quarters for credit fundraising from these clients on record.
- Ares: stated that "the institutional investor is not anxious. They're not allocating away from private credit. And in fact, I think they're looking at this as a huge opportunity to take advantage of a bizarre dislocation and bring liquidity into the market to capture excess return."
- KKR: received "meaningful inbound interest from institutions around our direct lending business" in the weeks leading up to their call, with clients viewing the dislocation as an "interesting entry point".

While acknowledging the reality of the outflows, managers assess their financial and strategic impact as minimal and manageable, saying that the growth in other areas far outweighs the softness in specific US wealth products.

- Ares: said its two private credit wealth products account for only 4.5% of its overall fee-paying AUM and that in a "very unlikely scenario" where if these funds experienced 5% quarterly redemptions for a full year with no new inflows, it would affect total firm FPAUM by only approximately 1% annually.
- Blue Owl: Net outflows from its two BDCs were less than 6 basis points of the firm's beginning-of-period AUM, with the funds collectively making up less than 17% of total AUM.
- KKR: Its private BDC footprint is "even smaller, around $3 billion of AUM or 0.4% of our AUM in total".

There is a consensus that the softness is cyclical and temporary. Managers are not altering their long-term plans for the wealth channel. Blackstone said it will continue to expand its wealth team and product suite at a "fairly rapid pace". Ares confirmed it has not changed its fundraising guidance.

- Blackstone, drawing a parallel to the earlier noise around his firm's real estate vehicle, stated, "We're going to get through this like we've always gotten through these moments, and the products will continue to grow".
- Blue Owl expects to "continue to be in a softer environment in wealth" but can maintain its margin targets regardless.

Managers defend the redemption gates (typically 5% of NAV per quarter) as a fundamental and necessary feature of the products, designed to protect investors from forced asset sales and match the illiquid nature of the underlying loans.

- Blackstone noted, "These caps on redemptions are not a bug. They're a feature of these products... The key question is, are you offering a premium in exchange for giving up this liquidity?"
- Blue Owl noted that for its OCIC fund, regular-way paydowns from underlying loans were almost $3 billion in the quarter, providing 3x coverage for the $1 billion in gross redemptions, before considering new inflows or credit lines.

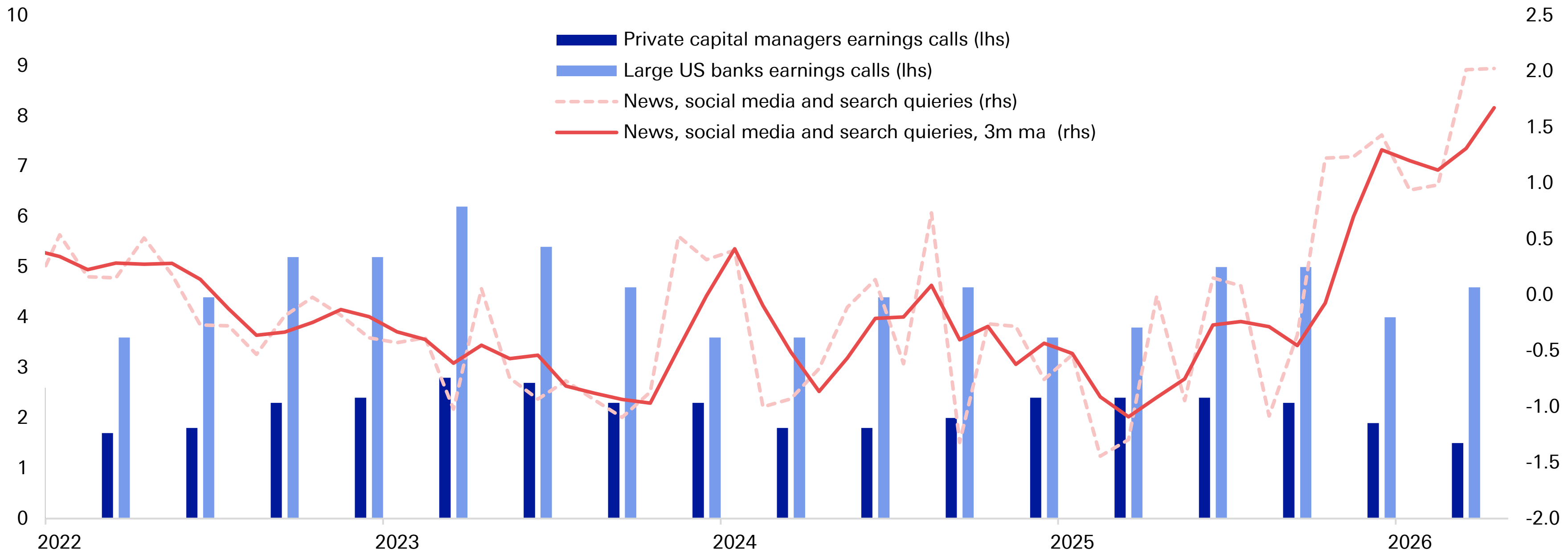
Product and channel diversification: Capital that might have gone to a US BDC can be seamlessly deployed elsewhere.

- Ares explicitly stated that any capital not deployed in its non-traded vehicles will be "taken up by other traded and institutional funds and SMAs with limited to no impact to our current year profitability". They also highlighted very strong inflows into their European direct lending and SME funds.

Source: Bloomberg Finance LP, Deutsche Bank.

When we asked our AI tools to contrast sentiment around private credit risks, it is clear that there is a big disconnect between sentiment in the media, which is the most negative it's been for a while, and private capital managers, who have been more concerned about this topic in 2023 when impact of rate hikes and recession risks were at the forefront. This is true for large US banks as well.

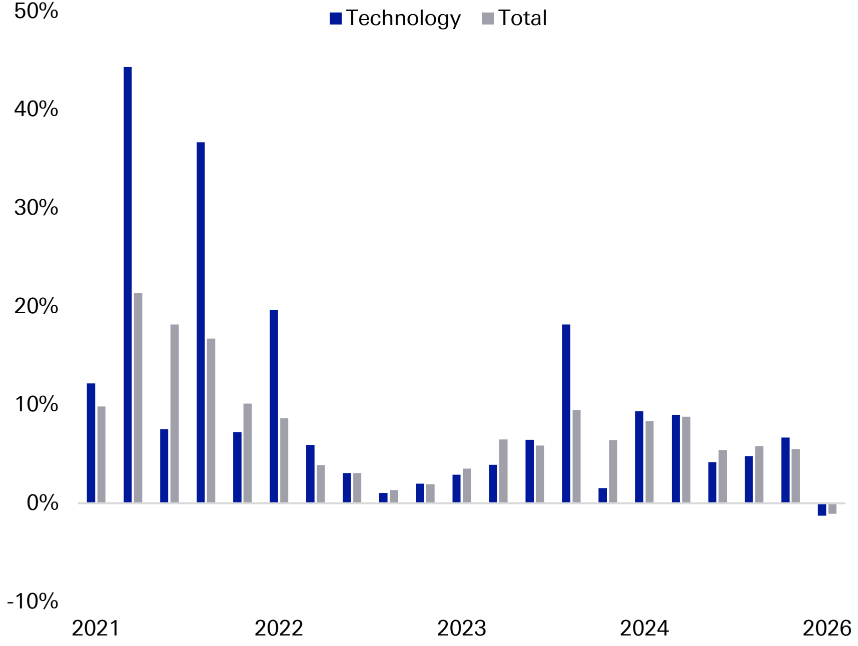
AI analysis of sentiment around private credit risk
(higher = more concerns; maximum of 10 for private capital and banks sentiment)



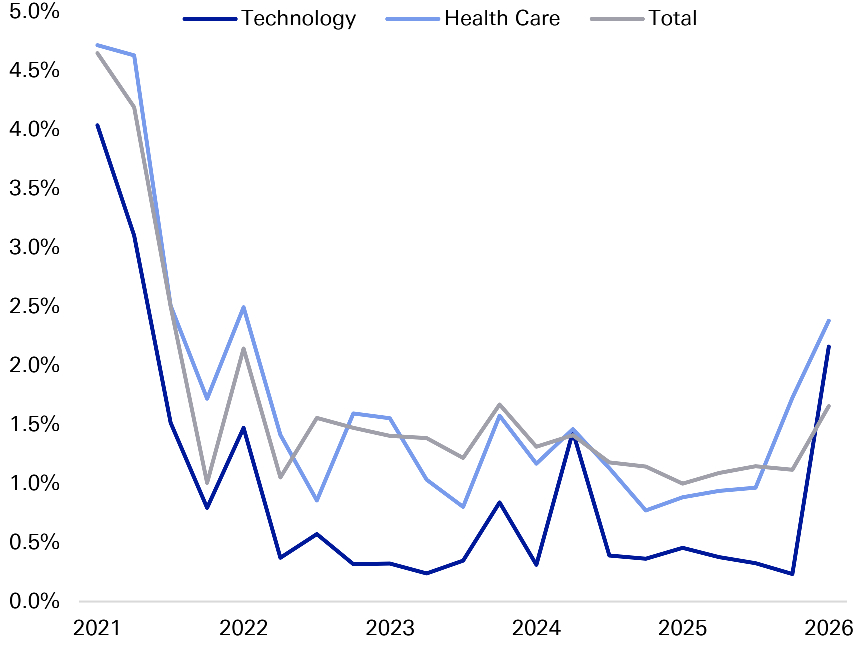
Source: Bloomberg Finance LP, Google Trends, Deutsche Bank.

While there has been a lot of focus on net outflows and rising default rates in BDCs, these metrics are worsening after quarters of strong growth and low credit stress economy-wide. Some decompression is reasonable as sentiment gives a pretext for some overdue write-downs, especially in tech. Nevertheless, the equity layer will need to deteriorate more to see big increases in overall default rates. Commentary from PE-driven managers suggests this is not an immediate concern

Growth of fair value of total assets of all selected BDCs *



Non accruals as % of assets at cost, all selected BDCs *

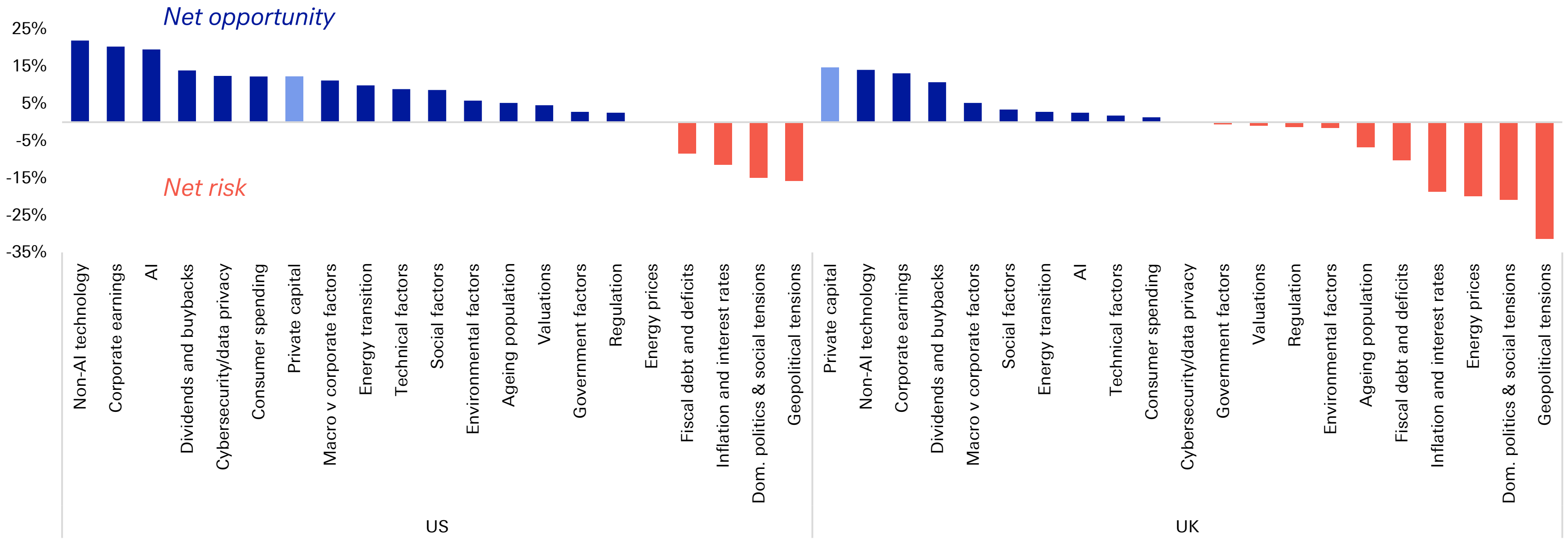


*Source: Bloomberg Finance LP, Deutsche Bank. * Note: as tracked by Bloomberg, total assets of c $420bn.*

Our survey of around 500 people in both the US and UK (a representative sample of the population) shows that the sentiment amongst the broader public towards private capital is sanguine. This is in contrast to market participants, which placed private capital risk third in our survey at the end of 2025 when asked about the biggest risks to market stability in 2026, which shows the impact of financial media sentiment.

For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026

Mean score ranging from 1 (significant opportunity) to -1 (significant risk)



Source: dbDataInsights, Deutsche Bank.

2. How serious is AI disruption risk for portfolios?

Still an equity question first. While services and tech make up c50% of BDC assets, we will need to see bigger markdowns by sponsors first to signal a lack of support from the equity cushion. Big managers assessment of their equity exposure shows this is not likely soon.

Managers are taking AI risk seriously, particularly within software, but generally view their direct exposure as limited and well-underwritten. They have quantified their exposure and stress-tested portfolios, concluding that the immediate risk to their credit-focused strategies is manageable:

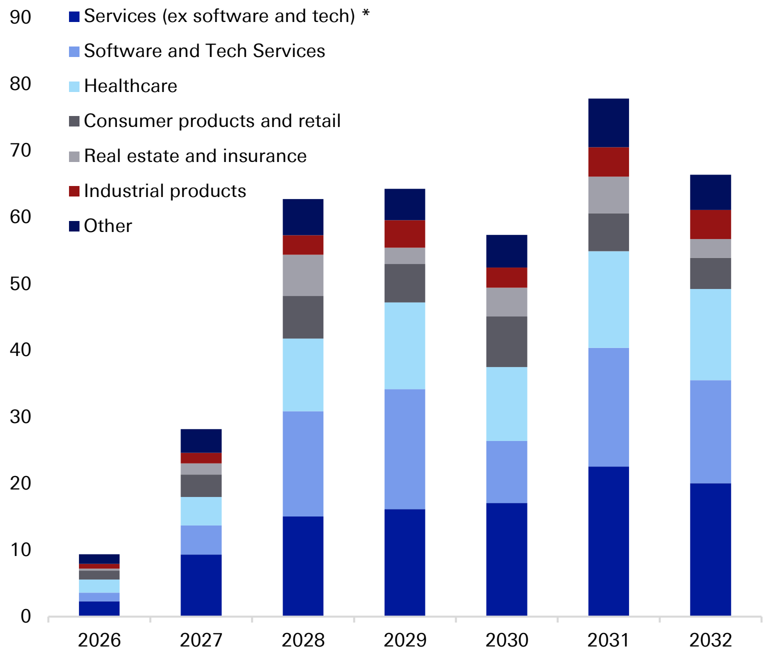
- **KKR:** Software represents **7%** of total AUM, **15%** in private equity, **5%** in credit, and **2.5%** in Global Atlantic. While public market weakness led to markdowns, the underlying software companies are seeing healthy single-digit revenue and EBITDA growth.
- **Ares:** Conducted a nine-week independent review of its software portfolio (6% of total AUM, <8% of private credit AUM). The study concluded the portfolio is "very well positioned," with **86%** classified as low-risk, **13%** as medium-risk, and only **1%** as high-risk of AI disruption. This high-risk portion represents less than 1% of the firm's total AUM.
- **TPG:** Has undertaken a "systematic review of the risks in and outside of software." In its TPG 8 fund, work showed that only **7%** of the remaining value fell into a "mitigate category" where material risk is perceived.
- **Blue Owl:** Acknowledges a "software maturity wall" coming in 2028-29 but stresses that its tech portfolio consists of very large companies (average EBITDA of $320 million) with significant equity cushions from sponsors. They view the current situation as an "equity question, not a debt question".

How are managers mitigating the risk?

Mitigation strategies include proactive portfolio re-underwriting, strategic divestment, internal AI adoption for productivity, and a thematic pivot toward AI-beneficiary assets.

- **Blackstone** is assessing companies based on their incumbency models, proprietary data, and workflow integration. **Ares** is focusing its senior lending on companies providing core operational software for regulated industries, avoiding more vulnerable areas like content generation.
- A cross-firm strategy is to invest heavily in the infrastructure enabling the AI revolution, mostly data centres. The need for data centres directly feeds into the need for more power. **Blackstone** is the most active private investor in the US utility sector.
- Leveraging AI in portfolio companies: **KKR** has deployed AI across over 150 portfolio companies to automate workflows and enhance products, using its operational team, Capstone, to ensure lessons are shared across the firm. **TPG** is investing to help its software portfolio companies leverage AI opportunities.

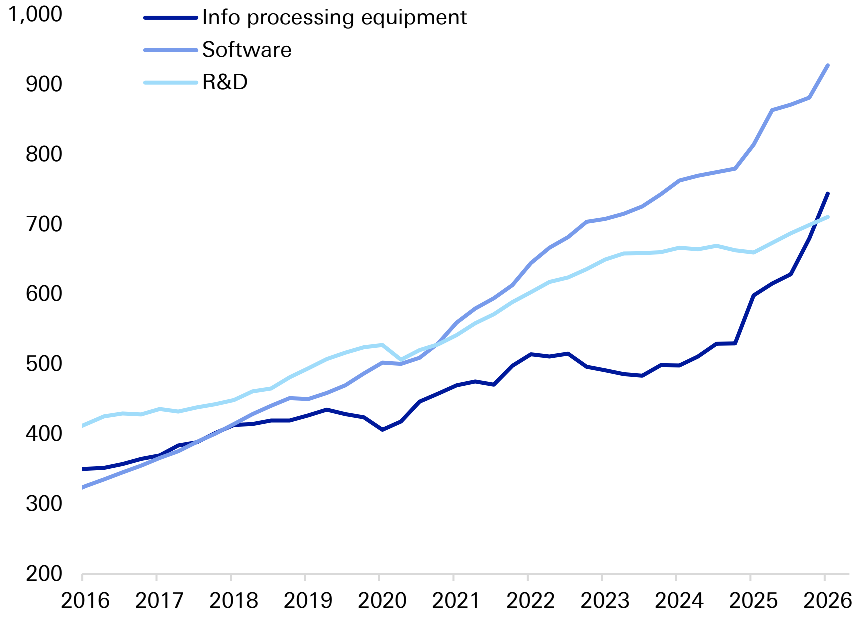
Selected BDCs, maturities by industry ($bn)



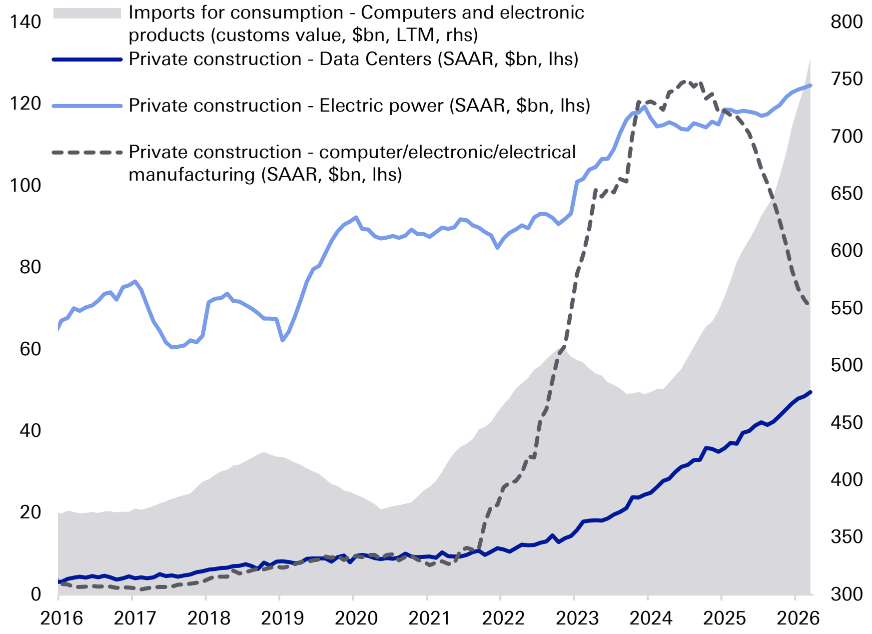
*Source: Bloomberg Finance LP, Deutsche Bank. * Note: services include industries such as industrial, consumer and financial services; BDCs as tracked by Bloomberg, total assets of c $420bn.*

In the meanwhile, managers are doubling down on infrastructure investments amidst investor appetite for the AI theme, and this is likely to further fuel AI build out. Power is amongst the bottleneck private capital firms are likely to tackle further.

Real private non-residential fixed investment (SAAR, bn 2017 chained dollars)



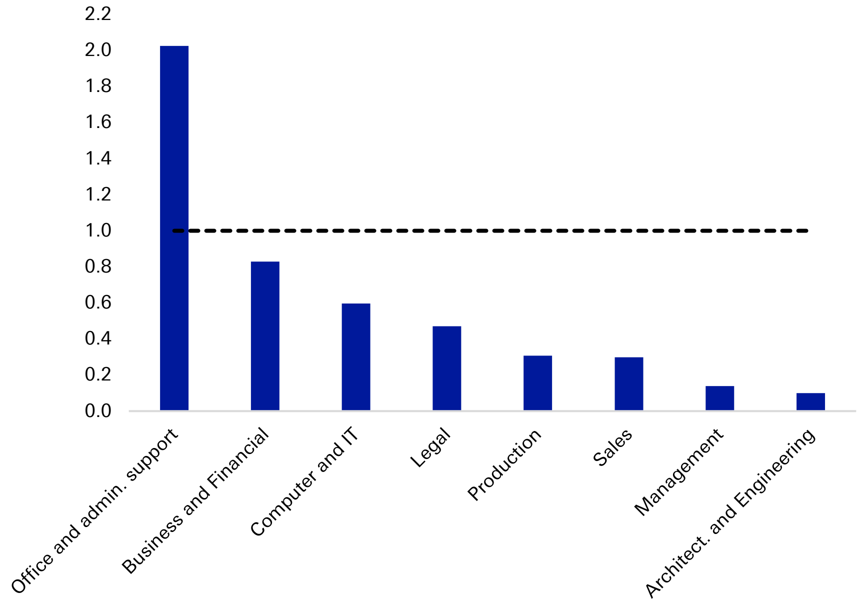
Data centre construction has run ahead of new (electric) power investments



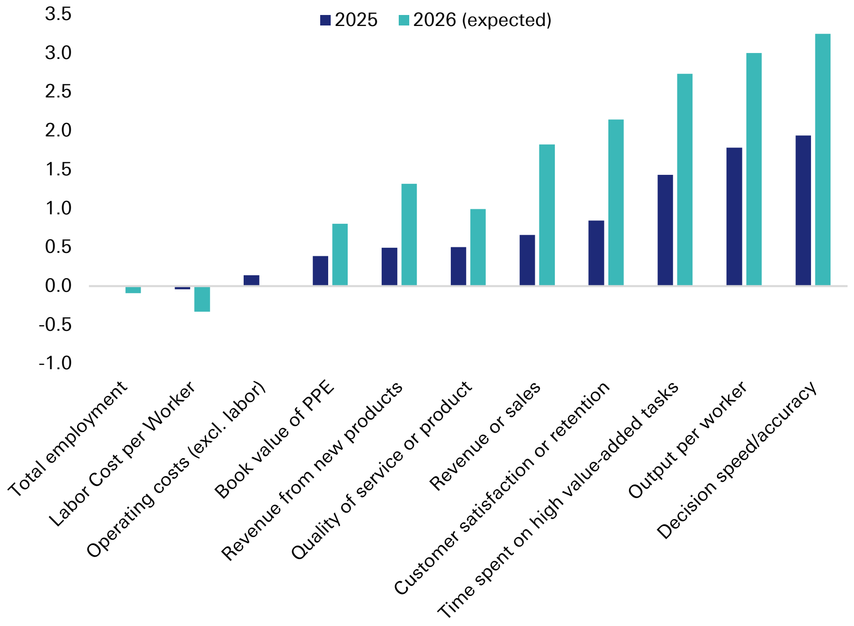
Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

For software, the wide range of potential outcomes of AI's impact means a more benign path than the media suggests may be plausible. Indeed, US CFOs expect the fastest growth in AI's impact in 2026 to be in, first, generating more sales and then some productivity gains and do not describe most corporate roles as outright replaceable or see a significantly negative impact on headcount.

CFO survey: AI negative exposure index by occupation group
ratio of replacement mentions to enhancement mentions, >1 indicates that AI is more frequently described as replacing rather than enhancing tasks



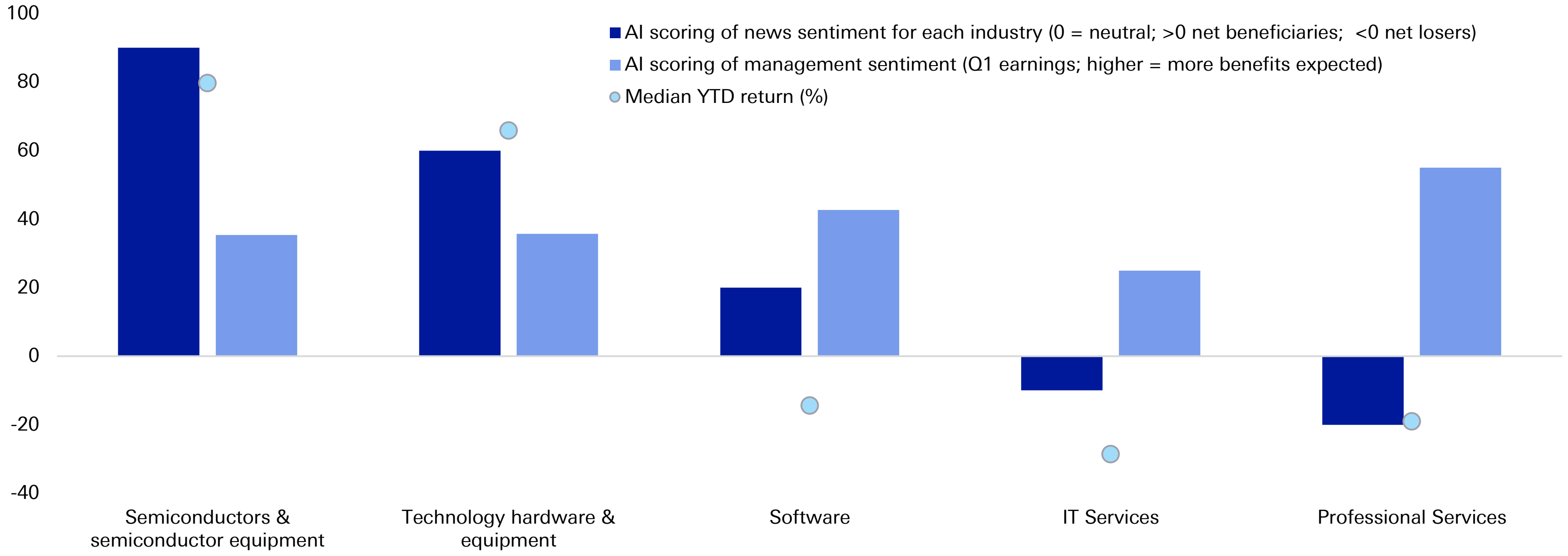
The impact of AI usage on firm outcomes
(%, averages across bucketed options)



Source: Bloomberg Finance LP, Richmond Fed CFO survey, Federal Reserve Bank of Atlanta "Artificial Intelligence, Productivity, and the Workforce: Evidence from Corporate Executives", Deutsche Bank.

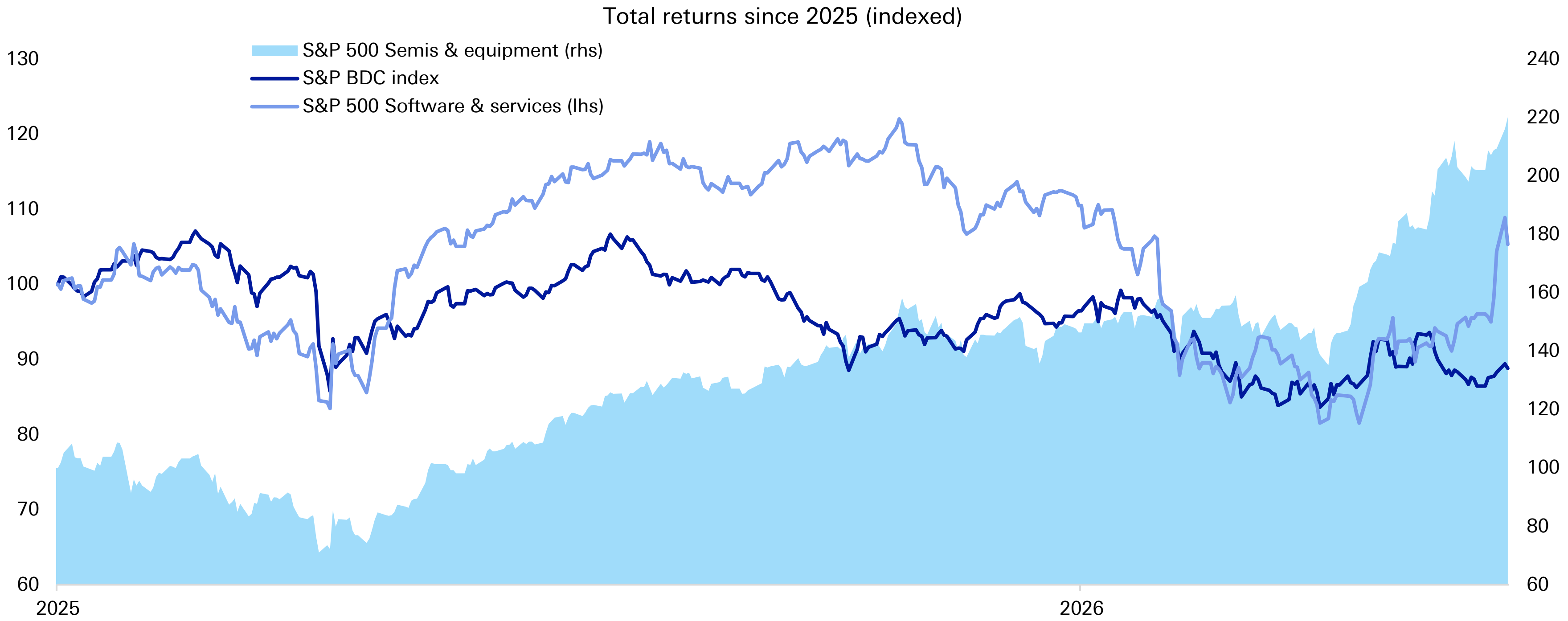
Moreover, our AI scoring of software risk sentiment again shows a big disconnect between what companies and media are saying about AI disruption risk, and, as the result, the potential mispricing in tech and services shares.

AI scoring of media and management sentiment around AI disruption risk for selected industries and their YTD performance (S&P 500 stocks)



Source: Bloomberg Finance LP, Google Trends, Deutsche Bank.

So there is still room for fears of AI doom in software to allay. This should return some confidence in the BDC market as well.



Source: Bloomberg Finance LP, Deutsche Bank.

3. Geopolitical volatility

Managers view the recent conflict in Iran as the latest in a series of "market-moving events" since 2020 that are challenging periods to navigate, but which do not derail long-term strategy. While sponsored dealmaking has been soft in recent months, it held up better ex. tech. Private capital is likely to further channel deployment into areas with thematic tailwinds like energy and security for now.

1. The message is that geopolitical shocks create near-term volatility but ultimately dealmaking recovers.
2. Managers view the conflict and broader geopolitical splintering as an accelerant for pre-existing investment themes such as defence and energy security.
3. Geopolitical uncertainty is a key factor impacting the monetization environment and the views are mixed.

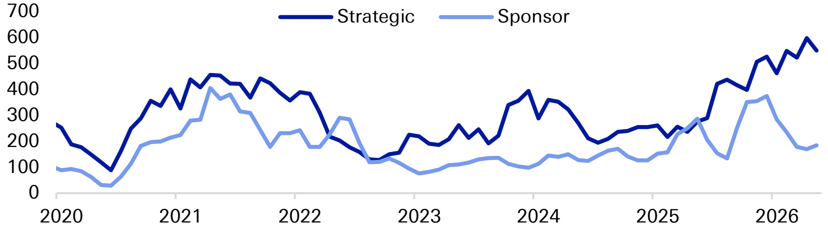
How serious is the impact on portfolios and markets?

- Managers assess the direct impact on portfolio fundamentals as limited, but acknowledge the significant indirect impact on market sentiment, deployment pacing, and monetization timelines. The primary effect is a slowdown in M&A and capital markets activity as buyers and sellers pause to evaluate the new landscape.
- The conflict in Iran is cited as a reason for a pause in transaction activity. **Ares** noted that "when you see things like the conflict in Iran, you get a little bit of a pause as everybody just evaluates." However, they also reported that their aggregate pipeline is at a record level and that the direct lending pipeline is re-engaging, drawing a parallel to a similar pause after tariff announcements in April of last year, which was followed by a record deployment year.
- The increase in oil prices reinforces the strategic value of portfolios tilted towards energy and infrastructure. **TPG's** climate platform is benefiting from the increased focus on energy security, with the firm noting "The Strait of Hormuz may be bad for many things, but it's good for our business here," as it drives demand for renewables.

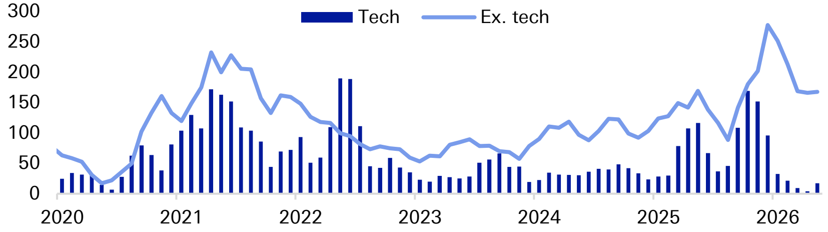
How are managers mitigating risk and finding opportunity?

- Managers repeatedly stress that their global and strategic diversification is the primary defence against localized geopolitical shocks.
- Managers are increasing exposure to physical assets and sectors benefiting from the focus on security and re-industrialization.
- In general, managers see the volatility as a chance to invest

US M&A activity has bifurcated between strength in corporate and weakness in sponsored deals…
($bn 3m sum)



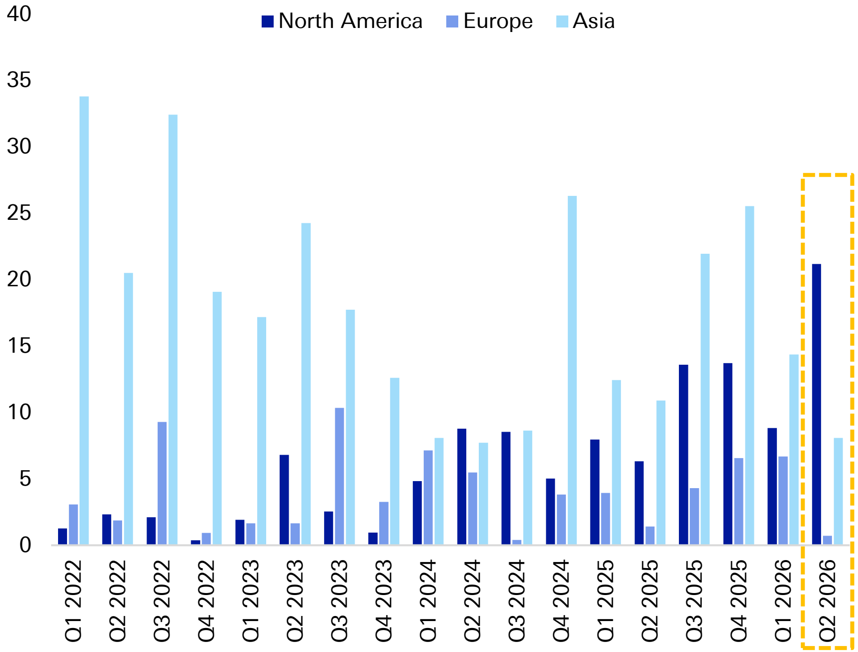
… but much of that PE weakness is tech-related. Other thematic sectors are in focus amidst all the dry powder
US sponsored M&A activity ($bn 3m sum)



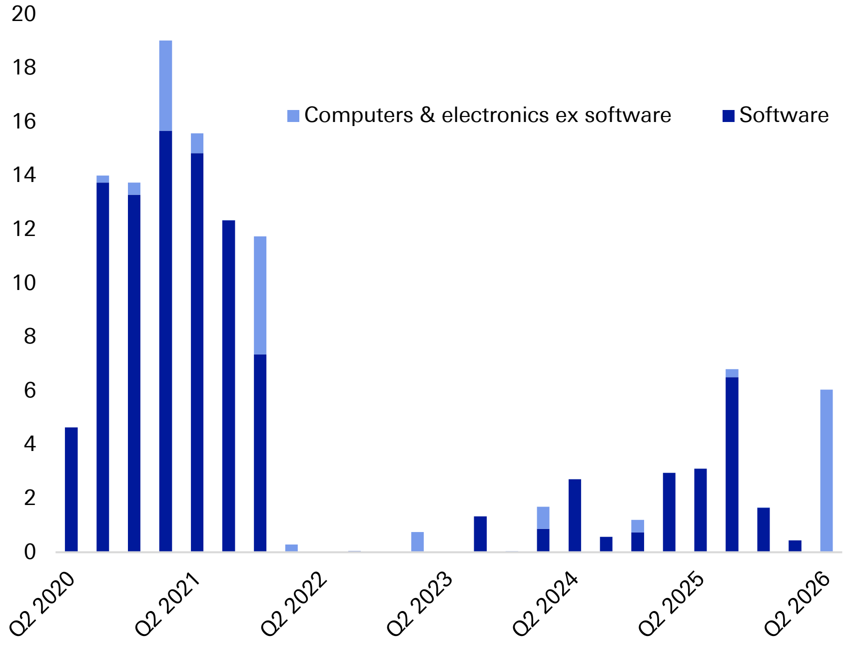
Source: Bloomberg Finance LP, Dealogic, Deutsche Bank.

For realisations, we expect IPO recovery momentum to continue through the rest of the year, with M&A activity also resilient. Although there are some near-term cooling risks, corporates have navigated a vast array of shocks and now are well-adapted to expect risks and volatility, supported by cash buffers and liquidity. This should keep a few options for sponsors to exit portfolio companies even if some market volatility persists. See more here.

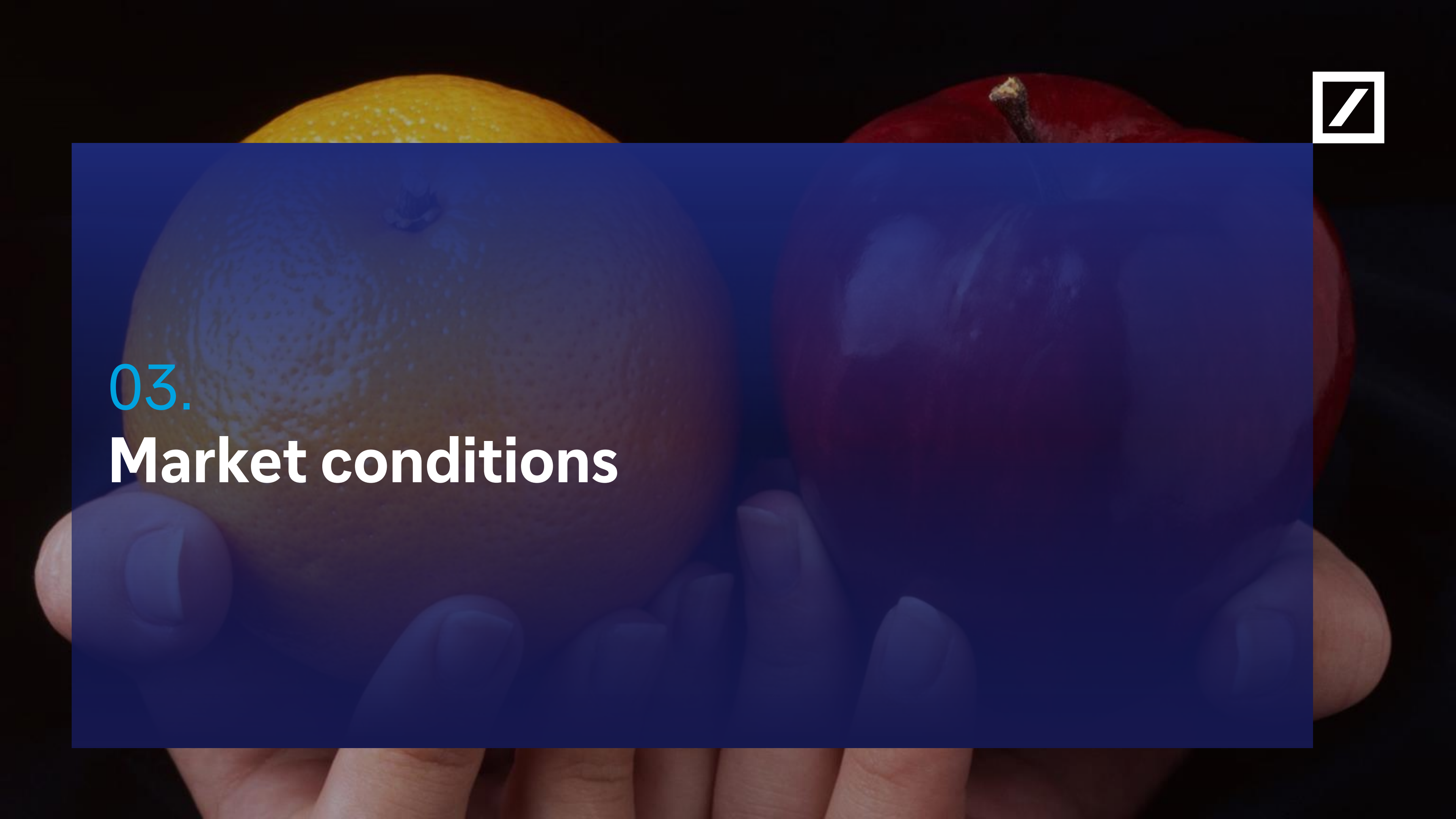
IPO deal values by region ($bn)*



North America IPOs, computer & electronics ($bn) *



*Source: Dealogic, Deutsche Bank. * Note: excluding financials, Q2 2026 is QTD as of May 18th.*

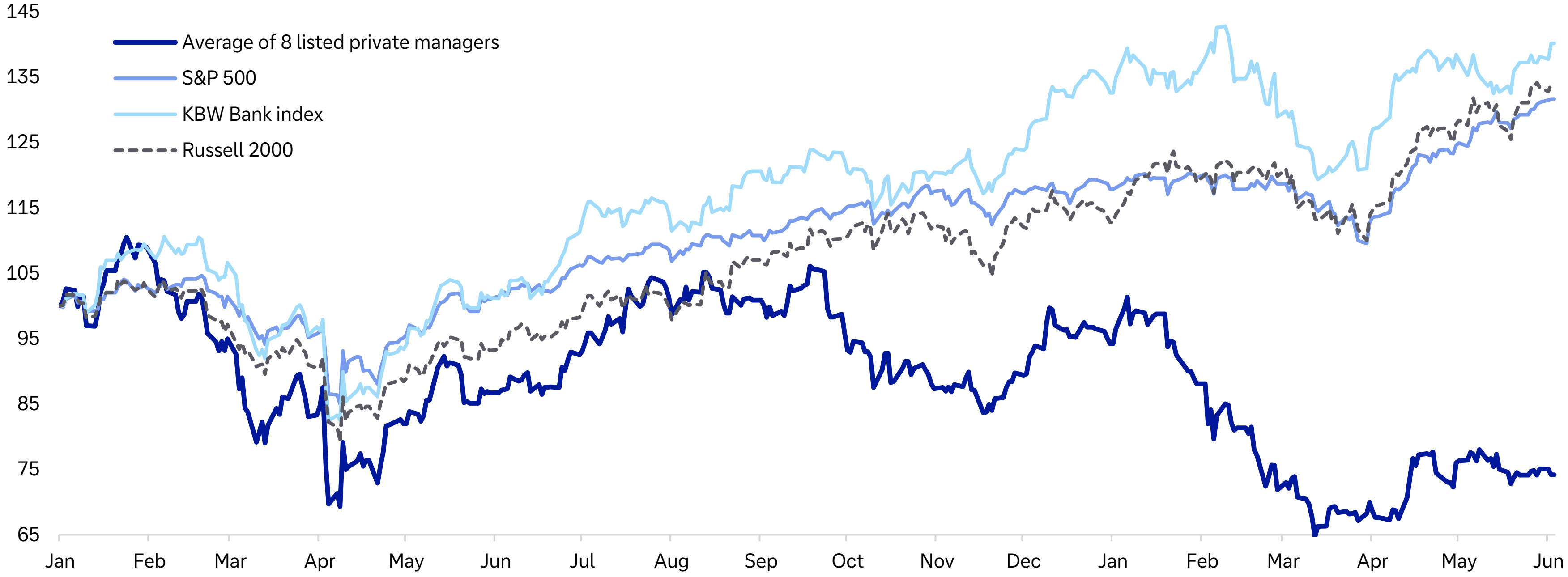


03.

Market conditions

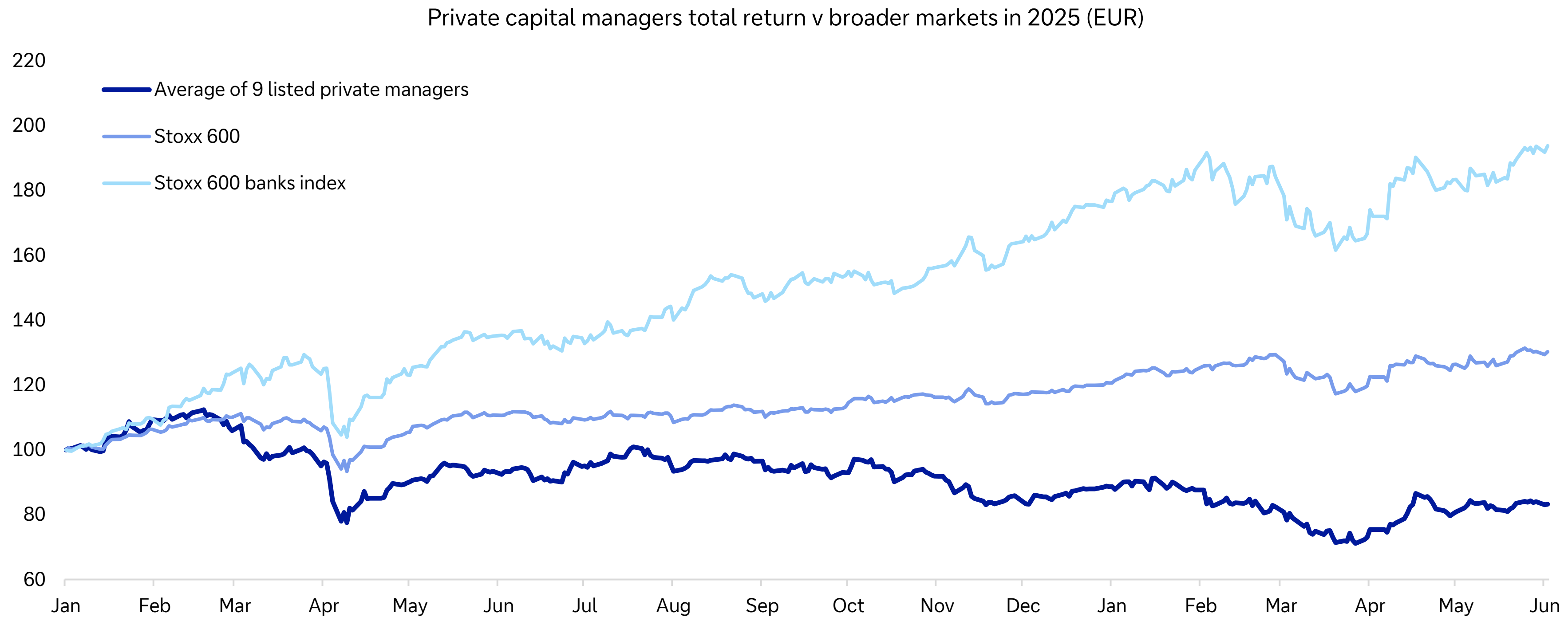
Shares of US-listed private capital managers are down -20% YTD amidst persistent negative sentiment and despite the recovery in the broader markets in Q2

Private capital managers total return performance relative to broader markets (USD)



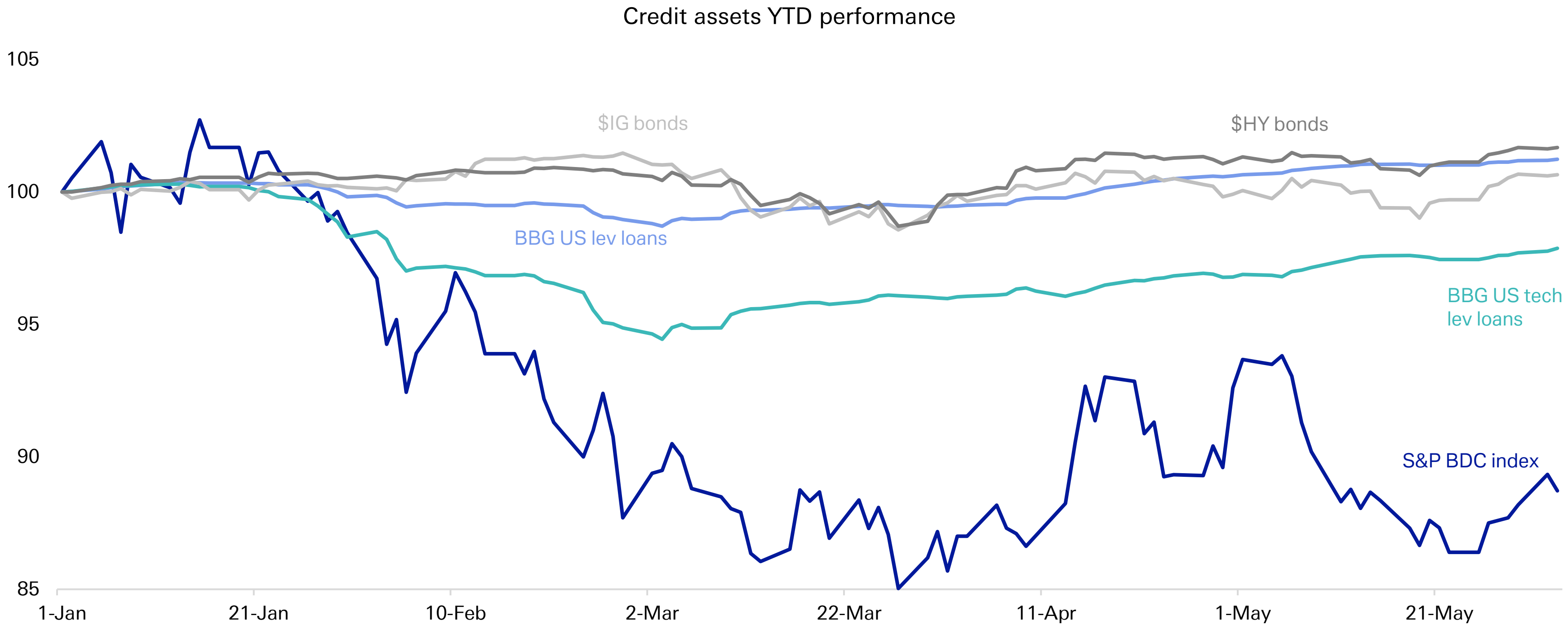
Source: Bloomberg Finance LP, Deutsche Bank.

The underperformance of private capital firms in Europe has been less stark this year (-5%)



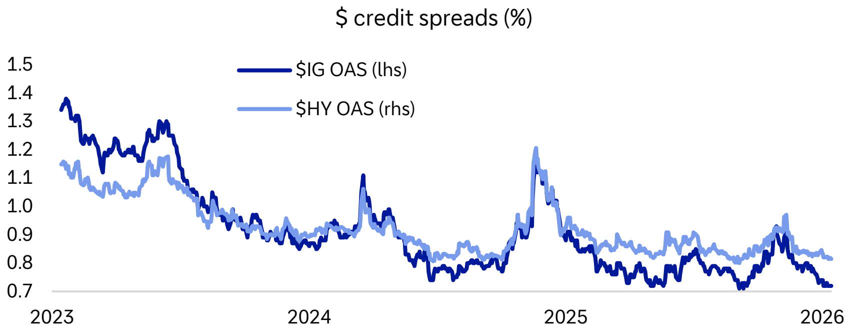
Source: Bloomberg Finance LP, Deutsche Bank.

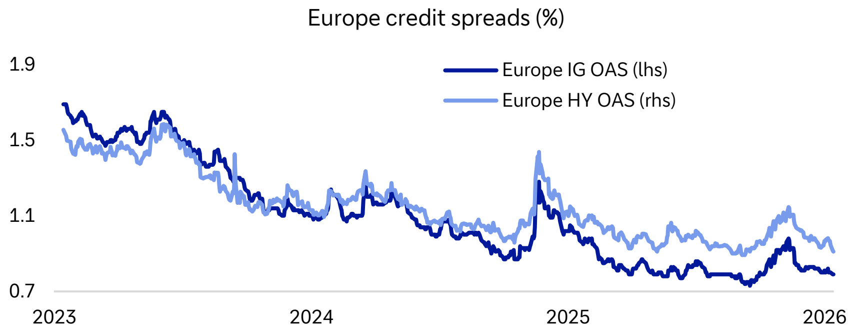
It's a similar story for BDC returns vs the broader credit markets

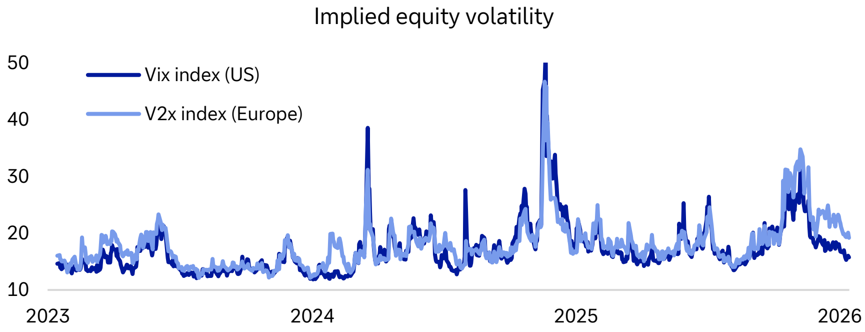


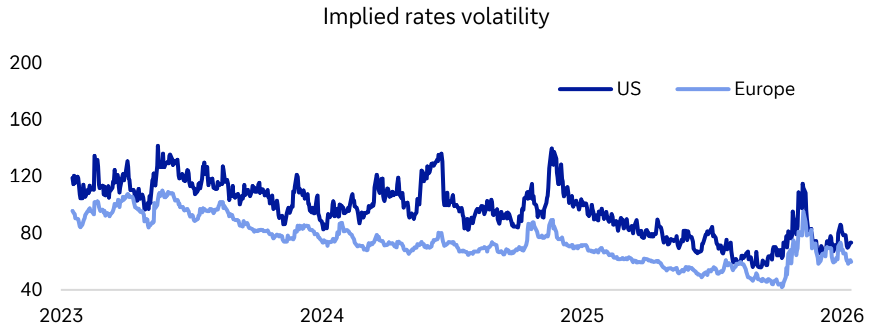
Source: Bloomberg Finance LP, Deutsche Bank.

Volatility is back to somewhat subdued in the US, with Europe's disadvantage amidst the Iran conflict more pronounced





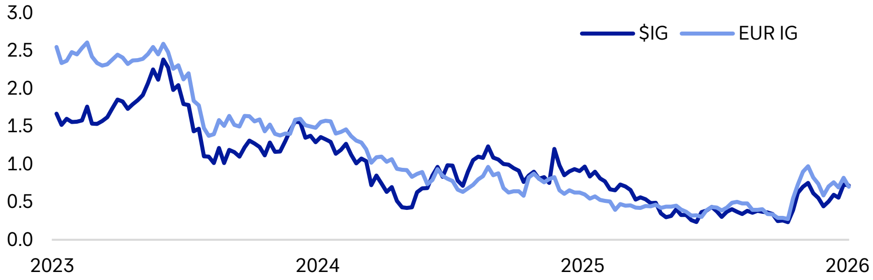




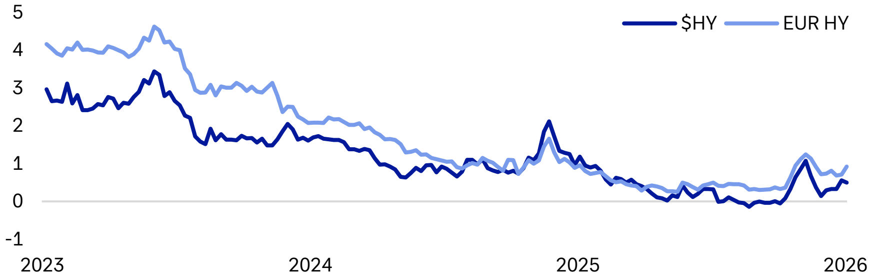
Source: Bloomberg Finance LP, Deutsche Bank.

Rising refinancing costs may deter some debt issuance but overall credit conditions remain favourable

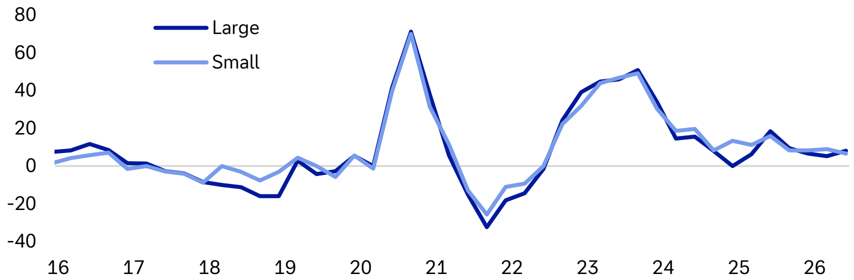
IG bonds - Refinancing proxy in %



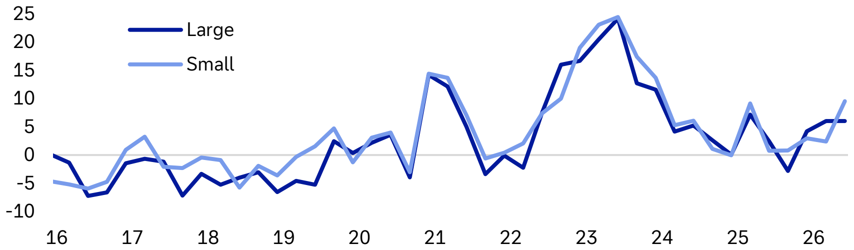
HY bonds - Refinancing proxy in %



Fed SLOOS: banks tightening standards for C&I loans to firms (%)

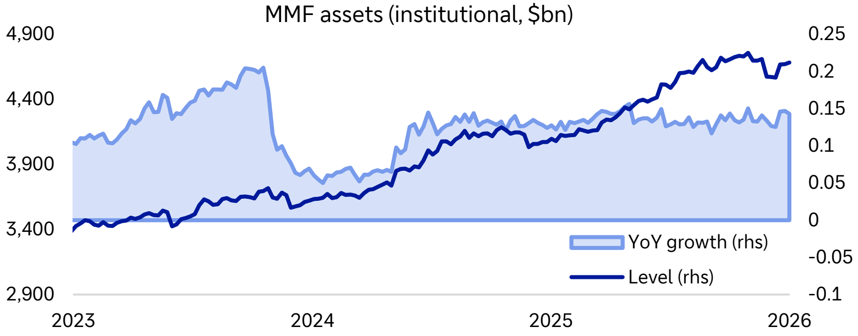


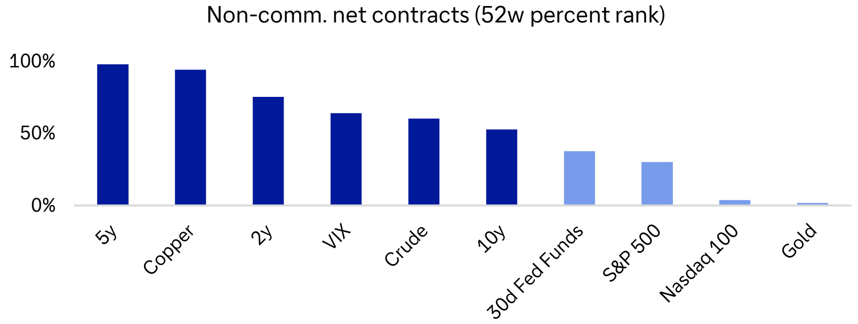
Euro Area bank lending survey: change in credit standards for business loans

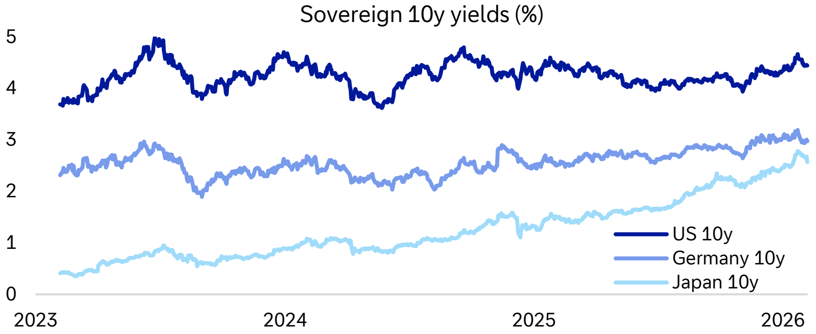


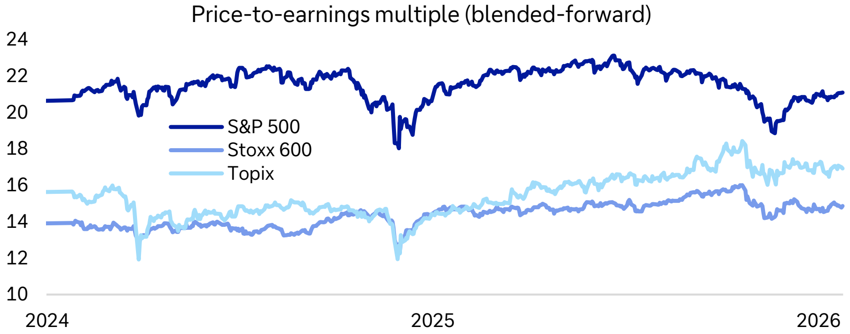
Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

The extent of upward pressure on sovereign yields is understandably a key topic for markets. While the US economy and markets have been accustomed to rates in the current range in the past couple of years, the picture is more negative in Europe and Asia.





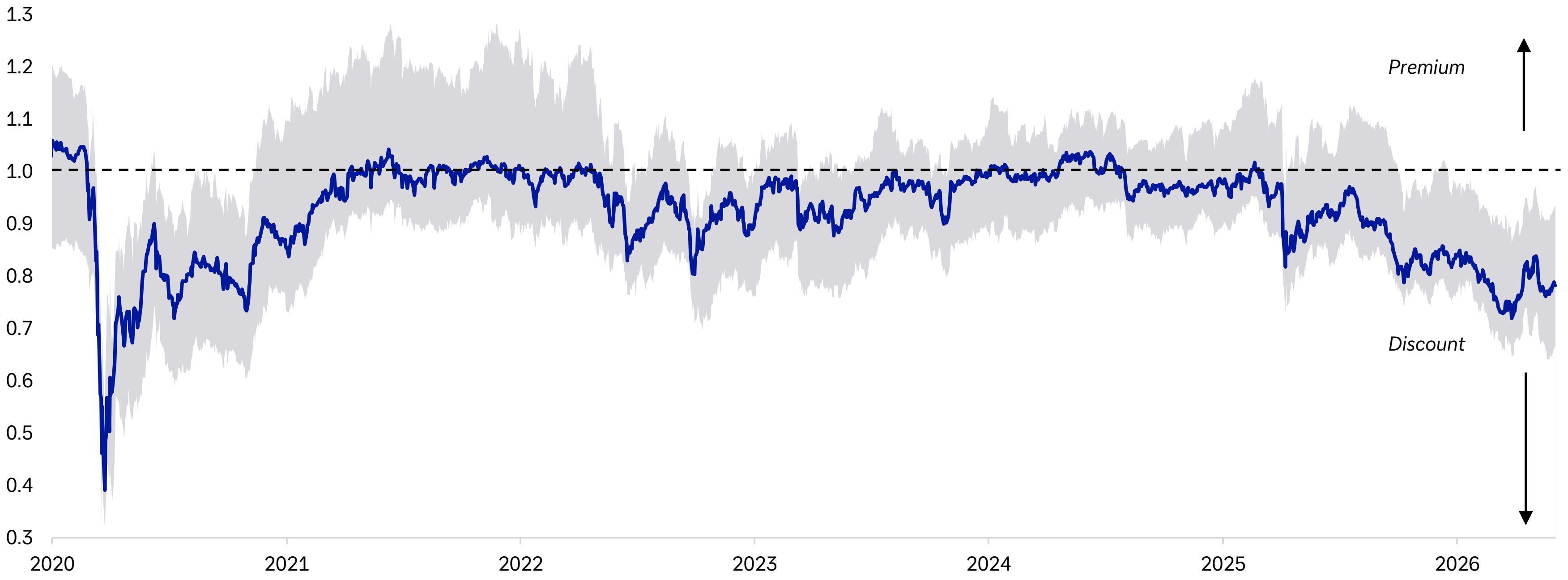




Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

The median BDCs discount to NAV is still wide and now stands at c 23%. Around 25% of BDC investments tend to be in tech, suggesting severe concerns beyond just AI disruption risks.

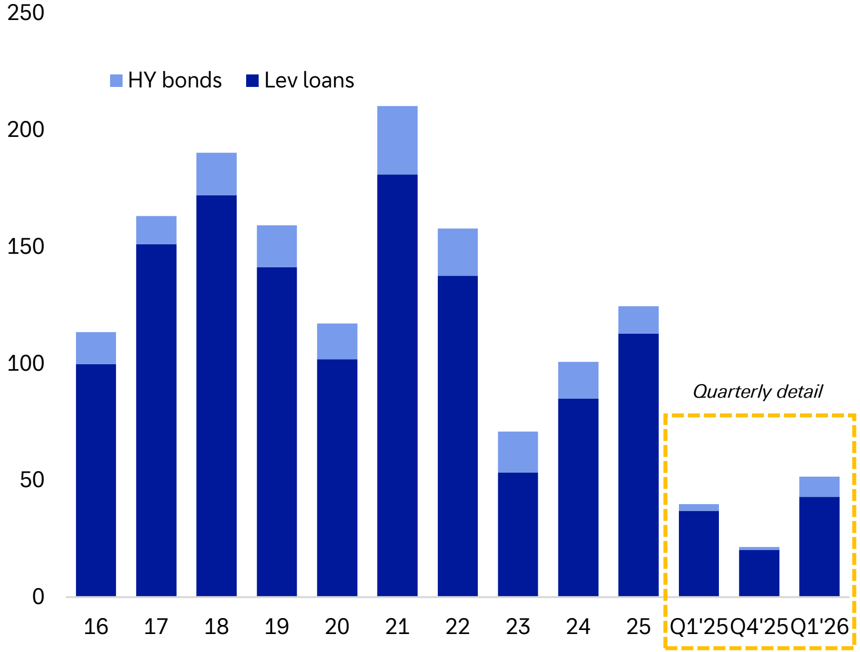
Median price-to-NAV ratio of funds in the S&P BDC index



Source: Bloomberg Finance LP, Deutsche Bank.

Q1 debt issuance for LBOs is already at 40% of the 2025 total. Higher yields and spreads decompression will help attract investor demand for more deals.

Debt issuance in North America and Europe for LBOs, $bn



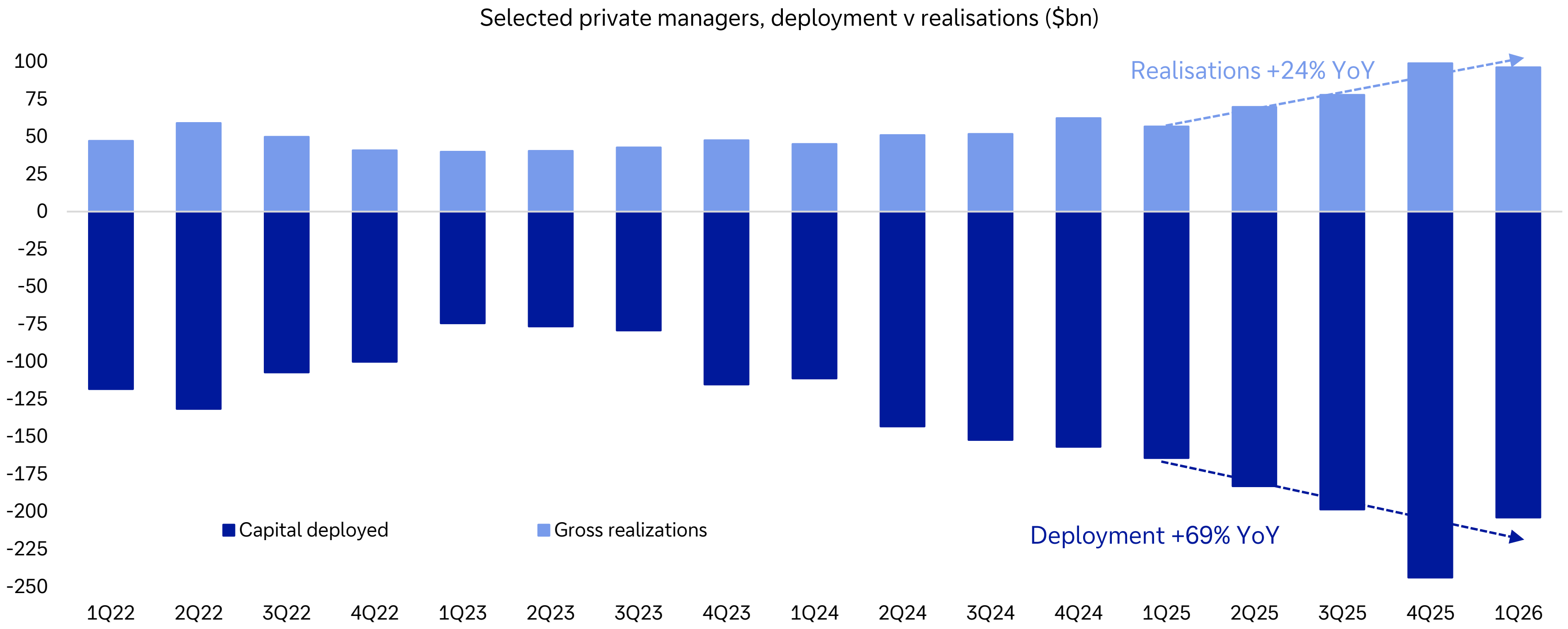
S&P UBS Leveraged Loan YTM index (%)

Source: White & Case, Bloomberg Finance LP, Deutsche Bank.

04.

Fundraising environment

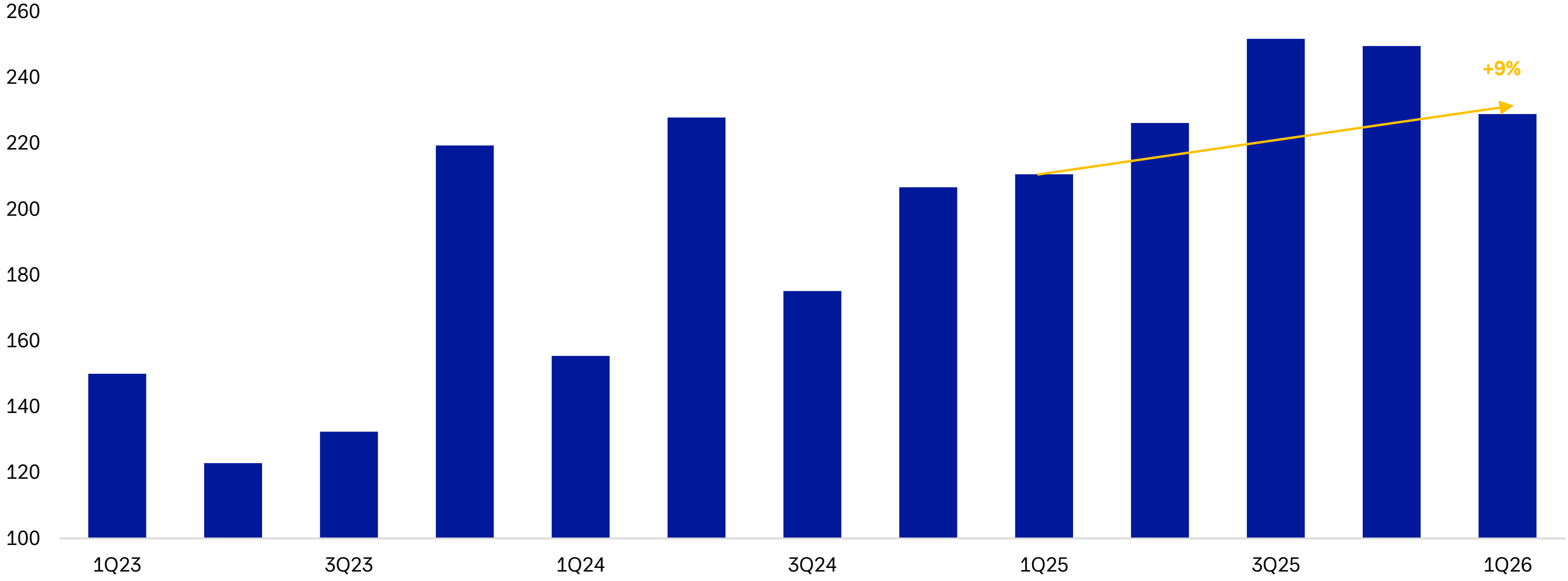
Despite a QoQ drop, both deployment and realisations have been solid YoY



*Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](), Deutsche Bank. * Note: Deutsche Bank company research forecasts.*

Aggregate fundraising growth continued in the negative territory. New fund launches will be the ultimate bellwether of LP sentiment and good progress on exits will help.

Fundraising by selected large, listed managers ($bn)

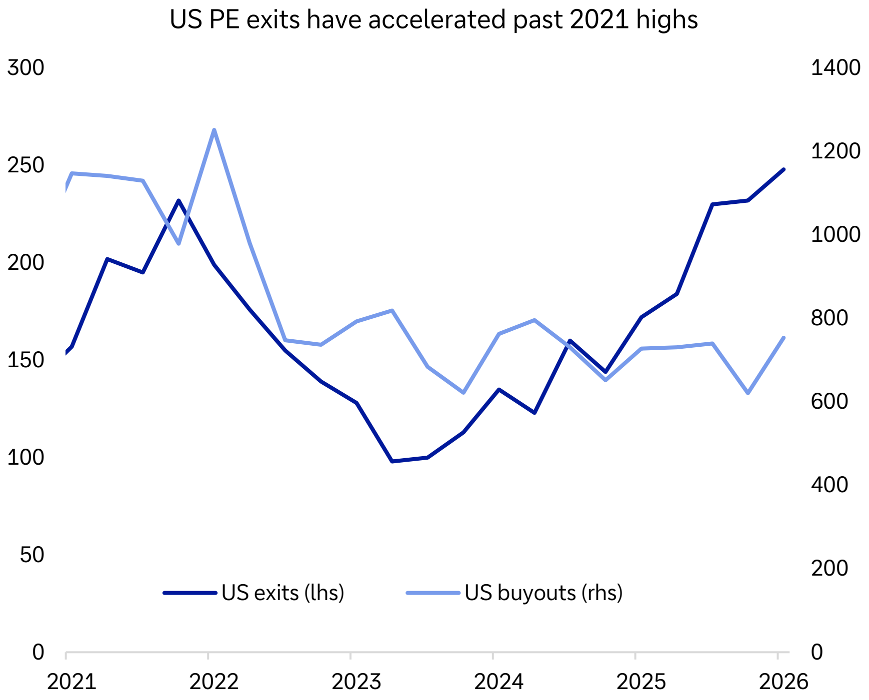


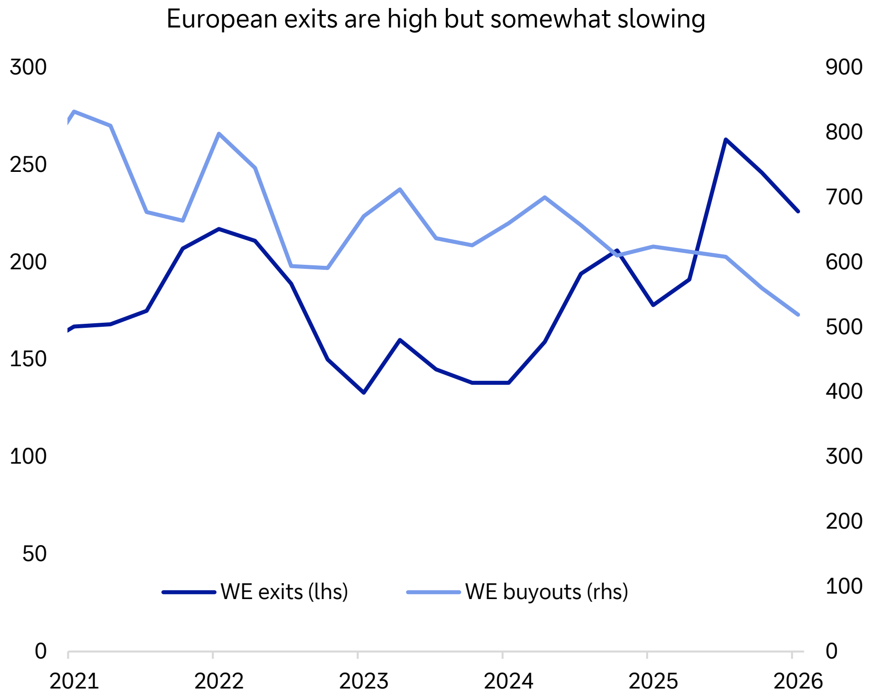
Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](), Deutsche Bank.

Gross retail fundraising (not taking into account redemptions) is down by a quarter. Managers emphasise that the issue is headline driven and concentrated, but the sentiment needs to improve significantly to restore strong growth.

*Source: Deutsche Bank company research – US Brokers, Asset Managers & Exchanges - Deutsche Bank Research, Bloomberg Finance LP, Deutsche Bank. * Note: gross, only funds with available Q1 data.*

Strengthening exit activity should assuage investor concerns about private equity and free up more capital to do deals

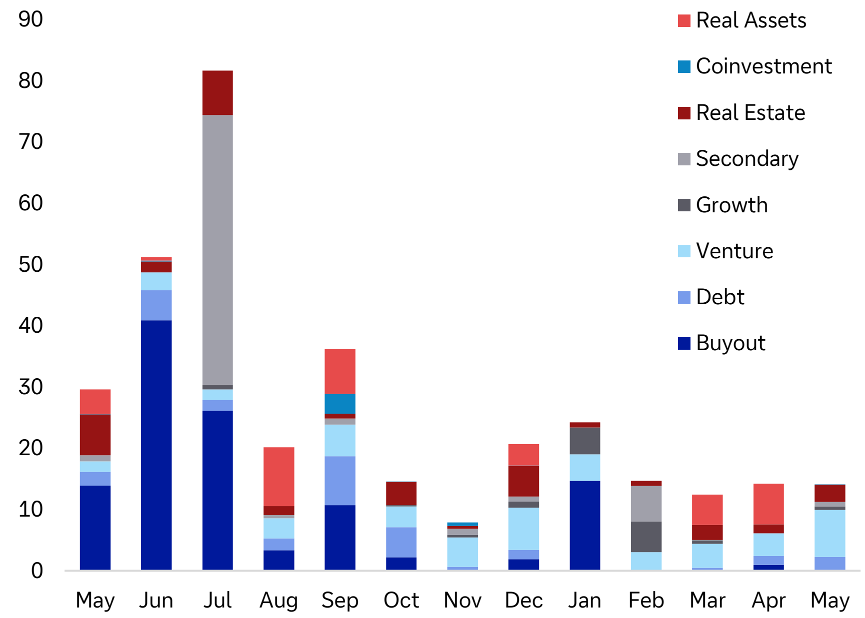




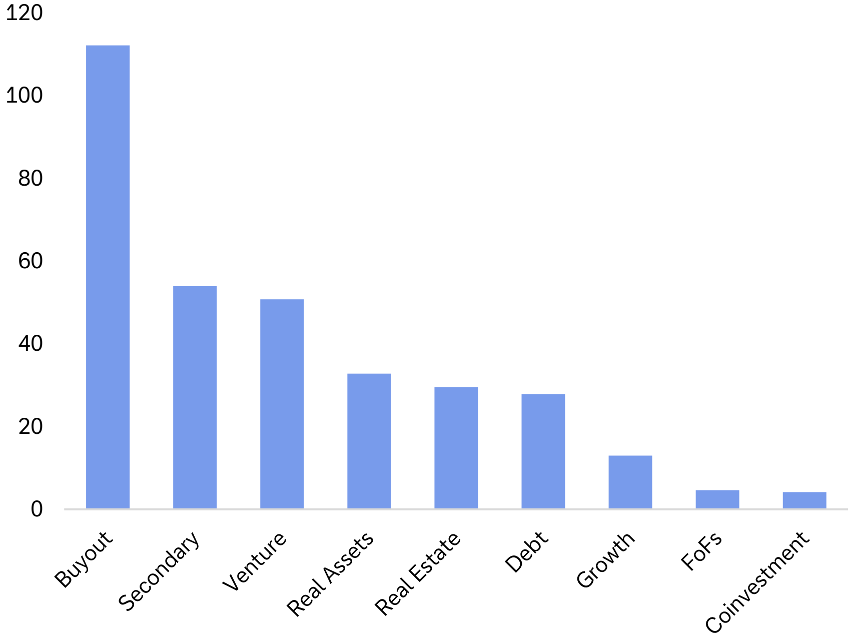
Source: White & Case, Deutsche Bank.

New fund launches – private equity gathering pace in recent months

Fundraising funds - target in $bn by strategy type and launch month, US and Europe

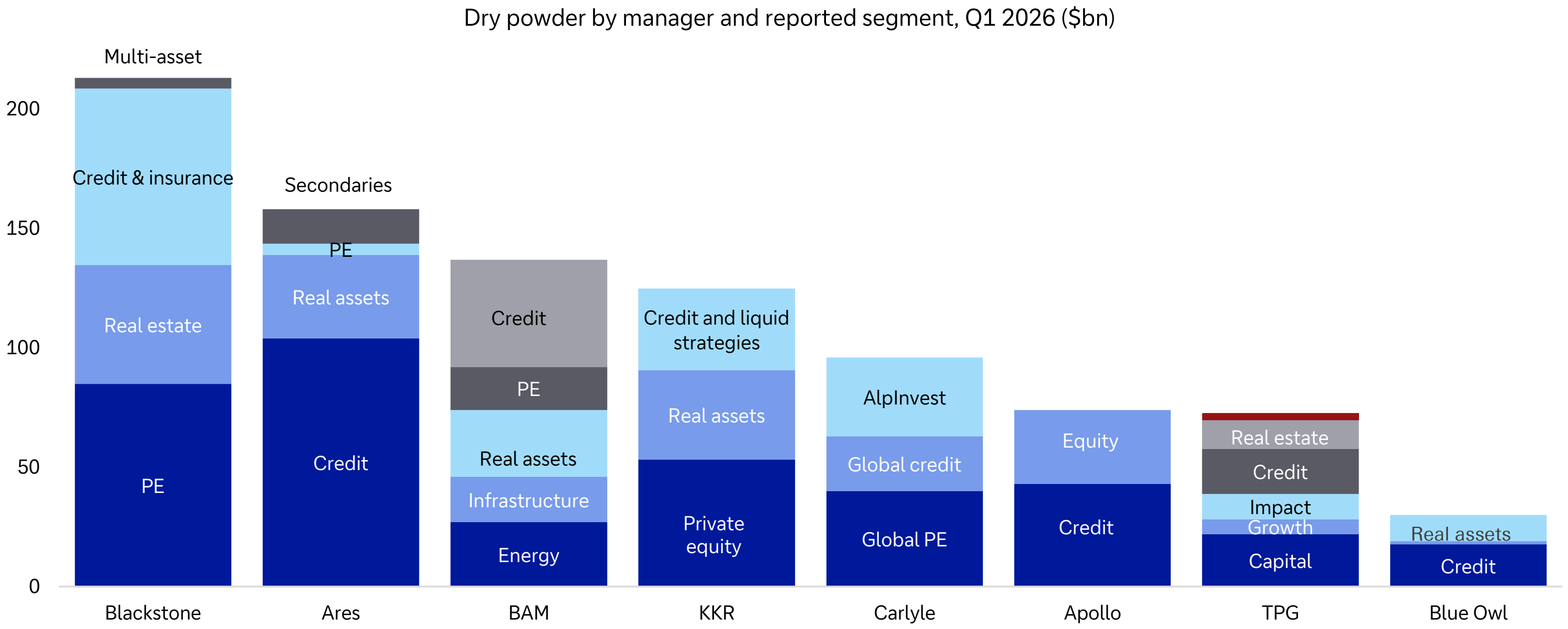


Fundraising funds in US and Europe, last 12m in $bn



Source: Bloomberg Finance LP, Deutsche Bank.

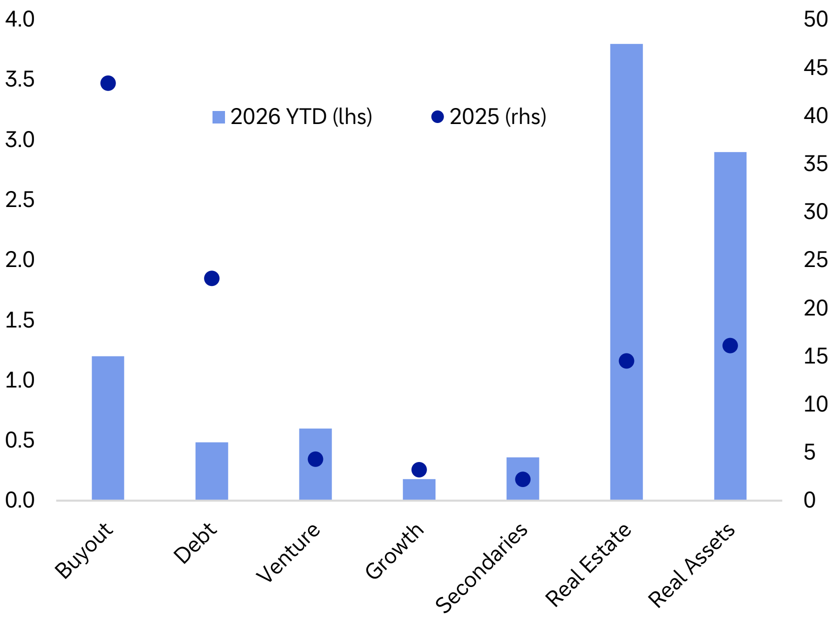
Dry powder by manager



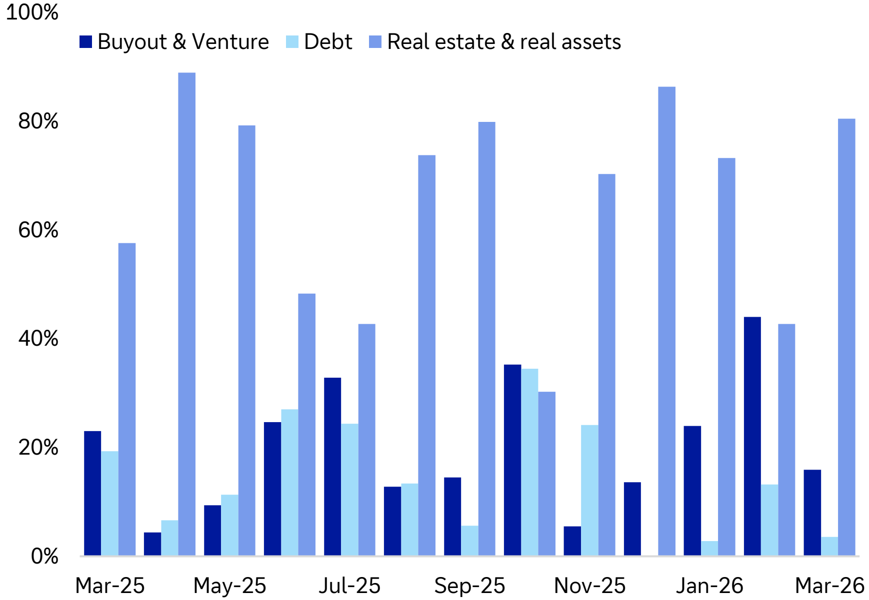
Source: Company reports, Deutsche Bank.

Recent disclosed LP commitments

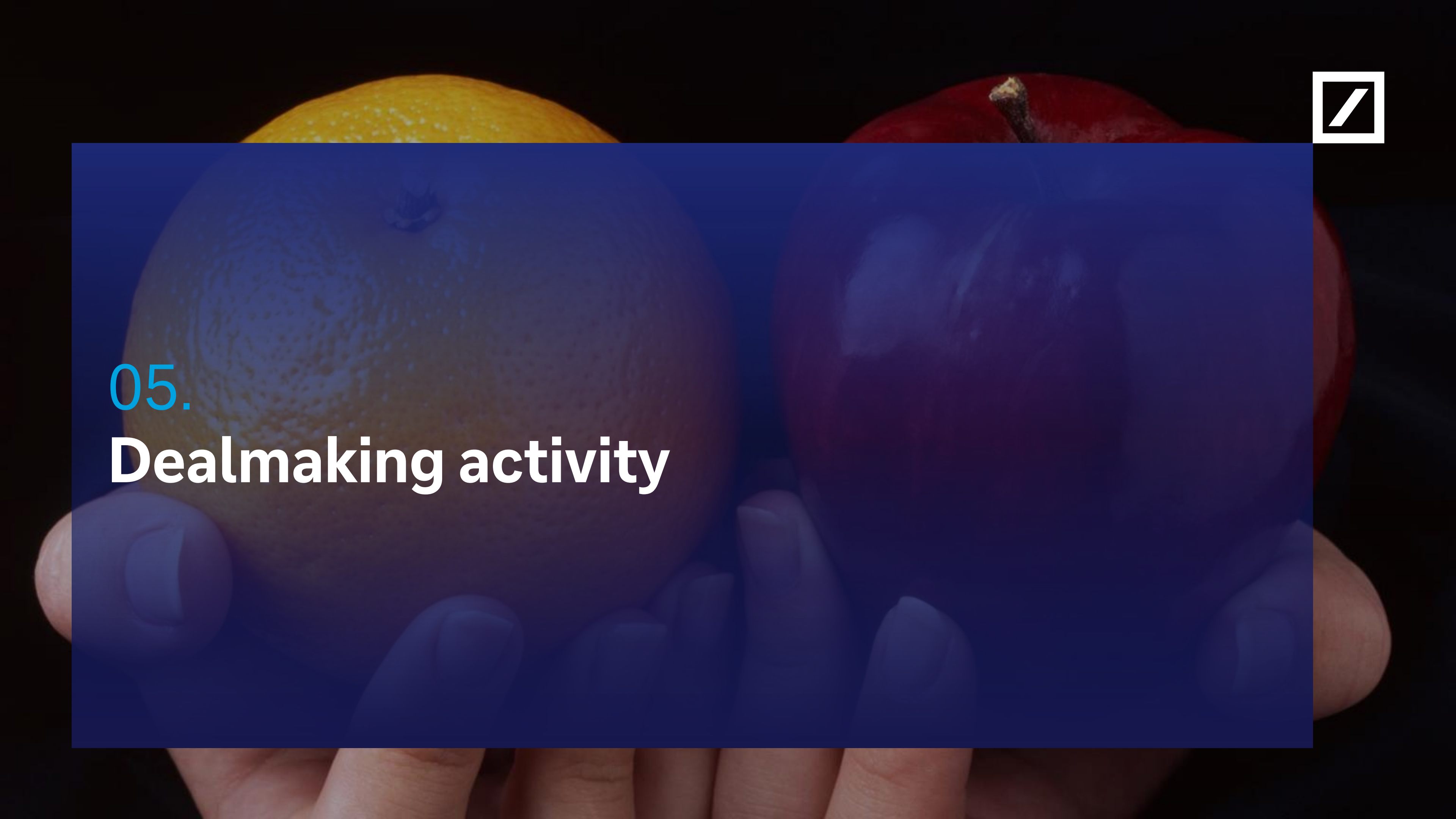
Disclosed LP commitments by strategy in $bn



Disclosed LP commitment targets by month and strategy, share of total for selected strategies

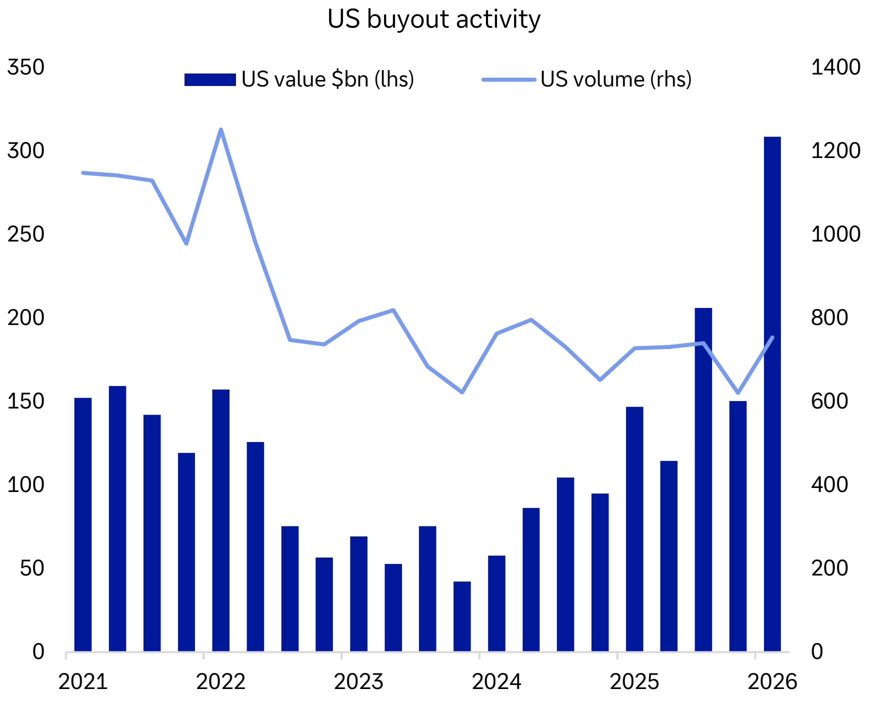


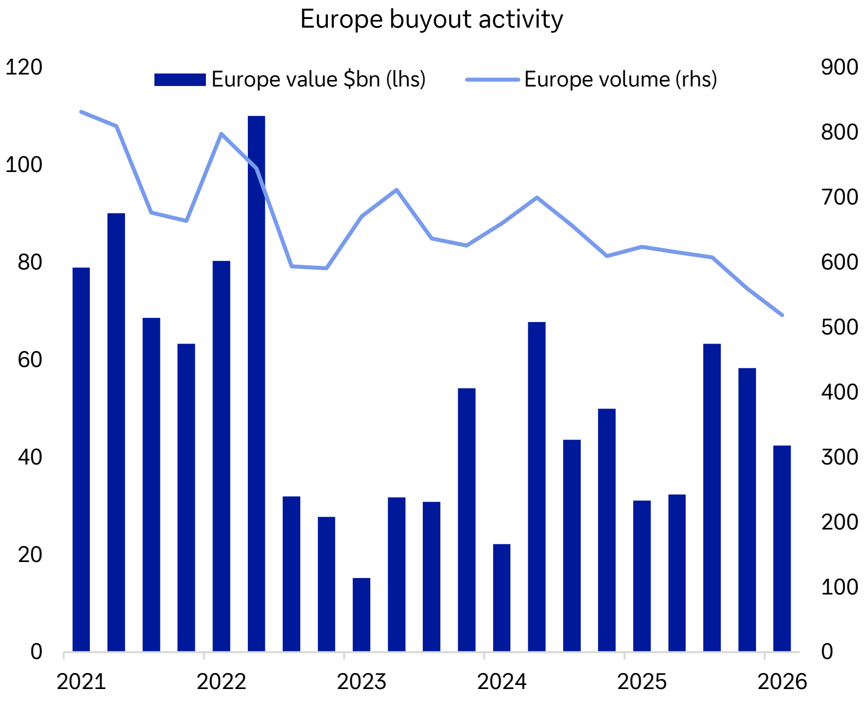
Source: Bloomberg Finance LP, Deutsche Bank.



05.
Dealmaking activity

Dealmaking activity in buyouts

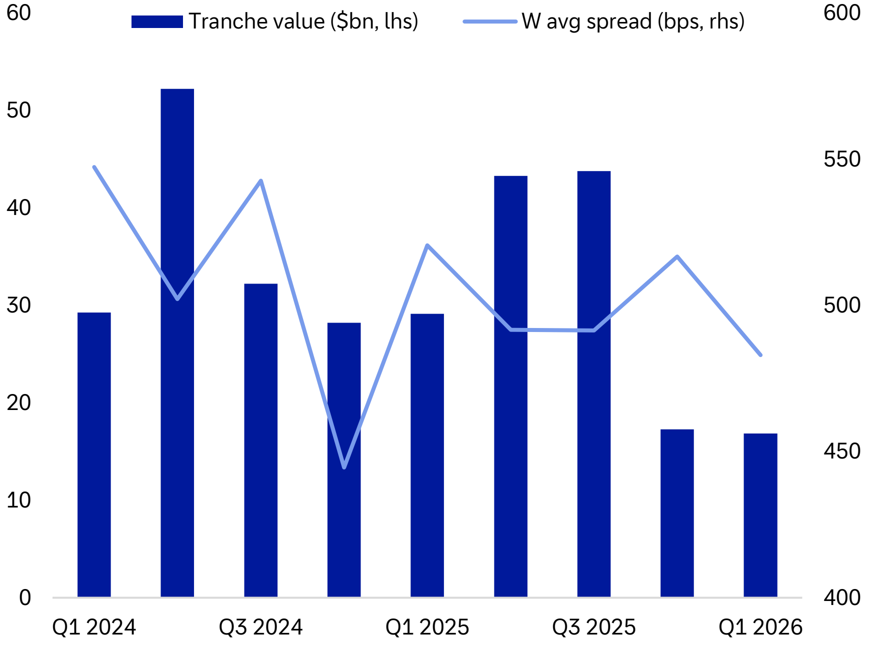




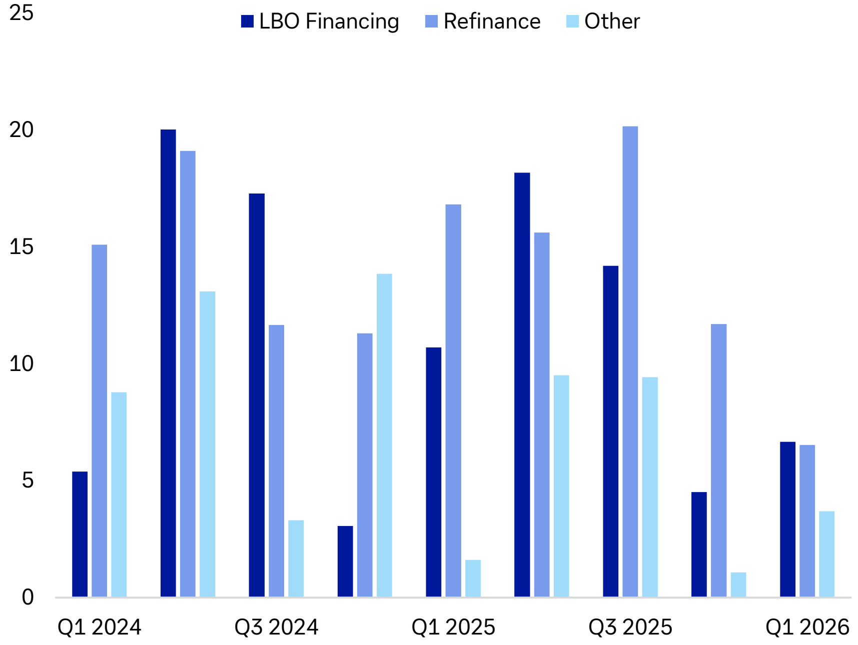
Source: White & Case, Deutsche Bank.

Origination in large private credit loans – loan volumes steady QoQ but down YoY as both LBOs and refinancing demand slowed

Loan volumes have been slowing

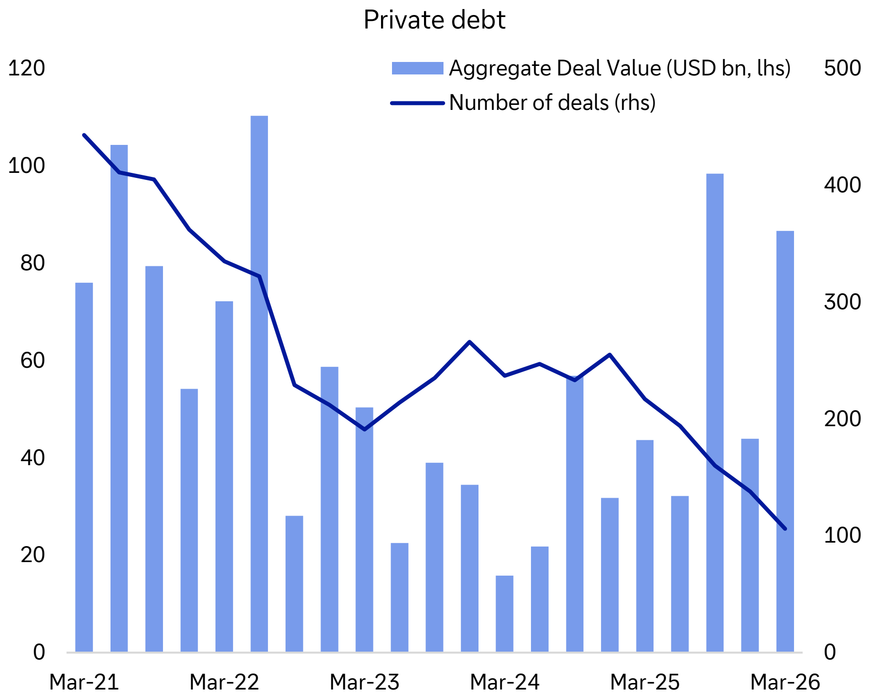


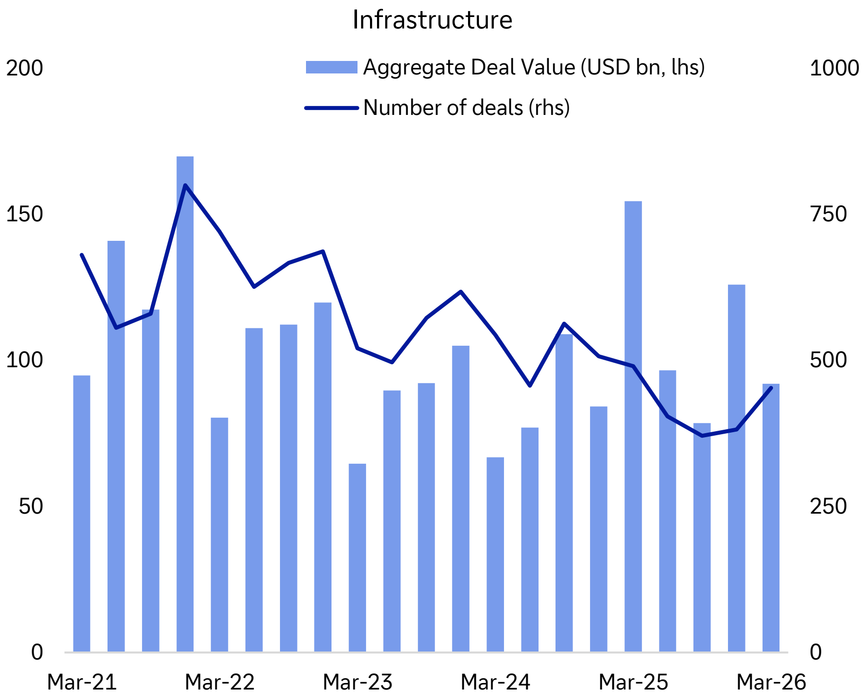
Loans by purpose ($bn)



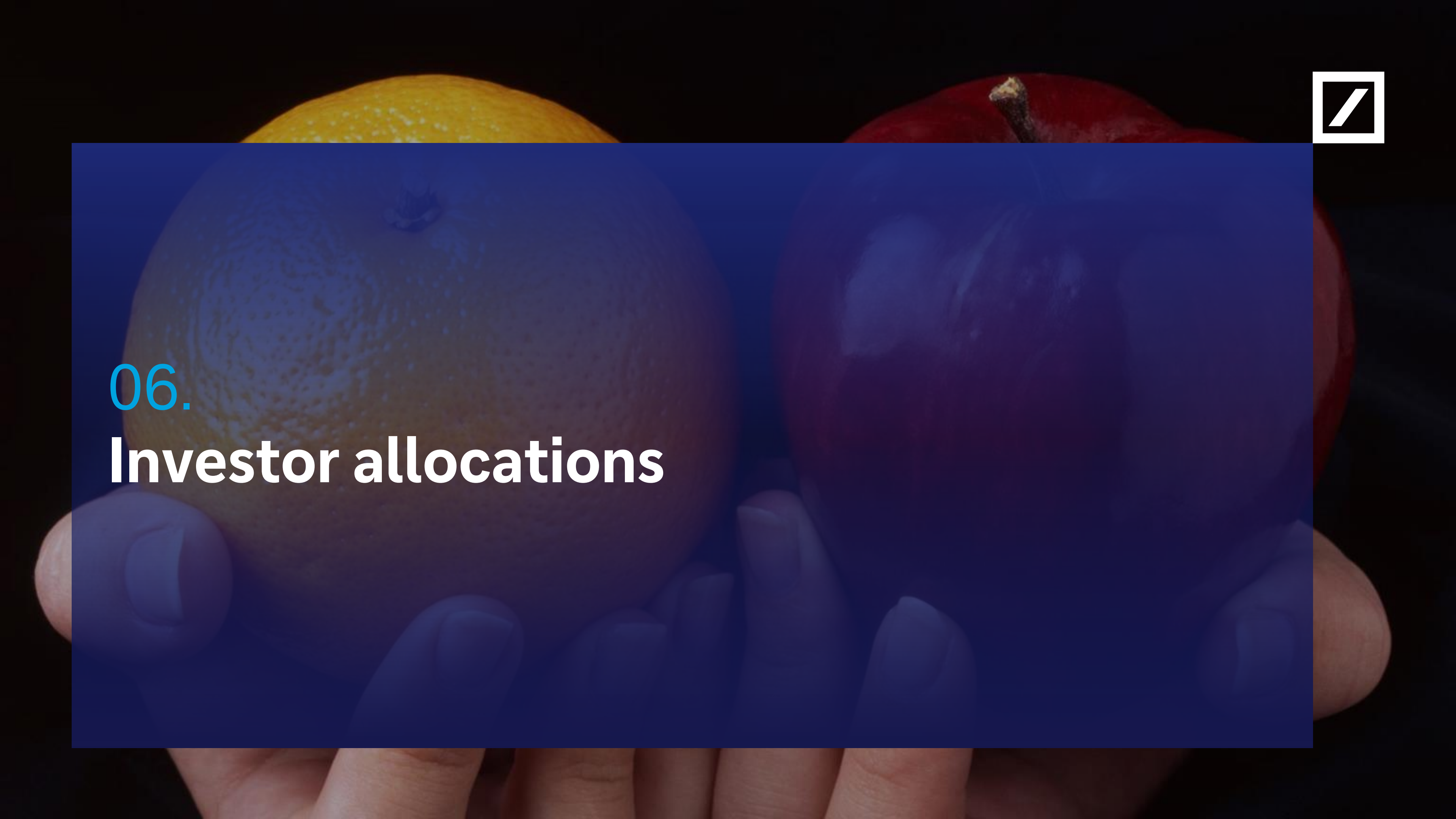
Source: Bloomberg Finance LP, Deutsche Bank.

Global dealmaking activity in private debt and infrastructure





Source: Preqin, Deutsche Bank.

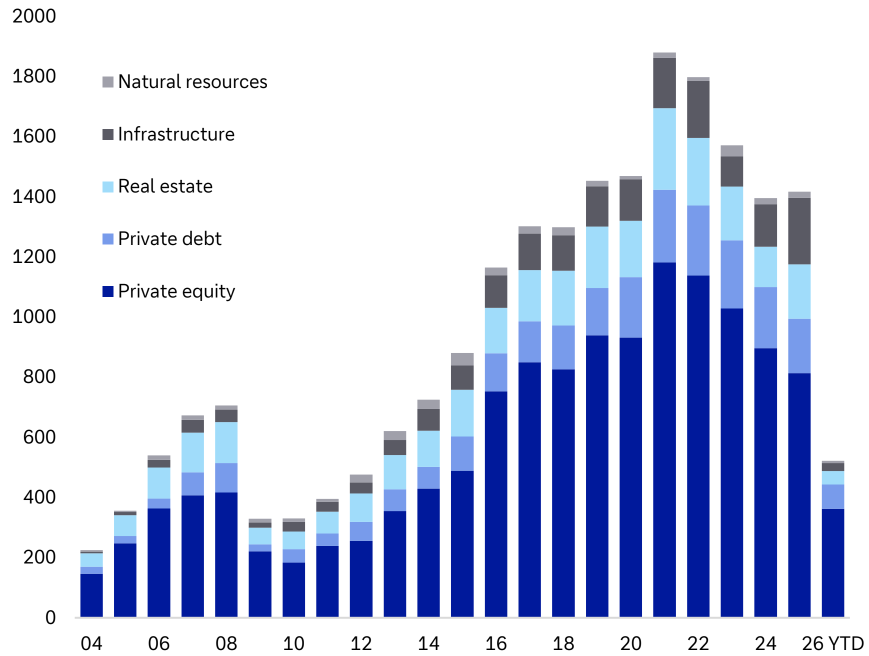


06.

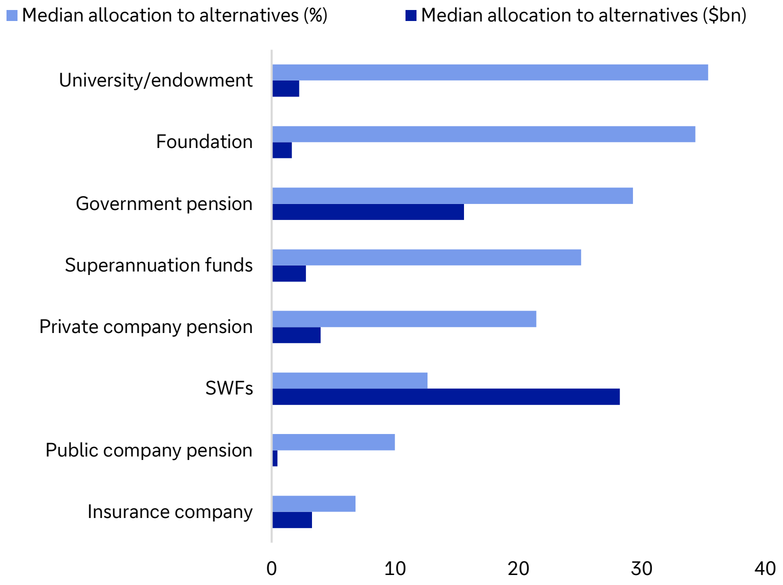
Investor allocations

Investor allocations

Private capital sum of funds raised ($bn)



Top 100 LPs in each category, allocations to alternatives



Source: Preqin, Bloomberg Finance LP, Deutsche Bank.

Appendix 1

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Deutsche Bank
Research

Chips: The Biggest AI Bottleneck

June 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Chips: The Biggest AI Bottleneck

Key points

1. Today's AI race is inherently a chip race. No country controls the overall chip supply chain in the short to medium term.

2. Key bottlenecks lie in advanced chips and specialized equipment, concentrated mainly in Taiwan and the Netherlands.

3. The US leads in advanced compute, but is constrained by its reliance on Taiwan's TSMC for chip manufacturing.

4. China's AI chips perform ~5x below US alternatives — a gap that could widen to 17x — though it is expanding in legacy node production. China maintains access to ~2.8mn H100-equivalent GPUs partly via remote cloud.

5. No single country can achieve full chip supply chain independence in the near term. The EU is on track for only 12% of global production by 2030, short of its 20% target.

6. The 2025 export control escalation — US chip controls alongside China's rare earth restrictions — illustrates how semiconductors and critical minerals have become interlinked economic tools.

7. The chip industry will bifurcate between AI leaders and laggards beyond 2026, impacting the ongoing memory shortage. Software efficiency gains, such as Google's TurboQuant, can rapidly reprice the hardware stack.

Key facts

~$1.5trn	in global semiconductor market revenue by 2030. AI will drive most of this.
60%	of global chip production was US-made in the 1970s, vs 11% by 2019.
40%	of Taiwan's chip supply chains should be reshored to the US, per Commerce Secretary Lutnick.
15 years	for the US to build a sustainable domestic chip manufacturing industry and packaging industry.
90%	of the world's most advanced semiconductors are produced in Taiwan.
11%	of US GDP is at risk of a Taiwan Strait disruption (SIA). A disruption could cost $5trn in year one (Bloomberg).
~80%	Chinese chips reach of Nvidia's A100 performance, but this gap is projected to widen significantly as US chips advance.
7 months	Chinese LLMS lag behind US counterparts.
2027	China's mature node capacity rises to 47% (vs. 34% in 2024). Taiwan's share would decline from 43% to 36%.
53%	of China's semiconductor exports were memory chips in Dec 2025.

Marion Laboure | Camilla Siazon | June 2026

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Deutsche Bank
Research Institute

2026: Steady, but AI & the Hawks are Circling (Software)

June 3, 2026

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Over the last year, we have received a healthy dose of macro tailwinds. The rapid buildout of AI infrastructure, Fed & ECB rate cuts (with inflation still north of 2%), the announcement and enactment of German fiscal stimulus, and record high equity prices and near-record tight credit spreads have helped to boost risk sentiment.

Yet despite this, the credit default picture is far more mixed than appreciated. Yes, US spec-grade default rates did decline from a peak of 5.1% last July to 4% today, as cyclical risks remained muted, but current US default rates are well above 20year median levels of 2.9%, with CCC defaults rising over the past year. In Europe, the lack of AI tailwinds meant defaults actually increased from 2.1% to 3.8% in March & 4.6% in April, easily above 20 year median levels, with elevated defaults in cyclical sectors and in the challenged economies of France/UK/Germany offsetting strength in Southern Europe.

And this is the main message of our piece: It is true we aren't expecting a major global default cycle to emerge. But equally, don't let the aggregate data and market prices make you think dispersion is going to disappear. **US & European default rates should remain above long-term averages through Q2'27**, with low recovery rates for those names which default, and a continued presence of distressed exchanges highlighting that many capital structures remain overleveraged, especially now as central banks shift more hawkishly.

In the US, we expect spec-grade default rates to slow from 4% today to 3.6% by YE'26, and then to climb slowly again to 4% by Q2'27. In essence, this is a message that the K-shaped US economy still lives. As we have written repeatedly since 2024, the US economy is effectively a "fire & ice" US economy. Perhaps today, we can change that label to an "AI & ice" US economy, given how narrow we believe the drivers of US growth to be. **On the positive side, US artificial intelligence tailwinds** feed back into very strong US corporate profits and a wealth effect that supports the high-income US consumer, and this is a major reason why our US credit-based recession model is only forecasting a 12% chance of a US downturn through Q1'27. It is very hard to see a spike in US defaults with overall US growth so solid.

But equally, a look at the "typical" US business today shows margin compression and demand expectations at near the worst rate going back to 2022. A more micro look at $HY coupons vs. $HY revenue growth rates & $HY debt to EV ratios tell a similar story. Default rates may not be alarming, but they are likely to stay more elevated than today's current spreads would imply.

And the tailwinds of AI adoption for aggregate growth are likely to be headwinds for spec-grade credit defaults through tighter financial conditions. This will eventually play itself out through two channels. The first will be more hawkish Fed policy and the second will be meaningful downgrades of US software names into CCC buckets, where investors have already demonstrated their impatience to provide flexibility to this rating cohort over the last few years.

The timing of this risk is admittedly uncertain, but it is growing more likely that a scenario could emerge in early 2027 where tangible AI disruption of software business models begins to damage software revenue and customer retention rates, at the same time Fed hikes emerge to eat up limited software free cash flow. This risk is getting priced in leveraged software names today, with 30% of US Software Lev Loans already trading with an implied CCC rating (vs. only 10% from the rating agencies). But a raft of software downgrades will have broader spillovers than appreciated. A quick spate of software downgrades is enough to push CCC buckets in US lev loan indices to 7.5%+, even assuming no broader sector risk. And further software stress is likely to increase liquidity pressure on BDCs, that could force the sale of leveraged loans and eventually force sizeable private credit loan markdowns. **With 80% of all $B rated debt financed itself in either the Leveraged Loan or Private Credit market, this is clearly our biggest risk to monitor.**

In Europe, we forecast spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. Europe doesn't have a K-shaped economy, but a generally weak one that is getting worse. Our European credit-based recession model is now somewhat above average, as banks tighten lending standards on borrowers at the same time profit growth is sluggish. But this isn't the primary reason we are turning more cautious on European defaults relative to last year. The biggest change is that the ECB is now compounding the weak Eurozone growth with at least two rate hikes likely in 2026.

This flow-through of tighter ECB policy into European spec-grade credit should make an impact. B rated European bond and loan maturity walls are well above historical averages over the next 2-3 years, with 37% of B-rated European HY bond issuers today already free cash flow negative. This would rise to 44% if total funding costs rise 50bps; a plausible scenario in a world where the ECB is now set to hike 50bps, while coupons on maturing bonds 2-3 years out, are 70 to 170bps below current market yields. Leveraged European businesses need lower rates before upcoming maturities, but it is now hard to see this materializing.

In the rest of the piece, we discuss which US & European sectors are most at risk of defaults, while assessing which rating buckets are appropriately or inappropriately priced for the risks ahead.

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Summary Bullets

Default Study: Steady, but AI & the Hawks are Circling (Software)

- **The main message of our piece:** It is true we aren't expecting a major global default cycle to emerge. But equally, don't let the aggregate data and market prices make you think dispersion is going to disappear. **US & European default rates should remain above long-term averages through Q2'27**, with continued pressure on global CCCs, and low recovery rates for those names which default.

- **In the US,** we expect spec-grade default rates to slow from 4% today to 3.6% by YE'26, and then to climb slowly again to 4% by Q2'27**.** In essence, this is a message that the K-shaped US economy still lives. Robust AI tailwinds will prevent a major spike in defaults, but businesses flagging margin compression & demand weakness ahead plus the prospect for further increases in $HY bond yields will prevent defaults from falling much as well.

- **Software downgrades are the largest risk for US spec-grade credit today**. If AI disruption shows up in weaker software fundamentals (rather than just lower multiples), software downgrades will fill up CCC buckets in US Lev Loan indices and force through meaningful private credit loan markdowns. With 80% of all B-rated debt in US spec-grade financed via private credit or leveraged loans, this risk is material, but timing of it is far from trivial.

- **In Europe**, we forecast spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. Europe doesn't have a K-shaped economy, but a generally weak one that is getting worse, with cyclical defaults rising, and upward pressure placed on defaults from sluggish growth in France, the UK & Germany. But a shift from expecting ECB cuts to ECB hikes is the key reason we are shifting more cautiously on Europe. 44% of European B rated HY bond issuers will be in negative free cash flow if total funding costs rise 50bps; a plausible scenario in a world where the ECB is set to hike 50bps, while coupons on maturing bonds 2-3 years out, are 70 to 170bps below current market yields.

- **Which US sectors could be most at risk for defaults?** The $HY sectors with the highest leverage currently are Transports, Media, and Telecoms. Telecoms are well flagged, with a high share of current distressed debt. Consumer Goods and Healthcare leverage is high in their respective histories and are likely to come under further pressure even in a mild shock. If the ongoing commodity shock impacts EBITDA margins more substantially and rising rates cause multiples to re-rate lower, Cyclical sectors such as Leisure, Transports, Retail, Tech, Capital Goods will come under greater pressure.

What's Priced into Credit?

- **Distressed debt risk is most concentrated in USD Spec-Grade markets,** particularly within Technology/Software and Telecom sectors. Over a third of distressed $HY debt is in Telecoms, with Technology the second largest risk at just under a sixth of distressed bonds. $Leveraged Loan distressed debt is dominated by Software (nearly two-fifths of distressed debt, three times its overall index weight), while **risk in €HY and €LL is more dispersed across both Cyclical and Non-Cyclical sectors**. Wireless, Healthcare, and Non-Electric Utilities lead among Non-Cyclicals, along with significant debt levels in Cyclical sectors like Consumer Cyc Services, Chemicals, and Packaging. In €LL, Trading Companies/Distributors and Healthcare each hold a 20% share of distressed loans, with Real Estate and Software also notable contributors.

- **$HY and $LL distress ratios started to diverge in October of last year** as investors became more wary of private credit risks. The level of distressed $LL issuers rose as AI-disruption became a theme early this year. Through all of this, $HY distress ratios have remained relatively resilient with more imminent risks priced into the loans market. €Spec-Grade distress ratios are elevated compared to post-European Debt Crisis lows, reflecting an expectation for moderate defaults moving forward.

- IG rating bands, as expected, continue to compensate buy-and-hold investors for all observed historic default periods. **Current BB & B spreads would compensate investors** for default risk in most time periods historically outside of protracted recessions. $CCCs are unattractive under any scenario other than among the best (lowest) outlooks for defaults seen in history with strong recovery rates.

- Assuming a 40% recovery rate, **current HY index spreads would not have compensated investors in 11 of the previous 40 cohorts**, which are all grouped around '85-'90 and the dot-com bubble burst. Even at a low recovery rate, BB investors would be compensated in nearly all cohorts, even if recovery rates were lower the average. For a buy-and-hold investor, current CCC spreads only compensate bond holders in very low default environments with strong recovery rates.

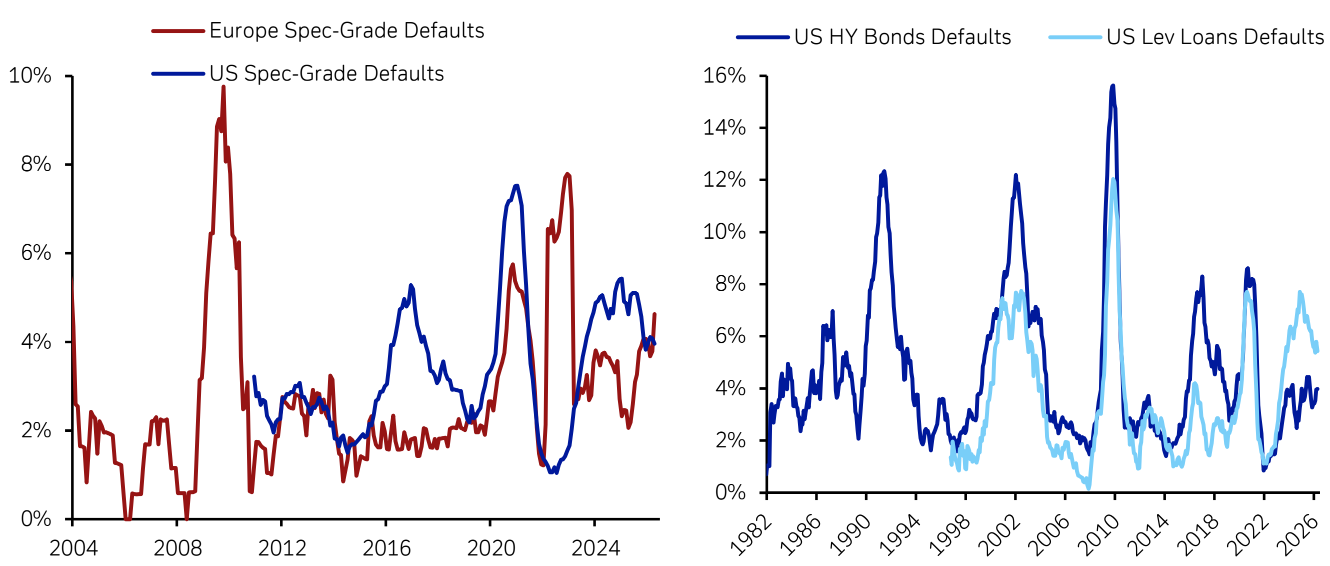
Data as of COB 05/28/2026

US spec-grade default rates

Better than expected, but no return to a low default world

US spec-grade default rates have made a steady, albeit unspectacular, move lower since we published our 2025 default study. Trailing 12m issuer-weighted US speculative-grade default rates (including distressed exchanges) are 4%, near the lows seen since 2023, and down from a peak rate of 5.1% reached last July. Nearly all of this improvement has come from the $-Lev Loan market, where default rates have dropped 120bps from last July. By contrast, $HY bond default rates have been far steadier, only dropping around 20bps from last July, and climbing again in recent months (Figure 1).

Figure 1: $-Speculative-grade, €-Speculative-grade, $HY and $-LL default rates (all include distressed exchanges, trailing 12m)



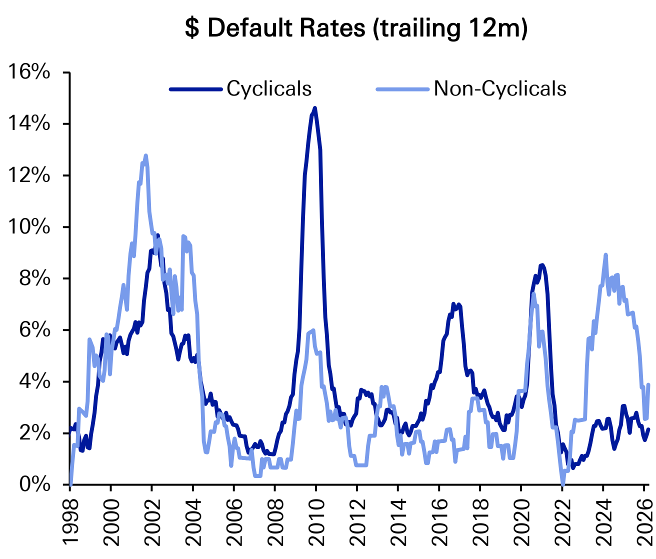
Source : Deutsche Bank, Moody's, S&P

The drop in US spec-grade default rates belies levels that are still fairly elevated in historical context. Case in point: The median & average US spec-grade default rate over the last 20 years is 2.9% & 3.6% respectively, vs. 4% today. Hence, US credit stress is actually higher than what many investors have grown accustomed to, especially in CCC-rated bonds & loans, where little leeway is being given for over-leveraged capital structures. But equally, default rates have fallen further than we predicted in our 2025 study, with our projection of 4.8% by Q2'26 proving too pessimistic.

Over the last year, US defaults have been nudged lower by a stronger US economy, led by continued AI tailwinds, but also an accommodating Federal Reserve, and certainly a Fed that was more accommodating than we had expected. In our view, a world of above-target inflation & severe policy uncertainty (due to US tariffs) meant the Fed would not provide rate cuts, before job cuts. However, the reality was different. The Fed cut rates 75bps despite US core PCE running at a steady 2.8% Y/Y pace and with little tangible realisation of meaningful job layoffs.

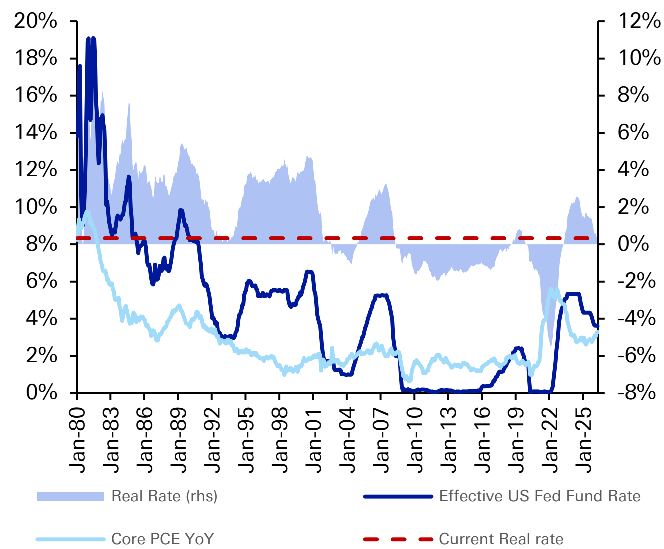
The result was certainly a better mix of cyclical & non-cyclical default rates than we anticipated. Fed insurance cuts and lower overall yields, plus past restructurings in the Telecom & Healthcare sector, allowed non-cyclical default rates to drop materially over the last year, while cyclical defaults remained unchanged as the US unemployment rate remained surprisingly steady in the current "low-hire, low-fire" business environment (Figure 2).

Figure 2: USD Non-Cyclical defaults slowed over the last year while Cyclical defaults were steady



Source : Deutsche Bank, S&P
*issuer-weighted

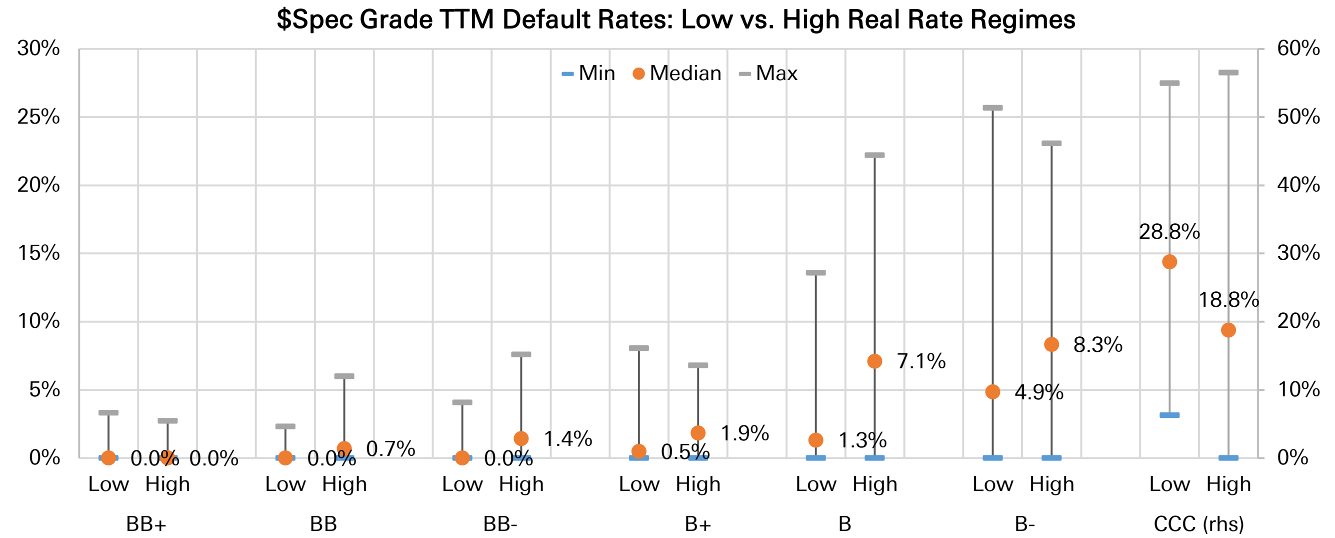
Figure 3: US real rates moved lower over the last year, as the Fed did cut before job cuts



Source : Deutsche Bank, Bloomberg Finance LP

In addition, real US interest rates continued their decline seen over the last 2 years (Figure 3). And as we wrote in our 2024 default study, the trajectory of real interest rates is critical for the trajectory of future default rates, particularly for B2 & B3 rated bonds and loans. In Figure 4 below, we produce a similar exercise, dividing up US economic history into low and high real rate regimes, as measured by months in which the difference between the effective Federal Funds Rate and Core Y/Y PCE is in its bottom third or top third percentile, since 1980. As Figure 4 illustrates, the median B2 & B3-rated annual default rate is far more elevated in a high real rate (7.1% & 8.3%) vs. a low real-rate regime (1.3% & 4.9%). And last year's data corroborates this trend. US B-rated default rates were well below recent averages, and certainly far closer to a low real-rate than high real-rate world (Fig 5). Simply put, Fed policy was a tailwind for Bs last year.

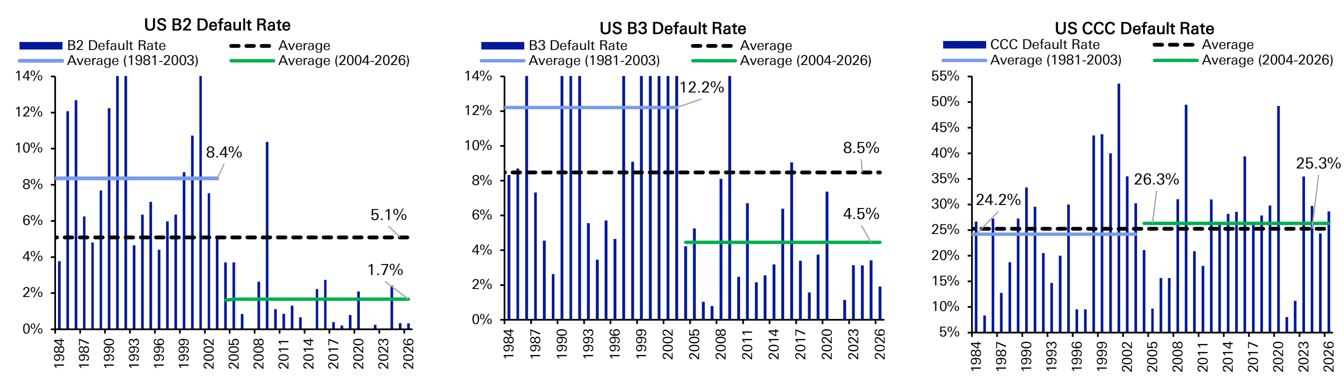
Figure 4: The Fed's next move matters. Real rates make a big difference in B2 & B3 default rate trends.



Source : Deutsche Bank, Bloomberg Finance LP, S&P, $-spec grade default rates are available from Dec'81 onwards. We define the High (Low) real rate regime as the months in which the effective Federal Funds rate - core Y/Y PCE is in the top (bottom) 1/3 of its distribution since 1980.

Note however that real rates have a negligible impact on CCC default rates, both historically, and also last year, when US CCC default rates were quite elevated, and in line with historical averages. This is certainly a cautionary tale for future US default trends. Easier Fed policy and/or macro tailwinds from a transformative AI buildout will not prevent overleveraged capital structures from restructuring in secularly challenged industries. We have already experienced this directly in sectors such as US Telecom & US Healthcare in recent years. But one can apply this framework to US Software, especially considering the market is pricing 30% of US Software Lev Loans today with an implied CCC rating (more on this later).

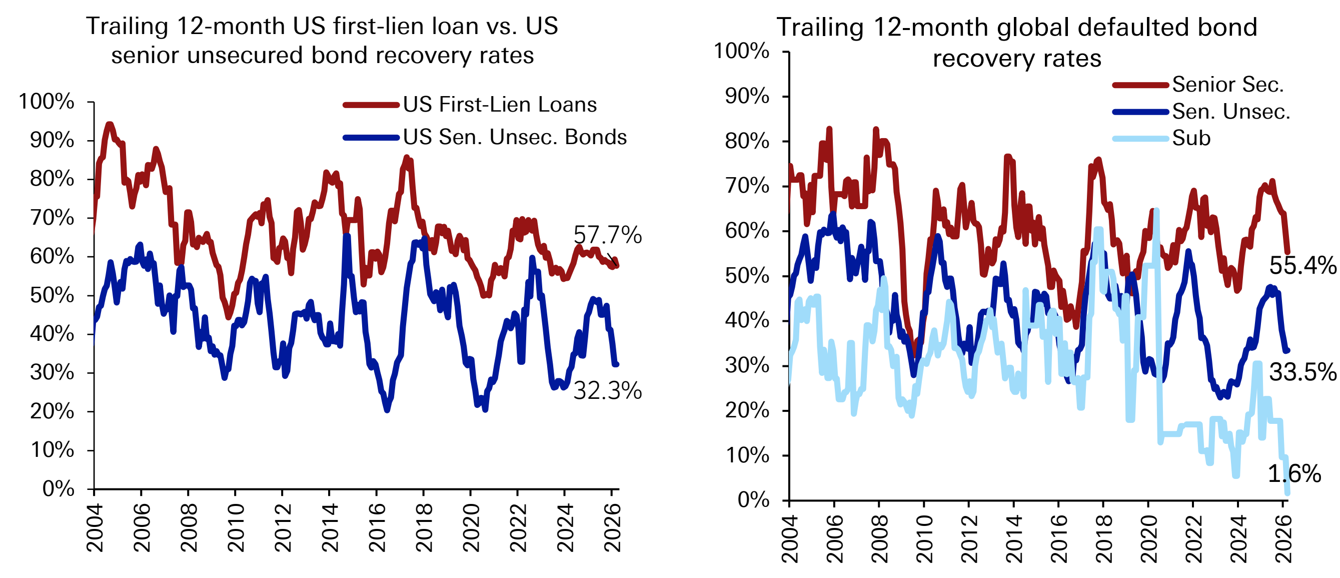
Figure 5: The Fed cutting with inflation at 2.8% and limited job cuts kept B default rates at very low levels. But CCC default rates remain elevated, a cautionary tale for leveraged issuers that run into financial trouble ahead.



Source : Deutsche Bank, S&P

Despite rate cuts and a drop in US defaults, our views on lower recoveries & the continued prevalence of distressed exchanged did play out. One can see in Figure 6 that trailing 12m US Lev Loan recovery rates are only around 58%, while US HY senior unsecured bond recovery rates are stuck at a very low 32%. The picture is similar on a global basis, and when considering subordinated spec-grade debt as well, where recovery rates are near 2% over the last year. Overleveraged capital structures, with often little subordinated debt cushions, and often high levels of intangible assets as collateral are unsurprisingly seeing poor recovery rates.

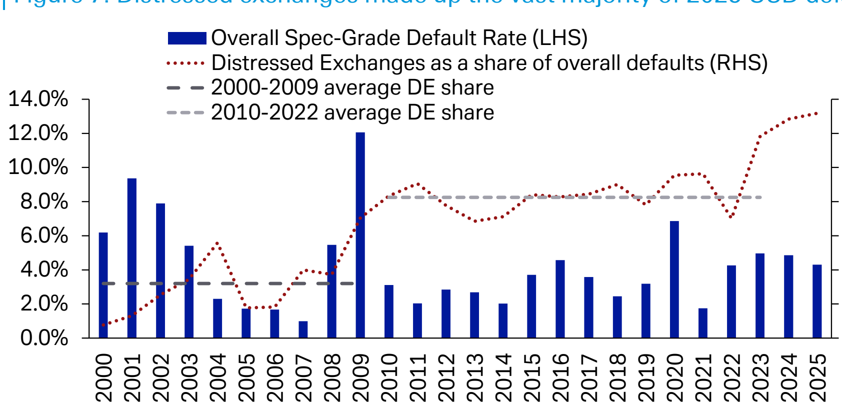
Figure 6: Default rates have been less severe than we expected. But recovery rates have been trending lower.



Source : Deutsche Bank, Moody's
*Measured by debt prices which are taken immediately prior to distressed exchanges or 30 days after non-distressed exchange defaults

On the distressed exchange point, while we have seen some slowdown in the monthly series we highlight in our Global Monthly Default Monitor, the overall prevalence of distressed exchanges remains. Distressed exchanges are an alternative to outright defaults or bankruptcies, in which creditors agree to restructure to continue receiving interest payments at the highest recovery rate possible, while the owners generally continue to retain their equity upside in the firm. In looking at data from Moody's, distressed exchanges made up 68% of overall global spec-grade defaults last year after being just under 60% of defaults in 2023. The post-GFC average until 2022 is just over 40% of defaults, while prior to 2009 LME activity was relatively rare with an annual average of around 16% (Figure 7).

Figure 7: Distressed exchanges made up the vast majority of 2025 USD defaults

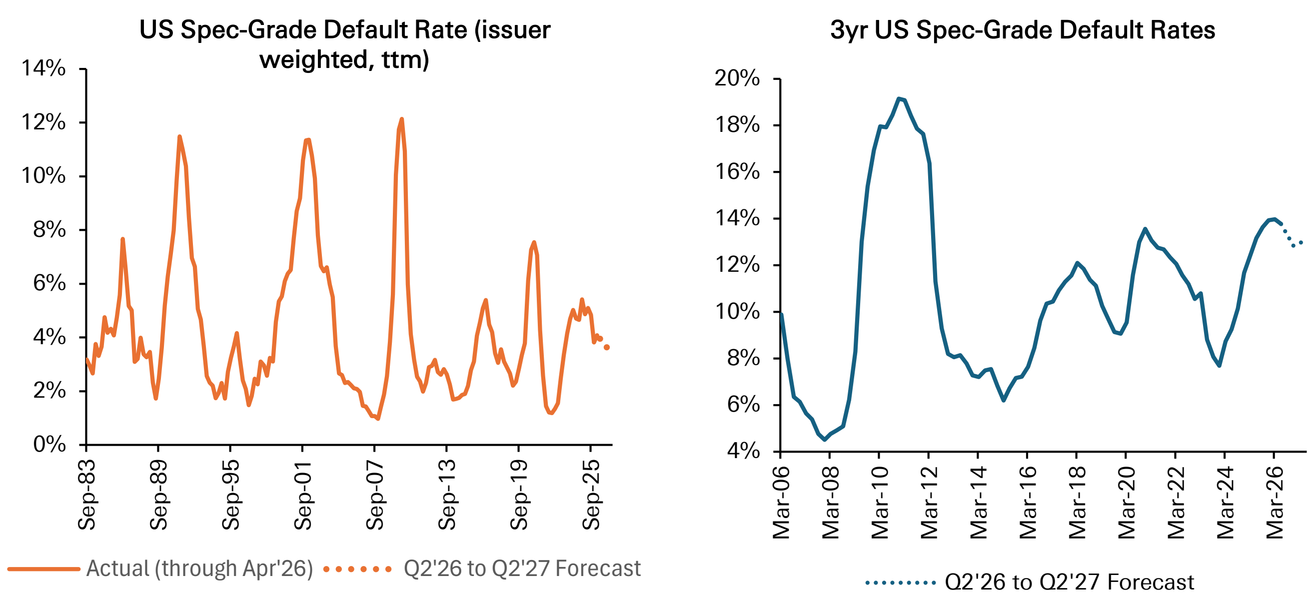


Source : Deutsche Bank, Moody's

What is the future outlook for US defaults?

We aren't forecasting a major US default cycle to emerge by the middle of 2027. But equally, the grind lower in defaults seen over the last year should abate; a return to say 2% defaults that were seen in past exuberant market cycles such as 2007, 2014 & 2021 is highly unlikely. Our forecast is for US spec-grade default rates to slow from 4% today to 3.6% by YE'26, and then to climb slowly again to 4% by Q2'27. If we are correct, the cumulative 3yr US spec-grade default rate will still be near some of the highest levels seen since the aftermath of the GFC (Figure 8).

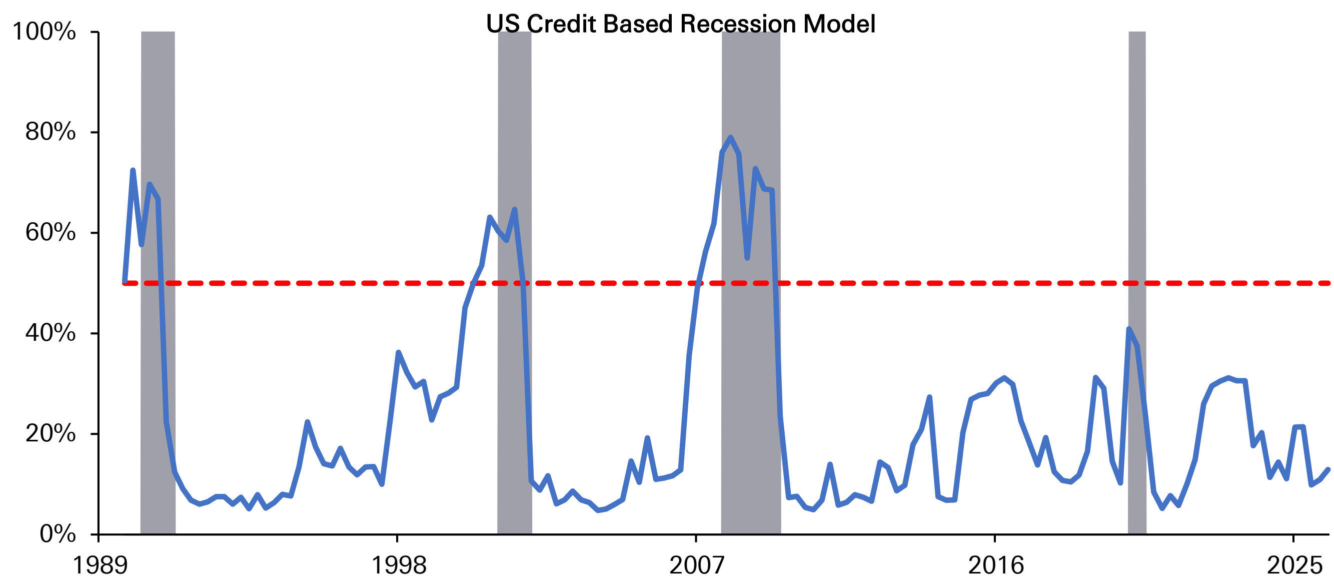
Figure 8: US spec-grade defaults should fall from 4.0% to 3.6% by YE'26, and then slowly climb to 4.0% by Q2'27. We don't see a broad default cycle. But we also don't see default rates collapsing, given higher rates and pockets of leverage. This is why 3yr cumulative default rates will still remain elevated.



Source : Deutsche Bank, Bloomberg Finance LP, S&P

In essence, this is a message that **the K-shaped US economy still lives**, and that US credit dispersion will persist. As we have written repeatedly since 2024, the US economy is effectively a "fire & ice" US economy. Perhaps today, we can change that label to an "AI & ice" US economy, given how narrow we believe the drivers of US growth to be. On the positive side, US artificial intelligence tailwinds feed back into very strong US corporate profits and a wealth effect that supports the high-income US consumer, and this is a major reason why our US credit-based recession model is still only forecasting a 12% chance of a US downturn through Q1'27. It is very hard to see a spike in US defaults with overall US growth so solid (Figure 9).

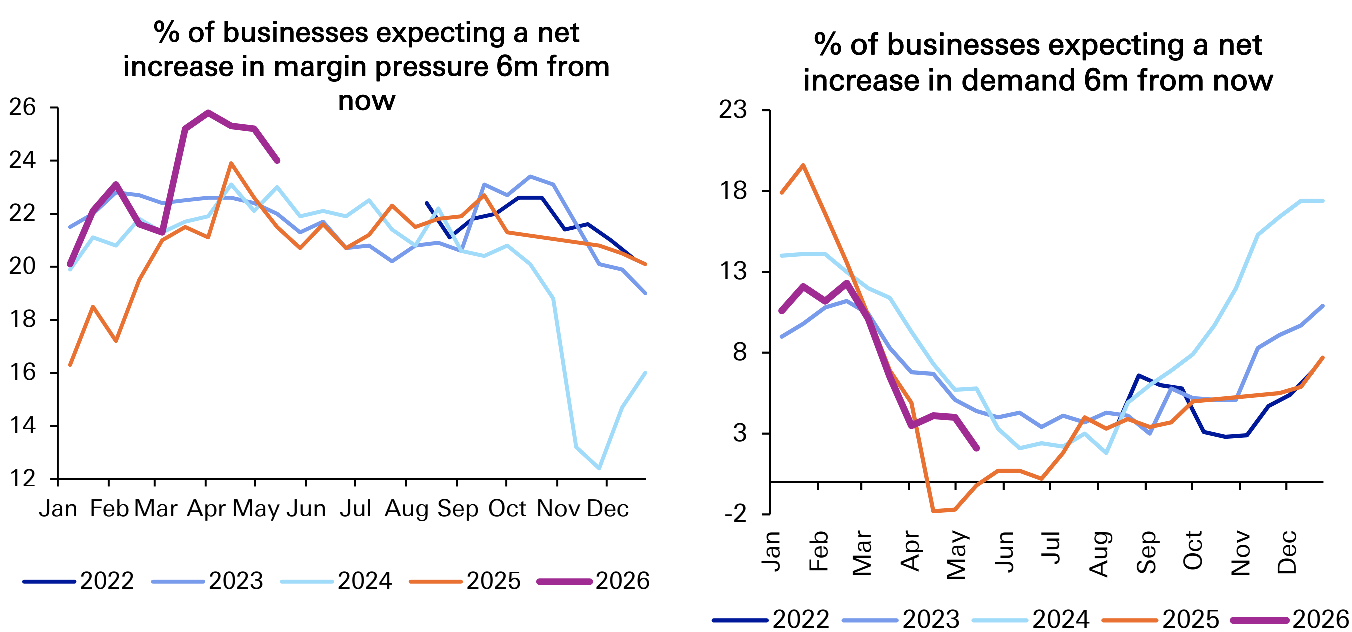
Figure 9: US cycle risks remain low, given robust corporate profit growth in particular.



*Source : Deutsche Bank, Bloomberg Finance LP, Haver, *Our US & EU credit indicators take an average of 0-12m forward recession probabilities across the following 4 indicators: 1) Y-Y change in GDP-based corporate leverage, 2) Y-Y change in GDP-based corporate interest coverage, 3) Y-Y change in US bank loan 30-89 day past due rates or Eurozone bank NPL rates, 4) SLOS Survey: % of banks tightening standards on small firm C&I loans or ECB BLS survey: % of banks tightening standards across business/mortgage/consumer loans.*

But equally, a look at the median US business, as proxied by the US Census Business Trends & Outlook Survey, paints a far different picture. US businesses today are forecasting future margin compression and future demand expectations at near the worst rate in 5 years (Figure 10). This would certainly imply higher $HY & $Lev Loan defaults, than what the aggregate macroeconomic data would imply.

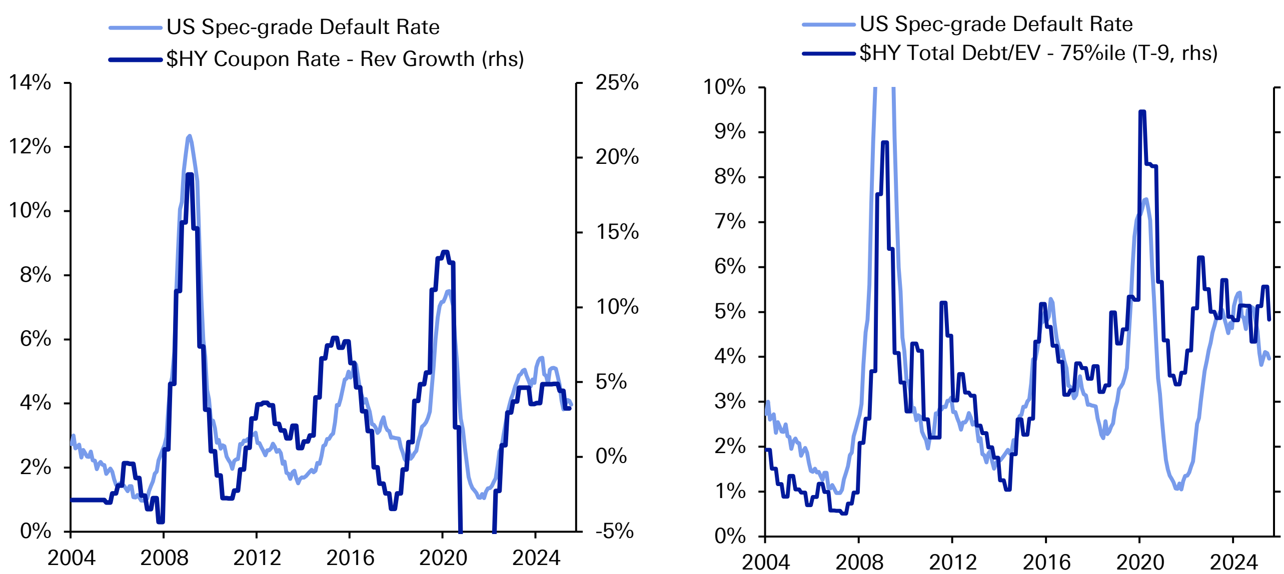
Figure 10: Aggregate economic data looks strong, but a look at the "typical" US business shows considerable worry over future margin pressure and future demand trends. The K-shaped economy still lives.



Source : Deutsche Bank, Business Trends & Outlook Survey

And **a couple of micro overlays also suggest US spec-grade defaults will remain steady**, but more elevated than what current spreads imply. The differential between $HY index coupons & $HY revenue growth rates has had a very strong relationship with contemporaneous defaults. Conceptually, if corporate debt service payments are rising faster than corporate income, it is harder for a company to service its debt. Currently, the $HY index coupon is 6.6% vs. a median company revenue growth rate of 3.4%, suggesting default rates will remain elevated, but should settle around 4.0%. Similarly, $HY corporate debt to EV ratios have strongly tracked future defaults, 9 months forward. This logic follows a Merton-Model style framework, where a sharp decline in equity valuations can signal the value of a firm's assets is at risk of declining below the value of a firm's debt, creating incentives for corporate owners/sponsors to restructure, to the detriment of creditors. In Figure 11, we compare the 75%ile of $HY debt to EV ratios vs. default rates, focusing on the 75%ile to capture the most leveraged public firms in the $HY universe. This analysis is also consistent with US spec-grade default rates settling around 4.0% by year-end.

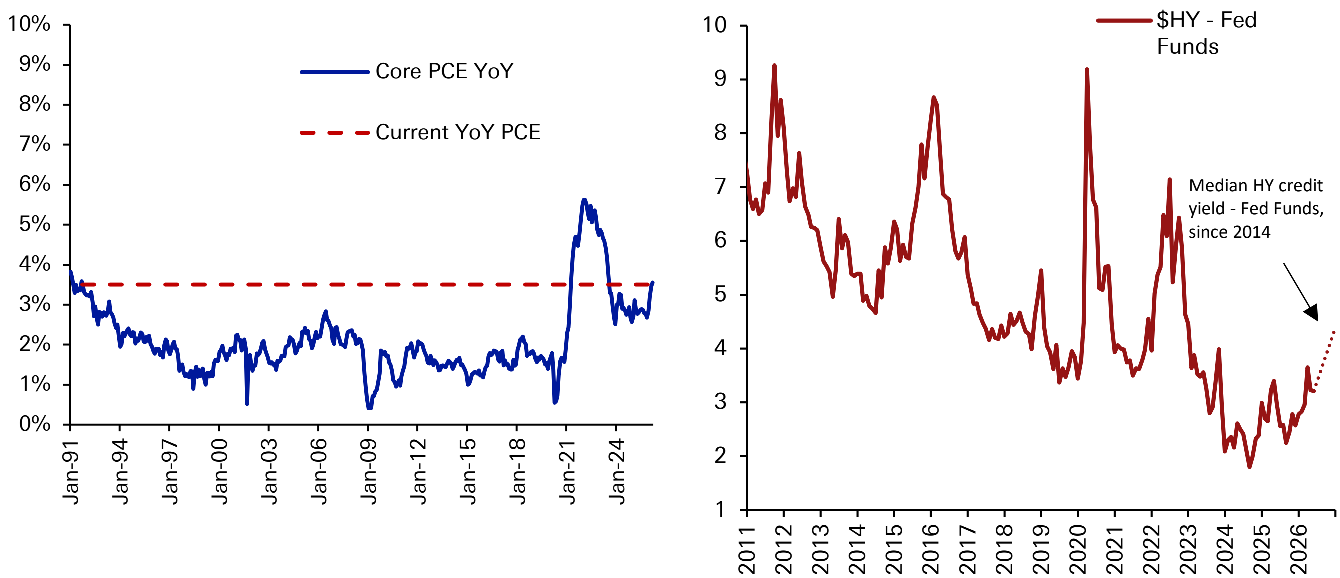
Figure 11: $HY coupon - revenue growth & $HY debt to EV ratios are largely consistent with our macro models forecasting 4% US spec-grade defaults over the next 9 months. Again, fundamentals do not suggest a major default cycle, but nor do they suggest a return to the low default rate world.



Source : Deutsche Bank, Bloomberg Finance LP, S&P

Lastly, **Fed policy is unlikely to be as accommodative** for US spec-grade issuers going forward, and the continued adoption of Artificial Intelligence is the reason why. US core PCE inflation is still running around a 3.5% Y/Y pace, which obviously reduces the odds of a Fed easing bias on its own. This is important considering overall $HY bond yields still trade 120bps too low relative to the current Fed Funds rate on an historical basis (Figure 12). In the context of low recession risks + a long Fed hold, $HY bond yields are likely to rise as US Treasury curves steepen. But even the risks of Fed hikes are starting to emerge, led by AI-associated capex and increasing memory chip shortages that are helping to keep US consumer inflation above target. The worst case scenario for US spec-grade credit markets and default rates would be Fed rate hikes starting at year-end 2026 or early 2027, at the same time that AI adoption is more tangibly disrupting the business model of floating-rate software issuers in the Lev Loan & Private Credit markets. This latter scenario is not in our base case, as we expect the Fed to resist rate hikes for as long as possible, but it is certainly the largest risk in our view, and one we explore more fully later in this piece.

Figure 12: US core inflation is well above target, and $HY almost looks like it is priced for Fed cuts (rather than hikes), with $HY bond yields currently trading 120bps expensive to the Fed Funds rate, since 2014.



Source : Deutsche Bank, Bloomberg Finance LP

European spec-grade default rates

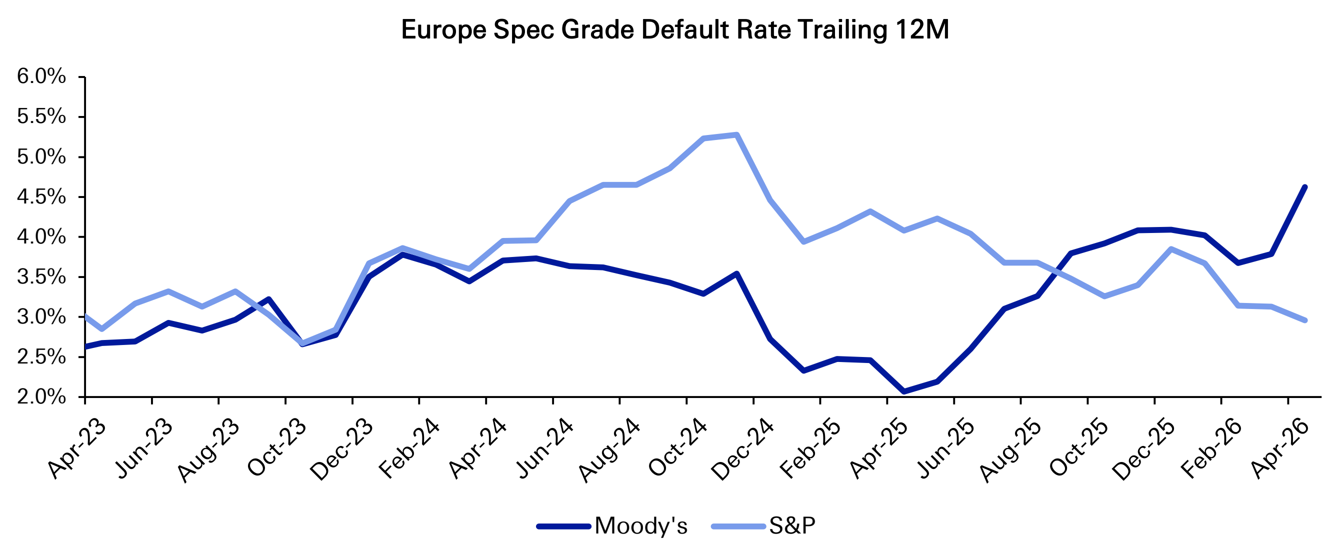
Surprise increase in defaults signals vulnerability to ECB hikes.

In contrast to the US, European spec-grade default rates have been climbing fairly aggressively over the last year. Moody's European spec-grade defaults have increased from 2.1% last year to 3.8% in March and 4.6% in April, after a particularly elevated month of April credit events that was the highest since Dec'23. European corporate default rates are now well above historical levels, with the median & average spec-grade default rate at only 2.3% & 2.9% over the past 20 years.

Our forecast last year for European default rates to stay stable, with levels around 2.3% by Q2'27, proved to be far too optimistic. What happened in Europe to push default rates so much higher?

The first item to note is that smaller sample sizes in Europe can make a large difference, when comparing default rates between different rating agencies. For example, S&P European spec-grade default rates have actually been declining gradually to around 3% in recent months (Figure 13)

Figure 13: European spec-grade default rates are diverging between Moody's & S&P



Source : Deutsche Bank, S&P, Moody's

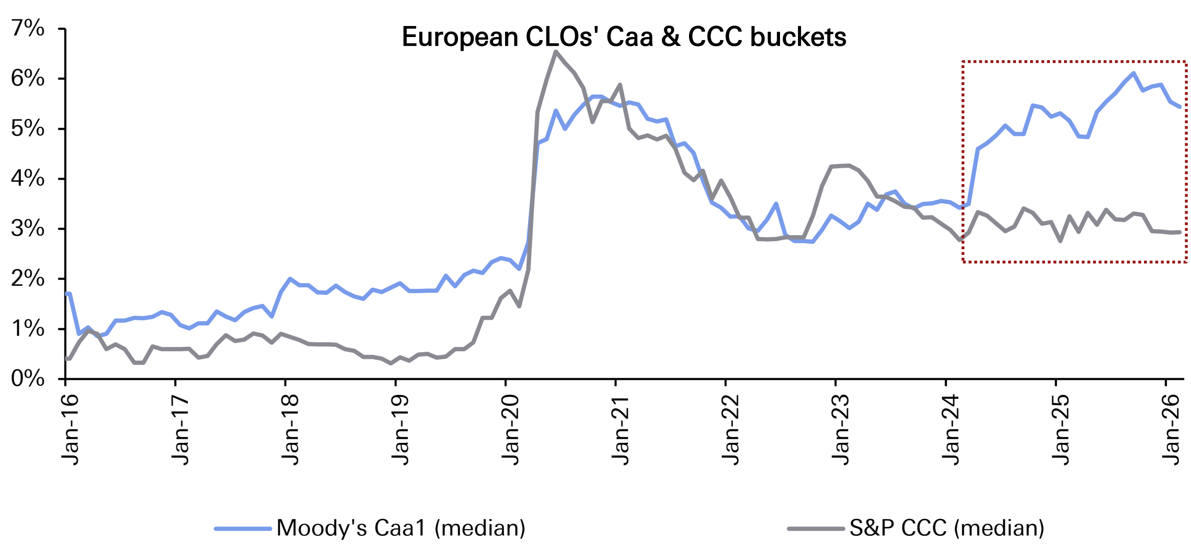
However, this is not necessarily a signal that European default rates will converge lower from here. On the contrary, one can make an argument that a higher Moody's default rate might be a leading signal for a higher future S&P default rate in the coming months. In our view, a higher Moody's European default rate is reflecting the fact that 1) Moody's takes a more conservative view than S&P with respect to what constitutes a credit default & 2) most defaults have been in the form of distressed exchanges, which Moody's is simply faster to recognize than S&P is.

On the former, we can see from our past work on European CLO CCC buckets, that Moody's is the more conservative agency in rating spec-grade bonds & loans (Figure 14). The median European CLO has a CCC bucket of 5.4% as rated by Moody's, but only 2.9% as rated by S&P. This reflects the fact that Moody's often rates securities on an expected loss basis (taking into account both the probability of default and future recovery rates), while S&P rates securities more on a probability of default basis.

In addition, the prevalence of distressed exchanges is driving a wedge between Moody's and S&P default as well. Moody's will often classify any distressed exchange as a default if 1) the lender takes a haircut & 2) the intent of the distressed exchange is to help the borrower avoid a future default. S&P by contrast will offer more flexibility, only classifying an exchange as a default if the haircut is sufficiently large & if a default would occur immediately, without the execution of the exchange.

Lastly, there is a simple timing element as well. Moody's will classify a distressed exchange as a default effectively from the announcement date, while S&P will only shift a security into default status after the legal exchange takes place (often 1-2 months later).

Figure 14: Moody's is generally the more conservative agency when it comes to assessing ratings & default risk in Europe, and one can see this starkly in European CLO CCC buckets today.

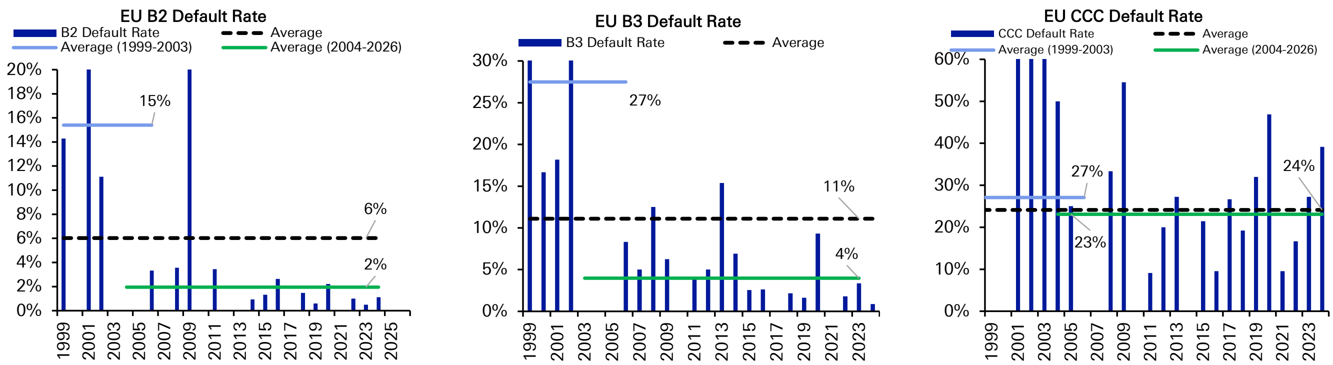


Source : Deutsche Bank, Intex, Pitchbook

There is no one answer as to which rating agency is more or less right from a conceptual standpoint. If one believes in a more benign economic scenario of low default rates and where distressed exchanges solve the problem of over-leveraged capital structures, then S&P will provide a better gauge of future credit risk. However, we have sympathy with the more conservative stance from Moody's given Figure 6 already showing very weak recovery rates, even in a non-recessionary environment, coupled with the fact that many distressed exchanges are re-defaulting in future years.

Removing ourselves from the question of which rating agency has the better grip on European credit risk, we can also make some more general comments on trends in European defaults. In a similar vein to the US market, almost all of the default pressure is being generated by European CCCs, where the patience for overleveraged capital structures in challenged industries is minimal (Figure 15). ECB rate cuts last year did help to limit the damage in Bs, but this will be called further into question as the ECB looks set to hike at least twice over the rest of 2026, thanks to the fallout of the Iran war and higher energy costs.

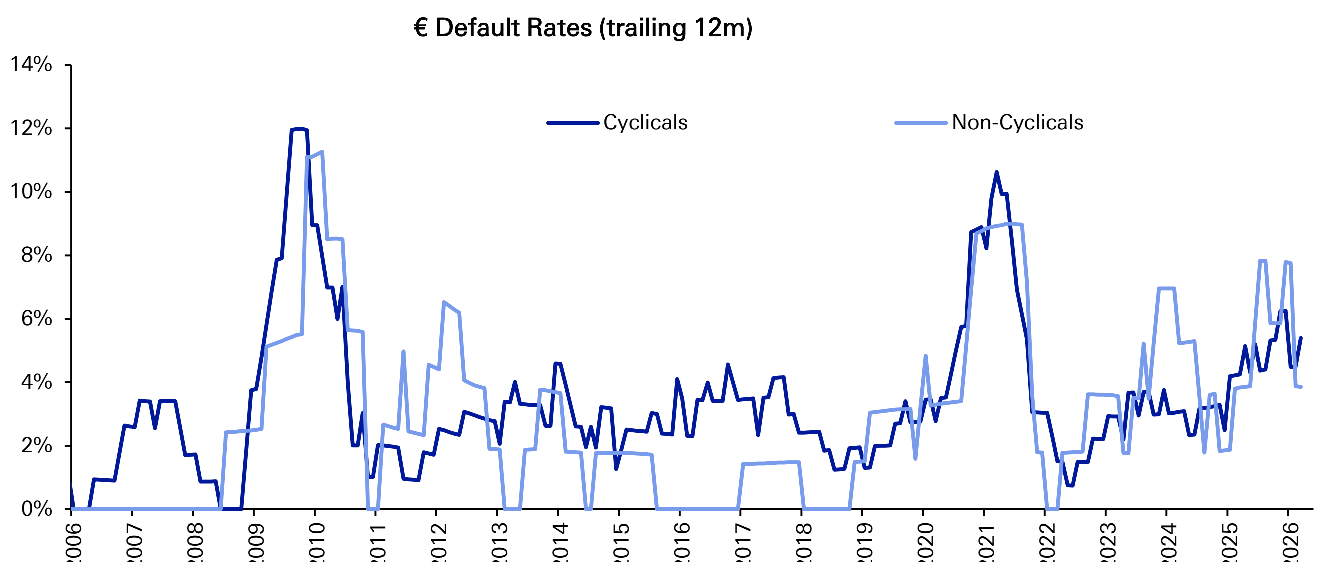
Figure 15: ECB rate cuts helped to keep B default rates limited. But even more starkly than in the US, European CCC default rates surged to above average levels. There is little patience in Europe for issuers who run into financial difficulty.



Source : Deutsche Bank, S&P

In addition, European cyclical risks remain elevated, and this is showing up in our data (Figure 16). While European non-cyclical default rates (3.9%) are similar to those of the US (3.9%), European cyclical default rates (5.4%) are now exceeding US cyclical default rates (2.1%) by a healthy margin.

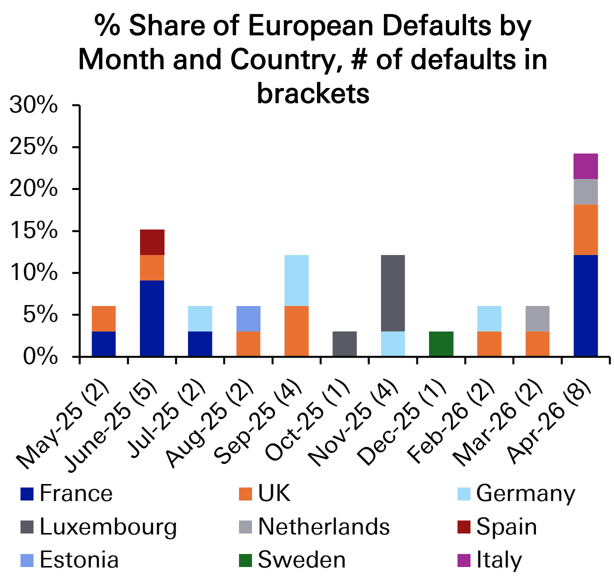
Figure 16: Unlike in the US, European cyclical default rates are starting to rise

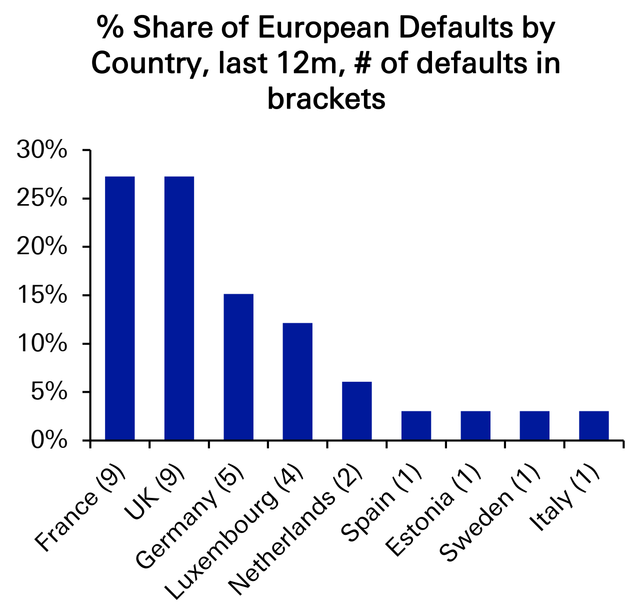


Source : Deutsche Bank, S&P

Finally, the weakest links of Europe from a country perspective are also showing up in our default rate (Figure 17). Over the last year, we have seen 9 defaults apiece from both France & the UK, 5 defaults from Germany, but only 1 apiece from Italy & Spain. This aligns with our prior work on rising bank NPLs in France & Germany, with still rapidly falling bank NPLs in Spain & Italy.

Figure 17: European defaults picked up in April, with most of the defaults over the last year coming from France, the UK & Germany.



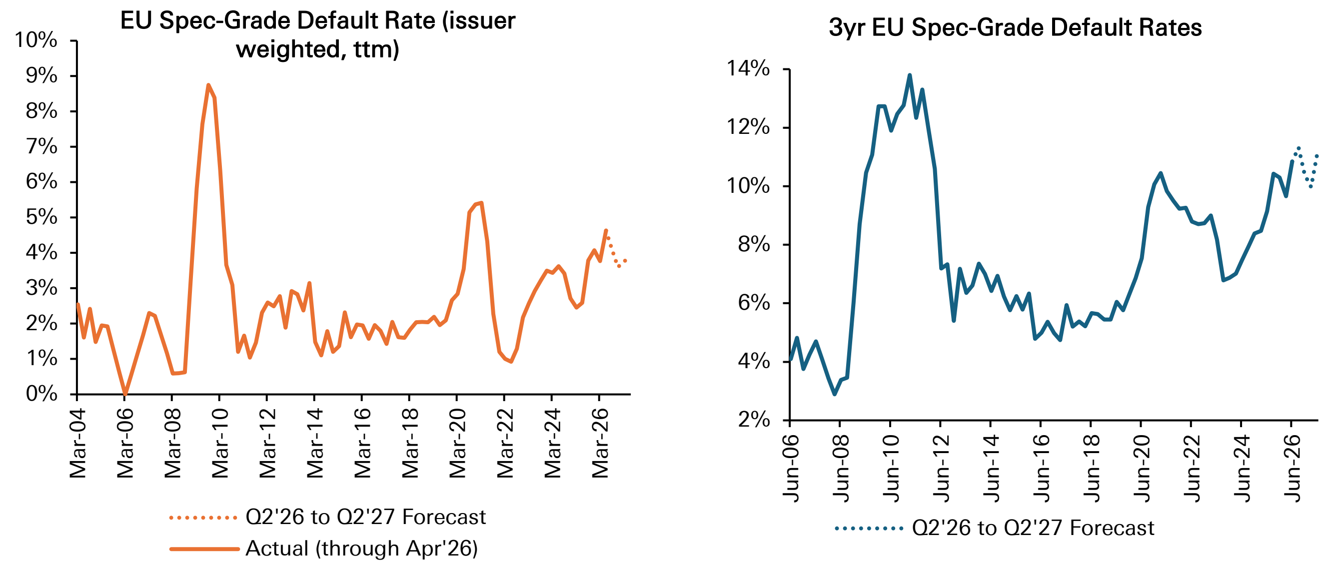


Source : Deutsche Bank, Moody's

What is the future outlook for European default rates?

A more measured pace of ECB rate hikes (June, then September) coupled with strength in Southern Europe should prevent a major spike in credit risk through the 1H'27, but default rates will remain above average, with the Iran being a major catalyst for this view. We forecast European spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. This would keep the cumulative 3 year European corporate default rates at very high levels historically (Figure 18).

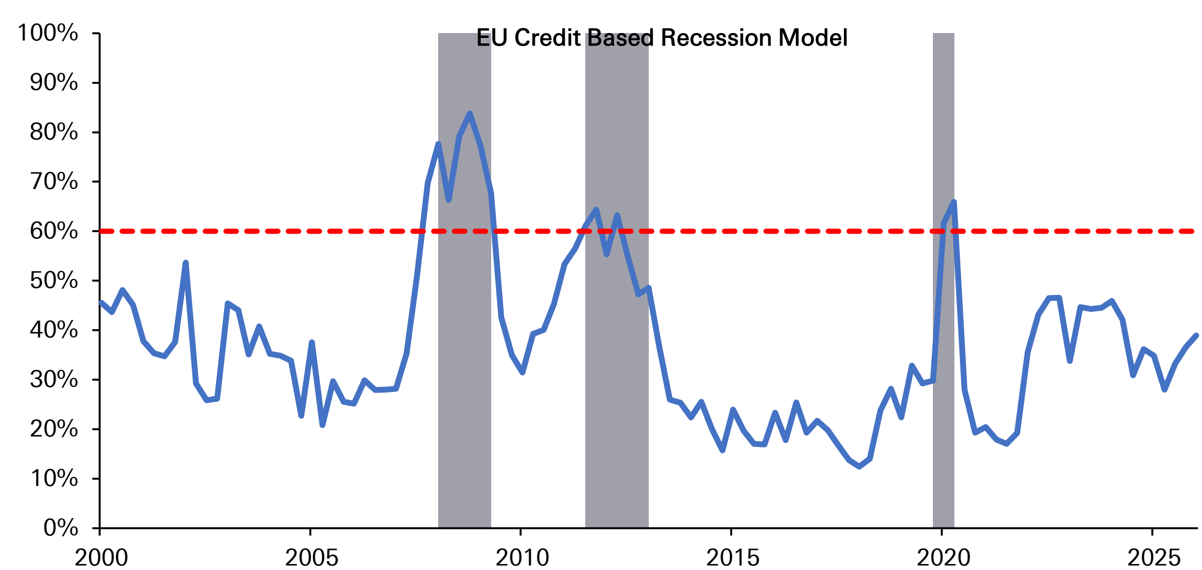
Figure 18: We forecast European spec-grade defaults to decline from April's very high 4.6%, but to settle at 3.6% by YE'26 & 3.9% by Q2'27. 3yr cumulative default rates will stay elevated.



Source : Deutsche Bank, Bloomberg Finance LP, S&P

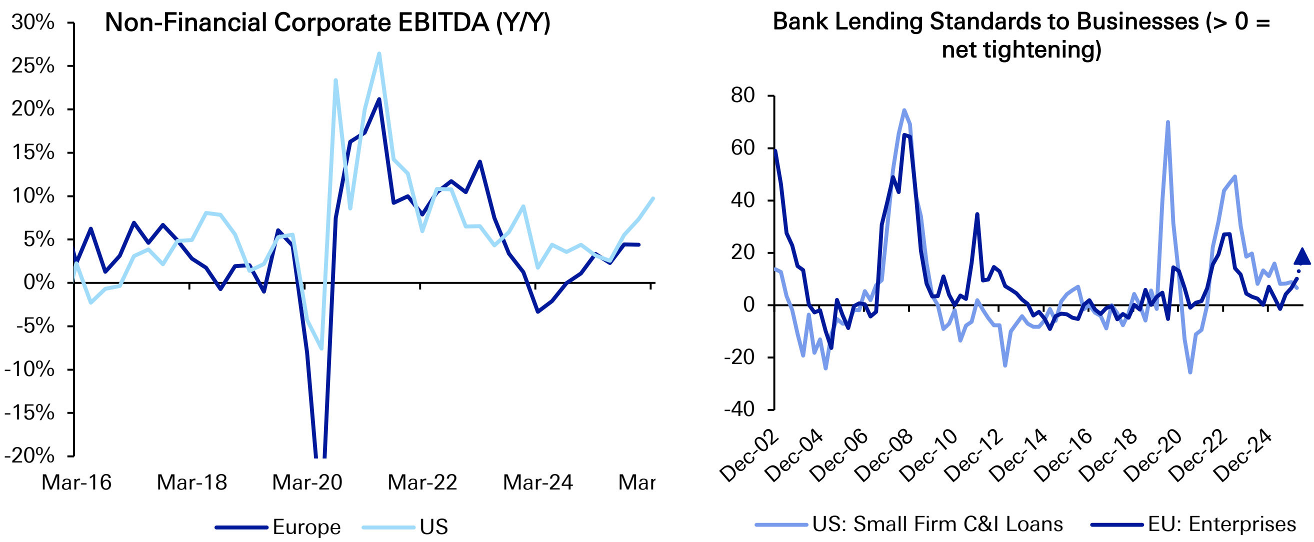
What is the reason for our view? Firstly, our in-house European credit based recession model is far from signalling an imminent downturn, but risks are rising and are now about 0.2 standard deviations above average, with more post-war data still to be captured from our model framework (Figure 19). Unlike the US, Europe does not have an AI tailwind boosting corporate profits. And in the critical bank transmission channel, European bank lending standards are now projected to be far tighter than what is taking place in the US (Figure 20).

Figure 19: European cycle risks aren't alarming, but are now somewhat above average.



*Source : Deutsche Bank, Bloomberg Finance LP, Haver, *Our US & EU credit indicators take an average of 0-12m forward recession probabilities across the following 4 indicators: 1) Y-Y change in GDP-based corporate leverage, 2) Y-Y change in GDP-based corporate interest coverage, 3) Y-Y change in US bank loan 30-89 day past due rates or Eurozone bank NPL rates, 4) SLOS Survey: % of banks tightening standards on small firm C&I loans or ECB BLS survey: % of banks tightening standards across business/mortgage/consumer loans.*

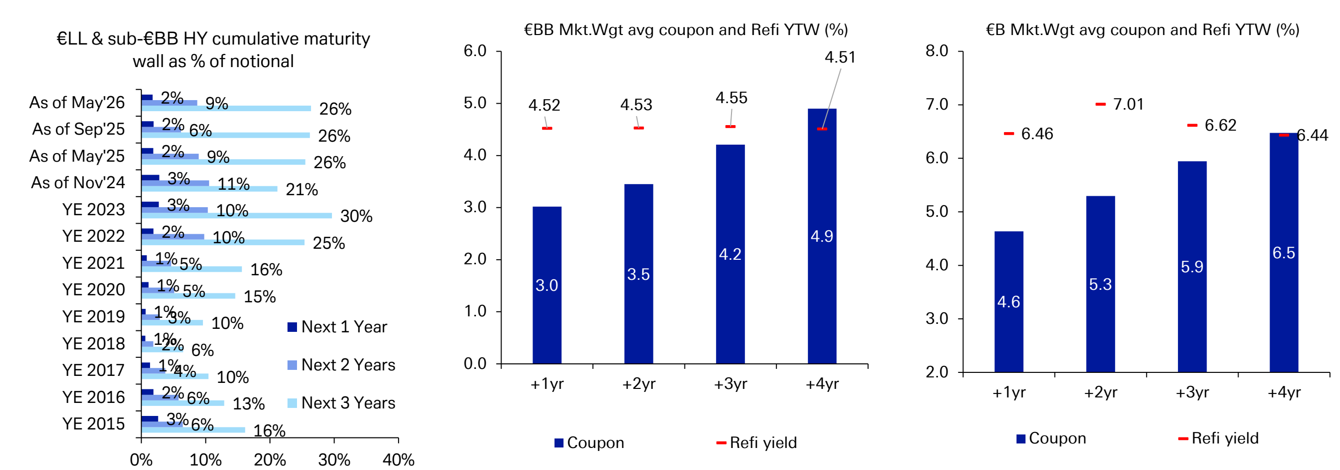
Figure 20: European corporate profits may be stalling out without the direct AI tailwinds being generated in the US. Meanwhile, European bank lending standards are now projected to tighten aggressively.



Source : Deutsche Bank, Haver, Bloomberg Finance LP

In addition, the recent shift in market expectation from ECB rate cuts to rate hikes is likely being underappreciated in terms of a market impact. We already know the maturity wall for lower-rated European spec-grade credits remains historically elevated over the next 2-3 years, especially when compared with pre-COVID levels. But in addition, B rated European companies on average would face a 70 to 170bps rise in funding costs, if these companies needed to refinance their maturing bonds in 2 to 3 years, at current market yields (Figure 21).

Figure 21: Leveraged European corporate maturity walls are far more elevated than pre-COVID levels. And the extra cost to refinance over the next 2 to 3 years will be material if the ECB isn't cutting anytime soon.

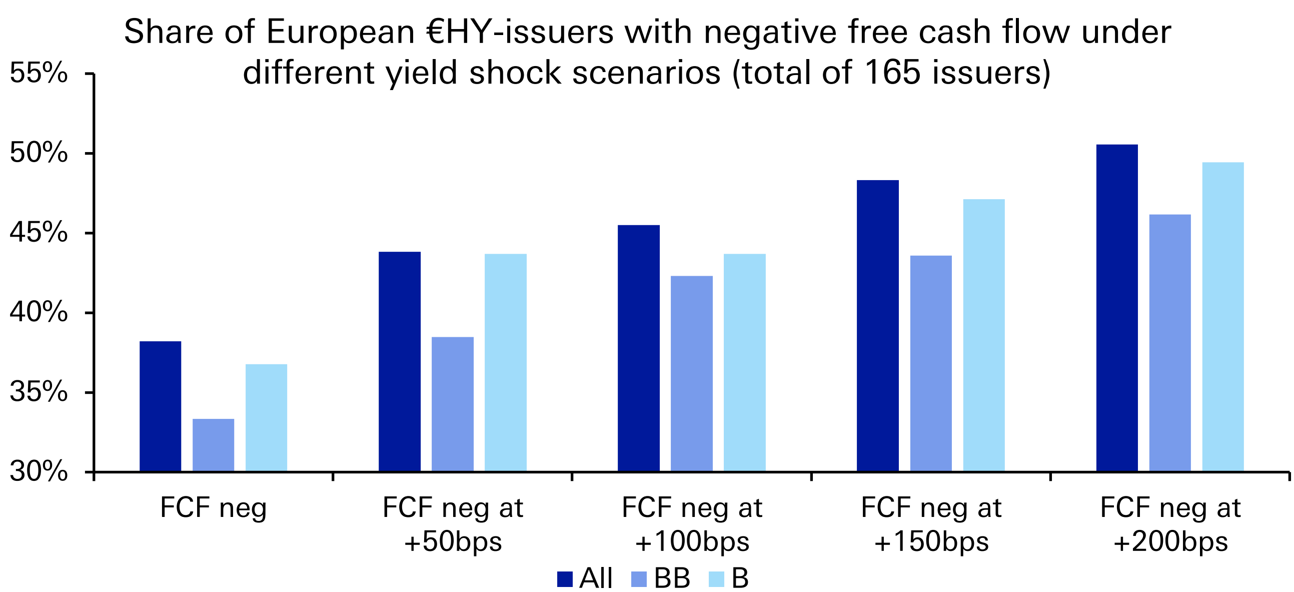


Source : Deutsche Bank, Pitchbook, Bloomberg Finance LP

This increase in funding costs is unlikely to be easily absorbed by lower-rated European HY firms. As Figure 22 illustrates below, we already face a scenario where 37% of B rated European HY issuers are currently in negative free cash flow status.

A simple 50bp increase in total funding costs would drive those percentages higher to 44%, and that is actually the biggest delta in our scenario analysis below, although clearly further increases in bond yields would drive the level of negative free cash flow to more concerning levels.

Figure 22: A simple 50bp shock in total funding costs could mean that around 45% of European B HY issuers are into negative free cash flow status



Source : Deutsche Bank, Cognitive Credit
Only Issuers from European Countries

Put simply, a small increase in funding costs can drive meaningful differences in European cash flow generation. And this is one reason we have become more bearish on Europe in recent months. Weak growth is not the sole factor of why our stance here has shifted. European spec-grade markets could face moderately weaker growth without higher defaults, with the ECB acting as a relative tailwind through insurance rate cuts. But as it stands today, the Iran war is likely to leave Europe with both growth and central bank headwinds at the same time.

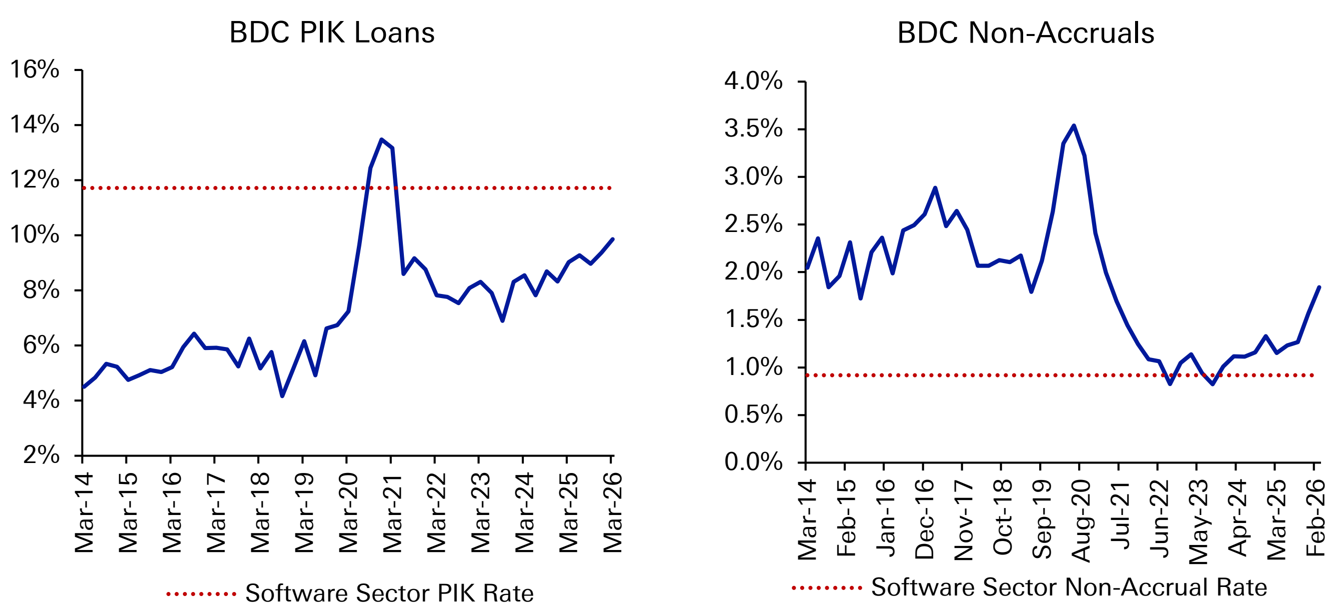
Tactical and structural factors to watch

A US Software glitch can materially tighten funding markets

As discussed in our abstract, the tailwinds of AI disruption for aggregate growth will likely become headwinds for spec-grade credit, through tighter financial conditions in private credit & leveraged loans. This will eventually play itself out through two channels. The first will be more hawkish Fed policy, but the second will be meaningful downgrades of US software names into CCC buckets, where investors have already demonstrated their impatience to provide flexibility to this rating cohort over the last few years. In essence, risks are rising that we could see a scenario in early 2027, where tangible AI disruption of software business models begins to damage software revenue and customer retention rates, at the same time actual Fed hikes eat up limited software free cash flow.

The implications of this scenario are not being priced. In private credit specifically, PIK interest usage (paying interest in kind, rather than in actual cash) is elevated both in aggregate and for software names, but non-accrual rates are still below averages, with plenty of room for software non-accrual rates to rise over the next few years to catch investors off-guard (Figure 23).

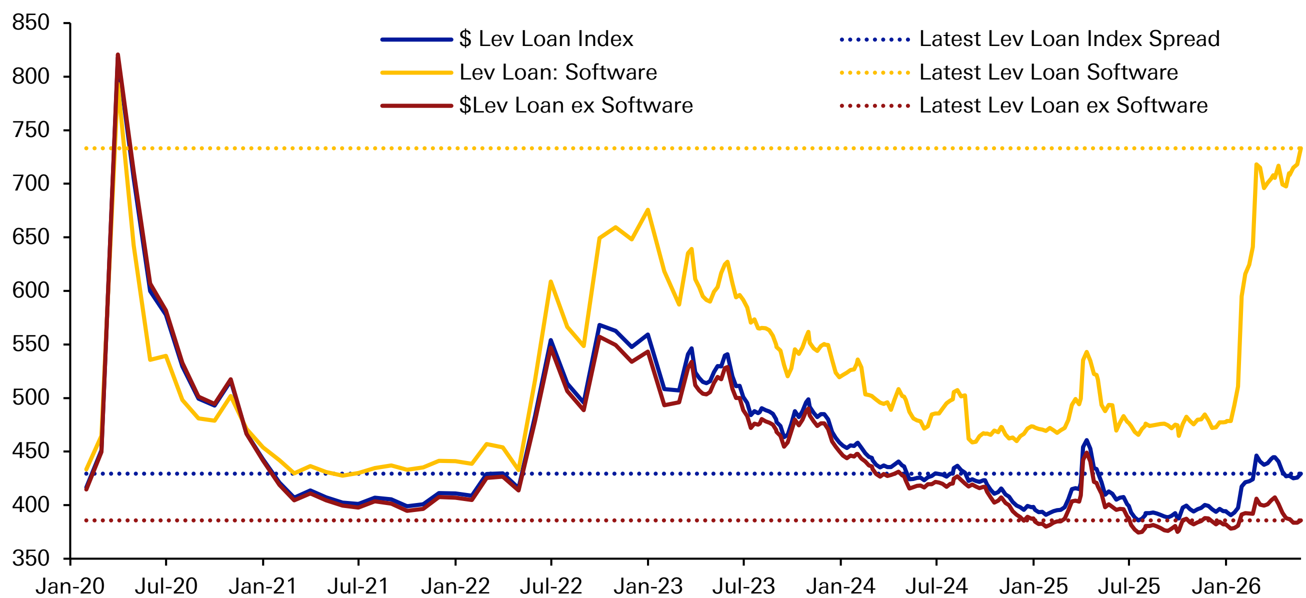
Figure 23: Private credit PIK rates and non-accrual rates are rising. But while software PIKs are high, software non-accruals are still well below average, and this will likely change over the next 1-2 years.



Source : Deutsche Bank, SEC EDGAR

And in the leveraged loan market, software risk has rapidly re-priced, but with an uncanny lack of spillovers to other parts of the market, in a way that makes the divergence between $HY energy and ex-energy in 2015 look tame (Figure 24). US Leveraged Loan software spreads are now trading at 733bps, at YTD wides, and trading at the widest levels since March 2020. However, software stress remains isolated, with ex-software spreads trading at 386bps, below 2021 levels, when central banks were still running historically easy monetary policy. This is an astonishing divergence between software & ex-software leveraged loan spreads, especially considering US software is the largest sector in both the US Loan and Private Credit markets. Future software stress, most tangibly in the form of rating downgrades, would cascade through broader spec-grade financial markets in a more disruptive fashion than is appreciated.

Figure 24: Software glitch? US Lev Loan software spreads are trading at 733bps, near the YTD wides



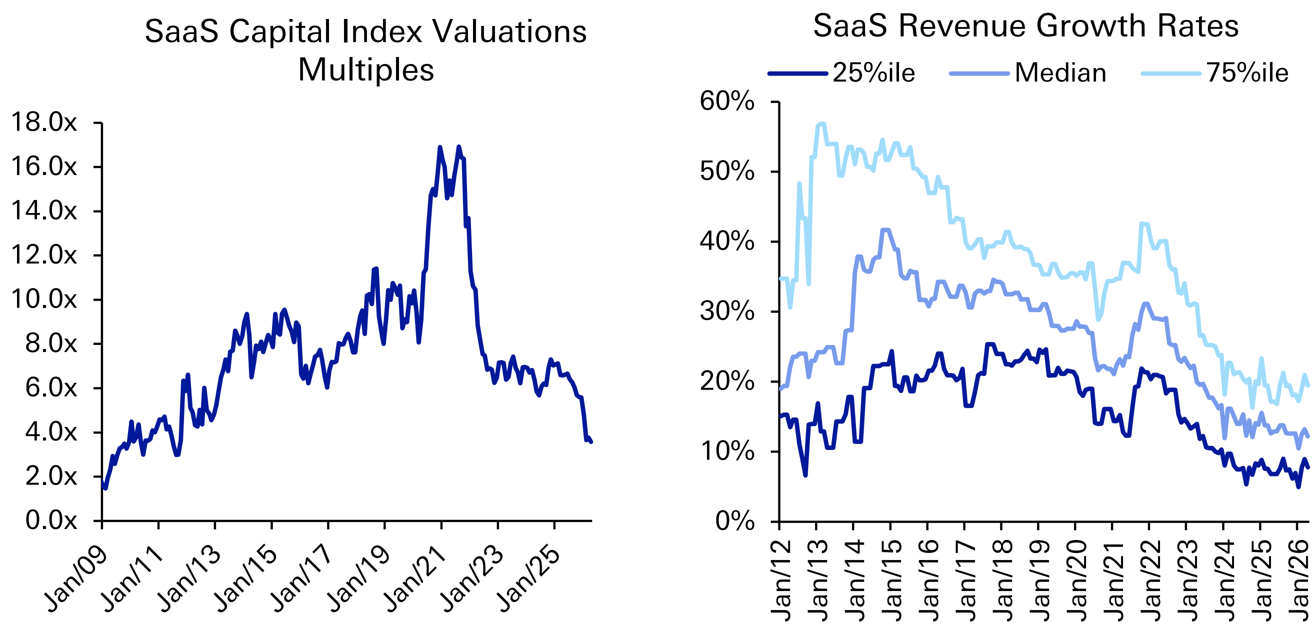
Source : Deutsche Bank, Pitchbook

It is clear loan investors remain concerned about the impact of AI disruption on the long-term viability of the software sector. AI disruption to Software comes in many flavors. First, it reduces clarity on future recurring revenue growth, as customers may decide to forgo renewing contracts. This is critical for the leveraged software space, since a large share of firms currently have negative EBITDA.

Second, AI will likely reduce white-collar employment over time, which cuts against the "per seat" model that many software firms employ to earn revenue. Third, it paradoxically could encourage many software companies today to over-invest in AI ("if you can't beat 'em, join 'em"), before knowing how successful AI models will eventually become, or how best to integrate them. This isn't a trivial concern, given that many SaaS firms have limited financial flexibility. An investment mistake today could prove costly later, if financing markets seize up.

Put simply, equity & credit investors are making a judgement call that a universe of highly leveraged B2 & B3 rated names, facing rather large maturity walls over the next 2-3 years (which need to be refinanced within the next 1-2 years), will have difficulty adapting to a world of increased AI adoption. And this is why software multiples to annualized recurring revenues have dropped nearly 50% since 2024, implying that LTVs for leveraged software firms are far higher than the 40-50% often cited by investors in the space (Figure 25).

Figure 25: Software multiples have collapsed 50% since 2024, implying LTVs are far higher than advertised. But the biggest catalyst for further software weakness will be when revenue growth is disrupted.

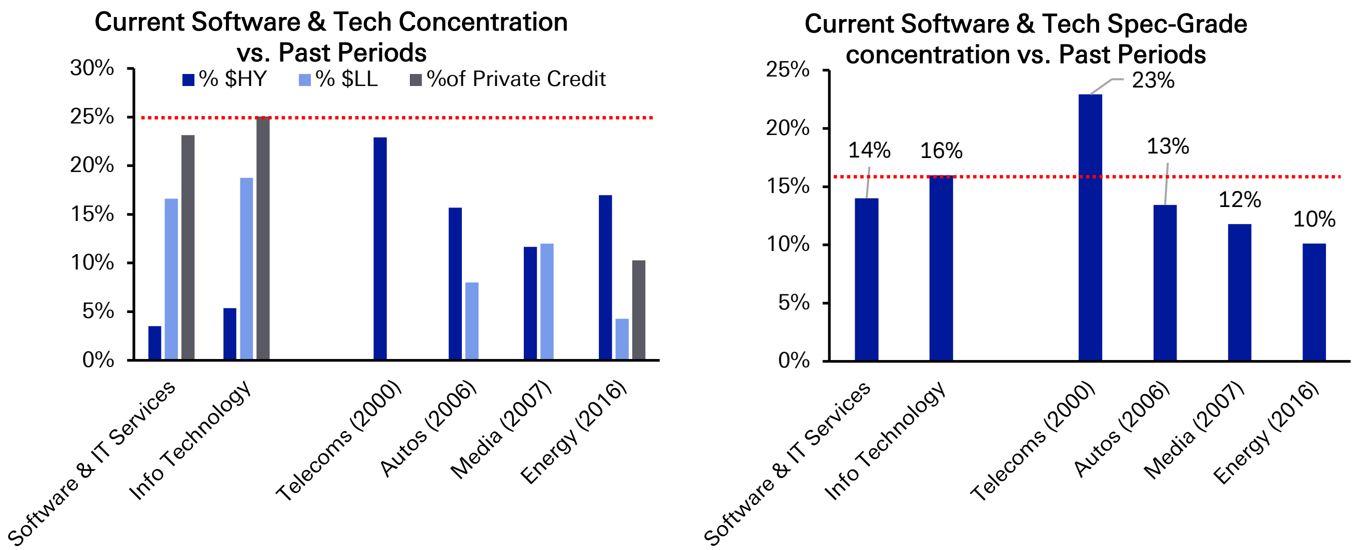


Source : Deutsche Bank, SaaS Capital Index

The challenge of course is timing: When will this AI disruption will actually show up in software company fundamentals? To-date, one can see in Figure 25 that while software multiples have collapsed, software revenue growth rates have remained stable. This is one reason why rating agencies and private credit funds (such as BDCs) are not reacting to the large drop in secondary prices within the leveraged loan market. The answer on the timing question will determine whether software modestly or materially tightens US spec-grade financing conditions over the next year.

But structurally, it is useful to consider the transmission channel of how software stress would spread in a manner that could matter meaningfully to both credit & multi-asset investors. The first point is that software is a very large and consequential sector in historical context. Software comprises a larger share of US spec-grade credit today than Shale Energy did in 2016. And in private credit specifically, software & software-related names comprise a greater share of the asset class than the Telecom sector did to $HY in 2000 (Figure 26).

Figure 26: The Software and Technology exposure within US Private Credit is larger than Telecom's exposure within $HY in 2000. Across all of $-spec-grade credit (HY, LL, Private Credit), software & tech exposure will be more impactful than Shale Energy in 2016



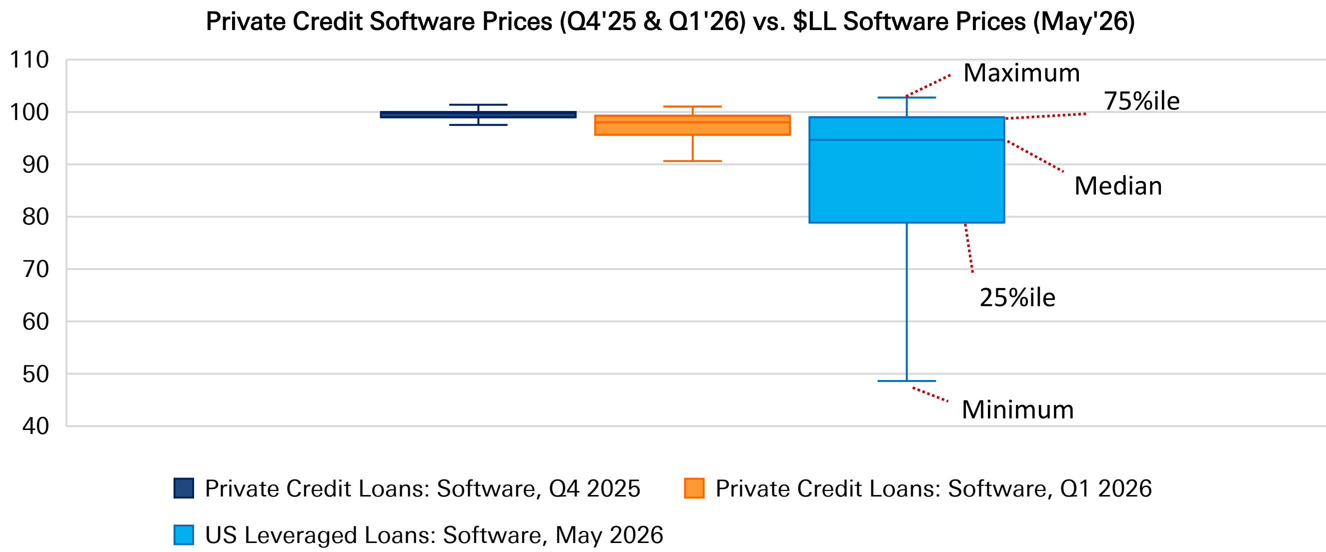
Source : Deutsche Bank, Bloomberg Finance LP, Pitchbook
**Private Credit estimate is based on projected investable assets relative to observable BDC sector concentrations*

But there is still inertia in the overall credit ecosystem that is preventing this software stress from propagating more broadly. To us, this inertia cannot last forever.

The first unstable equilibrium can be found in US private BDCs, where still stale software loan markets are leading to liquidity pressures and storing up future problems for the asset class. We can first see this within US private BDCs (Figure 27), where we analyse the 7 largest private & "semi-liquid" BDCs in the market today. After combing through their 10-K & 10-Q filings, we see that any software loan markdowns taking place are very gradual and ultimately insufficient (Figure 27).

For example, through the end of Q1'26, the median private credit software loan was still valued at a price of 98, while the 25th percentile of private credit software loans was still valued at 95.6. This is far more optimistic than what the very similar US leveraged loan market is signalling. The median US Lev Loan software name trades at 95.2, while the 25th percentile of US Lev Loan software names trades at 78.9.

Figure 27. The private credit market is pricing that AI will only have a minor impact on leveraged software names. The reality is more likely priced in Leveraged Loans, where the impact is more dispersed, but far more material*



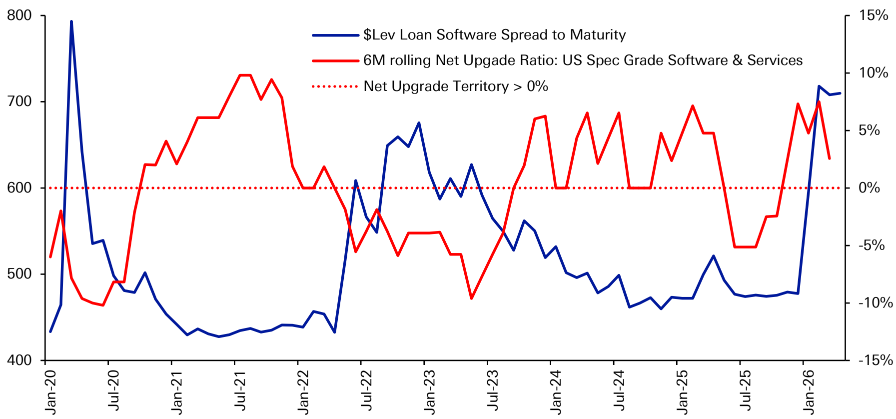
Source : Deutsche Bank, Bloomberg Finance LP, SEC EDGAR Form 10-Ks
**Private Credit Software loans were pulled from the 10-K and 10-Q filings of the following 7 private BDCs. Private Credit loan price is calculated as Fair Value/Par Value. US leveraged loan software prices are pulled from the Bloomberg US Leveraged Loan Index. The plots above exclude outliers that are more than ± 1.5x the interquartile range (75th - 25th percentile).*

This inertia in private credit market pricing is quite striking in the tails of the distribution. Careful readers will just have noted the 25th percentile of private credit software names are marked with a higher valuation than the median US Leveraged Loan software name. And when we look at distress ratios, the story is similar. Only 5.8% of private credit software loans were valued at a price less than $80 at the end of Q1, compared to 25.6% of US Lev Loan software loans today that are now trading below $80.

Ultimately, US private BDCs continue to price a very tight software distribution, in essence implying that AI will have a very broad, but ultimately, minor impact on the entire software space. We suspect the reality will entail a lot more dispersion, with a much fatter left tail for private credit software companies.

But the second unstable equilibrium is coming from the rating agencies. Despite a massive widening in software spreads, US software names are still currently in net upgrade territory over the last 12 months, as proxied by S&P (Figure 28). This disconnect is clearly unsustainable. Either software spreads will tighten notably from here, or software downgrades will pick up more broadly in earnest.

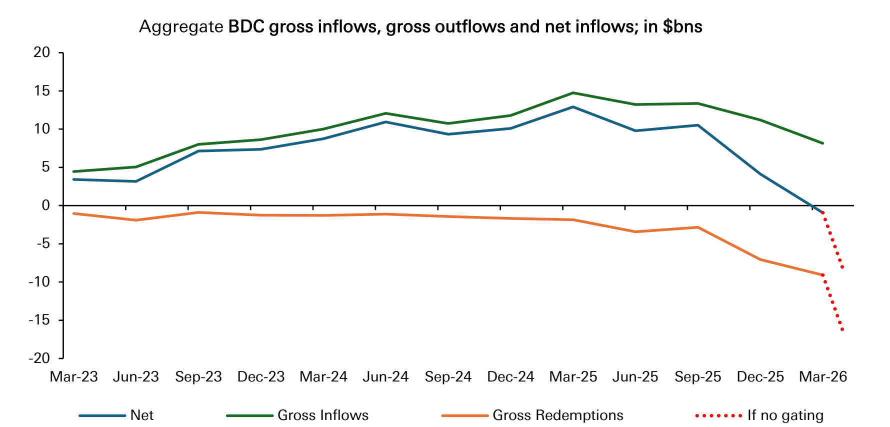
Figure 28: Software spreads have surged YTD, but rating agencies still have the spec-grade software sector in net upgrade territory. This divergence won't last for long.



Source : Deutsche Bank, Pitchbook

These unstable equilibriums will eventually prove problematic. The short-term problems from stale private credit software loan marks will continue to feed into US BDC liquidity pressures. Private BDC investors will continue to utilize their "first-mover advantage" and withdraw their capital, before private BDC software loans, and eventually NAVs, are marked lower in future quarters. We already experienced a wave of redemption requests in Q1, but we would expect new inflows to slow down dramatically from here as well, forcing BDCs to 1) increasingly issue more debt, 2) tap bank revolvers, and 3) sell leveraged loans to raise needed cash to meet growing net outflows. This part of the story is already in play, but will likely get worse in future quarters (Figure 29).

Figure 29: Continued divergence in private vs. public market pricing is likely to drive greater net outflows from BDCs in the quarters ahead, mainly as gross inflows into private BDCs now slow.

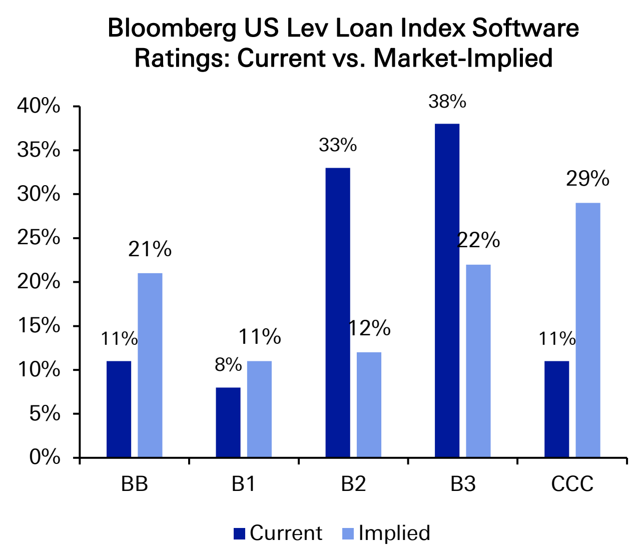


Source : Deutsche Bank, SEC EDGAR
Includes 7 private BDCs and 1 interval fund. Pulling from 10-Q and 8-K filings as of Q1 2026, and from PORT-P filings as of Q4 2025.

But an actual deterioration in software fundamentals is likely to be the catalyst from here that could meaningfully tighten financial conditions, as the US rating-agencies and third-party valuers assessing private credit fund portfolios quickly adjust their models. Again, the timing here is uncertain. But actual evidence of slowing revenue growth, lower free cash flow generation, and/or reduced customer retention ratios will reduce the ability of many software companies to refinance upcoming maturity walls, certainly increase rating agency downgrades.

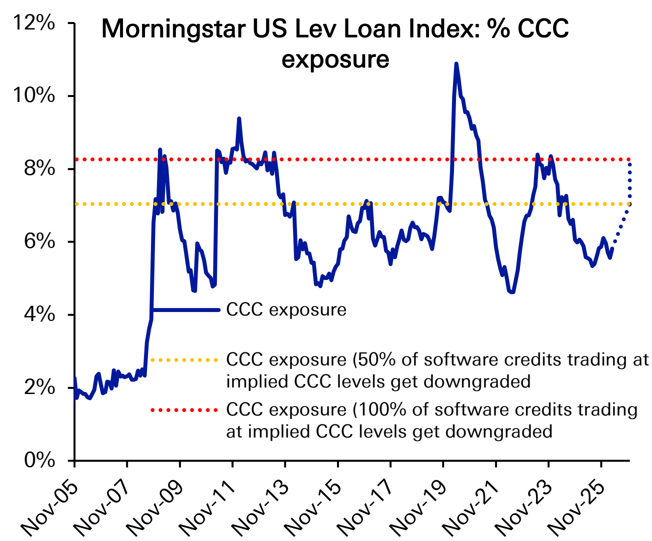
And if this fundamental pressure does emerge, the downgrade pressure on broader-spec grade credit could be significant. Based on current market pricing, we estimate that 29% of current US Leveraged Loan software names are trading with an implied CCC rating, vs. only 11% rated as CCC by the rating agencies (Figure 30). Given the large weight of software (~13%) in the US Lev Loan index, this could spike the share of CCCs in the Loan Index to 7-8.5% (from 5.8% today, Figure 31), on par with 2008-2009, 2011-2013, 2016-2017, 2020 & 2022-2023 levels (Note this 7-8.5% CCC basket is just looking at the software downgrade risk in isolation, without accounting for any additional downgrades hurting other sectors, whether from direct AI disruption of non-software sector business models to indirect AI disruption tightening financial conditions through tighter Fed policy and higher rates).

Figure 30: Market prices imply that nearly 30% of US software lev loans should be CCC-rated*



Source : Deutsche Bank, Bloomberg Finance LP, *market-implied ratings are calculated by comparing current individual $LL software prices vs the average price per $LL rating bucket. We assign a market implied software rating based on the average rating bucket price that is closest to the individual software loan price

Figure 31: If true, the share of CCCs in the US Lev Loan index will swell. That is the risk of the largest sector in the index facing AI disruption ahead.



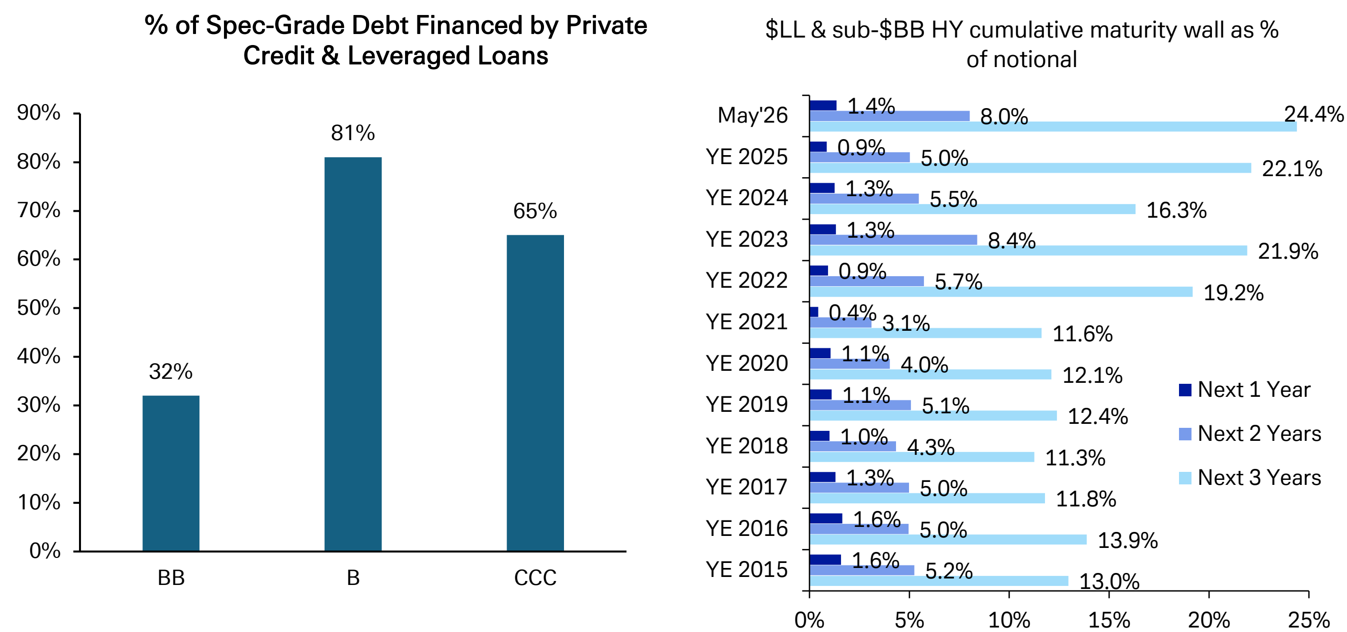
Source : Deutsche Bank, Pitchbook

And from the third-party valuer who assess the veracity of private credit loan marks, it is a very similar dynamic. Third party valuers effectively use calibration models to assess the "fair-value" of a private credit loan each quarter, using both market-wide macro inputs (such as traded loan index and sector spreads) and company-specific fundamentals to inform their view.

Currently, the dramatic widening in US Leveraged Loan software spreads is factoring into modest private credit loan markdowns today. But 1) the health of the broader Leveraged Loan Market (ex-software) & 2) the lack of any tangible impact on software fundamentals yet is limiting the amount of loan markdowns that private credit funds have to take. A raft of weak software earnings coupled with rating downgrades and rising CCC buckets in US Loan indices would lead both of these macro & company specific fundamental factors to deteriorate, instigating a far more rapid pace of private credit software loan markdowns than what we have experienced to-date.

To sum it up succinctly, we suspect software stress is likely to spread more broadly, within both private credit & leveraged loan markets. With 80% of all $B rated debt financed in either private credit or leveraged loans, the reality of still near record high T+2 & T+3 maturity walls for $B & CCC rated credit highlights that this is a risk not to forget (Figure 32).

Figure 32: 81% of all B-rated financing in US spec-grade credit comes from floating-rate Leveraged Loans or Private Credit. A problem if these markets face more pressure, given maturity walls 2 & 3 years out are at all-time highs.



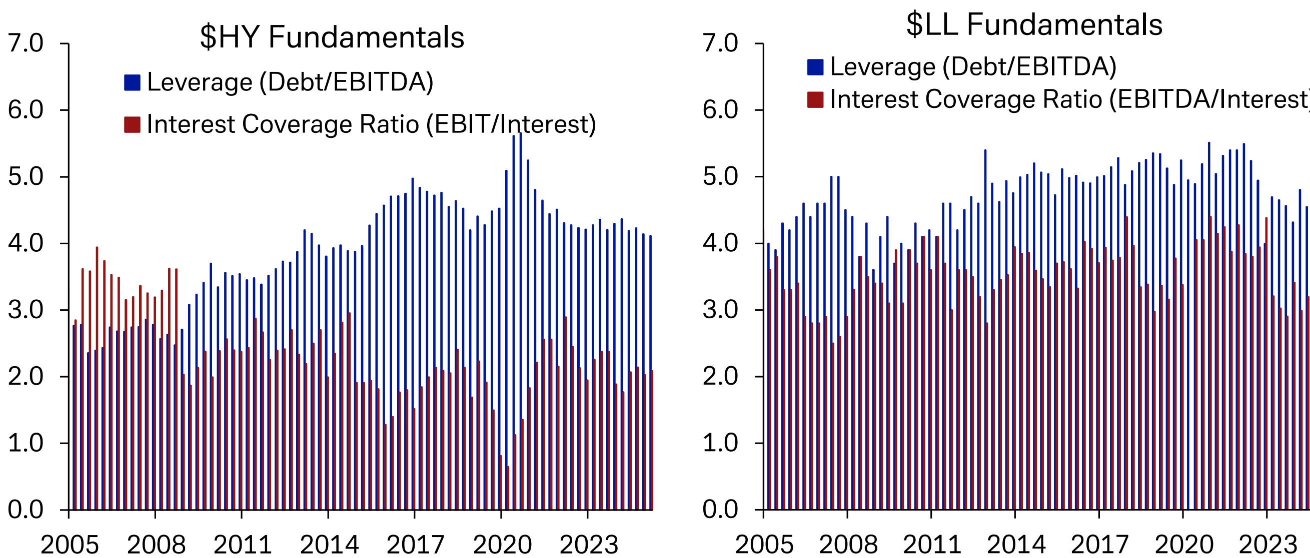
Source : Deutsche Bank, S&P Capital IQ Pro, Bloomberg Finance LP, Pitchbook

What markets/sectors are most exposed to upcoming risks?

As laid out above, **there are many intertwined risks that investors have to navigate in the months ahead**. If rising inflation leads to a more hawkish Fed, then the expectation for higher-for-longer financing costs could lead multiple contraction. Higher energy and commodity costs can also compress margins as consumers have less fiscal headroom than in the recent past. While AI-led US growth has powered the US economy, unless it broadens out there could be greater earnings growth dispersion.

$HY and $LL leverage had been rising in the years following the Financial Crisis until the post-Covid period. Since 2021, there has been a gradual easing in leverage. However, **even as leverage has gently eased, interest coverage ratios have remained quite contracted** over the last two years. This leaves spec-grade credit quite sensitive to higher rates moving forward. This is not a perfect comparison between markets given data constraints, but the trends are informative. The $HY data is the median issuer per quarter, while the Lev Loan data is an aggregation at the index level for new issuance.

Figure 33: USD HY and LL leverage continues to ease from Covid-era highs, but interest coverage ratios have not improved in a similar fashion

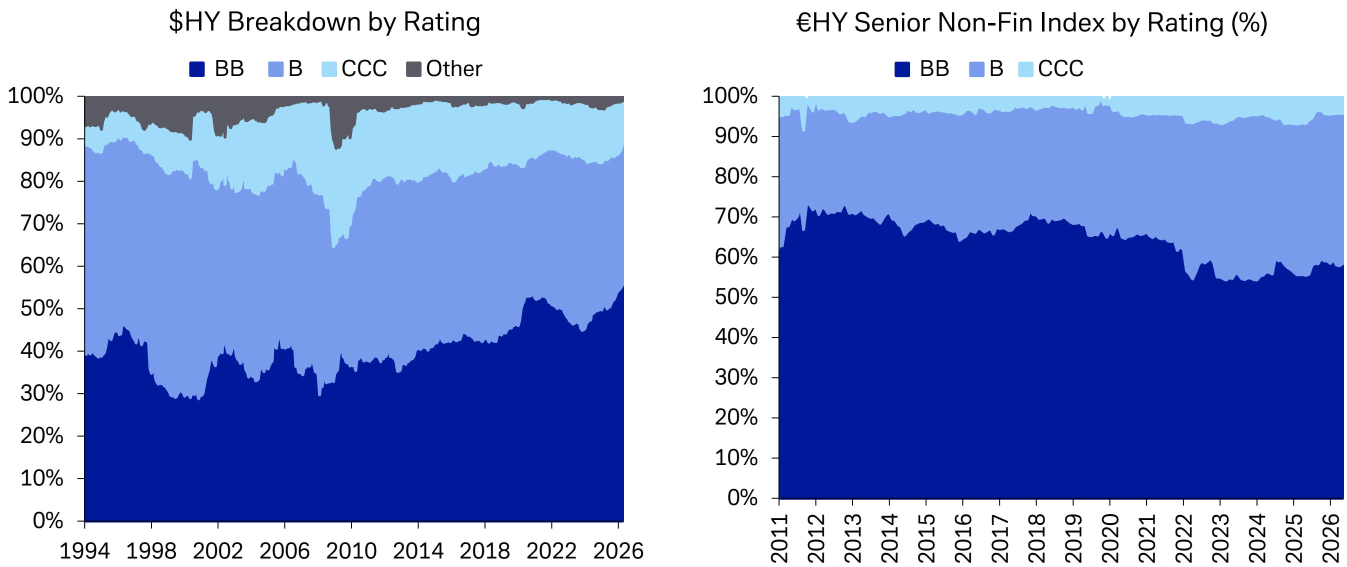


Source : Deutsche Bank, Bloomberg Finance L.P., Pitchbook LCD
$HY data uses the median issuer by quarter

Fixed-rate High Yield and floating-rate Leveraged Loan markets continue to grow further apart in average rating. $HY markets have further improved in rating quality over the last year. **$BBs are a record share of the $HY index at 56%**, while $CCCs now only make up 10% of the index. That is the lowest share of $CCC bonds since 2002. The improvement in $HY rating quality is due to multiple factors. One is that the HY market is looking to finance higher quality issuers. There is also a greater sum of subordinated-debt from $IG rated issuers. And lastly, there has been many recent large downgrades over the last two years that have introduced significant $BB debt into $HY.

€HY rating quality is slowly going the other way. Around 60% of Non-Financial seniors are rated €BB, which is down nearly 10pp from the highs in 2018. €CCCs still represent a smaller portion of this market with just 4% of notional value. As the level of €B debt grows, it introduces the potential for European defaults to move slightly higher structurally.

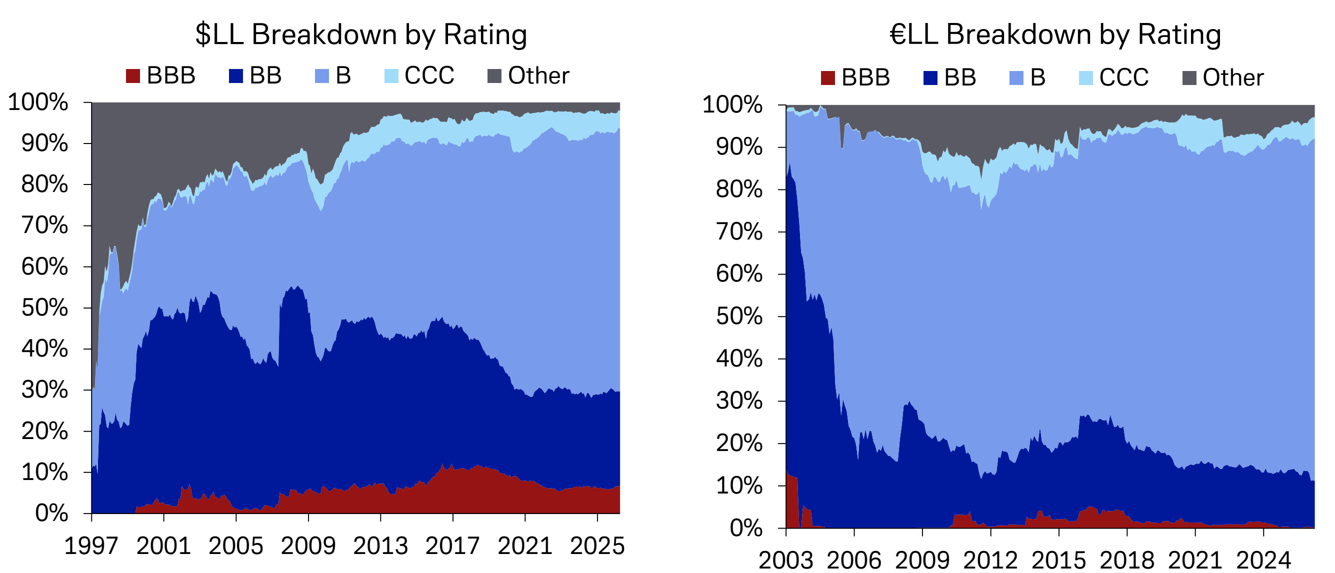
Figure 34: While $HY bond markets have steadily increased in credit quality over time, €Bs have become a more substantial part of €HY.



Source : Deutsche Bank, Bloomberg Finance L.P., ICE Indices, S&P

$ & € Leveraged Loan markets have become largely dominated by B-rated issuers. The share of $B Loans is up to 64% from 45% in 2015, having been just shy of two-thirds of the index since 2022. Meanwhile €B Loans are a record 81% of the market after being 66% of the market at the end of 2015. This shift of rating quality lower in loans relative to HY partly explains the divergent default rates between the two markets during the post-Covid cycle.

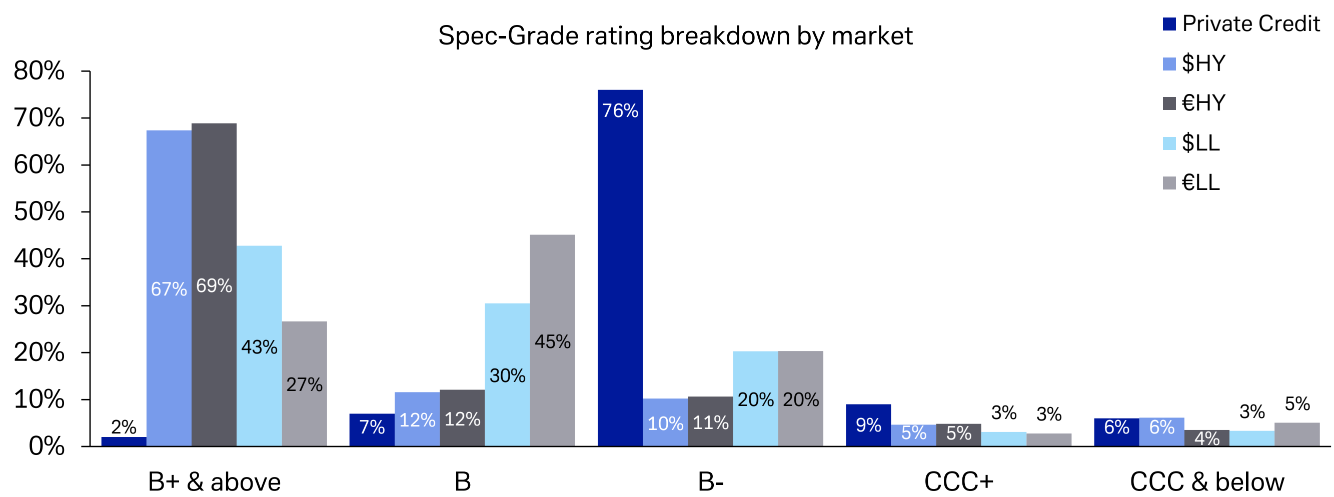
Figure 35: While Leveraged Loan indices have long been overweight B-rated debt, we are at a record share in EUR LL and just off the largest concentration in USD LL.



Source : Deutsche Bank, Bloomberg Finance L.P., ICE Indices

Looking ahead, **Private Credit is even more exposed** to acute risks. Both $ & € HY markets are over two-thirds rated B1 and above, with just over 20% B2 rated risk, and just around 10% of CCC-rated securities. $LL is better rated than €LL, with a plurality of debt rated B1 or better, while the largest share of €LL debt is rated B2. By contrast, **Private Credit is over 75% B3 rated**, just one notch away from a CCC-rating. Speaking of which, **Private Credit also has 15% CCC or worse rated debt**, while $HY markets is the closest public market with 11% of notional marked below a B3-rating.

Figure 36: Private Credit is lower rated than Lev Loan indices with an extreme concentration of B3 rated debt

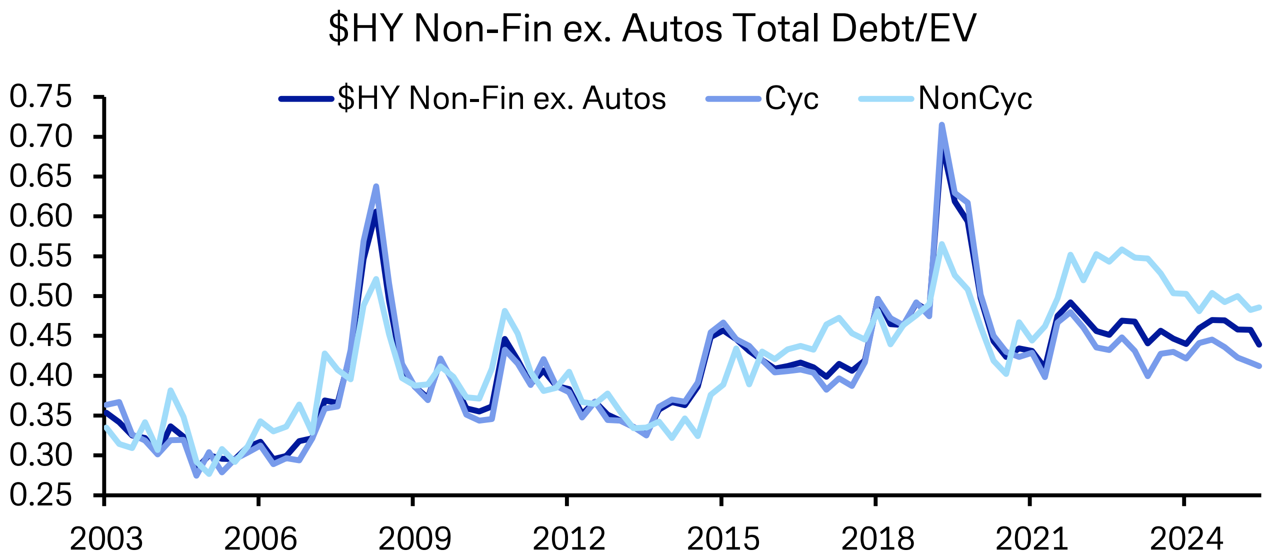


Source : Deutsche Bank, Bloomberg Finance L.P., Pitchbook, S&P Cap IQ

Which sectors are more exposed to defaults and fundamental risks?

We have found that debt to enterprise value ratios provide a better leading indicator of future downgrade and default rates than other leverage measures such as debt to EBITDA. This is based on a Merton Model framework, where falling equity valuations can create incentives to restructure the firm's capital structure, often to the detriment of credit holders. **Debt to EV ratios at the index level have been improving marginally over the last two quarters.** While the median $HY member may not necessarily be seeing the same multiple expansion that is evident in the S&P 500, $HY valuations have still been buoyant. Meanwhile debt levels have not risen substantially over the last few years. As such Debt/EV ratios are near the bottom of their range since the end of 2021, with Cyclical firms seeing more improvement than Non-Cyclicals.

Figure 37: The recent risk rally has caused $HY cyclical leverage to move toward post-Covid lows, even as Non-Cyclical leverage remains steady over the last year



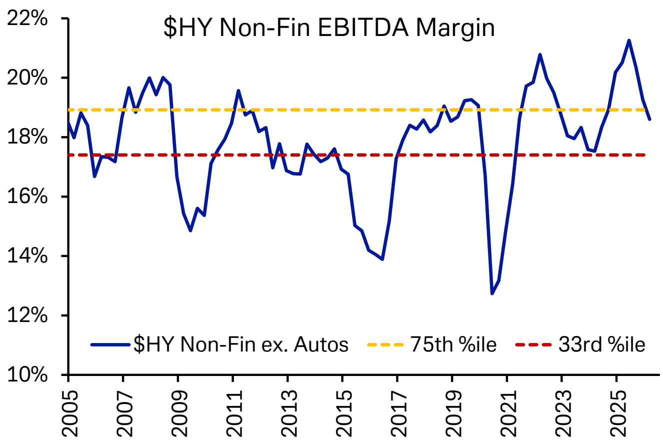
Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

Looking ahead to the next 12 month, **multiple risks can drive greater dispersion**. The energy shock and rising inflation worries are translating into higher US Treasury yields which can lead frothy multiples to re-rate lower. Rising energy and goods costs can also weigh on margins, which could lead to higher leverage for Spec-grade corporates in the coming quarters.

In order to model how these factors can effect leverage across sectors, we have created a scenario analysis in Figure 40 below. **The first scenario is one in which the Strait of Hormuz opens up shortly** and there is a smaller commodity shock, keeping margin compression moderate. We also would likely see less inflationary pressures under this scenario and therefore less multiple compression as the Fed does not grow as hawkish as they might otherwise. We grow current revenue by the median 12-month revenue expectation in each sector, and then estimate EBITDA growth based on the 75th %-ile observed profit margin since 2004. We then re-rate the sector at the 75th %-ile EV/EBITDA. If current multiples or margins are lower than the 75th %-ile historically, we just use the current levels.

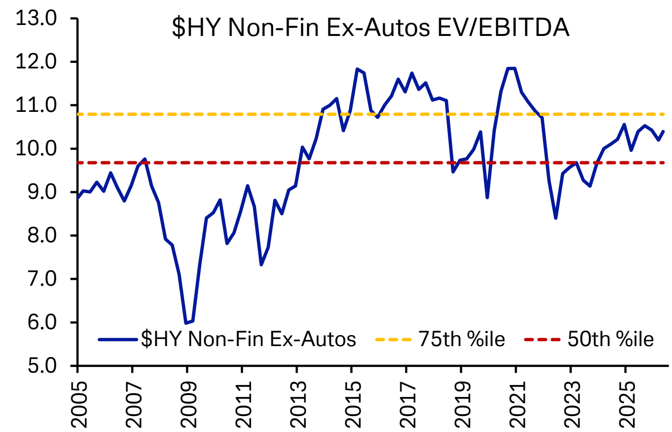
In the second scenario, there is a more sustained commodity shock, wherein oil prices are higher-for-longer, likely leading to higher yields on the back of higher inflation expectations and rate hikes. We still use the current revenue growth forecast but then contract multiples to their long-term median and margins to their 33rd %-ile unless they are already lower. For context, **Figures 38 & 39** shows what these levels look like relative to historic $HY equity multiples and EBITDA margins.

Figure 38: $HY EBITDA Margins have compressed over the last 2 quarters before any increase from the current commodity shock



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

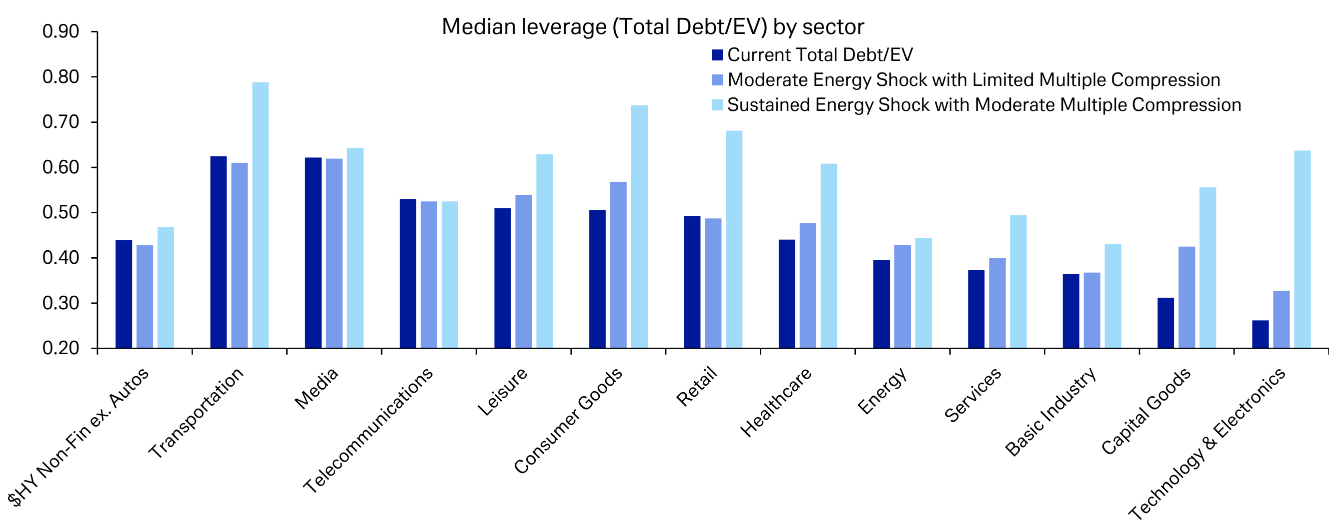
Figure 39: Equity multiples of the median $HY issuer are near the 75th %-ile historically, and had moved toward the median level during recent hiking cycles



Source :Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices

As we laid out above, $HY Cyclical sectors are starting out with better leverage metrics while some Non-Cyclical sectors have higher outright leverage or high leverage in their own history as seen in Figure 37 and Figure 40. **Telecoms, Consumer Goods and Healthcare** stand out as the main Non-Cyclical risks to watch, while **consumer-oriented Cyclical sectors and Transports** are also displaying higher leverage.

Figure 40: While Transports, Media, and Telcos represent the largest near term risks, a sustained energy shock could impact Consumer-oriented Cyclicals more

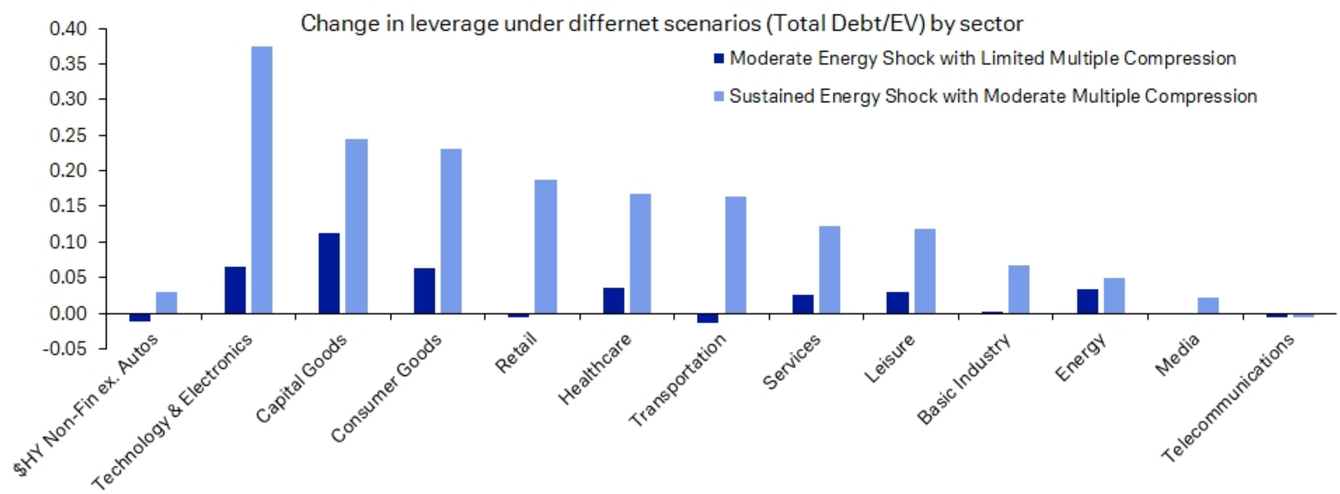


Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices
Moderate Energy Shock with Limited Multiple Compression = Margins compress to Min(current, 75th %ile since 2004) and Multiple compress to Min(current, 75th %ile since 2004) .
Sustained Energy Shock with Moderate Multiple Compression = Margins compress to Min(current, 33rd %ile since 2004) and Multiple compress to Min(current, Median since 2004)

Under our first scenario, the moderate commodity shock with limited multiple compression**, lowly-levered cyclicals see the biggest jump in leverage** on the back of both compressed margins and re-rated multiples. $HY Capital Goods, Technology, and Consumer Goods see the biggest jump in leverage.

Capital Goods and Technology currently enjoy relatively low leverage levels, but come under pressure quickly even in the more mild scenario. While there is some dispersion under the first scenario, the second more sustained price shock leads to a wider spread increase in Spec-Grade leverage across sectors. Nonetheless, the more adverse scenario still sees Cyclicals and Consumer-Oriented sectors underperform.

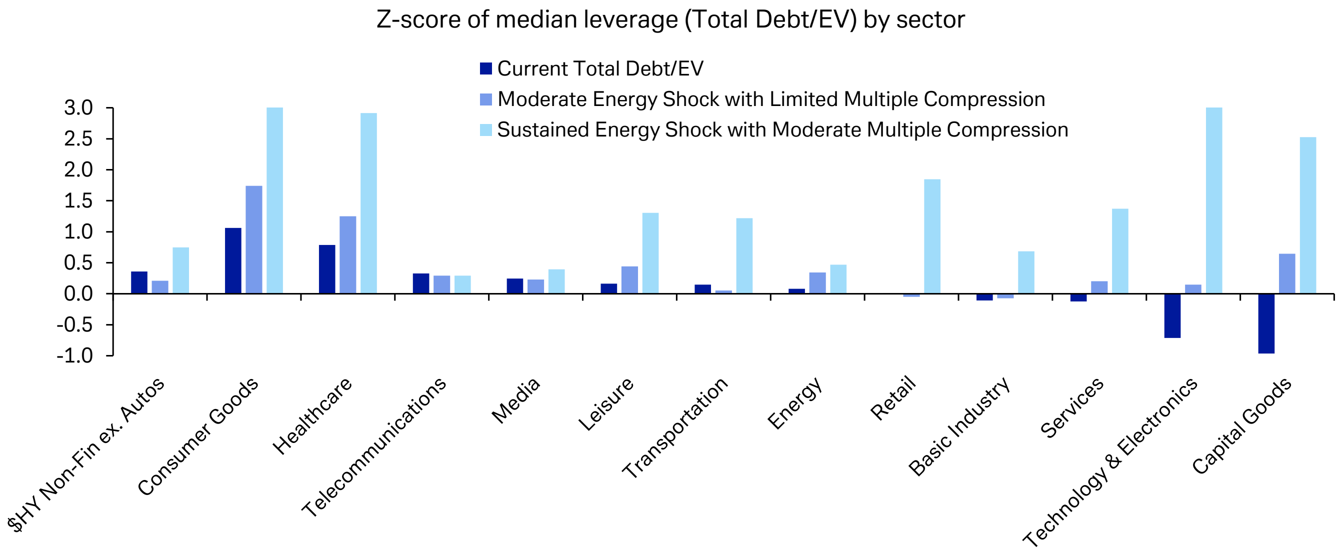
Figure 41: Technology, Cap Goods, and Consumer Goods would see the biggest impact from both a moderate and sustained energy shock and higher-for-longer rates compressing multiples



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices
Moderate Energy Shock with Limited Multiple Compression = Margins compress to Min(current, 75th %ile since 2004) and Multiple compress to Min(current, 75th %ile since 2004) .
Sustained Energy Shock with Moderate Multiple Compression = Margins compress to Min(current, 33rd %ile since 2004) and Multiple compress to Min(current, Median since 2004)

Viewing the scenario analysis in Z-score space offers a potential view into how risks may appear in the $LL and private-credit markets. Since we do not have visibility into how enterprise values are developing in these markets, tracking the trends in public HY markets can inform us on where risks are trending. Using the framework and scenario analysis above, **Non-Cyclicals are the main risks to monitor in the current environment**, with greatest risk in Consumer Goods, Healthcare, and Telecoms. **US Cyclicals would face much greater risks in adverse scenarios** if multiples contract and rising energy costs cause margins to compress substantially, though Healthcare and Consumer goods would still be large risks.

Figure 42: When looking at leverage in their own histories, Non-Cyclical sectors such as Consumer Goods, Healthcare, and Telcos are high, with the former two sectors seeing more stress under our adverse scenarios as well



Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE indices
Moderate Energy Shock with Limited Multiple Compression = Margins compress to Min(current, 75th %ile since 2004) and Multiple compress to Min(current, 75th %ile since 2004) .
Sustained Energy Shock with Moderate Multiple Compression = Margins compress to Min(current, 33rd %ile since 2004) and Multiple compress to Min(current, Median since 2004)

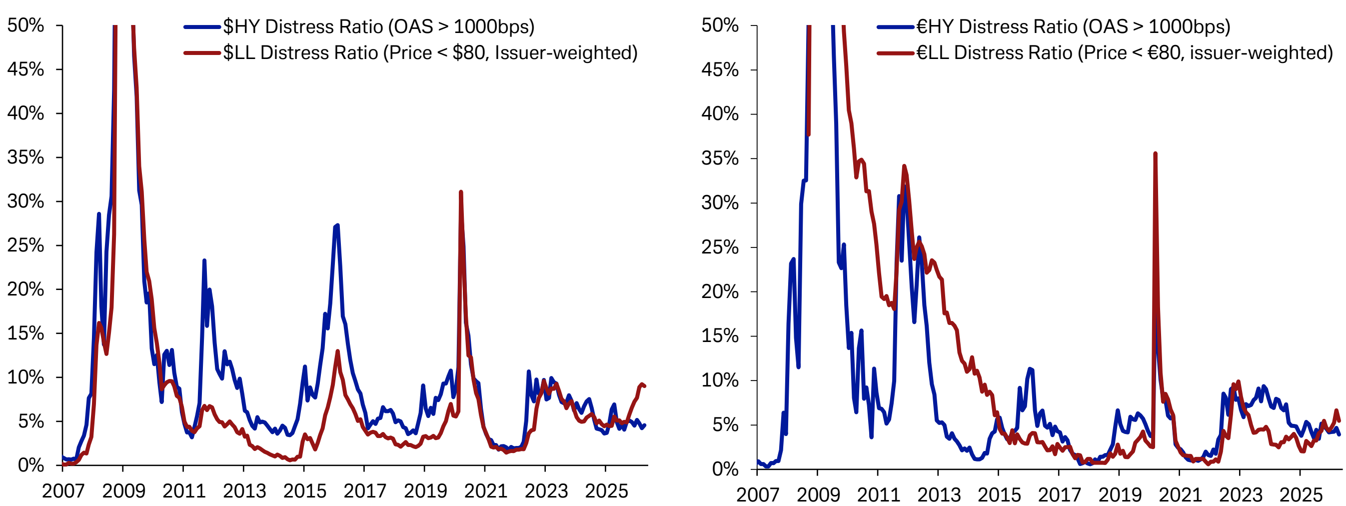
What's priced into credit?

LL distressed ratios are rising while HY ratios have moderated

How do our forecasts for US and EUR spec-grade default rates over the next 12 months compare to what investors are pricing into Leveraged Finance markets currently? We anticipate that US and European defaults will slow to 3.6% by year-end before climbing slowly back to 4% in the US and 3.9% in Europe by this time next year. **While overall index spreads are trading near historically tight levels, distress ratios are not similarly near their lows**. This reflects that investors have deemed pockets of debt quite risky and historically elevated distress ratios have been a good indicator of restructuring activity in 9-10 months.

$HY and $LL distress ratios started to diverge in October of last year. This coincides with investors becoming more wary of private credit risks following the events surrounding Tricolor and First Brands. The level of distressed issuers rose as AI-disruption became a theme early this year, sending some $LL Software prices significantly lower. Through all of this, $HY distress ratios have remained resilient near post-GFC lows, when discounting the 2021-22 period when there was ample fiscal & monetary support. **This reflects the nature of where defaults are likely to be concentrated in the coming months**, as LL markets currently have more of the structural risks we outlined above (more software and lower rated). €LL markets are smaller than the other spec-grade markets, but there is a similar dynamic of increasing floating-rate risk relative to €HY markets where distress ratios are pricing in less imminent default risk. Nonetheless, €Spec-Grade distress ratios are elevated relative to the post-European Debt Crisis lows, as investors expect moderate defaults moving forward.

Figure 43: HY index distressed ratios have decreased over the last few years as index quality improved and issuer stress became more localized; meanwhile LL distress ratios have trended higher in recent months due in part to percolating software and private credit worries

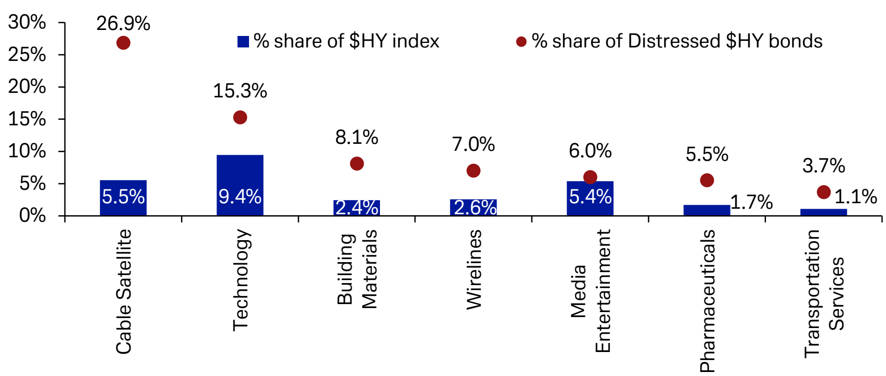


Source : Deutsche Bank, S&P, Bloomberg Finance LP, ICE Indices

USD Distressed Debt is highly concentrated, while there is greater dispersion in EUR markets

Delving into where distressed debt resides in each market, **risks are far more concentrated in USD Spec-Grade markets**. Technology/Software has the largest concentration of distressed debt, with significant Telecom risks a close second. Further away, there are signs that Consumer Cyclical as well as Healthcare risks are percolating. Within $HY bonds, Cable & Satellite (27%) makes up over a quarter of distressed debt. Adding in Wirelines (7%), means that Telecoms make up over a third of $HY debt trading wider than 1000bps. The recent software weakness weighs on $HY Technology (15%) which has the second largest sum of distressed bonds with Media (6%) the fifth largest but nearly in-line with the sector's overall weight in the index.

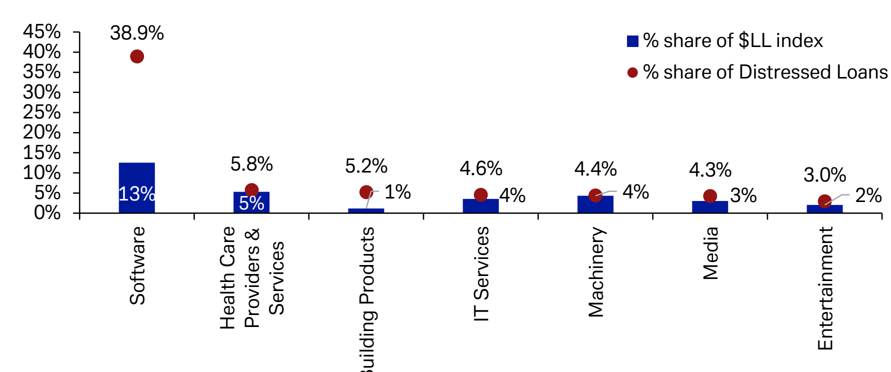
Figure 44: $HY Cable Satellite is the main sector with an outsized share of distressed debt followed by Technology.



Source : Deutsche Bank, Bloomberg Finance L.P.

In $Leveraged Loans, the Software sector remains the main story. It makes up roughly 13% of the $LL market and has nearly two-fifths of the distressed debt. At nearly 39%, its share of distressed loans is basically triple its weight in the index. Building Materials is the only other $LL sector with outsized risk priced in, but it is a much smaller sub-sector.

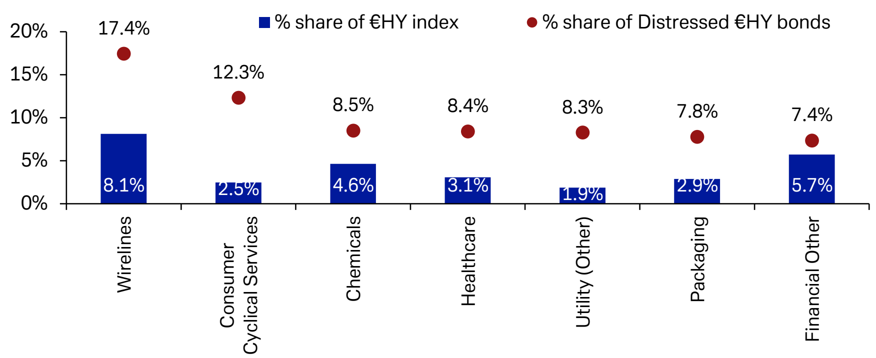
Figure 45: Software continues to have the largest share of distressed debt in the $Lev Loan market.



Source : Deutsche Bank, Pitchbook

Within €HY & €LL, the risks are far more dispersed. Risks are also being priced into both Cyclicals and Non-Cyclicals. Wireless issuers have over 17% of distressed bonds to lead €HY, with fellow Non-Cyclical sectors Healthcare (8%) and Non-Electric Utilities (8.3%) also in the top 5. On the Cyclical side, Consumer Cyc Services (12%), Chemicals (9%), and Packaging (8%) round out the other €HY subsectors with significant outsized priced-in default risk.

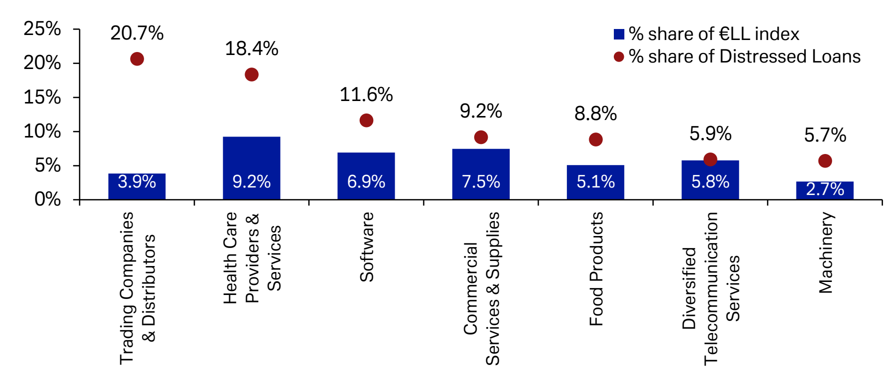
Figure 46: €HY distressed bonds are spread between Cyclical and Non-Cyclical issuers, with the greatest share in Wirelines



Source : Deutsche Bank, Bloomberg Finance L.P.

€Leveraged Loan distress debt is a much smaller market and the risks are more concentrated. Trading Companies and Distributors (21%), which highlights how soaring energy prices and supply chain issuers could be impacted by the conflict in the Middle East. Healthcare (18%) has a slightly lesser share of distressed loans as Trading Companies, with Software (11.6%) the next largest concentration.

Figure 47: €LL distressed loans are more concentrated in Trading Companies and Healthcare, with some Software exposure as well



Source : Deutsche Bank, Pitchbook

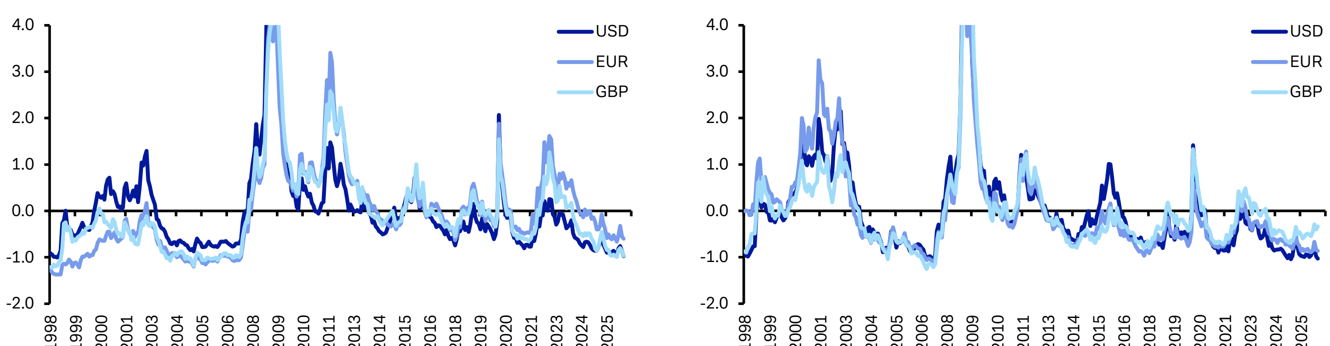
Non-financial (cash) spread implied default rates

This final section was the initial focus of these annual studies. The purpose was to assess the level of credit spreads required to compensate buy-and-hold HY investors for default risk. We also analyze what level of defaults are currently implied by spreads, and how much excess spread is being provided to investors under different default scenarios.

The analysis focuses on Non-Financials around the 5-year tenor. To achieve this, we include all senior bonds 4-6 years until maturity within the ICE indices. Having calculated the average spreads for our sample bond sets, we use three different recovery assumptions (0%, 20% and 40%) to calculate implied default rates. We then compare these implied default rates with the appropriate average five-year cumulative global default rate, as well as the worst five-year periods observed since 1981. This is also compared against the average and worst five-year level for the primary region of the various bond markets (Europe – EUR; UK – GBP; US – USD). We have highlighted in Figure 49 the cells where the implied default rate is lower than the average experience (darker blue) as well as those where the implied default rate is lower than the worst but higher than the average (lighter blue). The shading is based on regional default rates. This analysis should be viewed from the perspective of a longer-term buy-and-hold investor.

HY spreads across USD and EUR in particular are trading near multi-decade tights. The Q1 volatility stemming from AI-disruption fears and then the fallout of the US-Iran war has dissipated, with **spreads returning to near their long-term tights**. The solid economic growth that we referenced above as well as elevated yields have meant investors are looking through some of the percolating risk factors. Given the starting point for spreads, we do expect greater decompression and dispersion as margin pressures and higher-for-longer rates set in.

Figure 48: Z-scores of IG (left) and HY (right) credit spreads since 1998 highlight just how tight valuations currently are, especially USD and EUR HY markets



Source : Deutsche Bank, Bloomberg Finance LP, ICE Indices

With the current rich valuations in mind, we look at how spread implied defaults compare to past periods. As to be expected, IG-rated implied default rates are going to be well above even the worst observed defaults for all recovery assumptions. Current $IG spreads imply a 3.9% cumulative five-year default rate, assuming a 40% recovery, and a 6.4% default rate with 0% recovery. This premium is due to spreads also pricing in some liquidity and downgrade risks that are not captured by the implied default equation we lay out in the appendix.

Even as overall HY index spreads are near historical tights, current 5yr spreads are still pricing in an above average default rate across markets. This is even when we assume 40% recoveries. **Assuming higher recoveries, the implied 5yr cumulative HY default rate is around 20% or better across all three markets, which is still above the average cumulative five-year default rate in each region by a decent margin**. Though it remains well below the worst-case scenario defaults, which would approximate a recessionary environment in the next five years. Current **CCC spreads in both USD and EUR are too are too tight** to adequately compensate investors for average historical cumulative 5 year defaults assuming a 40% recovery rate. If recovery rates fall to near 20%, then not only do current CCC spreads not compensate investors but $HY and $B spreads are too tight to protect investors.

Figure 49: Non-Financial 5-year cumulative spread implied default rates based on different recovery assumptions

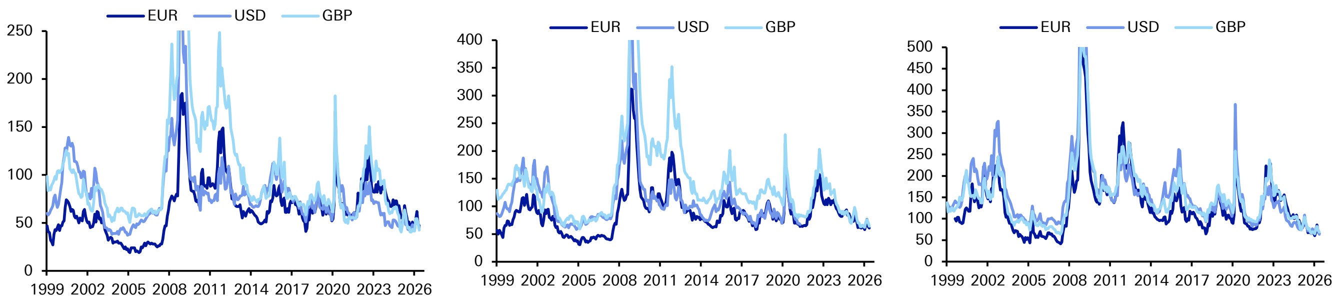
			Spread Implied 5yr Cumulative Default Rate			Actual 5yr Cumulative Default Rates			
						Global		Regional	
		5yr Spread	0% Recovery	20% Recovery	40% Recovery	Worst	Average	Worst	Average
EUR	IG	78	3.9%	4.8%	6.4%	2.9%	0.8%	2.4%	0.3%
	AA	44	2.2%	2.8%	3.7%	0.5%	0.1%	0.0%	0.0%
	A	64	3.2%	4.0%	5.3%	1.2%	0.3%	1.8%	0.2%
	BBB	90	4.4%	5.5%	7.3%	5.0%	1.6%	5.9%	1.0%
	HY	295	13.5%	16.6%	21.5%	32.5%	16.5%	38.8%	14.7%
	BB	214	10.0%	12.3%	16.0%	20.0%	7.9%	27.3%	7.3%
	B	382	17.2%	21.0%	27.0%	39.9%	19.3%	47.6%	17.5%
	CCC	753	29.5%	35.5%	44.2%	68.5%	50.5%	64.0%	41.5%
USD	IG	64	3.1%	3.8%	5.1%	2.9%	0.8%	2.4%	0.9%
	AA	29	1.4%	1.8%	2.3%	0.5%	0.1%	0.6%	0.1%
	A	46	2.2%	2.7%	3.6%	1.2%	0.3%	1.3%	0.3%
	BBB	79	3.8%	4.8%	6.3%	5.0%	1.6%	4.3%	1.6%
	HY	268	12.3%	15.2%	19.7%	32.5%	16.5%	30.9%	16.7%
	BB	167	7.9%	9.8%	12.8%	20.0%	7.9%	17.6%	7.8%
	B	327	14.9%	18.3%	23.6%	39.9%	19.3%	38.3%	19.0%
	CCC	746	30.1%	36.1%	44.9%	68.5%	50.5%	72.5%	52.3%
GBP	IG	79	3.9%	4.8%	6.4%	2.9%	0.8%	3.6%	0.4%
	AA	16	0.8%	1.0%	1.3%	0.5%	0.1%	0.0%	0.0%
	A	61	3.0%	3.8%	5.0%	1.2%	0.3%	3.2%	0.3%
	BBB	94	4.6%	5.7%	7.5%	5.0%	1.6%	7.3%	1.0%
	HY	402	17.8%	21.8%	27.9%	32.5%	16.5%	42.9%	16.1%
	BB	329	14.6%	17.9%	23.1%	20.0%	7.9%	30.0%	5.5%
	B	444	19.8%	24.1%	30.8%	39.9%	19.3%	42.9%	15.6%

Source : Deutsche Bank, ICE Indices, S&P
Note: Darker shading means implied default rate is lower than long-term global average. Lighter shading means implied default rate is lower than the worst historical global rate. As of 5/28/26

Non-financial (cash) default spread premiums (DSP)

Default spread premiums (DSP) provide us with a sense of the excess spread offered in credits over that which is required to compensate for different default scenarios. We start by looking at DSP histories by rating in the EUR, USD and GBP credit markets. They are calculated by subtracting the spread required to compensate for average (historical) default rates and assuming an average recovery rates (40%) from the index spread level. In Figure 50 we show this for IG ratings, where DSPs have always been positive.

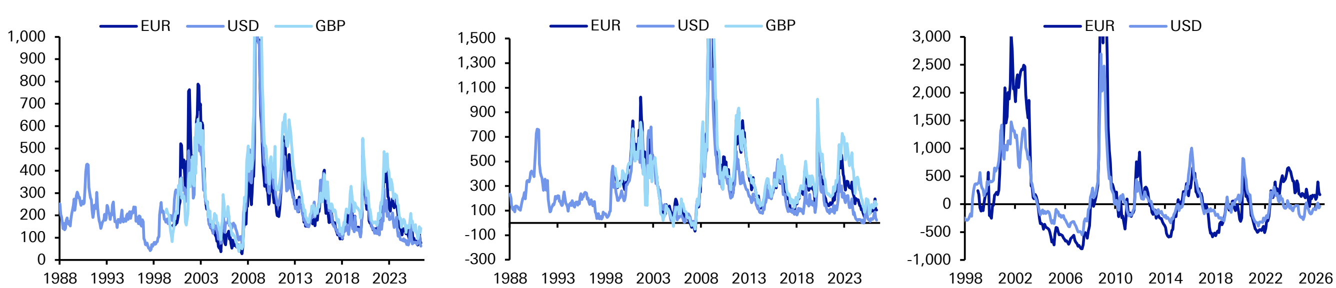
Figure 50: IG DSPs By currency and rating (bps) – AA (left), A (middle) and BBB (right)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE indices

In Figure 51, we show the same for ratings across HY. With spreads near historic tights, it follows that default spread premiums are near the lows of their long-term ranges. This indicates that **investors are being compensated for very little outside of default risk.** We appear to be in a period where liquidity, rating transition, and recovery uncertainty premia are being heavily discounted. The only ratings where DSPs are not at historic lows are EUR and USD CCCs. The former is a relatively small and idiosyncratic market, while the latter's DSP is still negative, meaning **$CCC spreads are not compensating for average default risk** and average recoveries at current levels for buy and hold investors.

Figure 51: HY DSPs by currency and rating (bps) – BB (left), B (middle) and CCC (right)



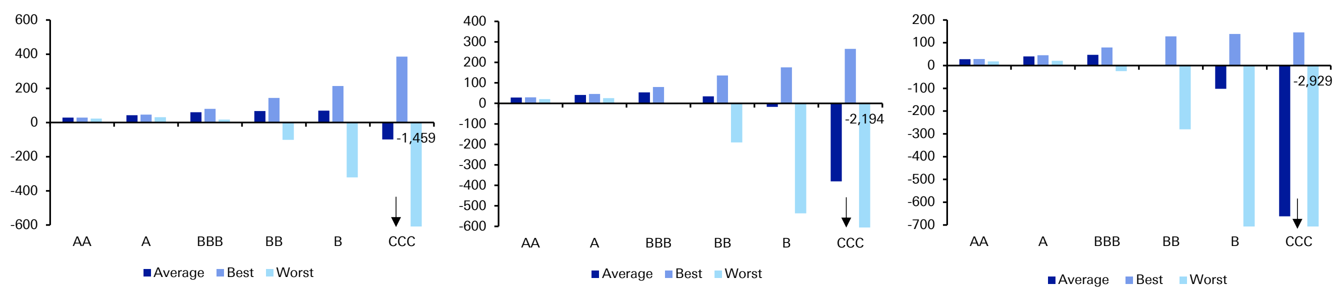
Source : Deutsche Bank, Markit Group, ICE indices

Focusing on the current market pricing, we consider **which rating buckets are offering the greatest DSP given different default and recovery scenarios**. Once again, this analysis is based around the five-year maturity point (focused on bonds with 4-6 years to maturity) and not using index spreads as a whole as we do in Figures 61 and 62. In addition to average default cycles, we also consider the worst historical cycle for each market as well as the most benign. Additionally, we introduce two more recovery assumptions (0% and 20% in addition to 40%). This framework essentially tries to assess the most attractive default-adjusted spread from a buy-and-hold perspective.

We start by looking at USD credit in Figure 52 and provide some of the key highlights below.

- Based on an average default cycle and 40% recovery, **single-Bs offer the widest DSP**, with BBBs seeing the largest premium if recovery levels were to fall.
- Under the worst-case default scenario, which would be a recessionary default cycle, IG generally offers the widest DSP under all recovery assumptions, with even BBs negative at 40% recovery.
- **CCCs only outperform from here under the most bullish scenarios**, wherein they offer a sizeable premium. But of course in the worst-case scenario, there is a substantial wipeout of returns. As laid out above, **current CCC DSPs are negative for an average historical default rate** even at a 40% recovery assumption.

Figure 52: USD non-financial default spread premiums for average, best (lowest) and worst (highest) 5-year cumulative default observations assuming 40% (left), 20% (middle) and 0% (right) recoveries

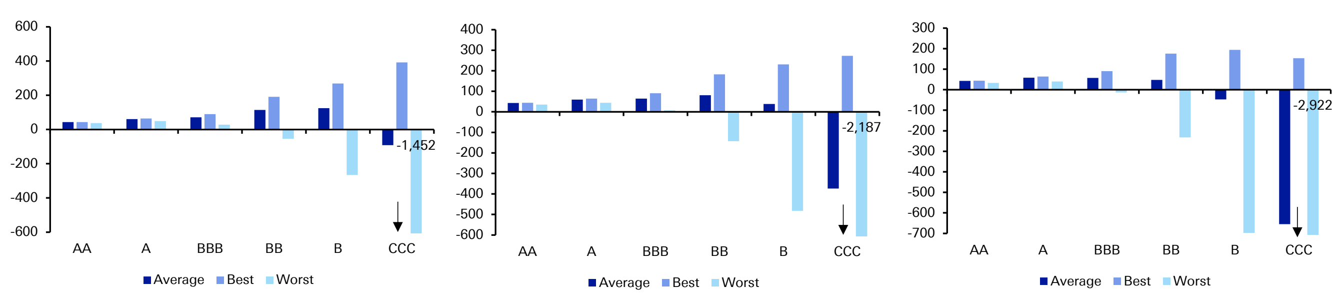


Source : Deutsche Bank, ICE indices

The EUR market analysis, laid out in Figure 53, has broadly similar conclusions.

- **Single-Bs provide the widest DSP** based on an average default cycle for the 40% recovery assumption scenario, single-As actually provide the largest premium if recoveries are 0%, while BBs offer the best at 20%. This speaks, in part, to how tight EUR BBBs trade.
- **CCC DSPs are negative even if recoveries are 40%** under average default scenarios, and grows substantially if recoveries are lower than that.

Figure 53: EUR non-financial default spread premiums for average, best (lowest) and worst (highest) 5-year cumulative default observations assuming 40% (left), 20% (middle) and 0% (right) recoveries



Source : Deutsche Bank, ICE indices

While we believe that defaults can moderate in the short-run, there are still risks we will see elevated cumulative defaults going forward. Especially if the Fed is not going to be easing policy in the short-to-medium term and there is potentially less fiscal support ahead if Congress becomes split in the latter half of President Trump's second term. Higher-quality, high yield bonds can offer a good level of buy-and-hold value despite potential spread volatility.

Cohort analysis

In this sub-section, we update our analysis assessing what's priced in to current spreads by comparing them with the spreads that would have been required to compensate for default based on all realized 5-year cohorts over the last 30+ years using different recovery assumptions. We look at this on a global basis.

Global cohorts

We start by looking at defaults globally with the analysis based only on non-financial issuers. We calculate spreads required to compensate for default based on each actual realised 5-year discreet cohort formed since 1981 (based on S&P data). Overall, this analysis is useful for showing where current spreads are across the board and in which periods through history these spreads would have (and would not have) compensated for the risk of default and by how much. We use both 40% and 0% recoveries, as exact recoveries by cohort are not available. This provides us with a compensation range if readers want to estimate for a different recovery rate.

In Figure 55, we show the spread required to compensate for each historical cohort across both IG and HY rating bands. We have drawn a box around each cell where the spread required to compensate for default is wider than the tightest current spreads (as shown in Figure 54) of the relevant rating band across currencies (USD, EUR, GBP).

Figure 54: Current Non-financial spreads (vs. governments) by rating (bps)

	Currency	IG	AA	A	BBB	HY	BB	B	CCC
Current	EUR	75	48	65	85	269	171	377	1,338
	USD	72	47	57	89	271	161	295	936
	GBP	81	44	70	94	474	236	426	
Wide since 2007	EUR	380	195	321	548	1,927	1,401	2,459	6,160
	USD	524	320	460	650	1,660	1,234	1,825	2,992
	GBP	360	205	315	542	2,585	2,167	3,520	
Tight since 2007	EUR	59	24	52	70	212	142	260	475
	USD	72	41	56	89	258	156	239	395
	GBP	76	31	66	88	173	127	215	

Source : Deutsche Bank, Bloomberg Finance LP, ICE Indices
Data as of 5/28/2026

When looking at global defaults, the IG analysis as always shows that current spread levels across all ratings are wider than the spread needed to compensate for each of the past 5-year default cycles. As discussed, this is because of the other risks priced into spreads. Though it speaks to how tight BBBs are trading, that at a 0% recovery assumption, current BBBs would not compensate default risks around the dot-com era.

The HY market provides far more interesting observations for investors. Of the 40 cohorts that we have data for, going back to 1982, current CCC spreads would have adequately compensated investors for nearly half (21) of the previous cohorts when assuming a 40% recovery. **For HY overall, current spreads would not have compensated for 11 previous cohorts**, which are grouped around '85-'90 and the dot-com bubble burst.

For a more extreme zero recovery assumption and with spreads near long-term tights, current valuations would only compensate investors for cohorts which saw historically few defaults. This would be the periods like the mid-90s, post-dot-com era, post-GFC, and the recent post-Covid cohort. These are also cohorts which benefitted from easier monetary policy. Only in 6 of 40 cohorts would CCCs have been adequately compensated for the ensuing default risks at current spreads if recoveries were zero.

While we are not forecasting a dire default cycle ahead, there are concrete risks to keep cumulative defaults materially above the best-case periods of the past where default rates bottomed out. Therefore this exercise indicates that **current CCC spreads are too optimistic.** B3s are also likely priced for perfection while higher-rated spec-grade debt can outperform if there is no recession on the horizon. While our modelled 3-year cumulative default rates are likely to stay higher than average, the lack of recessionary risks mean current cohorts are not likely to reach the extremes seen of the 90s or dot-com era.

Figure 55: Spreads required to compensate for default based on Global 5-year cumulative default rates by cohort (boxed numbers show past 5-year default periods where current spreads would not compensate for eventual default risk)

	40% Recovery								0% Recovery							
	IG	AA	A	BBB	HY	BB	B	CCC	IG	AA	A	BBB	HY	BB	B	CCC
1981	8	0	3	25	166	166	192		13	0	5	41	277	277	319	
1982	11	0	8	27	235	224	228	471	19	0	14	46	391	374	380	784
1983	12	6	3	32	214	181	213	655	19	11	5	53	356	301	355	1,091
1984	9	5	3	23	269	188	320	750	15	8	5	39	448	313	533	1,250
1985	5	5	6	5	275	172	379	360	8	8	9	8	458	287	632	600
1986	13	5	8	33	241	98	342	631	22	8	13	55	401	164	570	1,052
1987	22	5	15	52	275	141	289	921	37	8	25	86	459	235	482	1,535
1988	15	5	10	34	307	152	353	923	25	8	16	57	512	253	588	1,538
1989	10	0	0	36	353	186	403	1,108	17	0	0	60	589	311	672	1,846
1990	8	0	0	26	353	148	434	1,630	13	0	0	44	589	246	723	2,717
1991	5	0	3	13	276	70	365	1,031	9	0	5	22	460	116	608	1,719
1992	3	0	3	4	148	24	207	657	4	0	5	7	247	41	345	1,095
1993	2	0	3	3	132	49	196	462	4	0	5	6	221	82	326	769
1994	3	0	3	7	136	58	196	772	6	0	5	11	227	96	326	1,286
1995	8	0	0	21	144	86	173	831	13	0	0	35	240	143	288	1,385
1996	8	0	3	18	169	91	242	404	14	0	4	30	282	152	404	674
1997	15	0	7	29	265	144	369	1,292	25	0	12	48	442	240	614	2,154
1998	31	0	13	58	395	218	543	1,698	52	0	22	97	659	363	905	2,830
1999	32	0	12	61	491	268	648	1,891	54	0	21	102	819	446	1,079	3,152
2000	33	0	15	58	476	244	618	2,030	55	0	25	97	793	407	1,030	3,383
2001	35	0	10	62	438	205	557	2,205	59	0	17	104	730	342	928	3,675
2002	19	0	0	36	308	111	352	1,542	31	0	0	60	513	185	587	2,570
2003	4	0	0	8	161	48	162	1,000	7	0	0	13	268	80	270	1,667
2004	6	0	2	9	105	44	113	554	9	0	3	15	175	73	188	923
2005	13	0	2	23	157	85	191	556	22	0	3	38	262	141	319	927
2006	8	0	2	13	188	83	239	731	13	0	3	21	314	138	398	1,218
2007	6	0	0	11	218	93	279	908	11	0	0	19	363	154	465	1,513
2008	5	0	0	9	267	85	351	1,677	9	0	0	15	445	141	585	2,795
2009	2	0	0	3	263	49	298	2,017	3	0	0	4	438	81	497	3,361
2010	0	0	0	0	125	23	114	649	0	0	0	0	208	38	191	1,081
2011	1	0	2	0	121	38	133	791	1	0	4	0	202	63	222	1,319
2012	4	0	2	5	142	61	145	1,015	6	0	4	8	237	101	242	1,692
2013	2	0	0	3	156	46	177	933	4	0	0	6	261	77	295	1,555
2014	4	0	0	7	153	46	180	769	7	0	0	11	255	77	301	1,282
2015	4	0	0	6	166	47	195	922	7	0	0	11	277	79	325	1,536
2016	8	0	0	13	190	36	219	1,200	14	0	0	21	317	60	365	2,000
2017	5	0	0	8	161	27	171	989	8	0	0	13	268	45	284	1,648
2018	5	0	4	5	165	30	174	1,317	8	0	7	9	274	49	290	2,195
2019	4	0	0	6	180	36	196	1,292	7	0	0	10	300	59	327	2,153
2020	5	0	0	7	185	45	163	1,780	8	0	0	12	308	76	272	2,967
2021	3	0	0	4	132	29	109	513	5	0	0	7	221	49	181	855

Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Appendix A

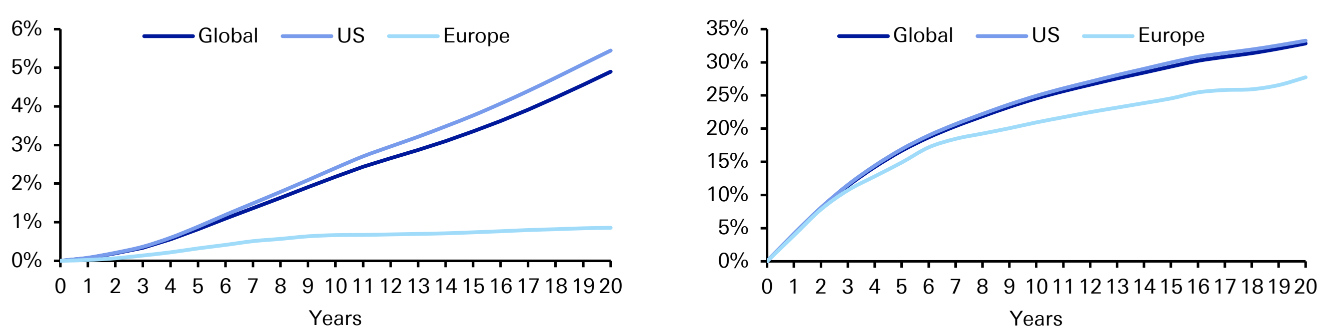
Methodology, raw data and calculations for HTM analysis

The spread required to compensate for default is calculated using the expected loss for each rating band given historical default and different assumed recoveries. The starting point for such calculations is the rating agencies' historical databases on default. For the purposes of this report, we have mainly used data from S&P, as it provides us the flexibility to focus purely on non-financial defaults as well as looking at defaults regionally, although much of our analysis focuses on the global numbers.

US vs. Europe cumulative default rates

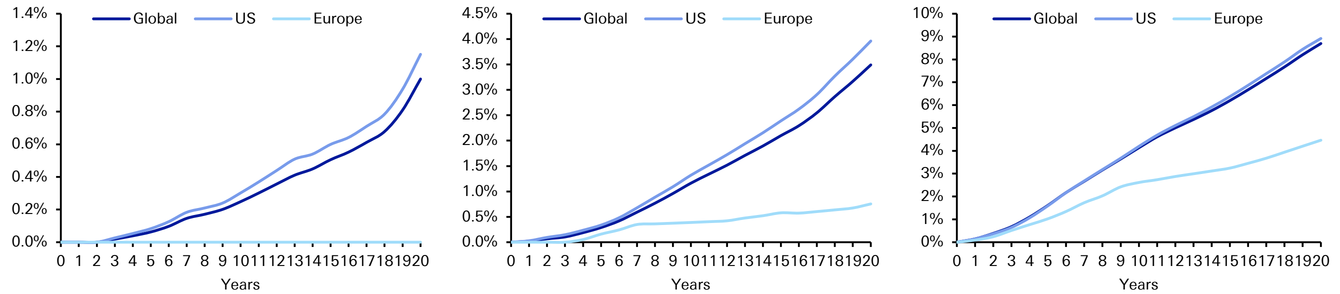
In Figure 56 - Figure 58 we show a number of charts comparing cumulative default rates for European issuers with those of US issuers. The main point here is that, in general, average historical European default rates have been lower. This is in large part due to the fact there is much less history for rated European issuers and less CCC rated issuers specifically. Based on the S&P database, there has only been more than 100 European IG rated issuers since 1995, and for HY there has only been more than 100 since 2004. Therefore, most of the relevant data for Europe occurs during the past decade or so of extremely low defaults, which probably helps explain the lower default rates relative to the more developed US market. This is also why we have focused on the global numbers for much of our analysis. Over time we would expect European markets to converge further to US markets.

Figure 56: S&P Average Cumulative IG (left) and HY (right) Default Rates



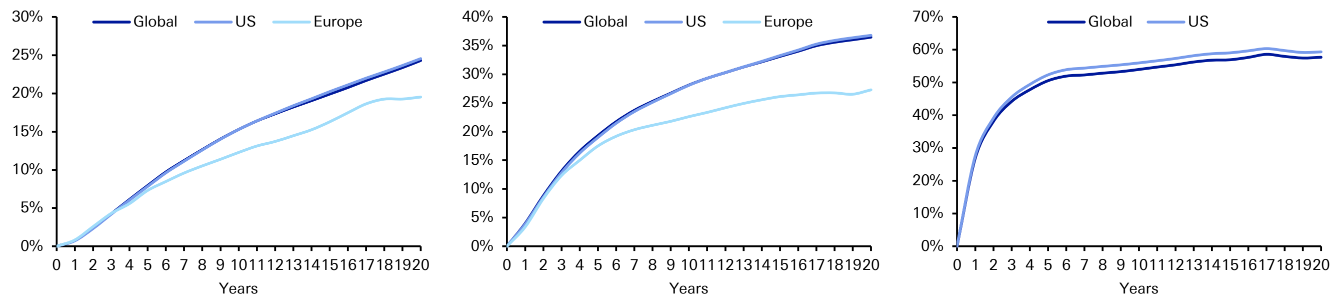
Source : Deutsche Bank, S&P

Figure 57: S&P Average Cumulative AA (left) A (middle) and BBB (right) Default Rates



Source : Deutsche Bank, S&P

Figure 58: S&P Average Cumulative BB (left) B (middle) and CCC (right) Default Rates



Source : Deutsche Bank, S&P

Calculating spreads required to compensate for default

The spreads calculated are spreads based on the maturity of the indices. That would imply we need to calculate the spreads required to compensate for cumulative default probabilities over that time period. The formula we use assumes constant instantaneous hazard rate, which is determined by the spread to that maturity. We then integrate the continuously compounded survival probability to get the cumulative survival probability over time. Thus, the formula can be used to calculate the spread from cumulative default probability or vice versa. We get the cumulative default probabilities from the S&P database and calculate the compensation spreads from them.

$$D = 1 - e^{-ST/(1-R)}$$

where D is the cumulative default probability over time T and S is the Spread to that maturity and R is the recovery

Solving for spread S,

$$S = -\frac{(1-R)}{T}\ln(1-D)$$

Some of our analysis uses different calculations where it makes more sense to look at the path of defaults, rather than simply assuming a constant instantaneous hazard rate.

Recovery rates

In terms of recovery, we have used a selection of different rates throughout the note, generally focusing on 0%, 20% and 40%. However, in Figure 59 (for comparison purposes) we publish a table showing average senior unsecured recovery rates for different rating bands. Recovery rates can fluctuate depending on different macroeconomic backdrops, which is why we include the range of recovery scenarios. Recoveries are a very important factor when trying to assess losses in the case of default and therefore calculate the spread required to compensate for default.

Figure 59: Moody's Senior Unsecured Recovery Rates by Rating (1983-2025)

	Year 1	Year 2	Year 3	Year 4	Year 5	Average
Aa	37.2%	39.0%	38.1%	44.0%	43.2%	40.3%
A	37.7%	46.2%	47.4%	46.8%	46.2%	44.9%
Baa	38.4%	40.3%	41.0%	41.3%	41.4%	40.5%
Ba	40.1%	40.3%	40.0%	40.0%	39.9%	40.1%
B	36.5%	36.4%	37.2%	37.9%	38.5%	37.3%
Caa-C	38.9%	39.0%	38.8%	38.8%	38.7%	38.8%
IG	38.2%	41.6%	42.7%	43.1%	43.0%	41.7%
HY	38.3%	38.2%	38.3%	38.5%	38.7%	38.4%
All	38.3%	38.4%	38.5%	38.8%	39.1%	38.6%

Source : Moody's

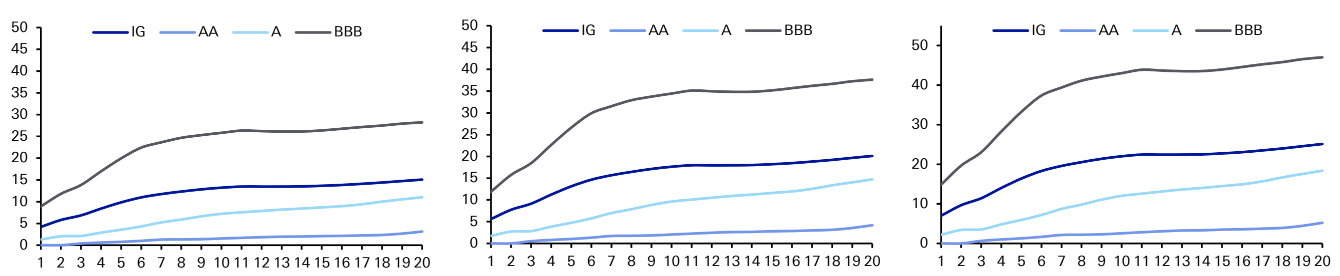
Spreads required to compensate for default

Figure 60: Spreads Required to Compensate for Default Assuming Different Recovery Levels

	40% Recovery								20% Recovery								0% Recovery							
Years	IG	AA	A	BBB	HY	BB	B	CCC	IG	AA	A	BBB	HY	BB	B	CCC	IG	AA	A	BBB	HY	BB	B	CCC
1	4	0	1	9	252	46	247	1,701	6	0	2	12	336	62	329	2,267	7	0	2	15	420	77	412	2,834
2	6	0	2	12	250	73	280	1,283	8	0	3	16	333	98	373	1,711	10	0	3	20	416	122	466	2,139
3	7	0	2	14	241	88	284	1,035	9	1	3	18	322	118	379	1,380	11	1	4	23	402	147	473	1,725
4	8	1	3	17	230	97	275	875	11	1	4	23	307	129	366	1,167	14	1	5	28	384	161	458	1,458
5	10	1	4	20	219	101	261	772	13	1	5	27	292	135	348	1,030	16	1	6	33	365	169	436	1,287
6	11	1	4	22	207	104	248	668	15	1	6	30	276	139	330	890	18	2	7	37	345	173	413	1,113
7	12	1	5	24	195	104	234	590	16	2	7	32	260	139	312	787	20	2	9	39	325	173	389	983
8	12	1	6	25	185	104	220	525	16	2	8	33	247	138	294	700	21	2	10	41	308	173	367	875
9	13	1	7	25	177	103	209	487	17	2	9	34	236	137	279	649	21	2	11	42	295	172	349	812
10	13	2	7	26	169	102	200	450	18	2	10	34	225	136	266	601	22	3	12	43	282	170	333	751
11	13	2	8	26	162	99	191	418	18	2	10	35	216	132	254	558	22	3	13	44	269	166	318	697
12	13	2	8	26	155	97	182	394	18	2	10	35	206	129	243	525	22	3	13	44	258	161	304	656
13	13	2	8	26	149	95	175	373	18	3	11	35	199	126	234	497	22	3	14	44	248	158	292	621
14	14	2	8	26	144	92	169	353	18	3	11	35	191	123	225	470	23	3	14	44	239	154	281	588
15	14	2	9	26	139	91	163	332	18	3	12	35	185	121	218	442	23	3	14	44	232	151	272	553
16	14	2	9	27	135	89	159	317	18	3	12	36	180	119	212	422	23	4	15	45	225	149	265	528
17	14	2	9	27	130	88	153	295	19	3	13	36	173	117	204	393	24	4	16	45	217	147	255	491
18	14	2	10	27	126	86	147	276	19	3	13	37	167	115	196	368	24	4	17	46	209	144	245	460
19	15	3	11	28	122	86	142	262	20	4	14	37	163	114	189	349	25	4	18	47	204	143	237	437
20	15	3	11	28	119	85	138	251	20	4	15	38	159	114	183	335	25	5	18	47	199	142	229	419

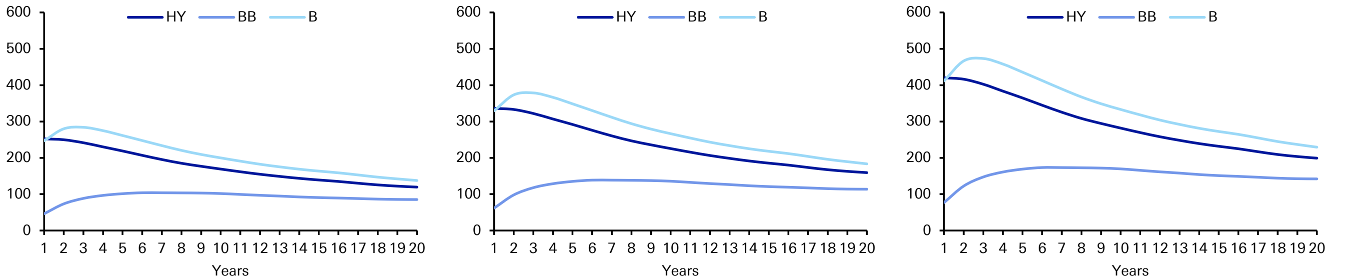
Source : Deutsche Bank, S&P

Figure 61: IG Spread Required to Compensate for Default Assuming a 40% (left), 20% (middle), 0% (right) Recovery



Source : Deutsche Bank, S&P

Figure 62: HY Spread Required to Compensate for Default Assuming a 40% (left), 20% (middle), 0% (right) Recovery



Source : Deutsche Bank, S&P

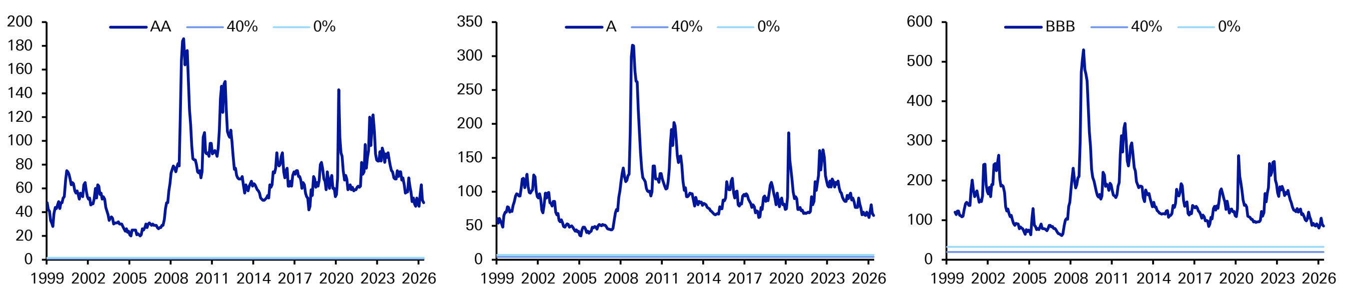
Credit spreads vs. spreads required to compensate for default

Whilst we have formed most of our conclusions in earlier sections of this note, the following section takes a more detailed graphical look at the spreads required to compensate for default probability assuming different recovery rates against historical index level spreads.

The charts focus on non-financial indices across the rating bands. We include charts for the EUR, USD, and GBP markets across both IG and HY. We initially show the spread histories compared with the spread required to compensate for average historical default based on the appropriate rating and index average life, assuming both a 40% and 0% recovery. We then show the DSP charts where we subtract the spread required to compensate for default from the actual trading levels. For this we only use the analysis where we assume a 40% recovery.

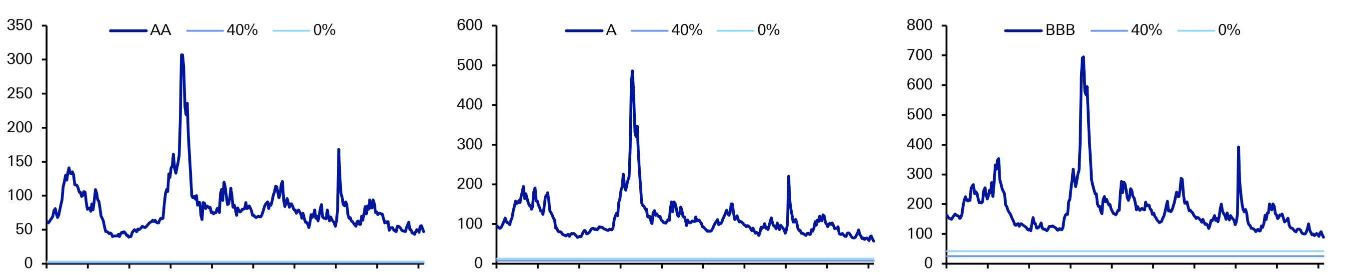
Investment Grade

Figure 63: EUR AA (left), A (middle) and BBB (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



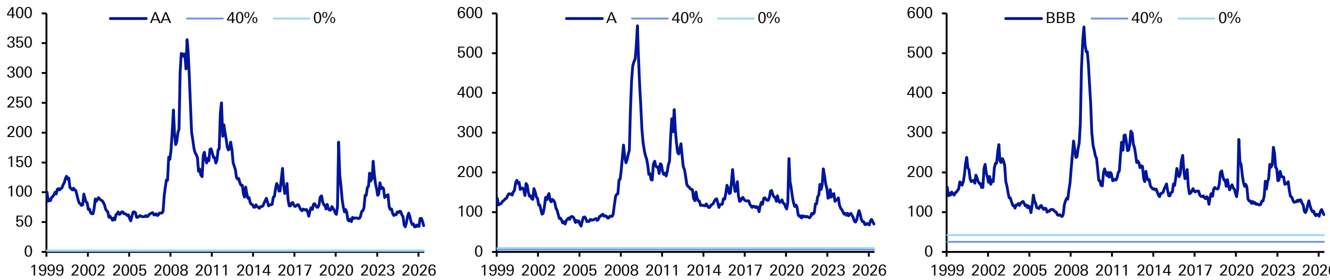
Source : Deutsche Bank, Markit Group

Figure 64: USD AA (left), A (middle) and BBB (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



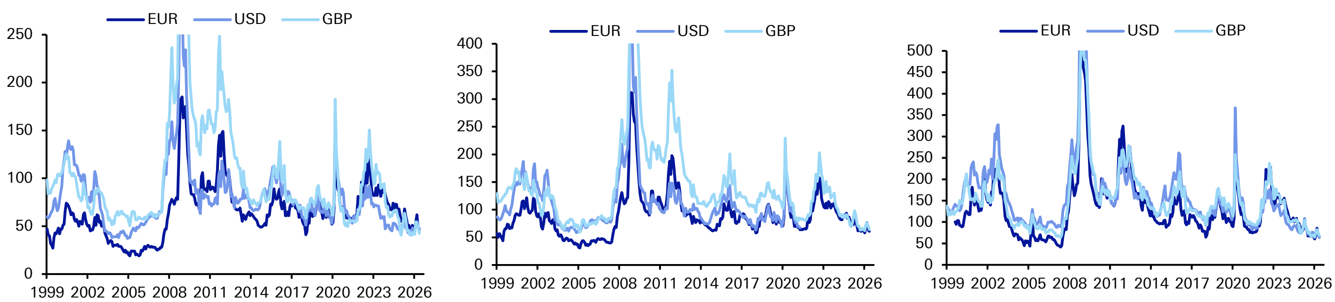
Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Figure 65: GBP AA (left), A (middle) and BBB (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

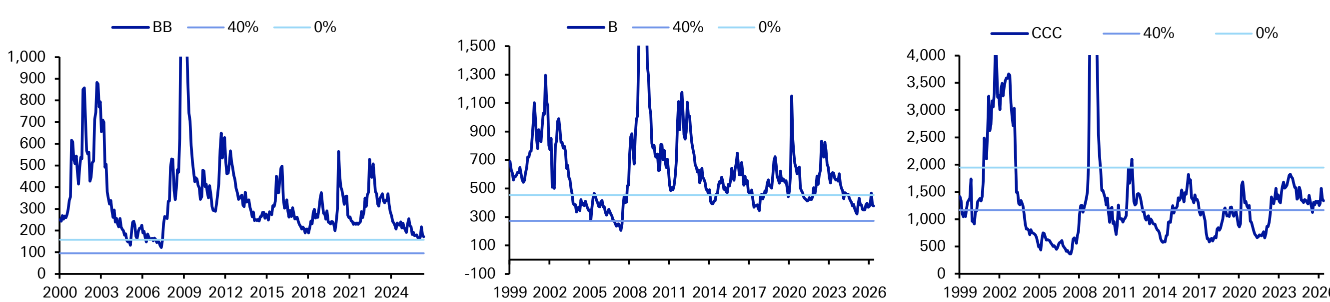
Figure 66: AA (left), A (middle) and BBB (right) Default Spread Premium (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

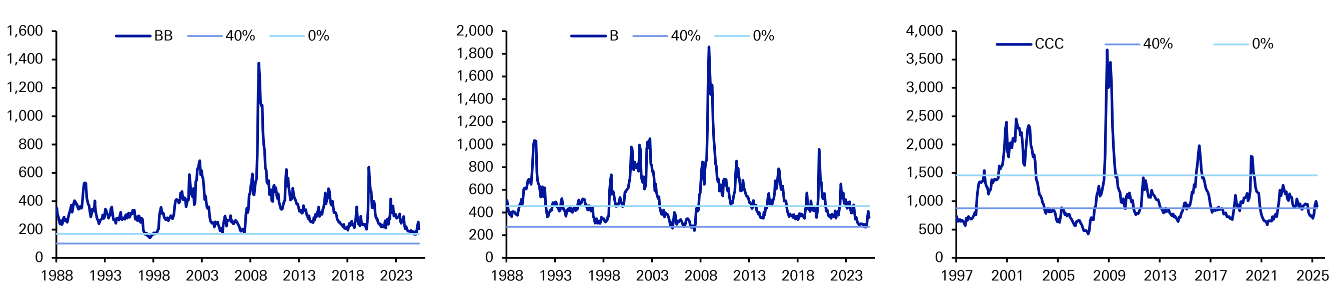
High Yield

Figure 67: EUR BB (left), B (middle) and CCC (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



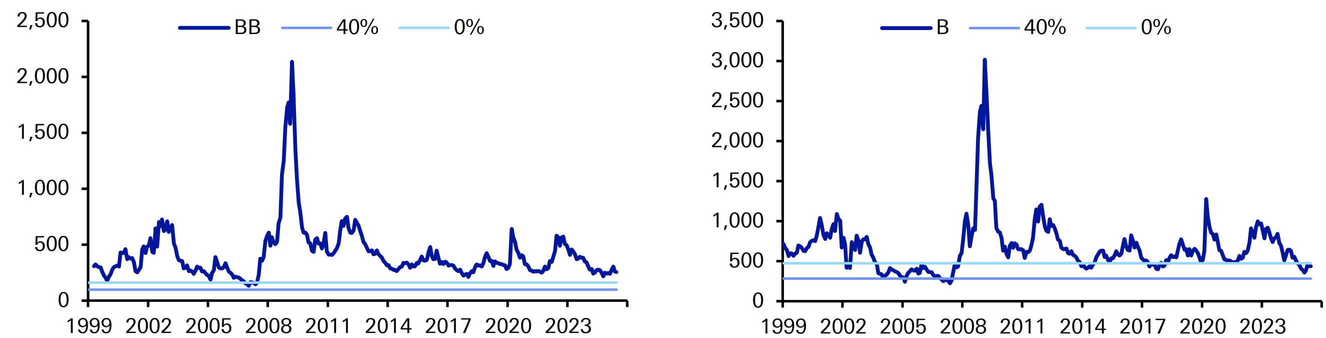
Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Figure 68: USD BB (left), B (middle) and CCC (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



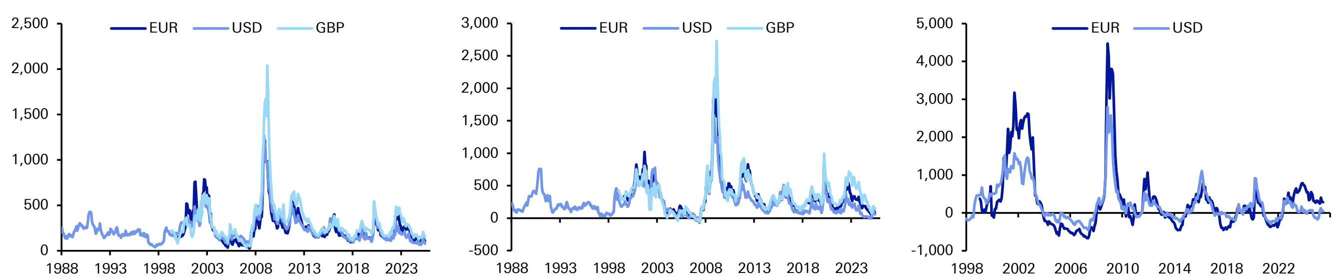
Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Figure 69: GBP BB (left) and B (right) Index Spread Histories vs. Default-Compensation Spreads (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Figure 70: BB (left), B (middle) and CCC (right) Default Spread Premium (bps)



Source : Deutsche Bank, S&P, Bloomberg Finance L.P., ICE Indices

Why additional spread is required

This piece has only ever aimed to look at the spread required to compensate for default. In effect, it is the first building block in assessing where credit should trade, as defaults are the only way a buy-and-hold investor actually loses money in credit relative to the risk-free investment. In reality, there are other important factors that should be taken into consideration. However, trying to put a number to this becomes increasingly subjective and open to interpretation. For those wanting to explore further, these are some of the factors we believe should be considered for those not buying and holding to maturity.

Liquidity

While credit has always tended to be less liquid than government bonds (there were countries where this was not the case post the sovereign crisis), it could be argued this issue remains important largely due to the evolving regulatory environment and significant intervention in recent years. We have written on a number of previous occasions that liquidity risk is a potentially bigger worry for investors than default risk and why investors with a mark-to-market requirement may require more compensation. The initial sharp sell-off in credit during the pandemic arguably highlighted these liquidity issues. Although with the Fed buying corporate bonds for the first time after this shock, this might lead many to think the ultimate downside trading liquidity issue has been permanently reduced. However, what happens in the next crisis if inflation is higher?

Return over risk-free alternative

Clearly, one does not invest in credit to only get a return equal to the risk-free (i.e., government bonds). We need additional spread for it to be worth holding credit on a risk-reward basis.

What is the risk-free rate these days?

The sovereign crisis severely compromised the notion of a risk-free rate to benchmark credit. However, in the years since the sovereign crisis we've gone from high sovereign risk and high yields to one where more and more government markets were trading with negative yields as aggressive central bank intervention dominates. So, investors have had to question the suitability of government credit benchmarks from both extremes over the past few years. This makes it very difficult to assess relative risk. If inflation returns more permanently, this adds yet another dimension.

Cost of credit investing

Clearly, the further you move along the credit curve, the more intensive you need to be in analysing credit. Therefore, the cost base is higher and requires a higher reward than simply compensating you for default risk. Management fees also have to be paid out of returns.

Ratings transition risk

Some investors can only own bonds down to a certain rating category and may be forced to sell if a bond is downgraded through this lower limit. Hence, we probably need a premium for risk of downward rating migration, especially in troubled times. However, this changes over time and investors are less forced sellers on downgrades below certain levels than they used to be.

Uncertainty and timing of recovery rate

In an event of default, the amount and timing of recovery is uncertain. We note that ultimate recoveries for unsecured senior bonds have averaged 47.4% since 1987. This is above the level of 34.8%, over the same period, that trading prices a month after default implies. The latter is often the recovery rate used in models.

Short-term performance targets

Investment grade rarely fails to compensate for default risk over the medium to long run, but investors are very much measured on a short-term basis. They therefore need some cushion to protect against mark-to-market volatility.

Quantifying all these factors above is outside the remit of this report. The buy-and-hold investor has less need to worry about them, though. We hope this report shows the crucial default building block that credit spreads are built around.

Structural US & European corporate default models

Figure 71: US Structural Corporate Default Model (1983-2022, quarterly)

	Coefficients	*t Stat*	*P-value*
Intercept	-0.177	-7.1	0.0%
US Non Financial Corporate Debt to GDP (T-9)	0.386	7.7	0.0%
US Non Financial Corporate Short-Term/Long-Term Debt (T-21)	0.001	2.9	0.4%
BBB Real Yields (T-6)	0.005	5.3	0.0%
DB US Credit Recession Indicator (T-15, squared)	0	7.4	0.0%

Source : Deutsche Bank, Bloomberg Finance LP

Figure 72: Europe Structural Corporate Default Model (1983-2022, quarterly)

	Coefficients	t Stat	P-value
Intercept	-0.05	-2.2	2.9%
EA-20: Non Financial Corporate Debt to GDP (T-3)	0.001	2.9	0.5%
DB Eurozone Credit Recession Indicator (T-9, squared)	0	6.5	0.0%
Real ECB Policy Rate (T-3)	0.136	1.9	6.7%

Source : Deutsche Bank, Bloomberg Finance LP

Appendix 1

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Financial companies see positive outlook despite geopolitics

June 2, 2026

Positive feedback from DB's 16th Global Financial Services Conference

US and European banks and asset managers have a broadly positive outlook for the months ahead as accelerating consumer spending and loan growth, along with AI-driven efficiencies, outweigh geopolitical disruption and higher energy prices, according to talks at Deutsche Bank's 16th annual Global Financial Services Conference in late May.

The event at Deutsche Bank Center in New York brought together industry leaders for two days of discussions focused on the key issues shaping the sector: the impact of higher interest rates, loan demand against a backdrop of geopolitical uncertainty, strong capital markets activity, and the implications of AI for both cost efficiency and revenue opportunities as well as threats. Broader themes included regulation and deregulation, the continued rise of private markets despite recent strains, M&A, and the future of financial services.

US Banks: leaning in

The overriding theme is that the US banks continue to perform very well against a strong backdrop for their key businesses. Consumer spending is accelerating even within discretionary areas, commercial loan activity is strong with good backlogs, and institutional businesses (ie investment banking and trading) are strong or robust. Credit quality remains stable or strong with office no longer a drag. Capital levels are strong and banks are leaning into lending and capital markets-related areas given excess levels and hope for regulatory capital relief.

European Banks: holding up well, for now

The fundamental backdrop for European banks remains supportive despite the Gulf conflict and its repercussions. Loan growth is holding up so far, though uneven across markets, while higher rates support margins, especially for banks with sizeable hedges. Furthermore, the outlook for fee income and capital markets revenues remains positive, though it is partially overshadowed by structural concerns around European competitiveness in light of less favourable regulation. AI is still seen as a net positive for European banks by allowing for new cost efficiencies, with flat to lower costs being the new gold standard.

Positively, asset quality remains strong and resilient, without showing cracks from geopolitics and higher energy prices so far. Lastly, due to high profitability, strong capital ratios, and a much-improved track record, M&A is back in focus, not only with respect to domestic bolt-ons and acquisitions of product factories to diversify revenues, but increasingly also through a more open-minded approach to cross-border deals, although the latter still carry higher execution risk.

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Alternative and Traditional Asset Managers: long-term structural growth vs more cautious near-term environment

Alternative managers expressed a uniformly bullish outlook on long-term structural growth, even as they navigate a somewhat more cautious near-term environment. The most prominent topic was the disruption in the private credit wealth channel, where managers attributed recent net outflows in their business development companies (BDCs) to market-wide sentiment rather than fund-specific performance, and remain confident that a diversified product approach and a focus on retirement solutions will drive long-term growth.

On deployment, most managers noted a slow start to the year but expect a significant re-acceleration in investment activity and M&A in the second half as deal pipelines rebuild. Fundraising remains a bright spot, with firms guiding for a strong year fueled by institutional demand and growth in insurance, as the insurance channel itself remains a key growth area for credit. Fee-related earnings margin expansion continues to be a central focus, with managers displaying confidence in their ability to improve margins through operating leverage, the scaling of new strategies, and efficiencies from technology like AI.

Traditional asset managers revealed a consistent focus on navigating secular headwinds by executing on key growth initiatives with tangible progress being noted. While active equity outflows persist for some, firms are seeing improved organic growth profiles driven by a clear strategic pivot toward higher-demand products including ETFs and private markets. Moreover, managers continue to advance their technological capabilities, leveraging AI-driven platforms and exploring innovations like tokenization to enhance distribution and improve the client experience, all while maintaining a sharp focus on expense management to achieve margin targets.

Market infrastructure: getting ready for the future and tokenization specifically

A pervasive theme across traditional exchanges and digital asset platforms was the strategic preparation for the tokenization of assets, with many viewing it as a significant future unlock for growth and a way to reduce friction across the ecosystem. Several management teams highlighted timelines and strategies for their tokenization efforts, while also expressing optimism around a wide variety of organic growth initiatives, such as launching new derivative products, leveraging cloud-based data offerings, and expanding into prediction markets to attract new client segments.

Digital asset infrastructure providers are also advancing the evolving market structure by providing tokenization services as well as crypto platform integration to traditional financial institutions. AI is being used more aggressively, not only in cost reduction from less efficient operational processes, but also increasingly across a wide array of revenue generation opportunities, including automated and AI agentic trading, which is featuring more prominently on retail trading platforms. This is driving a stronger secular growth backdrop for trading volumes for retail brokers and exchanges, which may further accelerate over time as trading hours gradually extend toward 24x7.

Information services providers: a more differentiated view of AI

Broadly, financial data providers have seen an increase in data usage as automated models, decision systems continuously ingest and analyze data, accelerated by usage of AI agents. AI tools also reduce friction and make datasets easier to discover, test, and integrate, supporting cross-selling and upselling. Companies are becoming more distribution channel-agnostic, with focus shifting from software solutions to data feeds that are embedded deeper into new workflows.

At a thematic level, AI increasingly shifts the competitive moat from pure data delivery platforms to companies that own curated, high-quality proprietary or contributory data, a key bottleneck for effective and token-efficient AI deployment. Incumbent providers are benefitting from trusted standards, brands, and data governance. Internally, AI is driving productivity gains across engineering, research, and product development, allowing organizations to increase product velocity and speed to market.

Appendix 1

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This Month in Geopolitics: June 2026

Key events, themes and trends to watch in the month ahead

June 2026

Dr. Helen Belopolsky | Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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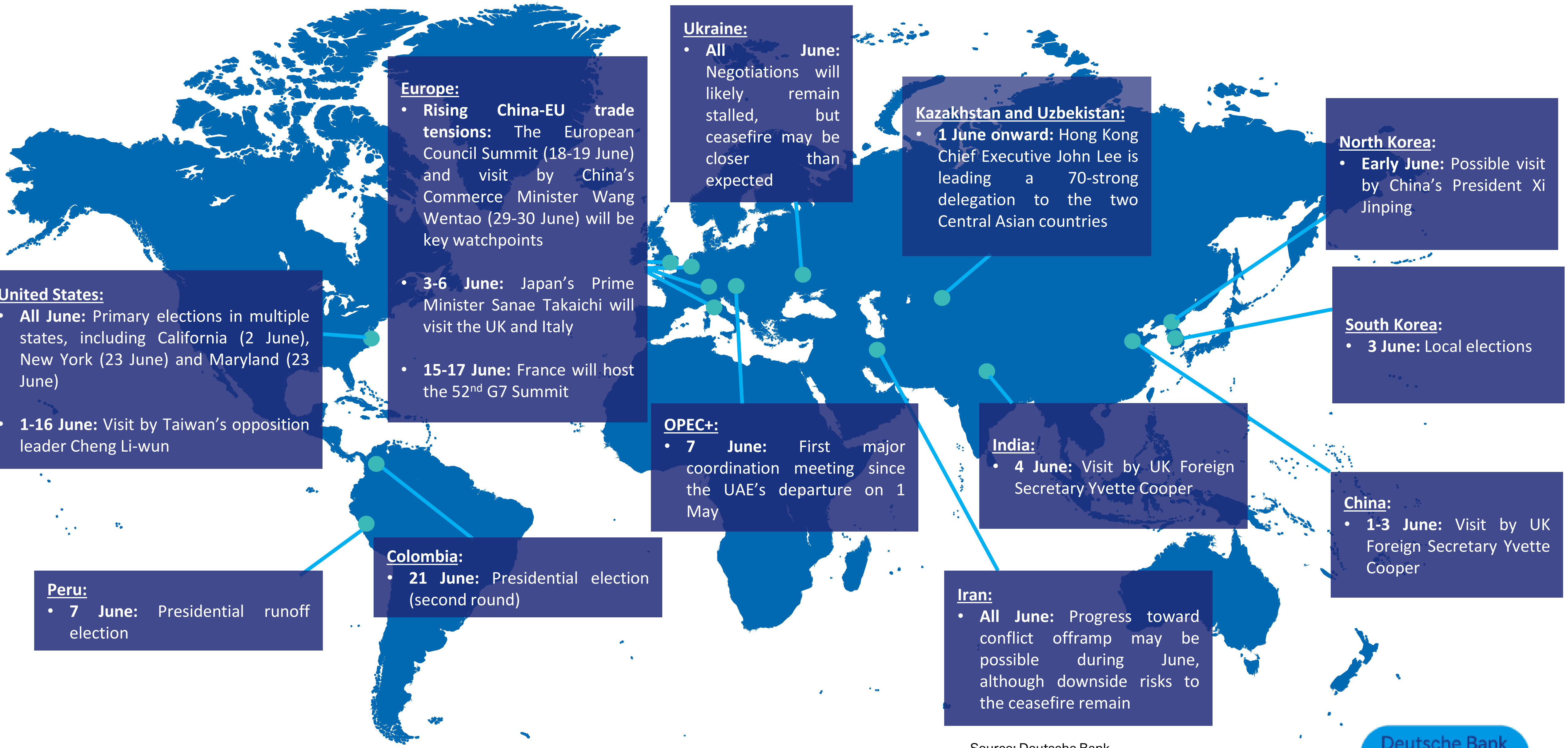
At a Glance: Geopolitical events to watch in **June 2026**

Our monthly Look-Ahead covers a *non-exhaustive* list of key events, issues and themes to watch in the month ahead. The below map intends to serve as an executive summary, providing a list of the issues at a glance. These are covered in more detail in subsequent slides.



United States:
- **All June:** Primary elections in multiple states, including California (2 June), New York (23 June) and Maryland (23 June)
- **1-16 June:** Visit by Taiwan's opposition leader Cheng Li-wun

Peru:
- **7 June:** Presidential runoff election

Colombia:
- **21 June:** Presidential election (second round)

Europe:
- **Rising China-EU trade tensions:** The European Council Summit (18-19 June) and visit by China's Commerce Minister Wang Wentao (29-30 June) will be key watchpoints
- **3-6 June:** Japan's Prime Minister Sanae Takaichi will visit the UK and Italy
- **15-17 June:** France will host the 52nd G7 Summit

Ukraine:
- **All June:** Negotiations will likely remain stalled, but ceasefire may be closer than expected

OPEC+:
- **7 June:** First major coordination meeting since the UAE's departure on 1 May

Kazakhstan and Uzbekistan:
- **1 June onward:** Hong Kong Chief Executive John Lee is leading a 70-strong delegation to the two Central Asian countries

India:
- **4 June:** Visit by UK Foreign Secretary Yvette Cooper

Iran:
- **All June:** Progress toward conflict offramp may be possible during June, although downside risks to the ceasefire remain

North Korea:
- **Early June:** Possible visit by China's President Xi Jinping

South Korea:
- **3 June:** Local elections

China:
- **1-3 June:** Visit by UK Foreign Secretary Yvette Cooper

Source: Deutsche Bank

Key events, issues and themes to watch in **June 2026**

Listed in chronological order

Russia-Ukraine war

While massive strikes continue and though trilateral negotiations are stalled as a result of the war in Iran, a number of recent developments suggest we may be nearer a ceasefire than many believe. As war fatigue mounts, the rising costs of protracted conflict may eclipse its diminishing returns, shifting the strategic calculus. The US brokered a three-day ceasefire from 9-11 May, tied to Russia's Victory Day commemorations. Russian President Putin publicly said the war may be "coming to an end". Diplomatic channels remain active.

A senior Ukrainian delegation recently travelled to Washington for renewed negotiations with US officials on possible frameworks for freezing the war. The talks focused on conditions for a dignified peace according to Kyiv. President Trump had previously targeted June for a framework agreement. Reporting suggests that Washington may increase pressure on Moscow and Kyiv if negotiations remain stalled into the summer.

Iran crisis (all June)

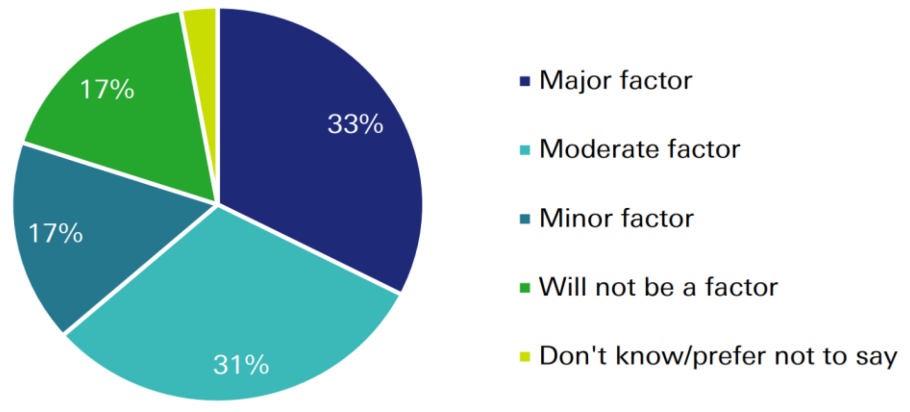
We maintain our baseline assessment that an offramp may be possible in the near term. Domestic and regional pressure on President Trump will likely continue to rise. Most US respondents in our recent polls indicated that the war in Iran and cost-of-living concerns will factor into their votes in the November midterm elections (see charts). The potential incremental benefits of the continued campaign will come up against increasing costs for the US, its allies in the region, and the global economy.

The deployment of additional military assets still provides the US with optionality to attempt to re-open the Strait of Hormuz or to raise pressure on Iran, although in our view, the likelihood of severe downside scenarios has diminished. Nevertheless, risks to the ceasefire remain, and we should expect Iran to ramp up its asymmetric tools and expand the area of conflict if fighting were to resume.

Primary elections in the US (all June)

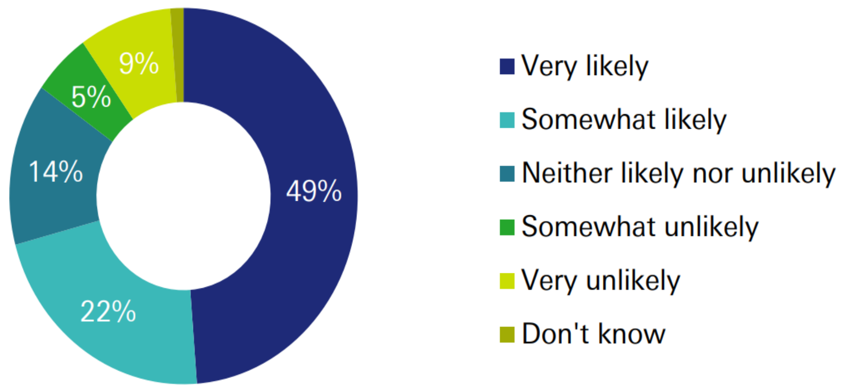
Numerous US states will hold primary elections to select candidates for the November mid-terms. Key primaries include California (2 June), New York (23 June), and Maryland (23 June). Other states with primaries include Iowa, Maine, Montana, Nevada, New Jersey, New Mexico, North Dakota, and South Dakota. Results can be an important signal of public sentiment about the current political agenda.

US participants were asked: In the upcoming midterms, to what extent will current US actions in Iran be a factor in your vote?



Source: dbDataInsights. Responses collected in April 2026 from an Online Nationally Representative sample of ~650 US respondents.

US participants were asked: How likely or unlikely is it that current cost-of-living concerns will influence your vote in the midterms?



Source: dbDataInsights. Responses collected in April 2026 from an Online Nationally Representative sample of ~650 US respondents.

Key events, issues and themes to watch in **June 2026**

Listed in chronological order

Rising China-EU trade tensions (all June)

The European Council Summit (18-19 June) and Chinese Commerce Minister Wang Wentao's visit to Brussels (29-30 June) will be key watchpoints. The EU is exploring new trade defences aimed at addressing perceived risks to EU industry from low-cost Chinese goods. European Commission President Ursula von der Leyen is expected to outline recommendations for new trade defence tools at the June summit. Beijing has warned that it could retaliate if the EU insists on new trade instruments. China's Commerce Minister Wang Wentao will reportedly visit Brussels in June for talks with EU trade chief Maroš Šefčovič in a bid to address rising tensions.

China's President Xi could visit North Korea (early June)

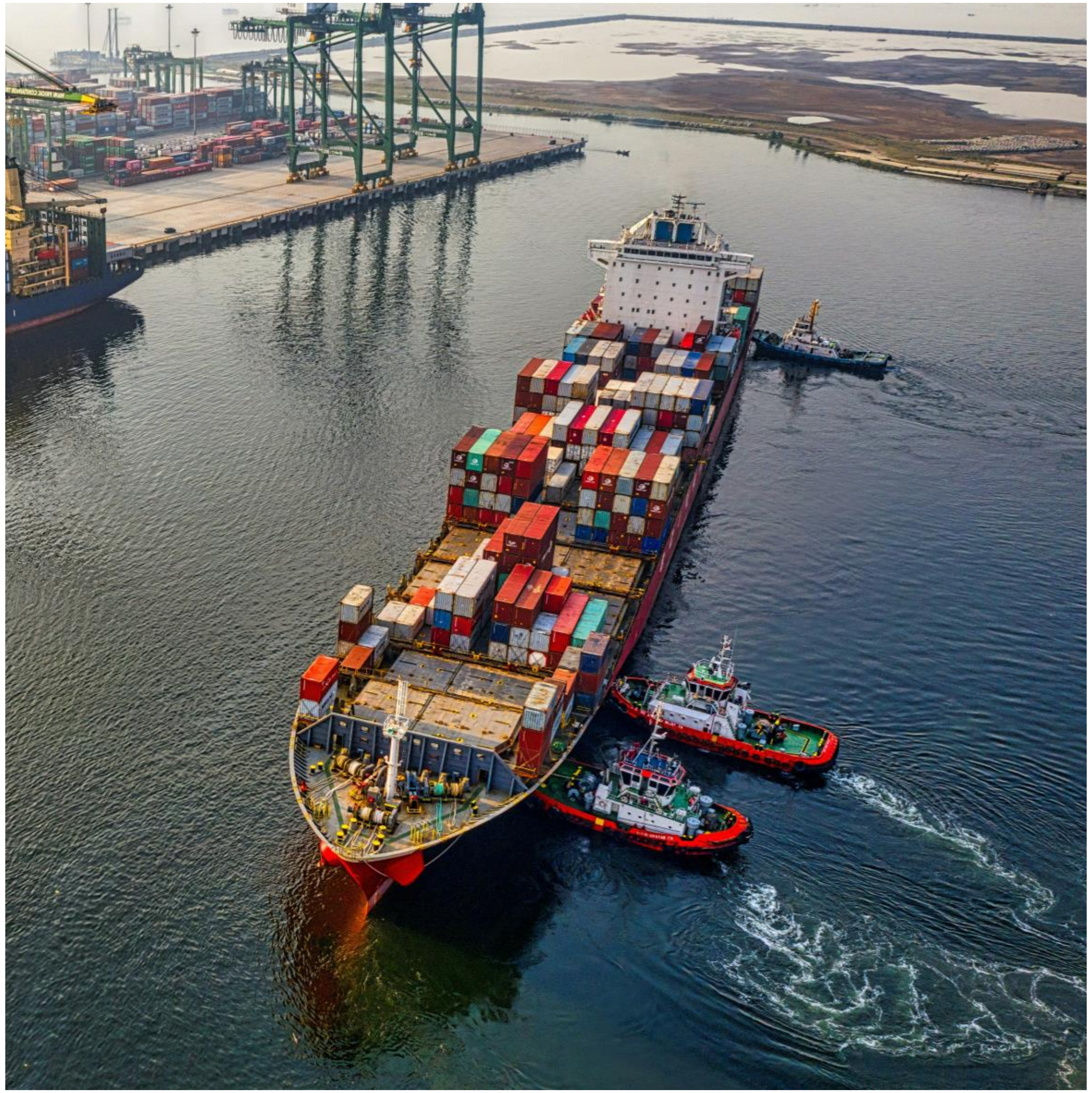
If confirmed, the visit may serve to accelerate efforts at renewed dialogue between the US and North Korea. Citing South Korean sources, media reports suggest Xi could soon visit Pyongyang, potentially to discuss bilateral ties and possibly to help revive US-North Korea talks. This is in line with our February report, which highlighted the possibility of renewed dialogue between President Trump and North Korean leader Kim Jong-un (see North Korea: A New Strategic Calculus).

UK Foreign Secretary Yvette Cooper is visiting China (1-3 June) and India (4 June)

The visits likely intend to strengthen dialogue and bilateral economic cooperation. Foreign Secretary Yvette Cooper's itinerary includes a stop in Beijing for talks with her Chinese counterpart Wang Yi, followed by business meetings in Shenzhen. Both sides will also hold the 11th China-UK Strategic Dialogue during the visit. Cooper will then travel to India where she will reportedly meet with her counterpart S. Jaishankar, as well as with other officials and members of the business community.

Taiwan's opposition leader Cheng Li-wun is visiting the US (1-16 June)

The visit could seek to shore up lines of communication between the opposition and Washington ahead of Taiwan's 2028 presidential election. Cheng Li-wun, the chair of Taiwan's opposition Kuomintang Party is visiting the US for meetings with officials, policymakers (including Republican and Democrat members of Congress) and academics. The visit follows in the wake of Cheng's April meeting with President Xi Jinping in Beijing, as well as ongoing uncertainty over the future of a US$14bn arms sale package to Taiwan.



Source: william william on Unsplash

Key events, issues and themes to watch in **June 2026**

Listed in chronological order

Hong Kong delegation is visiting Kazakhstan and Uzbekistan (1 June onward)

Hong Kong's Chief Executive John Lee is heading a visit by a 70-strong delegation to the two Central Asian countries. This is reportedly the largest delegation assembled during Lee's term in office, reflecting the growing significance of Central Asia amid Hong Kong's efforts to diversify trade and investment ties. The delegation reportedly includes business representatives from both Hong Kong and mainland China.

Japan's Prime Minister Takaichi to Visit the UK and Italy (3-6 June)

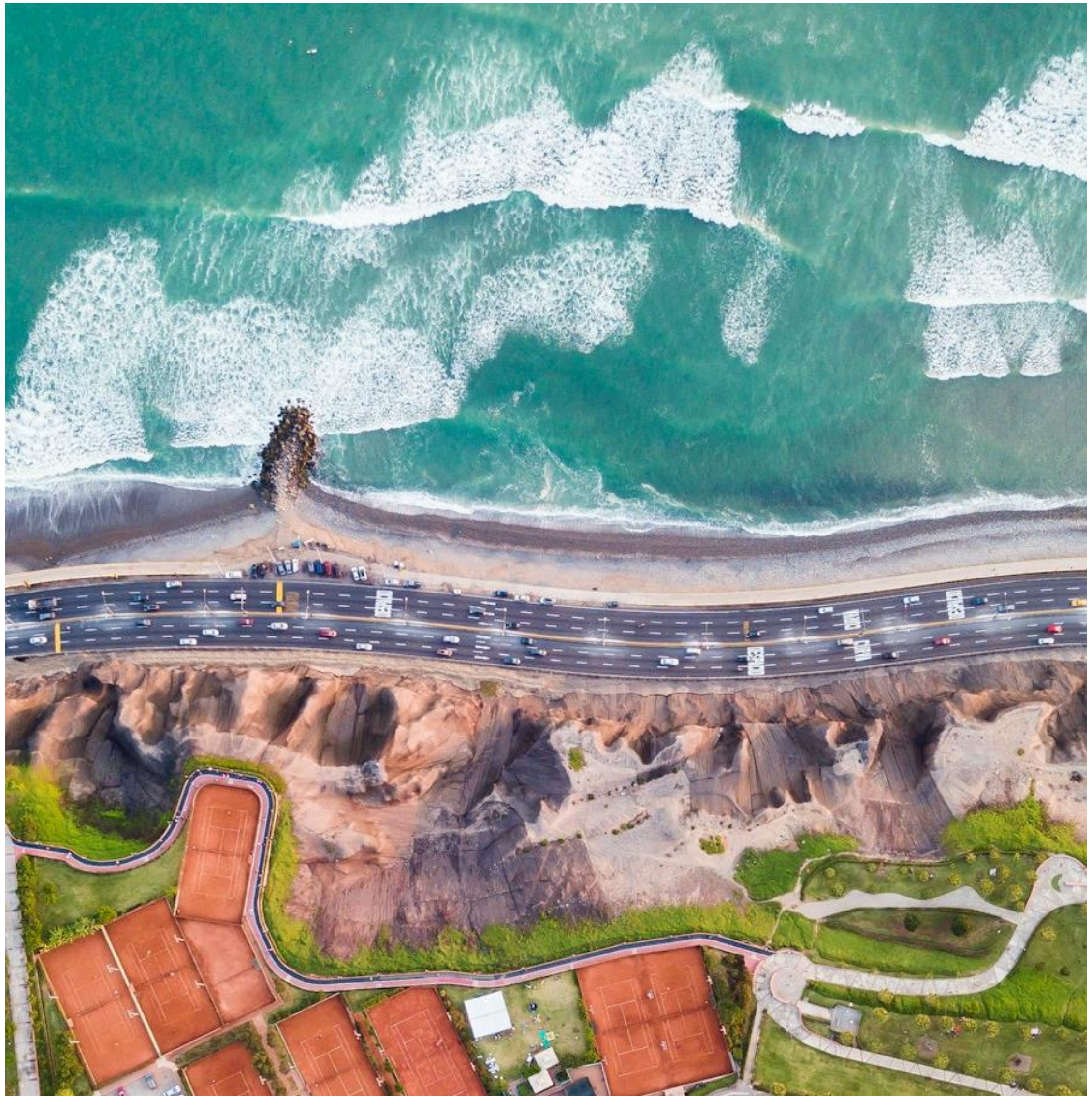
The trips are aimed at strengthening bilateral cooperation with key European partners, particularly on security and defence. The agenda reportedly includes discussions on economic security, a joint next-generation fighter jet program, Indo-Pacific security, and defence industrial cooperation. The visit reinforces a growing Japan-Europe security and technology partnership. It also signals closer coordination on economic security and supply chains.

Local elections in South Korea (3 June)

Recent polls place the ruling Democratic Party in the lead, ahead of the opposition People Power Party. The election will see 227 local governments, 4,000 local council seats and 16 mayoral and gubernatorial positions contested.

OPEC+ Ministerial Meeting (7 June)

This meeting will be particularly important given the impact of the continuing crisis in the Middle East on OPEC+ decisions. It will be the first major coordination meeting after the decision by the UAE to leave OPEC, which took effect on 1 May. The group is operating with a reduced and politically more fragmented membership. To watch will be signals on whether the remaining members can still agree on production quotas and supply discipline.



Source: Willian Justen de Vasconcellos on Unsplash

Key events, issues and themes to watch in **June 2026**

Listed in chronological order

Presidential runoff election in Peru (7 June)

The Peruvian presidential runoff election is expected to take place on 7 June, after the initial round of voting failed to produce a winner. This election is particularly significant for markets due to its potential impact on copper and mining policy, Chinese investment, fiscal policy, and constitutional reform risks. The outcome of the election could influence future foreign investment and the regulatory environment for these critical industries.

52nd G7 Summit (15-17 June)

The summit will seek to rebuild multilateral cooperation and find "points of convergence despite disagreements." It will be hosted by French President Emmanuel Macron in Évian-les-Bains. Key themes for the summit include: continued coordination on the Russia-Ukraine war and European security, economic stability amid trade fragmentation, critical minerals and supply chain resilience, energy security and transition, and AI governance. The summit takes place against the backdrop of significant geopolitical volatility and will undertake the challenging task of finding alignment.

Presidential election in Colombia second round (21 June)

The current president, Gustavo Petro, is constitutionally ineligible for re-election, making this election a pivotal moment for Colombia's political direction. Lawyer Abelardo de la Espriella won the first round of Colombia's presidential election on 31 May, ahead of senator Iván Cepeda. Both candidates will contest the second round on 21 June. The elections are crucial for the balance of power in Latin America, given Colombia's historical role as a key US partner in security and drug trafficking, despite recent strains in the relationship under President Petro's administration.

Additionally, Colombia's stance on China's growing economic presence in the region, particularly in energy and mining, will be a significant factor shaped by the new leadership.

Source: Arnaud Jaegers on Unsplash

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Investors see megatrends as an opportunity

May 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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We asked investors to explain why they are positive about investing in megatrends.

Our AI analysis of hundreds of responses is visualised in the word cloud below

This chartbook follows our report explaining how our new model analysing the impact of megatrends on economies and markets

We thank dbDataInsights for their contribution to this report

Source: dbDataInsights, Deutsche Bank.

Contents

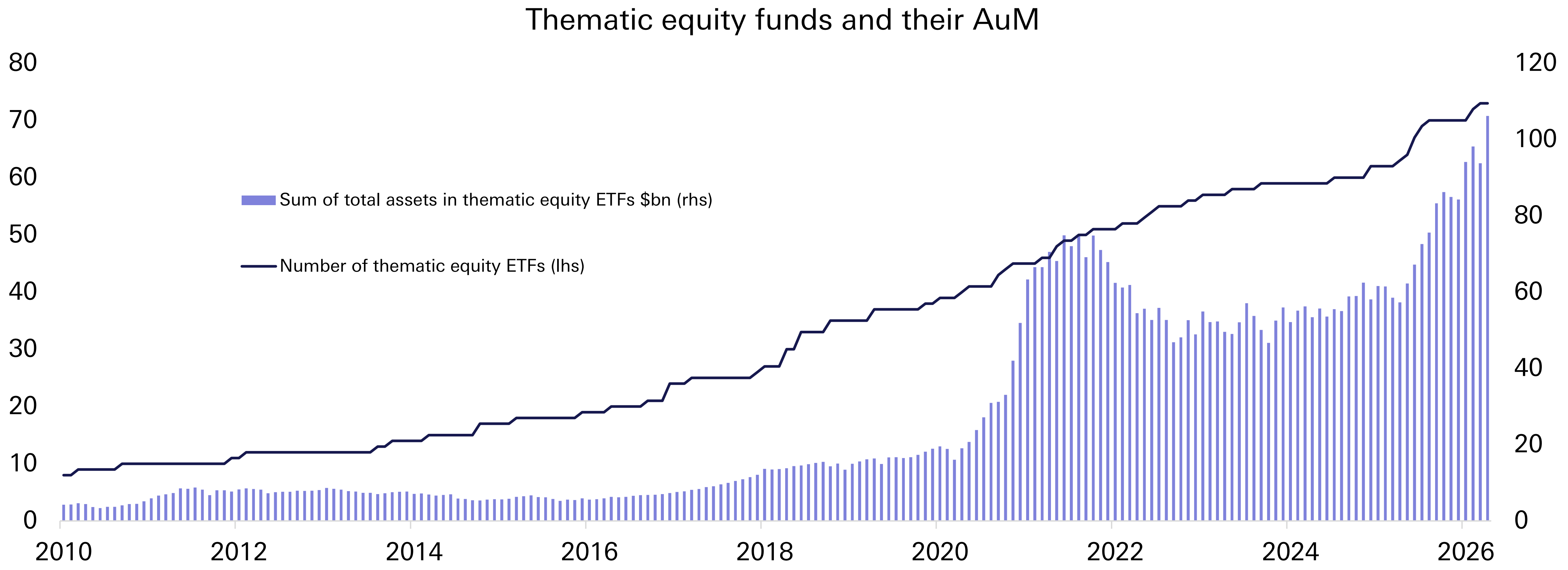
3. The rise of thematic investing
4. Risks and opportunities
5. Megatrend-driven changes to portfolios
7. Risk taking is in vogue
10. How investors react to uncertainty
12. Reacting to geopolitical uncertainty
13. What investors think about individual megatrends
   - Technology
   - Sovereign deficits
   - Geopolitics & globalisation
   - Domestic politics & social discontent
   - Demography
   - Energy disruption & transition

Our survey polled around 500 people in both the US and UK. The sample was designed to be representative of the population – not just a selection of investment professionals.

Thematic investing has been particularly strong in the 2020s

AI has also catalysed a rush of money flowing into thematic funds

The rise of thematic investing

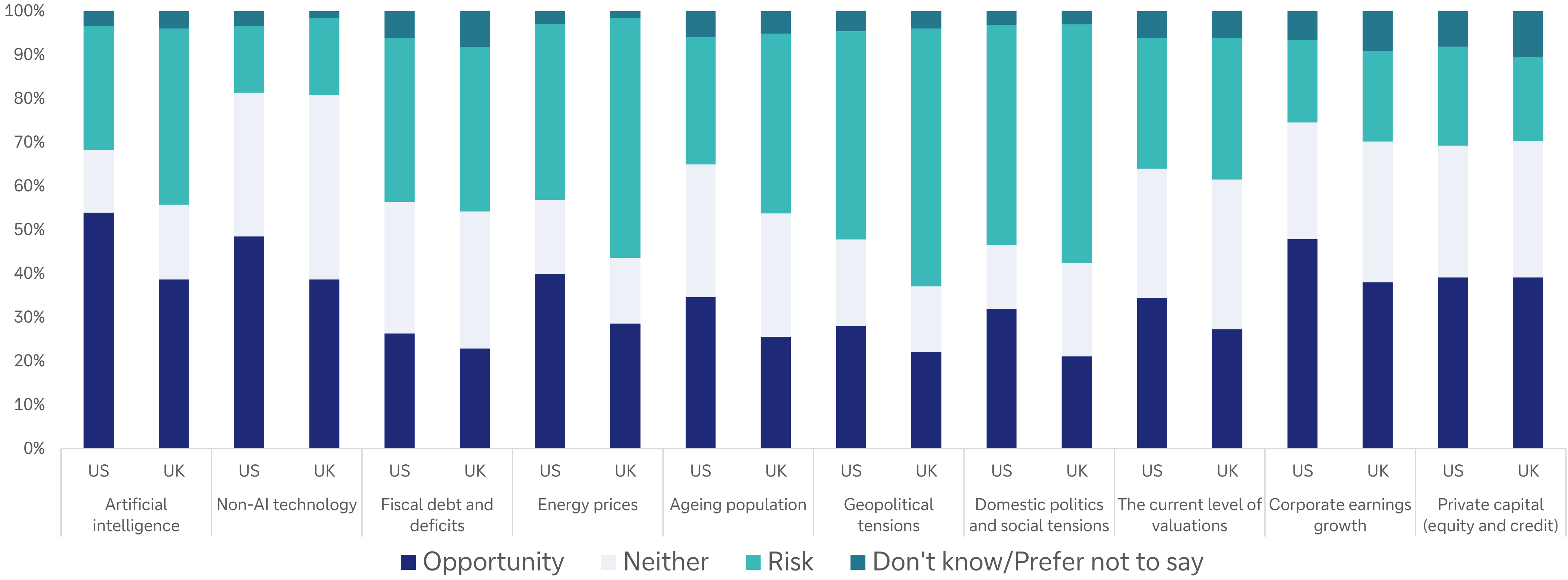


Source: Bloomberg Finance L.P., Deutsche Bank.

Tech – both AI and non-AI – are the strongest themes for investors

The similarity in the popularity of both AI and corporate earnings growth suggests that investors are happy to take a long-term position on AI regardless of the current capex requirements

"For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026."

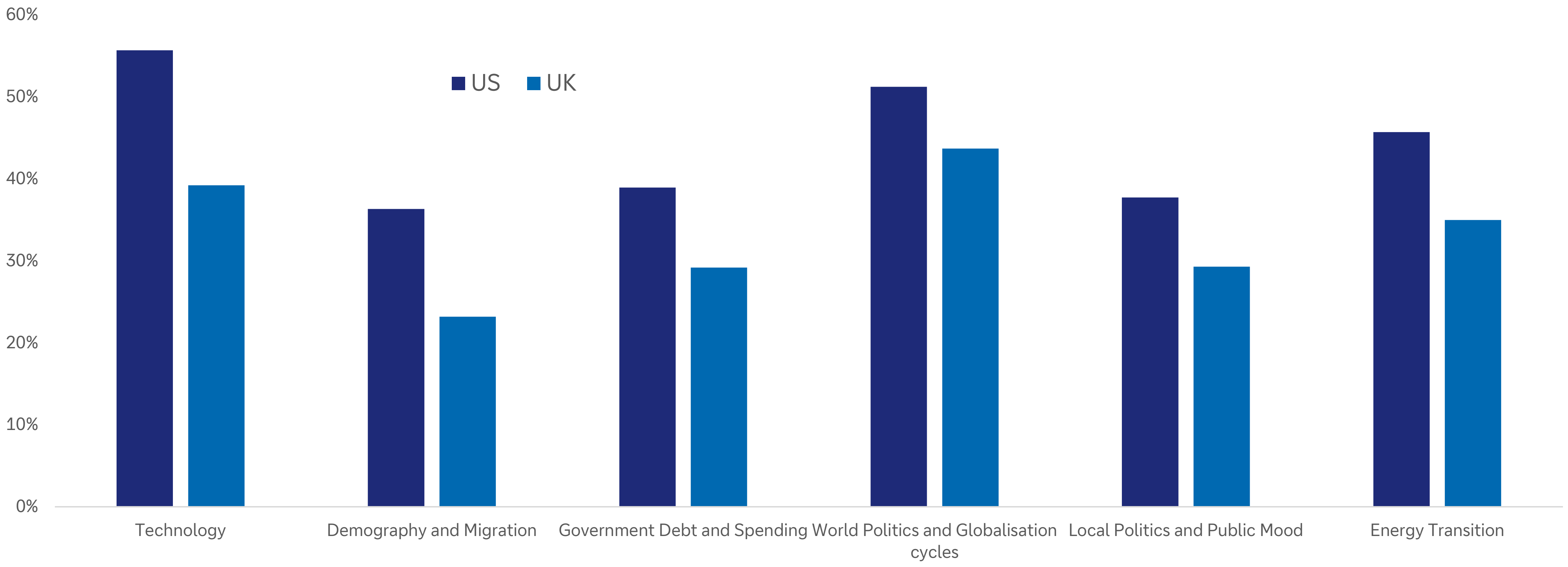


Source: dbDataInsights, Deutsche Bank.

Active management: Investor attention to megatrends seems to contradict the narrative that everyone is piling into passive indices.

Americans are more likely to actively manage their portfolios for megatrends, particularly for technology

Proportion of investors who have made a change to their investment portfolio in the last three years as a result of the following megatrends

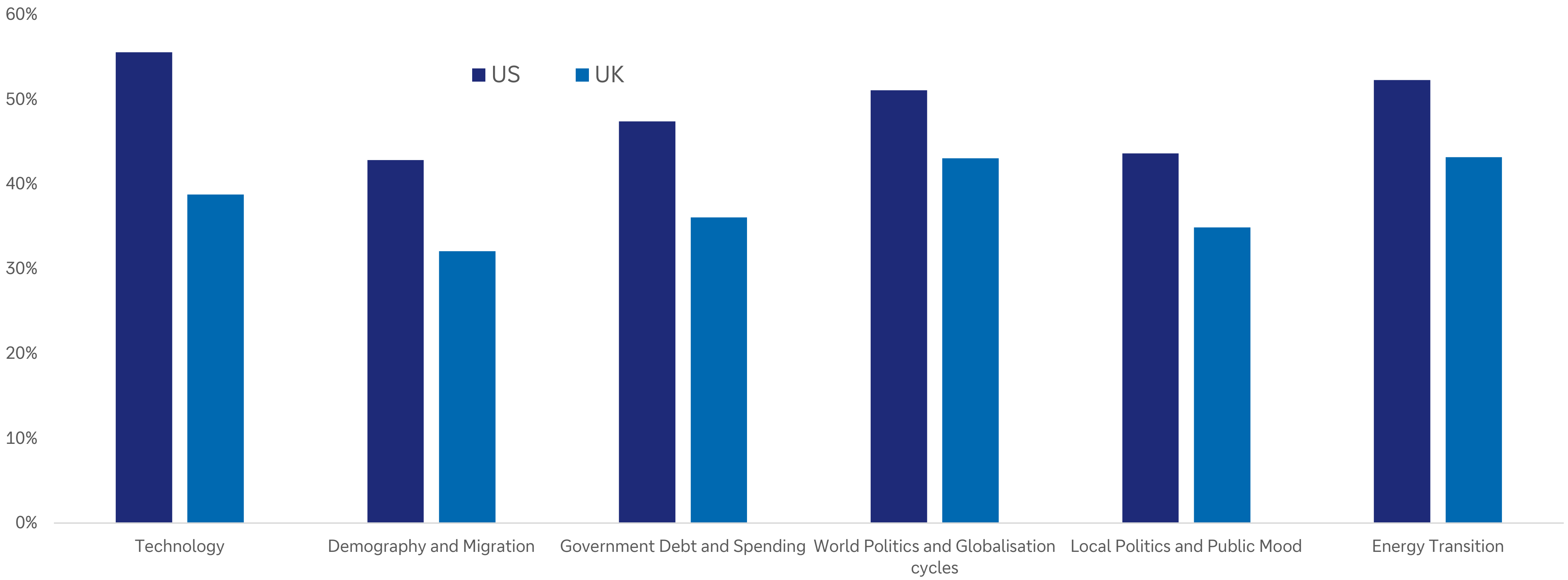


Source: dbDataInsights, Deutsche Bank.

Active management: The volatility of the last two years perhaps a key reason why investors continue to expect to make investment changes over the coming 12 months

No doubt volatility of global politics feeds into the decisions; however, the pace of AI adoption is also key

Proportion of investors who are 'likely' or 'very likely' to change their investment portfolio in the next 12 months as a result of the following megatrends

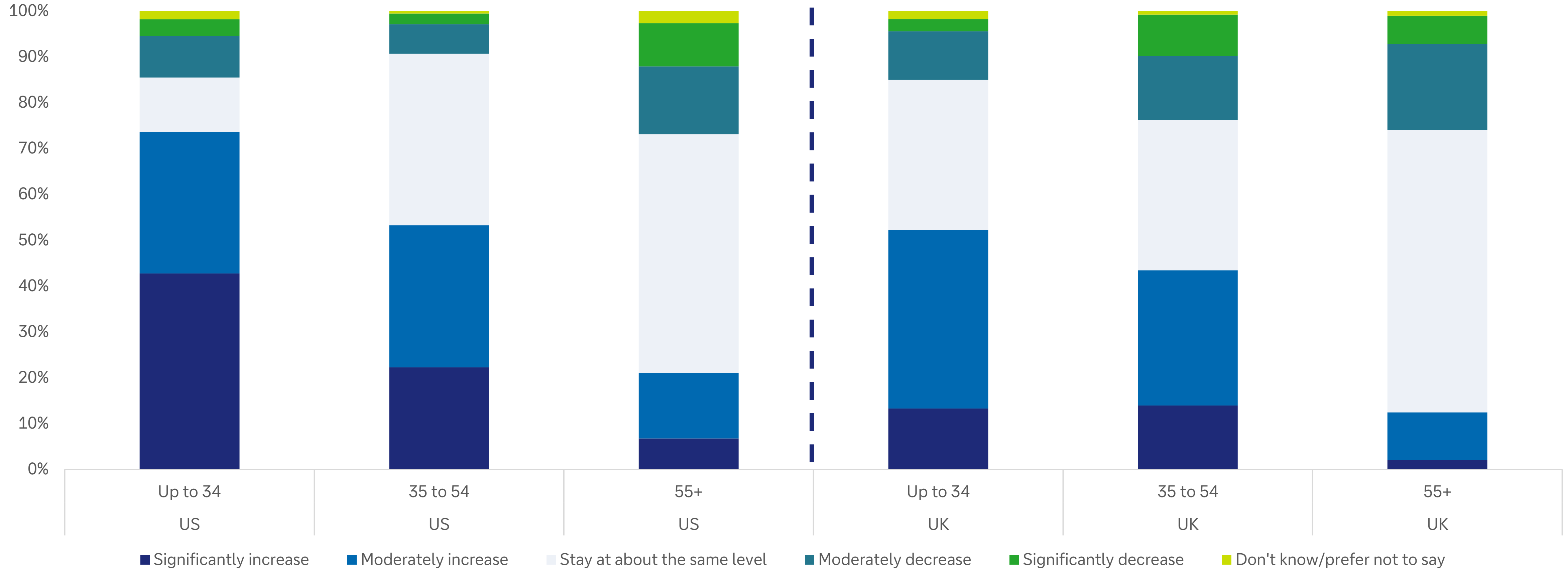


Source: dbDataInsights, Deutsche Bank.

Risk is back: Far more people expect to *increase* their risk taking over the coming 12 months

Despite the four major international shocks that have affected markets this decade, investors appear undeterred. Partly, this is due to enthusiasm about AI, but it may also be partly due to consistent way in which equity markets have rebounded after sharp shocks.

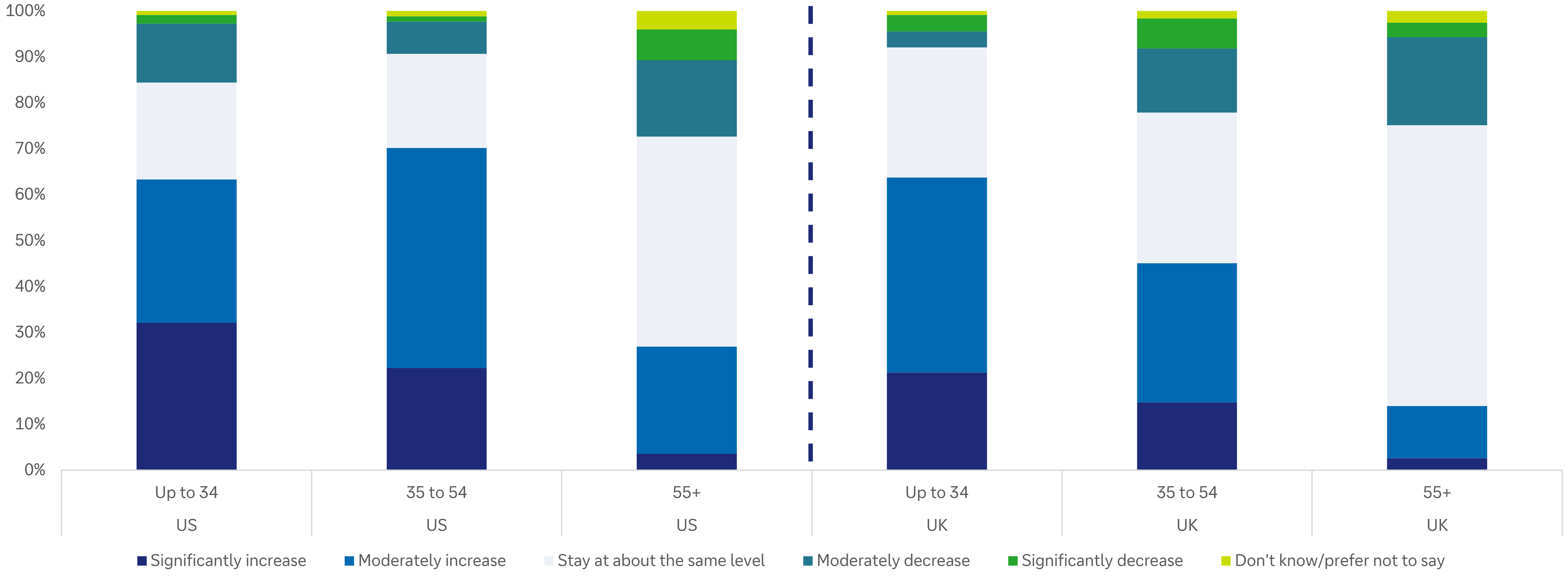
Over the next 1 year, do you expect your level of risk-taking in relation to your main investment portfolio to...



Source: dbDataInsights, Deutsche Bank.

Risk on: Three years out, expectations for risk taking are even stronger …

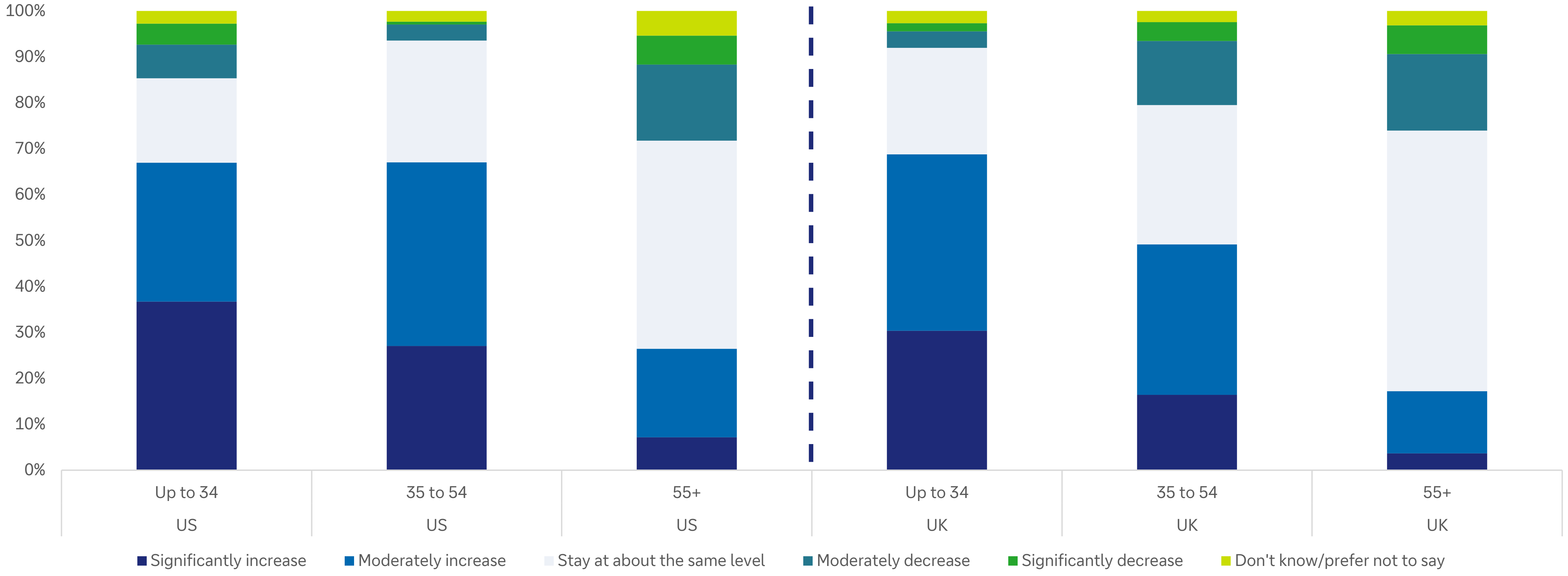
Over the next 3 years, do you expect your level of risk-taking in relation to your main investment portfolio to…



Source: dbDataInsights, Deutsche Bank.

… before risk tolerance plateaus into a 5-year timeframe

Over the next 5 years, do you expect your level of risk-taking in relation to your main investment portfolio to…

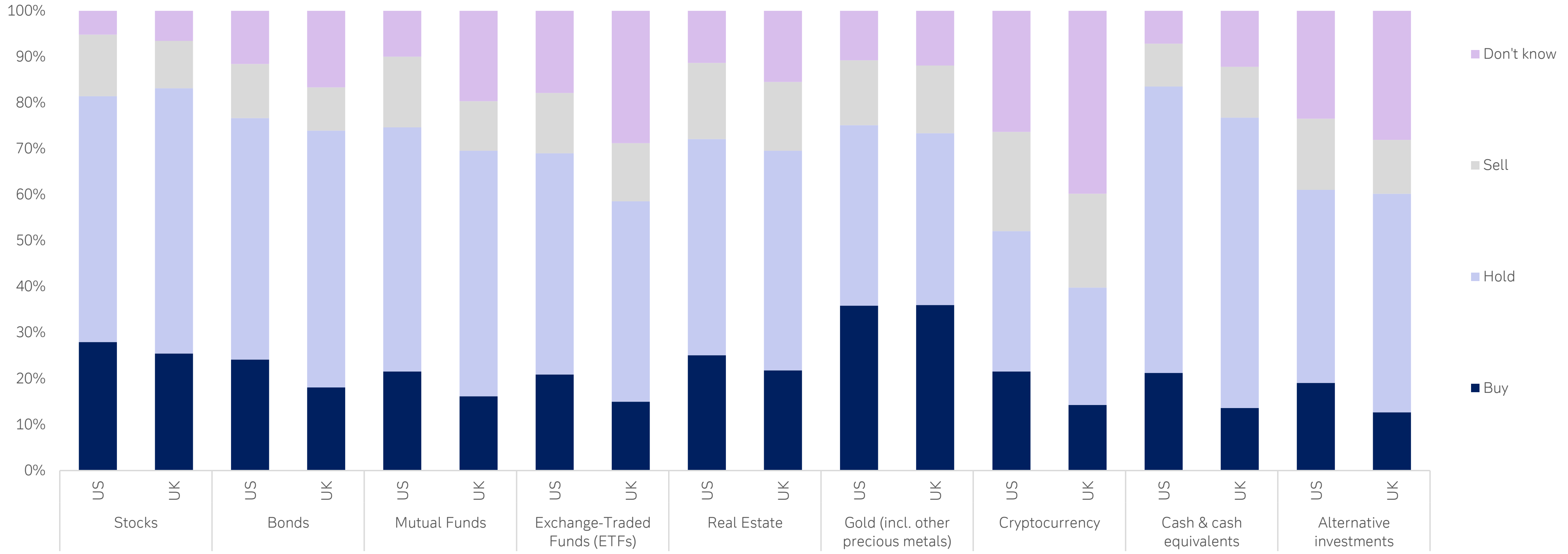


Source: dbDataInsights, Deutsche Bank.

What investors do when things are uncertain

The market volatility of the 2020s has been a great litmus test for how investors react to uncertainty. It is unexpected to see that equities are seen as a stronger 'buy' option than bonds

"In any period of economic uncertainty, what do you think is the best strategy for each of the following investment asset categories?"

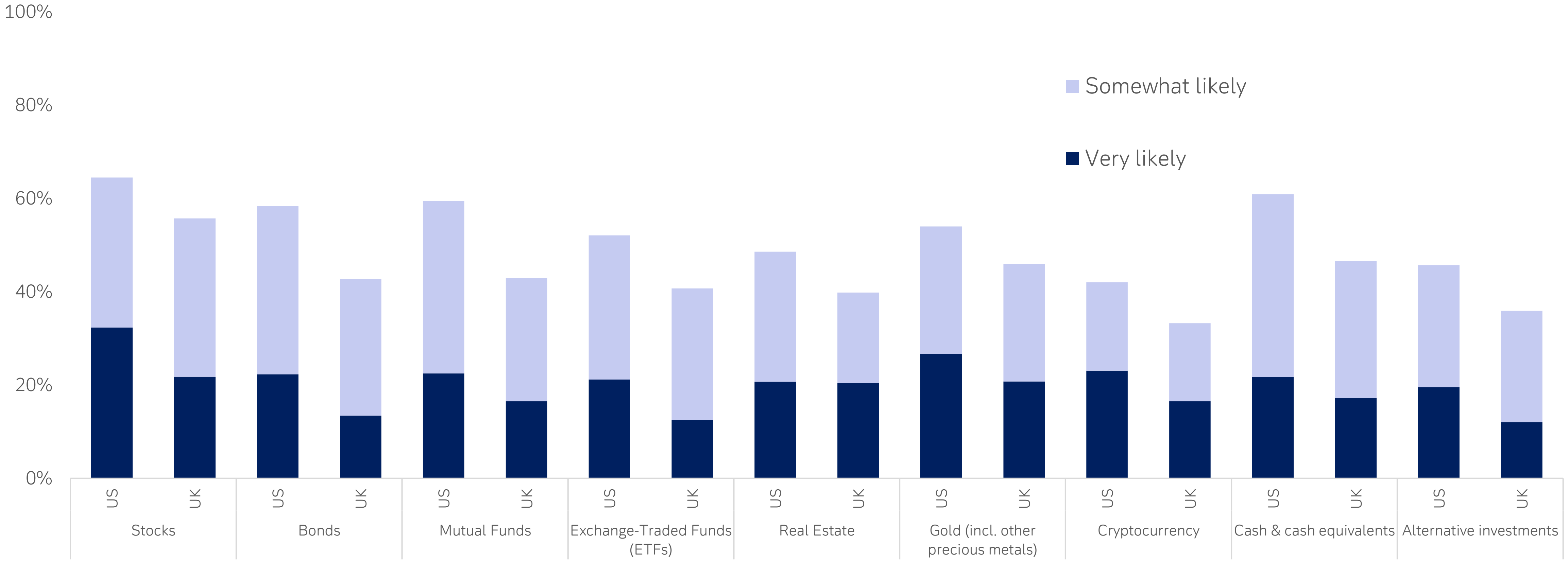


Source: dbDataInsights, Deutsche Bank.

Domestic politics can be highly correlated with economic growth …

… but the response from individual investors may be more muted than many expect

"If your country's government were to change in the near future, how likely or unlikely would you be to invest in each of the following asset categories?"

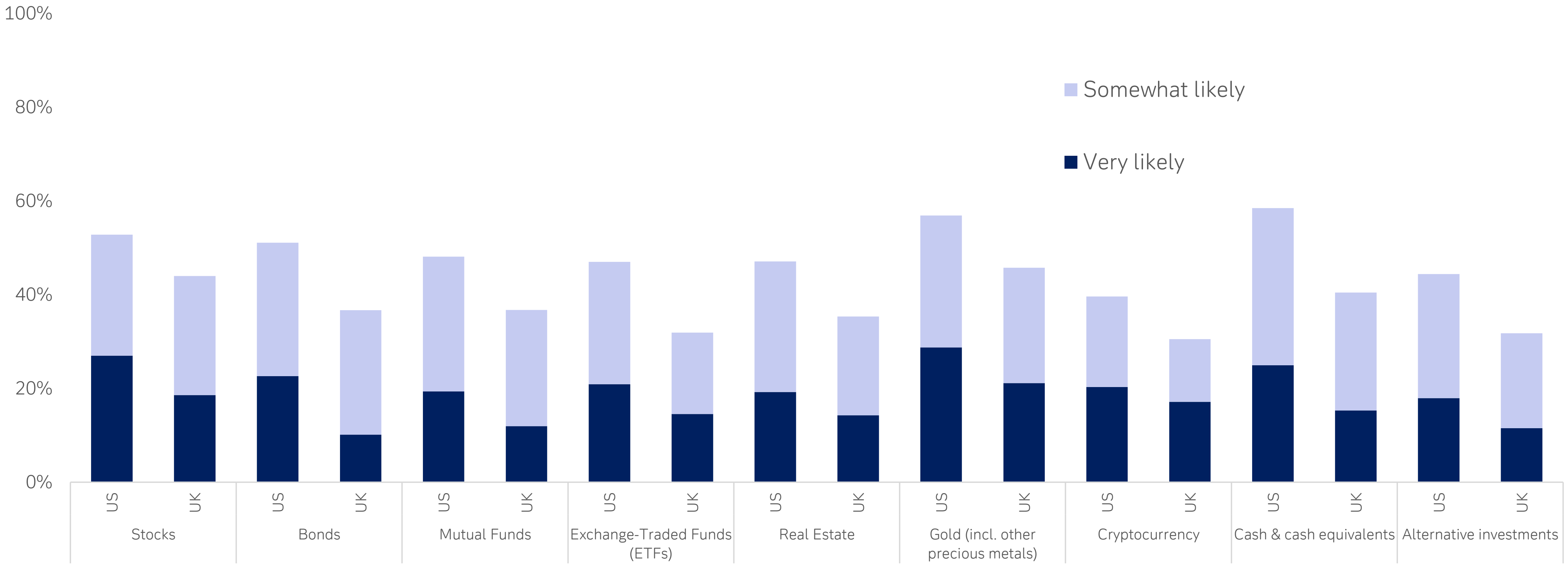


Source: dbDataInsights, Deutsche Bank.

Reacting to geopolitical uncertainty

Gold and cash are no surprise, but it is interesting to see that so many investors see equities as a geopolitical hedge. That may be another reaction to the rapid bounce-backs in equity markets this decade

"If tensions escalated between the US government and its major trading partners, how likely or unlikely would you be to invest in each of the following asset categories?"



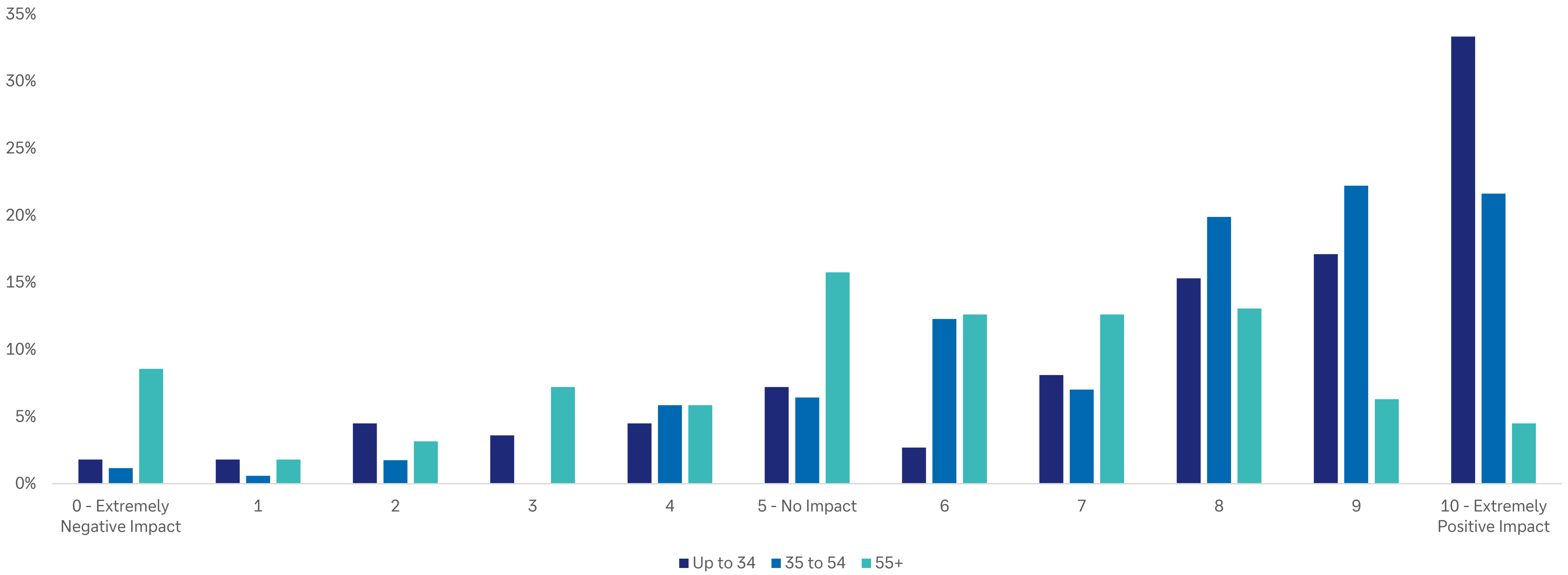
Source: dbDataInsights, Deutsche Bank.

What investors think about individual megatrends

Technology

Overwhelming optimism is the theme. The question is whether the returns will come from the specific assets to which investors are exposed, or from other more nuanced and diversified sources.

Investors' anticipated impact of technology on their main investment portfolio over the next 3 years (0-10)

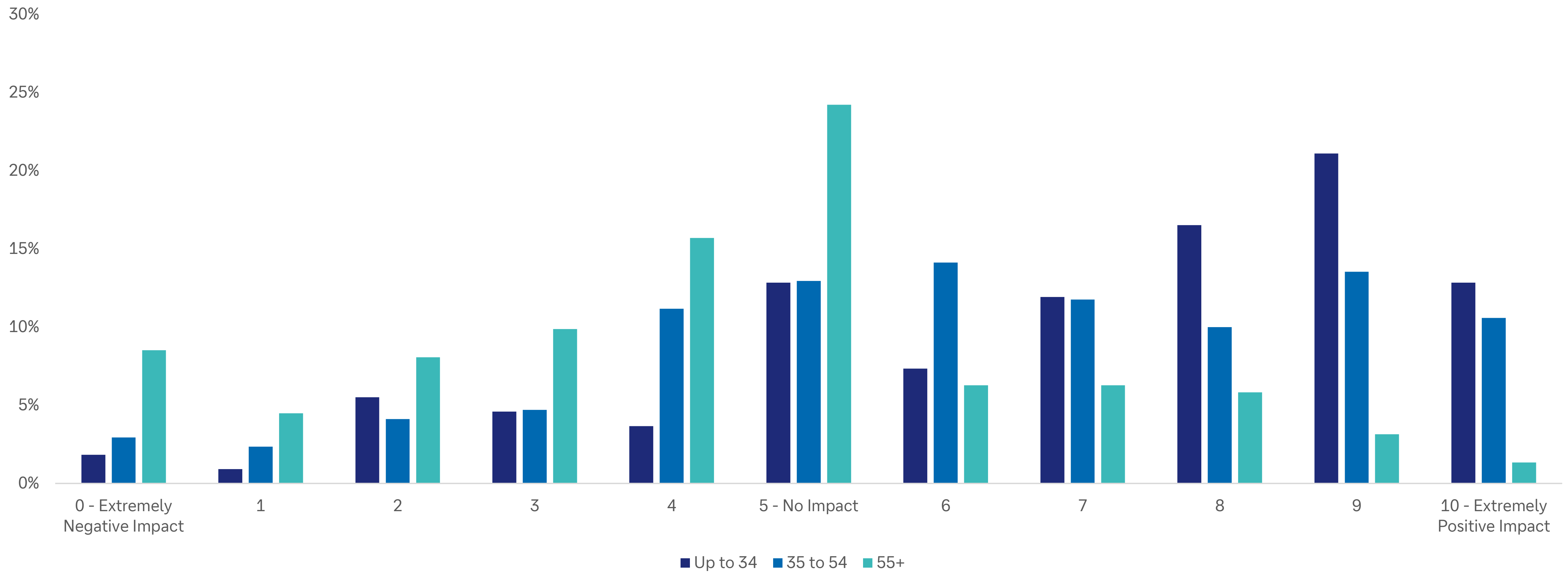


Source: dbDataInsights, Deutsche Bank.

Sovereign deficits

People are more optimistic than we expected about the impact of sovereign deficits. This may reflect an opinion that, somehow, governments will find a way to muddle through their current problems.

Investors' anticipated impact of sovereign deficits on their main investment portfolio over the next 3 years (0-10)

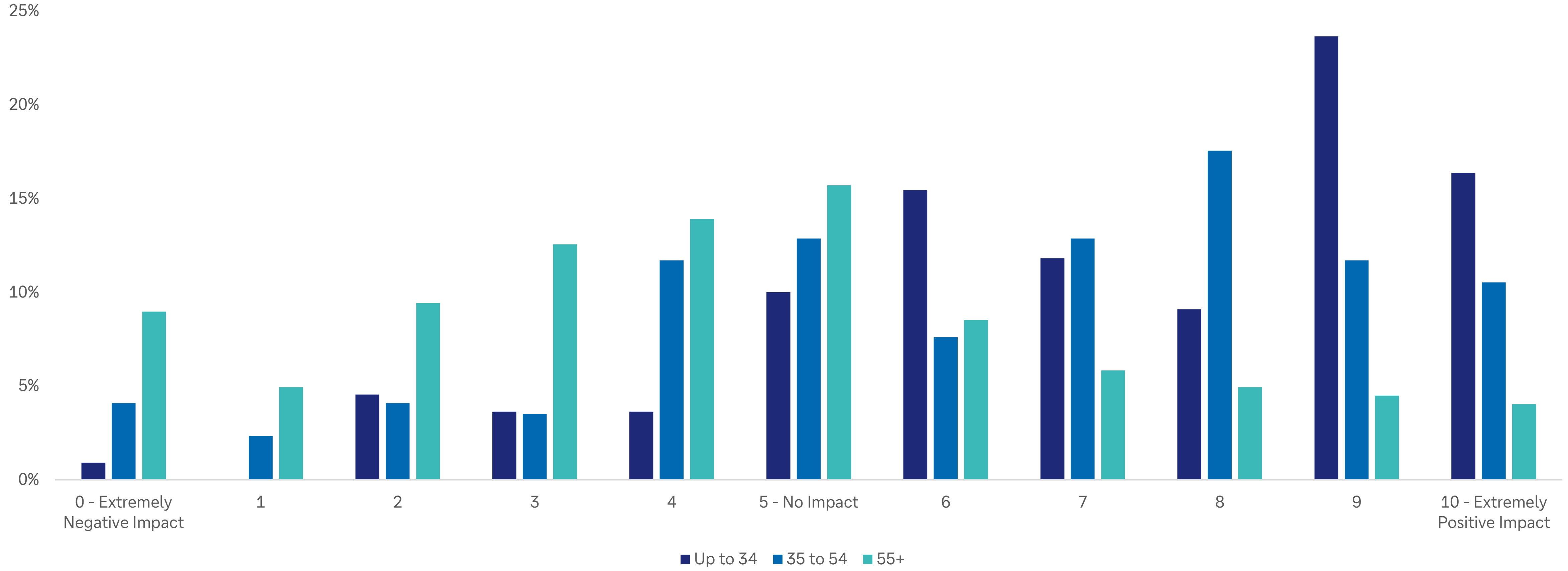


Source: dbDataInsights, Deutsche Bank.

Geopolitics & Globalisation

Perhaps the biggest surprise was that people generally expect geopolitics to have a positive impact on their investments. There are some strong arguments for this, particularly as trade has proved resilient to the recent tariff wars. However, it seems that most people expect current geopolitical conflict to resolve in their country's favour.

Investors' anticipated impact of geopolitics & globalisation on their main investment portfolio over the next 3 years (0-10)

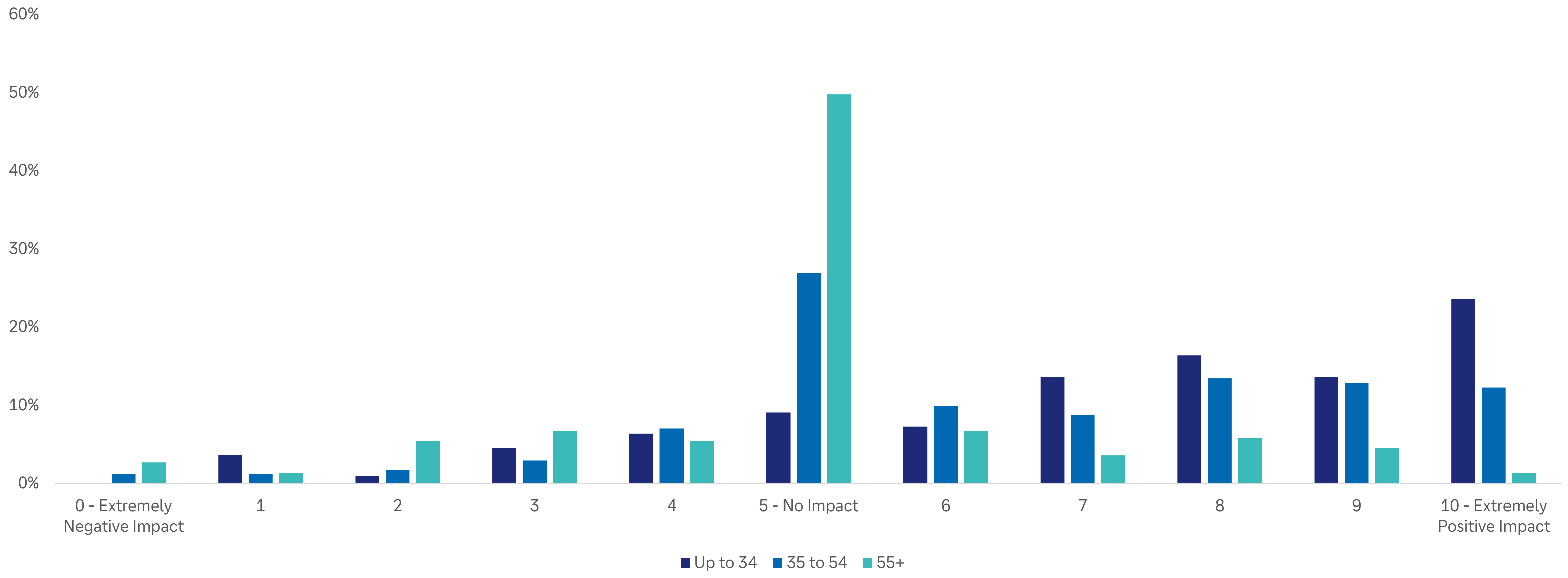


Source: dbDataInsights, Deutsche Bank.

Domestic politics & social unrest

Our model shows that domestic politics is heavily correlated with economic growth; however, investors do not rate it as a key investment factor.

Investors' anticipated impact of domestic politics & social unrest on their main investment portfolio over the next 3 years (0-10)

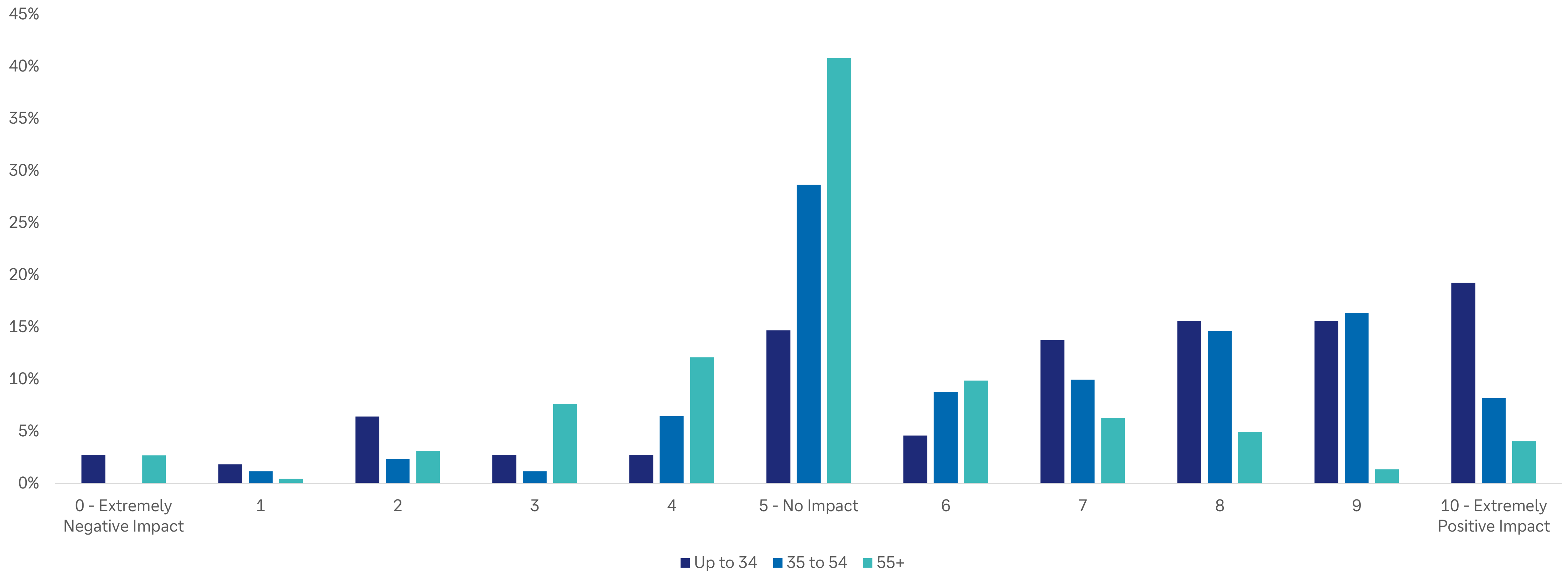


Source: dbDataInsights, Deutsche Bank.

Demography

Far more people are optimistic about demography. This also aligns with the positive tilt for sovereign deficits (which are deeply intertwined). To be fair, there are some underappreciated positive aspects to demography such as increasing healthspans and the consumer power of older people relative to those in the past.

Investors' anticipated impact of demography on their main investment portfolio over the next 3 years (0-10)

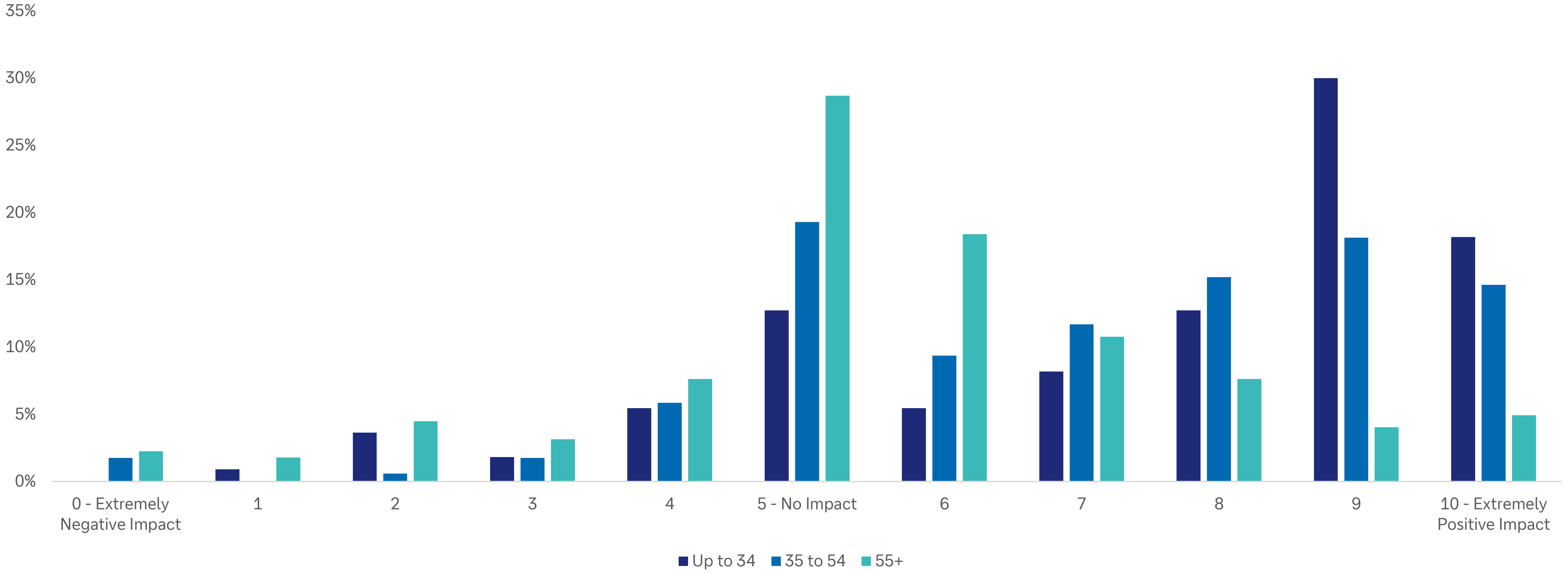


Source: dbDataInsights, Deutsche Bank.

Energy disruption & transition

There have been booms and busts in various alternative energy assets over the last decade. However, there is a strong realisation that energy (from all sources) will become increasingly important in an age of both AI and geopolitical weaponisation.

Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years (0-10)



Source: dbDataInsights, Deutsche Bank.

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Key takeaways from the Deutsche Bank Global Auto, Mobility & Robotics Conference

May 22, 2026

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Key takeaways from the Deutsche Bank Global Auto, Mobility & Robotics Conference

Held at the Deutsche Bank Center in New York, the recent Global Autos, Mobility & Robotics Conference brought together industry leaders for two days of discussions on innovation, strategy, and the next generation of solutions and capabilities. The event provided a comprehensive view of how the sector is evolving – highlighting both near-term resilience and the structural shifts reshaping the industry. From changing demand dynamics and geopolitical pressures to the rise of new technologies and growth areas, the conversations underscore a sector in transition. These key conference themes capture the core insights shaping the outlook for autos, mobility, and robotics today.

Resilience for now, but risks building

Auto markets in Europe and the US remain relatively resilient, with demand holding up better than expected in the near term. However, visibility is limited and companies are increasingly cautious about the outlook. Geopolitical tensions – particularly in the Middle East – are a key risk, with the potential to push up costs and weigh on demand. While impacts so far have been contained, a more challenging environment is expected if pressures persist into 2027.

Costs rising, pricing power limited

Cost pressures are building across energy, raw materials, and logistics, and are starting to flow through supply chains. While companies have partly offset this through hedging and cost controls, pricing power remains limited – especially in the US and Europe. Rather than raising prices directly, firms are relying on product mix and model upgrades, creating ongoing tension between protecting margins and sustaining demand.

Geopolitics driving cost pressures

Direct exposure to the Middle East remains limited for most companies, but the conflict is beginning to impact costs – particularly in energy and logistics. While hedging has softened the near-term effect, pressures are expected to build into 2027 as protections roll off. Early signs of supply chain inflation are emerging, with higher costs gradually passed through the system, often with a lag, and companies stepping up efficiency measures to offset the impact on margins.

New growth areas emerging beyond autos

Suppliers are increasingly looking beyond automotive markets to drive future growth, targeting areas such as data centres, energy systems, and defence. While these segments remain small today, they are seen as important medium-term opportunities. Continued investment and partnerships are expected to gradually scale these businesses and diversify revenue streams over time.

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China reshaping global competition

Chinese automakers are rapidly expanding internationally, gaining share across Europe, Southeast Asia, and South America. This is reshaping competitive dynamics and forcing established players to adapt through localisation, pricing changes, and new partnerships. For suppliers, aligning with Chinese exporters presents growth opportunities, but also intensifies competition and pricing pressure in global markets.

EV momentum moderating

EV growth is continuing but at a slower pace, as consumer demand shifts toward hybrids and traditional vehicles. In response, companies are taking a more balanced approach to powertrain strategies and scaling back the pace of EV investment. This reflects a more pragmatic, demand-led approach, with hybrids benefiting from extended lifecycles and a broader mix of technologies gaining prominence.

Efficiency and restructuring a key focus

Companies are stepping up efforts to improve efficiency, including restructuring programmes, underutilized plant closures, and footprint reductions. These measures aim to protect margins in a more uncertain environment. At the same time, adoption of automation and AI is accelerating, as firms seek to streamline operations and build more flexible, cost-efficient business models.

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ChatIPO: Quick take on OpenAI's record-breaking IPO plans

May 21, 2026

Authors

Adrian Cox
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Putting a giant share sale in context

The news that OpenAI is preparing to file for an initial public offering very shortly suggests that the maker of ChatGPT aims to strike while the iron is hot.

The poster child for AI plans to make a confidential filing with regulators as soon as tomorrow and aims to go public as early as September, the Wall Street Journal reported late yesterday, citing people familiar with the matter.

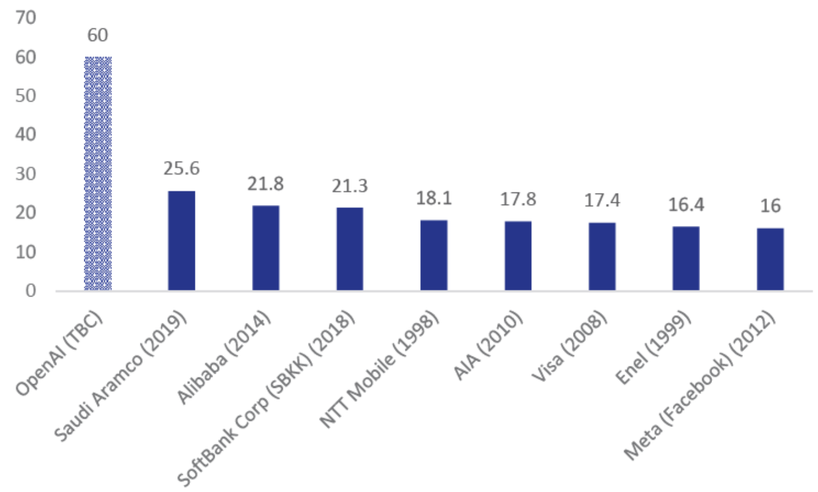
With an anticipated surge in IPOs this year, including of its fast-growing rival Anthropic, there will be a scramble to make the most of investor appetite for direct exposure to pure-play AI companies in the public markets.

Retail investors currently have to pick companies with AI offerings in other parts of the supply chain, either in infrastructure, such as semiconductor makers or cloud providers, or applications that sit on top of the foundation models. Shares in the Magnificent Seven tech companies have quadrupled since OpenAI launched ChatGPT on November 30, 2022.

OpenAI was valued at $852bn after completing its latest funding round at the end of March. Reports have suggested it could be planning to raise $60bn in an IPO that would value it at more than $1trn. That would be the biggest IPO ever, twice as big as Saudi Aramco's IPO, which was valued at $25.6bn in 2019, assuming no one else gets there first.

Even so, it has yet to be seen how public markets will value OpenAI and its peers once they open up their financial statements to scrutiny and explain the still little-understood economics of their business models.

Figure 1: A $60bn IPO would be more than double Saudi Aramco's record ($bn)



Source: Investopedia, media reports, Deutsche Bank Research

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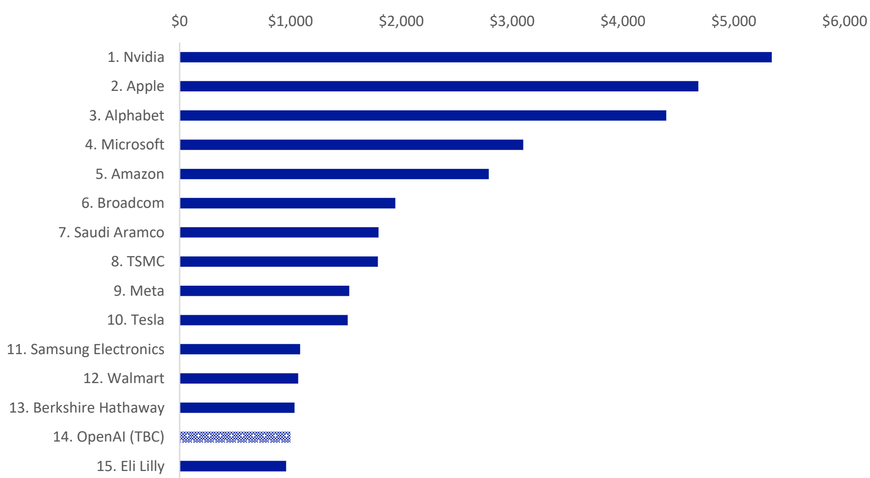
A trillion-dollar valuation would also make OpenAI, which has yet to make a profit and will reportedly make $30bn in annualised revenue this month, the 14th biggest company in the world by market capitalisation.

That would be just behind Berkshire Hathaway, which made revenues of over $370bn and net earnings attributable to shareholders of $67bn last year, and just ahead of Eli Lilly, whose sales topped $65bn and profit was $21bn.

The most valuable company is Nvidia, perhaps the nearest thing investors currently have to a large pure-play AI investment, whose shares have surged more than 13 times since the launch of ChatGPT to give it a market capitalisation of $5.4trn.

Figure 2: A $1trn valuation would make OpenAI the world's 14th biggest company ($bn)

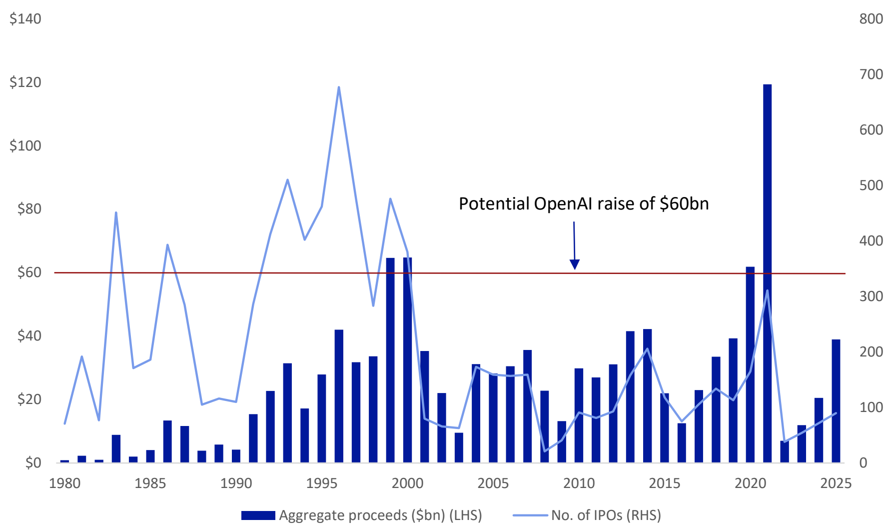


Source :Bloomberg Finance LP, Deutsche Bank Research

OpenAI is in a race with Anthropic, which overtook it in sales last month and is on track to generate $40bn in annual recurring revenue this month, according to data from The Information. The maker of Claude may be looking to raise more than $60bn in its own IPO this year, media reports say, even while it is simultaneously in talks with investors about raising cash at a valuation of $900bn – nudging it ahead of OpenAI.

A $60bn IPO would also be larger on its own than the total amount raised in US IPOs in all but four years since 1980, according to data from the University of Florida. It would barely fall short of the $65bn raised in both 1999 and 2000, and the $62bn raised in 2020, and would be half of the record-breaking $119bn raised in 2021.

Figure 3: A $60bn IPO would be just short of three of the four biggest years for US IPOs since 1980



Source : Jay Ritter, University of Florida, Deutsche Bank Research

While there may be concerns about the capacity of the market to absorb a number of IPOs valued at several hundred billion dollars this year, they would slot into an US stock market worth about $70trn overall. That is five times larger in nominal terms than it was even at the peak of the dot-com bubble in the late 1990s. At that time, there was an average of almost 500 IPOs a year, compared with about 120 this decade, and they are typically now coming to market at a more mature stage.

Last week, AI chipmaker Cerebras raised $6.4bn, including the overallotment option, in the largest IPO this year. The shares jumped by two thirds on its first day of trading, giving it a market value of roughly $67bn.

OpenAI has been preparing the groundwork for an IPO in recent months, streamlining its offering, focusing more on paying enterprise customers and clarifying its partnership with Microsoft. Earlier this week it won a long-running legal case brought by Elon Musk.

See the Deutsche Bank Research Institute for recent relevant research, including Will Claude's AI finance agents change your life? and AI, Iran and the Gulf 101. The website is available to all so feel free to share.

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Megatrends: AI vs the decade's structural headwinds

May 20, 2026

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Executive summary

We have never been more concerned about the six megatrends that impact economies and markets than we are today. Indeed, our new AI-powered megatrend model shows that the world faces severe headwinds from several overlapping megatrends that, post-WWII, has only been seen during the 1970s oil crises and the onset of the 2008 financial crisis.

In this piece, we explain our new megatrend model and how it uses AI analysis to quantify qualitative signals, filters our proprietary human-created data from dbDataInsights, and layers it all beside traditional data series. In all, we take almost 100 data points and track how each of the six global megatrends have waxed and waned quarterly over the last 70 years. In particular, we focus on the impact of these trends on GDP growth, equity markets and other key economic outcomes.

We then use our model to estimate how the most important megatrends will develop over the rest of the decade. We conclude that the defining feature of the next 5 years for developed economies will be the push-pull between technology and sovereign deficits. In other words, can the productivity benefits of AI technology outweigh the burden of growing sovereign deficits? All this will take place against the backdrop of several other megatrends which will both complement and work against us.

There is some good news. Our model shows that the current economic impact of technology is growing. And history shows that our technology indicator is correlated with growing productivity growth. Thus, how AI is adopted across corporates and economies will determine whether or not developed economies can overcome the drag of the megatrends working against us.

Finally, we look at the investment consequences of the current trajectory of the six megatrends. We outline a series of medium-term investment concepts that investors should consider, and then present a matrix that looks at several key asset classes. We also explain that traditional haven assets are not as effective as they were.

Megatrends are the biggest forces that shape markets and economies over time. Yet, they confound investors because they are rarely observable on any one day's trading. But what is clear from our model is that we are on the cusp of major historical change. We are certain that this transformation will demand more of us than has other episodes of profound shifts in the economic regime.

We thank the dbDataInsights team for their input into this publication.

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2. Interesting statistics on a page

- The other tech boom: In 2021, our technology indicator hit its highest point since the 1990s as lockdowns spurred massive tech investment.
- Through most of the 1960s, 70s, 80s, and 90s, technological improvements were associated with lower inflation. Although that relationship has reversed over the last decade, we put this down to non-tech factors (such as covid stimulus programmes). We expect greater tech improvement to lead to lower inflation in the medium term.
- Post-WWII, US real economic growth over any ten year period has averaged 3.8% pa.
- Since 1790, the US had 93 years where 10yr GDP growth was above 5% pa. In contrast, there were only 21 years where 10yr GDP growth was lower than 1.5% pa.
- 46% of Americans think today's investment environment is more certain than it was in the 2010s. Only 20% think it is more uncertain.
- 32% of Britons think today's investment environment is more certain than it was in the 1960s. Only 22% think it is more uncertain.
- 68% of the world's current GDP is generated by 12 countries in the G20 with declining working age populations.
- The role of domestic politics & social discontent may be underestimated by investors. Indeed, our megatrend indicator for this peaked in the 1990s - something that helped boost the tech boom. Today, it remains sharply negative.
- Support for populist left has surged over the last three years. In the 10 largest European countries, the populist left has similar support levels to centrist parties. The populist right enjoys the highest aggregate support.
- 67% of Americans believe the technology megatrend will help their investment portfolio over the next three years compared with 18% who think it will have a negative impact.
- Since 1990, the amount of GDP generated per kilogram of oil equivalent has quadrupled in $PPP terms
- Americans are far more positive than are Britons about the impact of each of the six megatrends on their investment portfolio over the next three years.
- 45% of Americans think the trends in their domestic politics will have a positive impact on their investment portfolio over the next three years. In contrast, only 39% of Britons think the same.
- Indicators of domestic politics & social discontent tend to be well correlated with GDP growth.
- Over the last 20 years, M&A has become correlated with megatrends.
- Since 2020, the combination of six key haven assets have failed (positive correlation with equities) over more time periods than they have worked. In particular, havens failed to protect against risk downturns during four big shocks of the 2020s (covid, rate rises, tariffs, Iran).

3. A new model to better understand megatrends

The precursor to the internet, Arpanet, was created in the 1960s. But despite this system, and its successors, being well known for some time as academic and government tools, it took three decades before a sudden shift in the technology megatrend catalysed the commercialisation of the internet and caused the associated market boom in the 1990s.

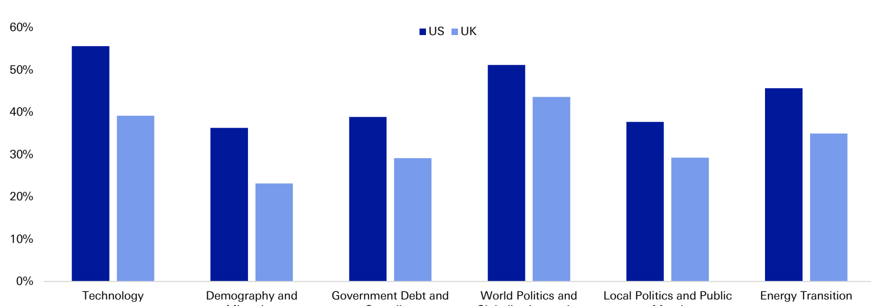
Megatrends are usually dinner party conversation because there is so much subjectivity and human bias involved. But now, AI can help us add the quantitative rigour needed to examine how megatrends affect markets and economies. Indeed, our megatrend model is only possible because AI allows us to quantify qualitative indicators, layer them alongside traditional data series and, most importantly, put an objective structure around the whole thing.

Megatrends have recently risen to prominence as our model shows that over the last 3 years (as interest rates have normalised) megatrends have become a key defining force in economies and markets. This has not been the case since before the 2008 financial crisis. As such, we expect them to define the rest of the decade and, as we will explain, we are deeply concerned at their current trajectory.

What investors think

Our recent dbdataInsights survey of investor attitudes towards megatrends shows that they are beginning to understand that megatrends have a growing influence on their portfolios. This one-off survey of US and UK investors complements our monthly surveys of a representative sample of the US and UK populations that feeds into our model. The following chart shows that investors in both the US and UK have made changes to their portfolios over the last three years as a result of all six megatrends.

Figure 1: Proportion of investors who have made a change to their investment portfolio in the last three years as a result of the following megatrends



Source : dbDataInsights, Deutsche Bank.

What exactly is a megatrend?

To us, a megatrend is a long-term, transformative force that shapes economies and markets. It can be seen in the data over multiple business cycles and has a proven ability to catalyse structural change. It alters how we live, work, interact, produce, and consume. It disrupts business models, create new industries, and reshapes global power dynamics.

Unlike exogenous shocks, megatrends are predictable to a degree. While their exact trajectory and impact can be uncertain, the underlying forces driving megatrends are often identifiable and can be analysed.

What our megatrend model does differently

Unique human-created data is critical in AI analysis to avoid 'samey' output. Hence, we complement our qualitative signals with data from years' worth of our proprietary surveys. All this is layered alongside traditional datasets that track economies and markets. Taken together, we can then use AI to draw out themes by connecting datasets from different industries and macro groups.

The end result gives us a structured framework for analysing how the world's megatrends are changing and how they are impacting markets and economies – specifically: GDP growth, equity prices, inflation, and interest rates.

Megatrends impact society on a different timeline to their impact on companies, economies and markets

The time gap between when a megatrend is talked about and when it actually impacts markets and economies can be very wide. Electricity, railroads, the internet, and demography are all obvious examples. Investment decisions that are too early are wrong. Our model aims to fix that by addressing megatrends in terms of their observable impact on market and economies.

As such, different aspects of our model can be either: 1) leading indicators of economic and market impacts or, 2) corporate, market or economic indicators themselves.

"Megatrends are back" – after decades in the wilderness, they will likely be the driving factor to 2030 and beyond

In 2015, our 'geopolitics & globalisation' megatrend indicator dropped heavily. That came as the shortage index fell, trade policy uncertainty rose, and the Kilian index was low. Just 12 months later, the couplet of Brexit and Trump triggered the end of unfettered globalisation and upended the way economies and market see risk.

Over the last four decades, people have either ignored or oversimplified megatrends:

- 1990s: Investors assume the once-in-a-century tech boom was solely responsible for everything that happened (spoiler alert: it wasn't, as we discuss later).
- 2000s: people treat the worst financial crisis in 70 years as an event in-and-of-itself that caused widespread problems, rather than being caused/augmented/amplified by many other trends.
- 2010s: QE and low rates mean nothing else mattered.
- 2020s: The many shocks of Covid, Ukraine, tariffs, Iran etc. are seen as one-off exogenous events and not part of the normal megatrends shaping economies
- The point here is that investors have largely treated most of the last 40 years as either being driven by unique, unlucky events or by low interest rates. As a result, there is very little institutional memory about how megatrends

influence markets and economies in 'normal' times. Investors are simply not used to analysing and incorporating megatrends in their investment decisions.

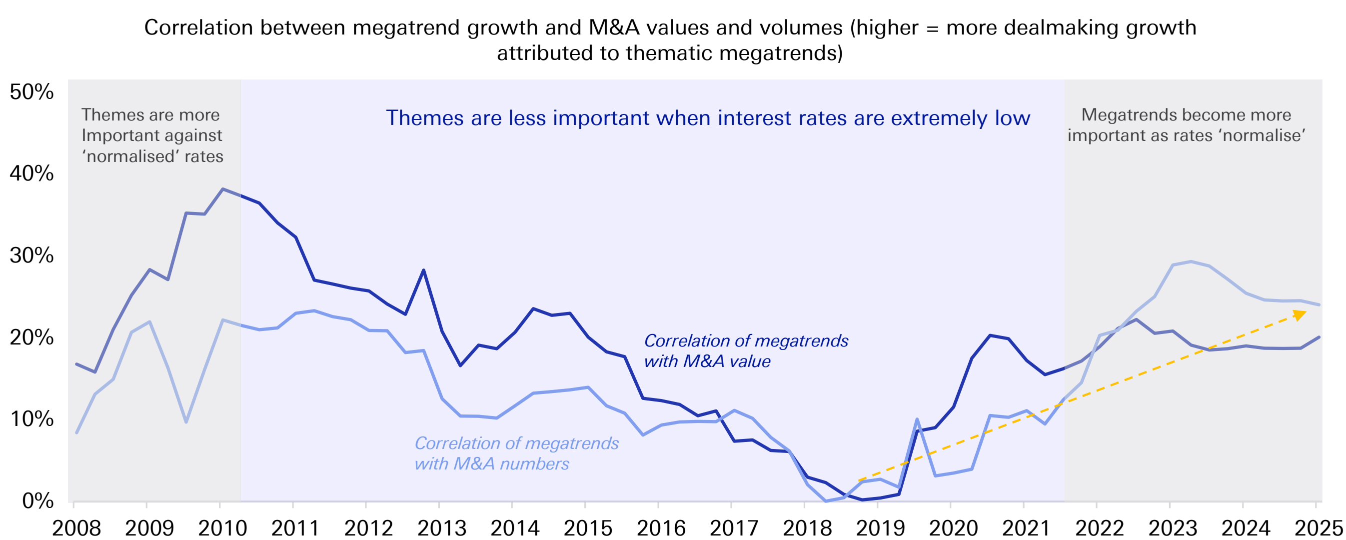
Now that we have moved out of the era of both very low interest rates and the covid aftermath, megatrends should have a more profound and obvious impact on markets and economies.

Greater volatility will amplify megatrends

Our megatrend indicators tell us to expect more volatility – in geopolitics, society, markets and, thus, corporates. Indeed, we have already seen how various sudden events this decade have led to supply shocks and these are increasingly driving an economic and market response – both negative and positive. On the negative side, they have pushed up inflation and constrained output. On the positive side, they have encouraged countries and corporates to pursue greater resilience.

This expectation of greater volatility comes right as the world settles into an era of more 'normal' interest rates. This is critical as low interest rates (and the aftermath of the aggressive rises in 2022) have been the defining feature of economies and markets for the last 15 years. Investors have become complacent as a result as the effects of low interest rates are sometimes very hard to see (our model shows it clearly). One of our favourite ways to see how interest rates have affected economies and markets is to compare megatrends to the thing that neatly intersects both these things – M&A.

Figure 2: M&A has become increasingly correlated with the growing impact of megatrends



*Source : Dealogic, Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank. * Note: rolling 10yr R^2*

Investors still underestimate how normalised interest rates are propelling megatrends to the forefront of economies and markets. Our analysis shows that everyone should pay more attention.

4. The megatrends and their impact on economies and markets

"There is no reason for any individual to have a computer in their home," – so goes the infamous quote supposedly uttered in 1977 by the founder of Digital Equipment Corporation. That same year, Apple unveiled the Apple II computer and the rest is history ...

Our megatrend model is based on the analysis of almost 100 data series (including AI quantification of qualitative indicators, and human-created survey data). These have helped us create a quarterly indicator for each of the six megatrends that shows how it positively or negatively impacts markets and economies over time.

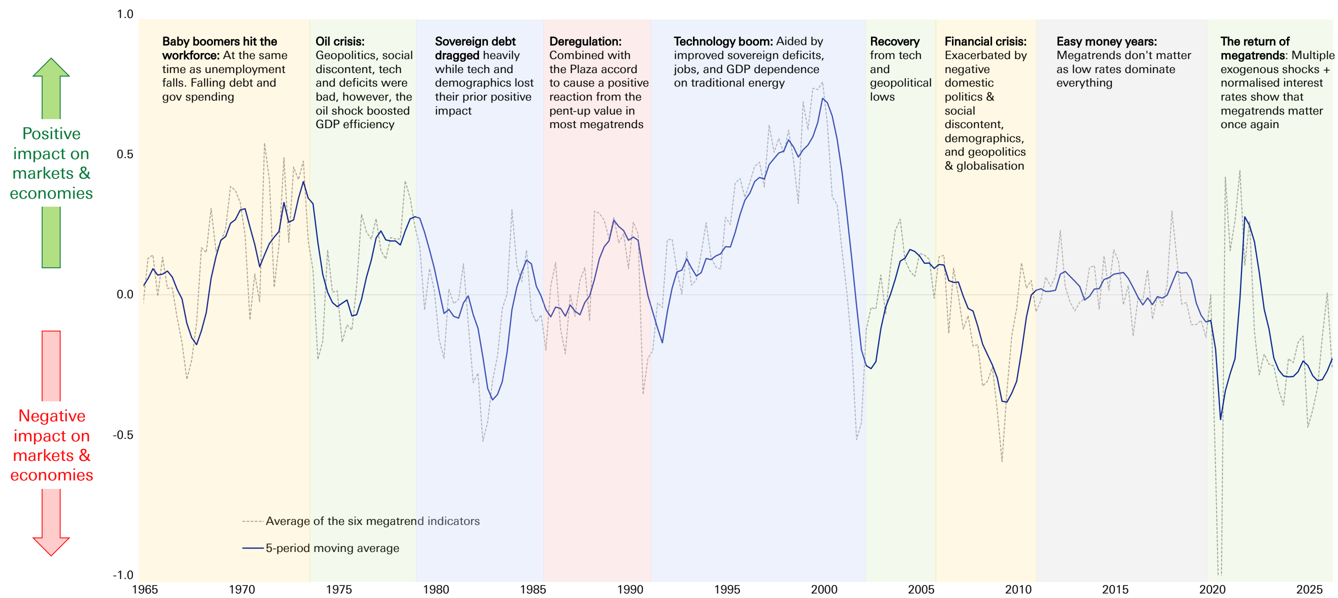
Below are the six megatrends that our analysis indicates should impact markets and economies. We summarise our view on each individually in the appendix.

- Technology – The 'adoption gap' is crucial to our analysis. This is the time period between when a technology is invented and when it has an actual impact on productivity, GDP etc. Our indicators predict that AI will turbocharge productivity and push markets and economies to new highs.
- Sovereign debt – This is the most important negative megatrend. The worsening deficit situation is being exacerbated by demographic decline. Yet, we are cognisant that some periods of high debt in the past have been followed by periods of high growth when other megatrends align.
- Geopolitics & globalisation – There are many negative signals, however, deteriorating international relations does not necessarily mean a linear negative impact on economies. Trade in goods can be maintained via reorganised supply chains. Arguably more important is the trade in ideas across borders which is permanently accelerating due to ever greater and quicker connectivity.
- Domestic politics & social discontent – Rising discontent is causing voters to support more extreme politicians with policies that may address short-term issues at the expense of longer-term macro prosperity. This trend will continue to drag on economies and markets
- Demography – This goes beyond an ageing population and immigration. Our indicators show that other factors are very important including, the incentive for young people to form households, have children, and participate in the economy.
- Energy disruption & transition – Our most complex megatrend: Energy security and diversification are key. However, we also see that although energy disruption causes near-term pain, it can lead to long-term positive effects. From this point of view, rising prices can incentivise processes that are more productive per unit of energy.

The current impact of the six megatrends on markets and economies

Our series of almost 100 datapoints eventually aggregate into one measure for each megatrend (standardised via z-scores) and our model examines these quarterly for almost the last 70 years. We can then aggregate these six megatrends into one indicator to see how the collective impact of megatrends overall is impacting markets and economies. The following chart shows that when our indicator is positive it means that megatrends, in aggregate, are having a positive impact on economies and markets. When it is negative, the reverse is true.

Figure 3: The combined impact of megatrends on economies and markets



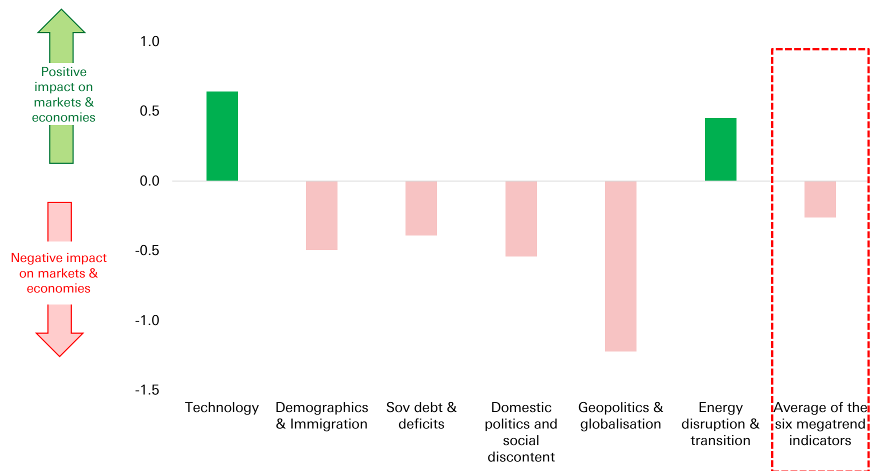
Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

There are many important takeaways from the above chart. Just a few include: 1) Today, the world faces severe pressure from the negative effects of megatrends, 2) The impact of megatrends on markets and economies can turn quickly, even if the underlying trend moves slowly, 3) Note the period during the 2010s (the era of low interest rates) where megatrends didn't matter.

Which megatrends are having a positive or negative impact on the world?

The following chart shows the position of our current megatrend indicators as of the latest quarter. Of course, any given quarter can have its own volatility, which is why we use a rolling average in the above aggregated chart.

Figure 4: The impact of megatrends on economies and markets (at March 2026)



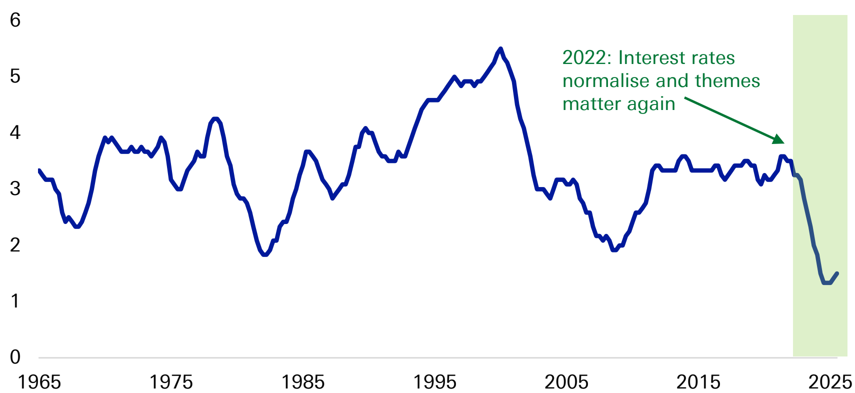
Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

5. The big concern today

Our aggregate megatrend indicator (Figure 3) tends to be more positive than negative. Indeed, it has spent twice as much time in positive territory than negative territory.

Yet, today, the indicator is deeply negative. Adding to that concern is that the number of megatrends working positively is historically low – this can be seen in the following chart. The two similar occurrences of these lows were associated with severe downturns (note that these figures are based on a 12-quarter moving average in order to minimise the noise from quarterly events).

Figure 5: Number of megatrends (out of 6) that are having a positive influence on markets and economies (12qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Megatrends are not independent from each other …

A typical flaw in many megatrend analyses is that they address the various themes independently because the overlap between them is just 'too hard' to figure out. Our model adjusts for this by accepting there is overlap between some of the data series that impact each theme. In addition, we standardise all the results so that the six megatrends can be compared for their impact and then aggregated. As such we, as investors, can see how important any combination of themes is at any point in time.

… nor are they consistent

Megatrends do not have a constant impact on the world, and they do not grow at a steady rate. Rather, their impact on economies and markets waxes and wanes over time. The following example shows how.

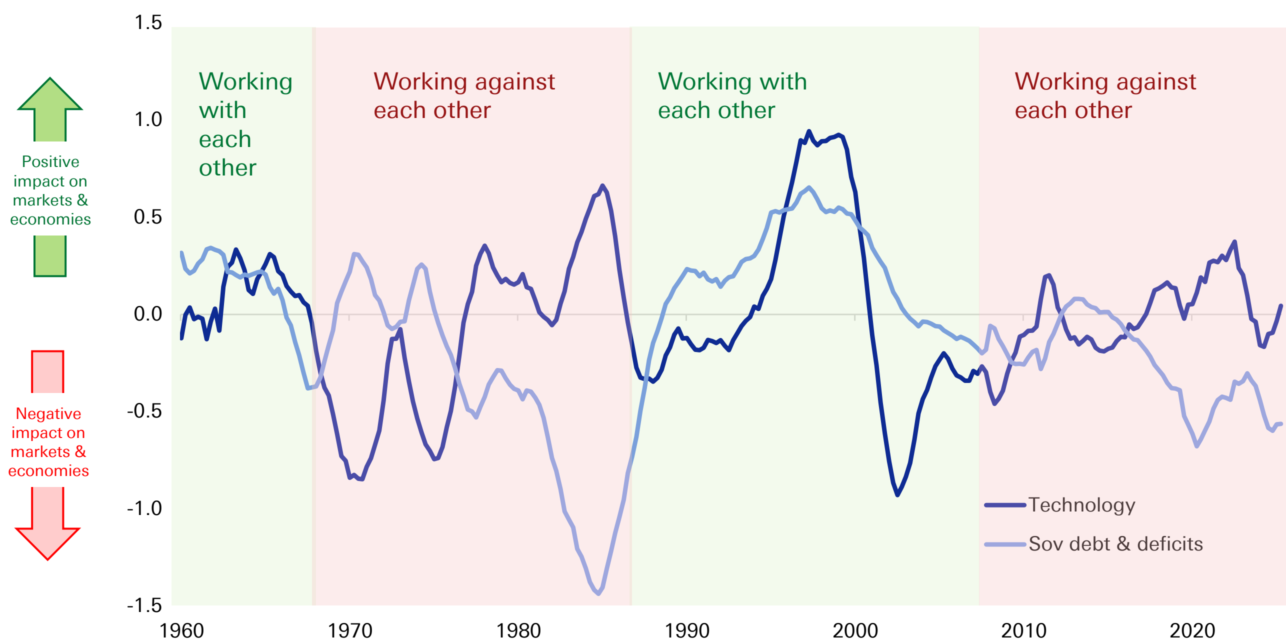
An example: Why the 1990s were so good

Figure 3 categorises each of the eras above by the 'typical' reason cited for them. However, this is far too simple. That is because different megatrends work in different phases – at times they complement each other, at other times they work against each other.

Take the 1990s. This decade was such a good period for markets and economies not just because of the tech boom. It was also a period where most of the megatrends were positive most of the time. Thus, relatively stable politics aided policy, the workforce was generally strong, and the background of all this reduced concerns about sovereign debt. The combination of several strong megatrends provided the fuel for the technology boom.

To illustrate, let's show just two of the trends. The following chart shows the interaction between the technology megatrend and the sovereign deficits megatrend. The green shading shows the periods when the two have worked together and the red shading shows the periods where the two have worked against each other.

Figure 6: Megatrend indicators for 'technology' and 'sovereign deficits' (12q average)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

The implication for the 1990s should be clear. Technology was racing ahead with the proliferation of the internet at the same time that growth in US government spending was low or negative, debt was falling as a proportion of GDP, and the 10-year term premium was falling. The combined result turbocharged markets.

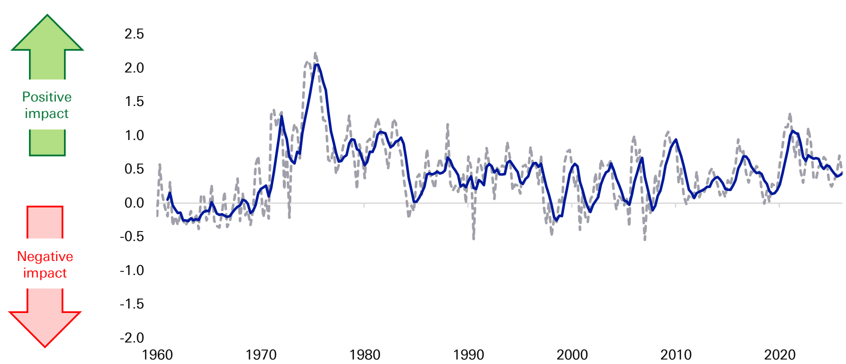
In other words, the positive impact of technology was amplified by the positive effect of the improving sovereign fiscal position. And that is before considering that three other megatrends were also making a positive contribution (geopolitics & globalisation, domestic politics & social discontent, and the energy transition). In fact, only the demographic megatrend worked against things. This was driven by rising house prices and falling growth in household numbers, which more than offset the positive economic impact of rising immigration.

In an equal and opposite way, the tech bust in the early 2000s did not impact the world by itself. Indeed, the bursting of the tech bubble coincided with 9/11, war, and recession. It was a time when the housing prices accelerated, the job market was tough, real wages ran negative, and the labour participation rate fell.

Another example: Lessons from the 1970s oil crises

The oil crises were overwhelmingly bad for society and the economy. However, during the 1970s, our megatrend indicator showed a mixed response. That is because the negatives of geopolitics, social discontent, technology and sovereign debt were offset somewhat by a huge increase in the energy megatrend. As a result, the world started finding new ways to create energy away from the dependence on the Middle East. Since 1990, the amount of GDP generated by one kilogram of oil equivalent has quadrupled in $PPP terms (World Bank). The chart below shows how the energy disruption megatrend has remained broadly positive over time.

Figure 7: The impact of energy disruption & transition on markets and economies (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Today's tech boom may be different from that in the 1990s

The background to today's technology boom is very different from that of the 1990s. Whereas the 1990s version was complemented by the positive effects of most of the other five megatrends, today's is not. Most megatrends are working against economies and markets. Therefore, if a technology boom is going to help bail out the malaise in the world economy it may have to work even harder to overpower the other megatrends.

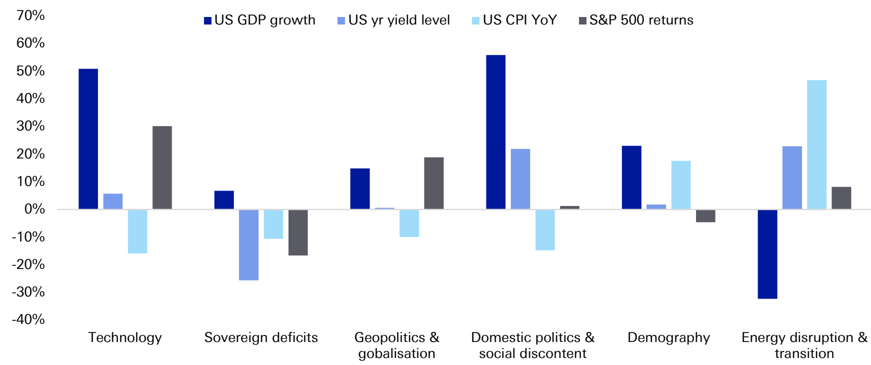
So, is it possible for AI to be significantly more impactful than was the growth of the internet in the 1990s? Our analysis shows that not only is it possible but also it is the most likely outcome. This is extremely fortunate for the global economy as several other megatrends are working against the world and need to be overcome in order for us to avoid a slip into economic malaise.

How important are our megatrend indicators for markets and economies?

The main point to tracking the six key megatrends is to assess their impact on key markets and economic variables. The following table shows the correlation of the megatrends and four key factors: US GDP growth, the S&P 500, 10yr treasury yields, and US interest rates. These calculations start, at the latest, from 1962 and in most cases earlier depending on data availability.

The following chart shows the 'all-time' (since 1960) correlation between our megatrend indicators and the four economic and market factors.

Figure 8: Full sample correlations between our megatrend indicators and the four economic and market factors (Q1 1960 to Q1 2026)



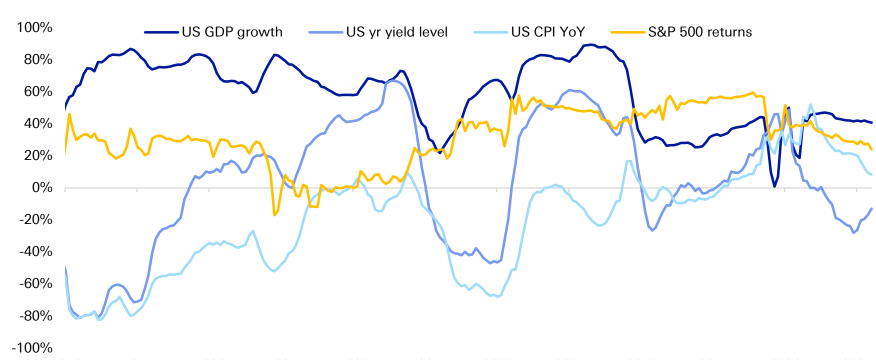
Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

There are two points to note

1. Big numbers – each of the four market and economic variables is materially correlated with several megatrends.
2. In any given period, a megatrend may be more or less correlated with the four market and economic variables than the above 'all-time' table indicates.

Take, for example, technology. This is generally very important for GDP throughout all periods. However, its effect on the other three variables has alternated between being very important and less important. The following chart shows how this has changed over time. In particular, the overall correlation with 10-year treasury yields is low, however, it is wrong to dismiss the relationship – as the following chart shows, technology and treasuries have had a very important relationship throughout most of the last 60 years.

Figure 9: Rolling 10yr correlations of our technology megatrend indicator with the big four indicators



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

6. Where our model shows megatrends are taking the world

The status quo is almost certainly wrong

Many corporates and investors today say they are preparing for volatility but, actually, they assume the status quo. They assume average economic growth of 1-3%, ongoing stability in debt markets, inflation that is elevated but manageable, and average stock market returns.

This status quo is almost certainly wrong. Our megatrend model shows that, as we look to 2030 and beyond, one of two scenarios for the developed world is likely, and it all depends on technology, in particular AI. Indeed, the backdrop of deteriorating sovereign debt and the demographic overhang means that technology is the most important, most likely (and perhaps only) way that the world can fight and win against this and other negative megatrends.

We must be honest and say that AI can go one of two ways – either it will fulfil its promise (as we believe) and turbocharge productivity, or it will fizzle into an average tech with merely incremental impact (we exclude the tail risk that AI goes rogue and destroys the world). In other words, either sovereign debt and poor demographics will become an overwhelming burden, or, technology will overcome it and push economies and markets to new highs.

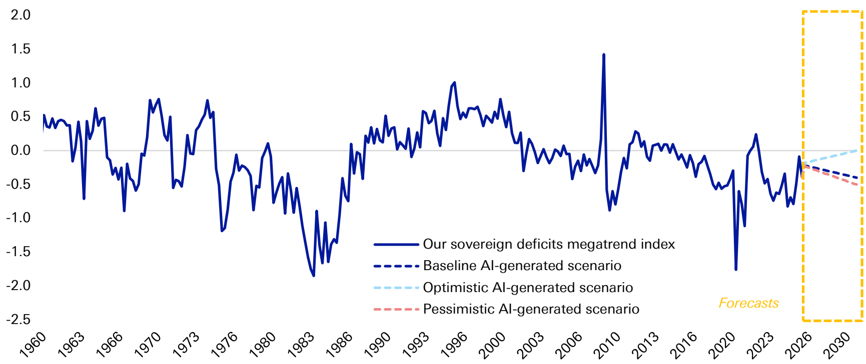
So, investors and corporates should prepare for either a productivity boom or a severe, prolonged downturn. And regardless of which scenario wins, because AI is now at the beginning of the J-curve of adoption, the coming few years will be punctuated by upheaval.

What tech is fighting against

Of all the six megatrends, the most worrying is sovereign deficits. In many developed countries, the pressure of already-large debt piles is likely to grow as the projection for budget deficits remains deeply concerning. Indeed, US debt is expected to rise from about 100% of GDP to 120% by 2036 (CBO).

The following chart shows how our model expects the sovereign debt megatrend to deteriorate over the rest of the decade.

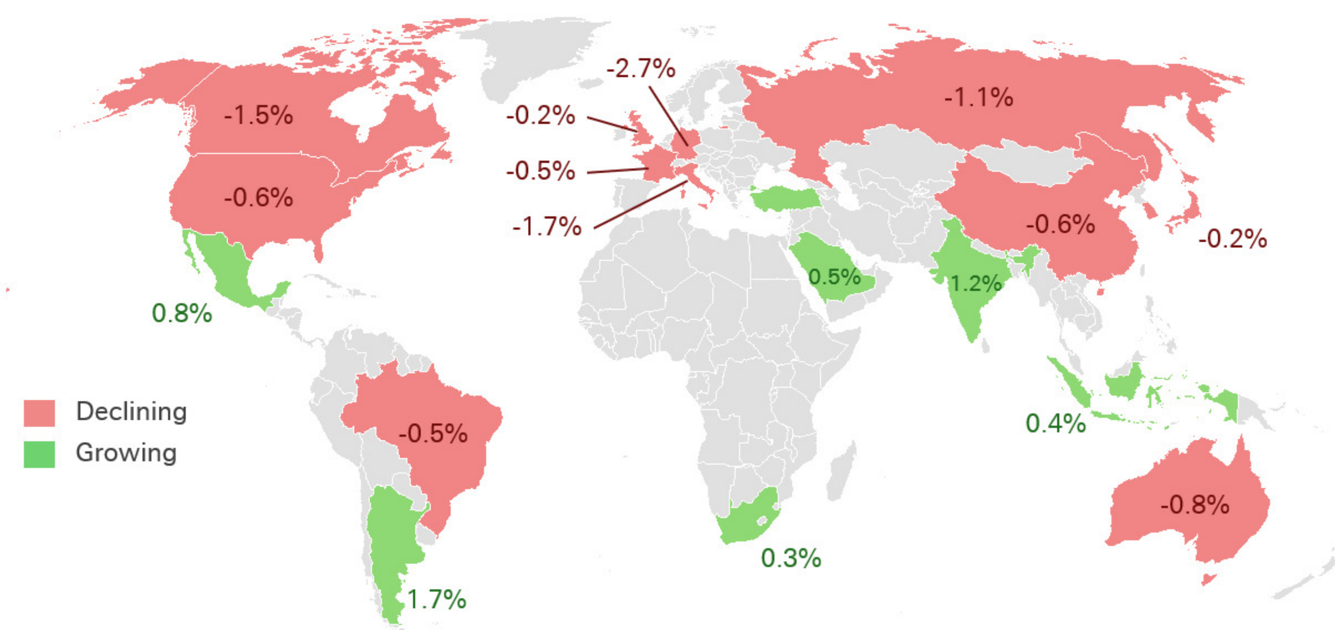
Figure 10: Our sovereign deficits megatrend index with AI-generated scenarios to 2030



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

A key driver behind the sovereign debt problem for most developed countries is poor demographics. The following chart shows the extent of the problem.

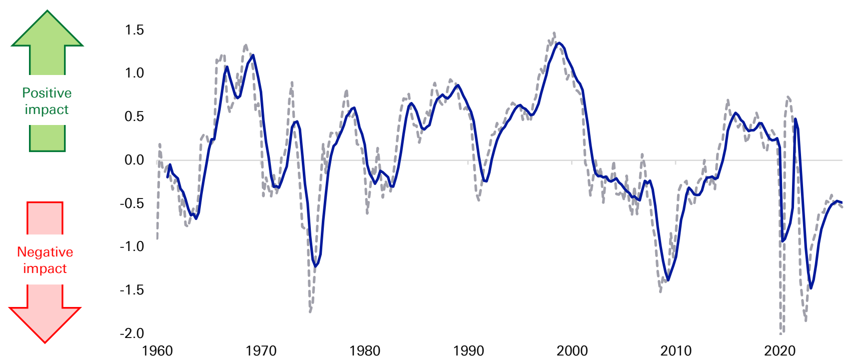
Figure 11: Growth in G20 countries' working age (20-64) population between 2025 and 2030 (as a % of the total population)



Source : UN, Deutsche Bank.

Demographic problems are well known. What is less well understood is how the megatrend of domestic politics and social discontent may also worsen the sovereign debt megatrend. Indeed, we measure social discontent based on many measures including, housing affordability, real wage growth, labour share of income and more.

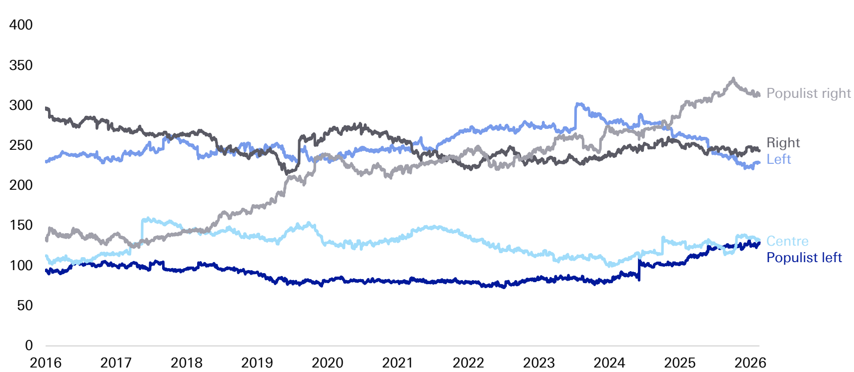
Figure 12: Our domestic politics & social discontent megatrend index



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Growing social discontent raises the risk that people will vote for populist politicians (from any side of politics) that may enact policies that add to debt today without regard for the long-term consequences. Of course, some politicians that are described as "populist" can exact positive, constructive policies, however, we fear that many typically do not. The following chart shows how populist support in Europe has increased over the last decade.

Figure 13: Sum of support for each category of current political parties across the 10 largest European countries

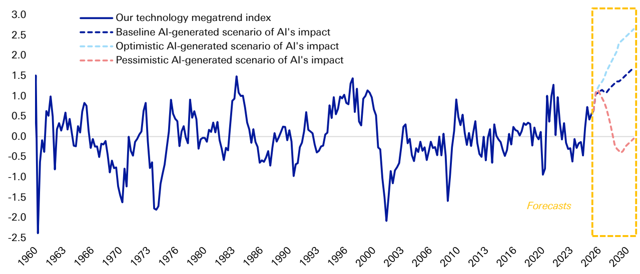


Source : Politico, dbDataInsights, Deutsche Bank.

Can tech outweigh the worsening trend of sovereign debt?

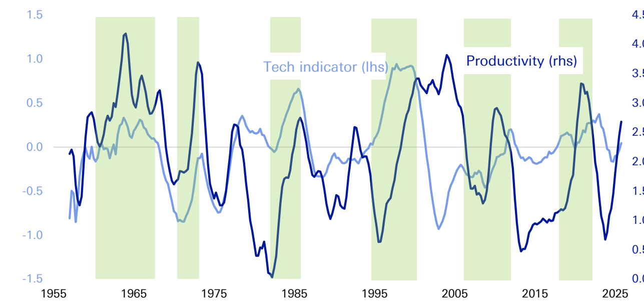
Yes, it can but the impact of AI on individuals, corporates, and economies has to accelerate from here. Our model shows this is the most likely outcome per the left-hand chart below.

Figure 14: Our technology megatrend index with AI-generated scenarios to 2030



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 15: Technology megatrend indicator v productivity growth (12qtr moving avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

The sharp turn up in our technology megatrend indicator is encouraging because this indicator, over time, has given us a reasonable idea of the direction of productivity growth – the key thing that will drive economies and markets forward. In general, whenever our technology indicator has accelerated to a peak, productivity growth has accelerated sharply. Notably, in the 1990s, it accelerated to 3-4% for consistent periods of time. There one big exception – in the late 1970s during the oil crisis and heady geopolitical tensions.

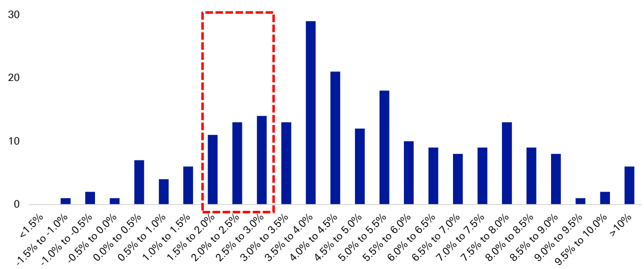
Our baseline expectation (per the above left chart) is that by 2030, our technology indicator will accelerate beyond levels seen during the 1990s technology boom. That implies we may see sustained productivity growth even greater than the 3-4% seen during prior peaks of our tech megatrend indicator.

History tells us that a bad sovereign debt position can be overcome

The "Long Depression" in the US in the mid-1870s came just a decade after the Civil War and, more immediately, followed a financial crisis and investor fear at the overbuilding of railroads. A lack of government intervention exacerbated the economic malaise and unemployment likely topped double digits. The sense of despondency was palpable. However, towards the end of the decade, the benefits of the railway technology began to boost the economy. At the same time, several other trends in agriculture and sovereign debt moved in a positive direction and resulted in startling economic growth.

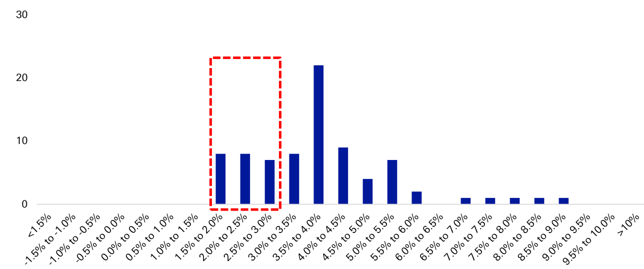
When we look at the history of the US economy, big jumps in growth are not unusual. In fact, they are quite prevalent. The following charts break down US economic growth into buckets. The red box surrounds the range that we usually expect GDP growth to sit in. But even these are infrequent compared with the jumps we have experienced.

Figure 16: Frequency of US GDP growth (1790-present, rolling 10yr, annualised)



Source : Finaeon, Deutsche Bank.

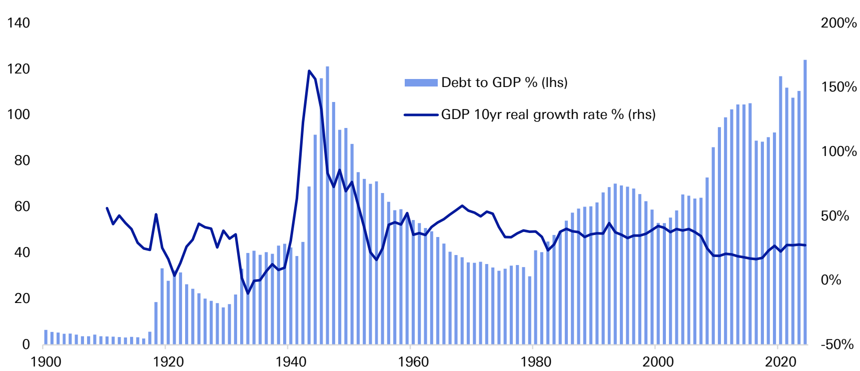
Figure 17: Frequency of US GDP growth (post-WWII, rolling 10yr, annualised)



Source : Finaeon, Deutsche Bank.

There are some critical instances through history where high sovereign debt levels were followed by periods of strong economic growth. One of the most notable is in the US after the second world war. At the war's conclusion, the US embarked on a 35-year period of deleveraging. And yet, this was a period of relatively strong economic growth – as the chart below shows.

Figure 18: US debt-to-GDP and GDP growth rate



Source : Finaeon, Deutsche Bank. (Note: Current figures differ from the CBO figures, however, this longer-term data series is used here to show the trend)

A key reason why economic growth was strong in the decades after the Second World War, despite high debt levels, was that other megatrends were working in a positive direction. Notably, globalisation grew quickly, domestic politics were reasonable, social discontent was low, and there was a positive impact from energy technologies.

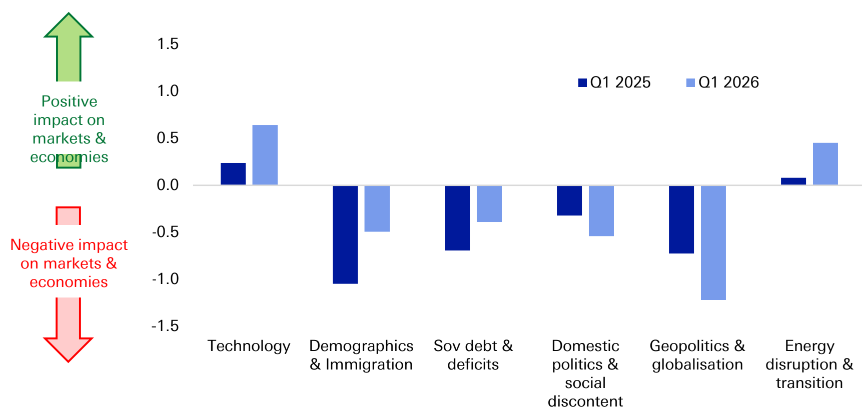
Today, most megatrends show worrying signs which is why we are almost entirely dependent on AI to bail out the world economy. So where will the US, and other developed economies, end up over the coming ten years? Given the wide variety of outcomes from AI, it is unlikely that we will see the 'normal' growth rate of 2-3% pa. Our model suggests we are more likely to see higher growth with a lesser probability of low, stagnant growth. It all depends on the megatrends and particularly technology.

Some cause to be optimistic in the near term

We do not put too much weight on the quarter-to-quarter movements in our megatrend indicators – there is frequently a lot of noise. However, we are conscious that the overall indicator has recently stabilised after its heavy drops over the last few years. This has been helped by a surge in the indicators for technology (driven by AI applications and data centre investment) and the energy transition (driven by a surge in renewables and commodity prices that have incentivised a lower dependence on energy for GDP growth).

In addition, the following chart shows that four of the six megatrend indicators have improved over the last 12 months.

Figure 19: Change in megatrend indicators over the last 12 months



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

When we take a step back, our megatrend indicator is positive in 60% of the quarters since 1956 (the beginning of our data). And when it is positive it tends to have a strong effect. In fact, the total positive impact of megatrends is double the negative impact.

What should corporates do?

Despite the volatility ahead, corporates must not let themselves be dissuaded from investing heavily in AI and other platforms for their business (whether physical or intangible). That is because as the volatility settles over the next few years, the AI-powered productivity boom may very well send economies to new highs. To take advantage, corporates must buy or build the right tech, physical assets, and organisational structure.

There has never been a better time for corporates to invest in AI and related infrastructure. Corporates in the S&P 500 alone are sitting on over $1.2tn of cash on their balance sheets while European companies in the Stoxx 600 sit on $1tn. Meanwhile, private capital has over $2tn of dry powder, so there is almost $5tn sitting in institutional money market funds. And that is before considering that debt financing options remain cheap and easily accessible.

7. Investment consequences

As our megatrend framework helps us look into the medium term, we can use it to identify some specific trends that we believe will help frame investment decisions today:

- The volatility caused by exogenous shocks in the future will be greater than the volatility from these shocks in the past.
- Stockpickers should benefit as stock prices move more in line with company fundamentals.
- Continued decoupling of the US and China despite sporadic pauses for ‘strategic stability’.
- Unpredictable sovereign debt sell-offs – particularly triggered by exogenous shocks and investor emotion.
- Haven assets could become risk assets – treasuries and the dollar, in particular, will become more volatile if the sovereign deficits megatrend deteriorates further.
- Private markets may become more attractive as investors look to avoid volatility.
- Uncorrelated assets assess will likely become increasingly popular (and expensive) as volatility and risk hedges (such as music rights, sports assets etc).
- Hedging will be difficult if haven assets lose their reliability.
- US over RoW based on:
  - Strong customer culture
  - Tech and capital markets leadership, and the institutional depth to support both
  - Demography is stronger than other DMs + China
  - Other countries now have to allocate resources to develop things that were formerly underwritten by the US, such as defence.
- Advantaged companies typically have:
  - A defendable moat rather than efficient supply chains
  - Natural geopolitical and currency hedges
  - Prioritise asset efficiency over size
- Capex-heavy business models could become just as attractive as asset-lite businesses
- A releveraging cycle should benefit companies
- Companies will tend to pursue transformational M&A rather than the ‘hopeful’ M&A they implemented during the era of low interest rates
- Return on equity will likely be more important than profit growth
- Companies may boost returns by spinning-off underperforming divisions
- We are cautious on markets where state-run enterprises play an important role

- Stocks with higher credit quality to outperform whilst the refinancing needs of highly-levered stocks should be closely watched

The following table sets out more specific views based on the current direction of megatrends:

Figure 20: Our medium-term thematic views by asset class

Asset class / Horizon	3+ years
Equities	Long US stocks in (1) knowledge-intensive industries like TMT and healthcare and (2) cyclical consumer industries; EU countries with stronger growth and less concern about sovereign debt to outperform; European firms with strong know-how to relatively outperform; US and US-aligned stock markets to outperform non-aligned markets; negative on oil and gas stocks; defence stocks to outperform; equities performance across countries and cyclicals v defensives to align along demographic dynamics; healthcare to benefit most clearly
Credit	Trade-exposed materials and industrial companies to face margin pressures from tariffs and trade realignment; this could widen HY vs IG gap; Spreads to normalise further in both the US and Europe but relatively more so in Europe; Spread widening will be back-stopped by excess liquidity
Yields	US yields will remain among least negatively affected major economies when it comes to inflation and debt concerns; Sovereign yields of countries with ageing population and no credible fiscal plan for greater dependency ratios to come under volatility; otherwise lower yields in countries where ageing is more "natural" (EM)
Commodities	Polarisation of commodities / trade along axes of influence will benefit producers in choke-point countries where diversification is impractical; economic policy, demographics and tech to be a net negative for oil prices and energy stocks; renewable energy will grow but associated stocks may struggle. A wave of public-to-private deals is possible
Alternatives	Private capital (PC) as a major beneficiary of greater concerns about aging population and pension shortfalls; PC will be key in financing strategically important industries given likely lack of red tape cutting as well as budgetary constraints in many major countries. This will be especially true in Europe (energy, infrastructure); A cycle turn for private credit will be coming; but private capital's momentum is unlikely to stop; it will become ubiquitous in institutional portfolios, with a rise in allocations likely

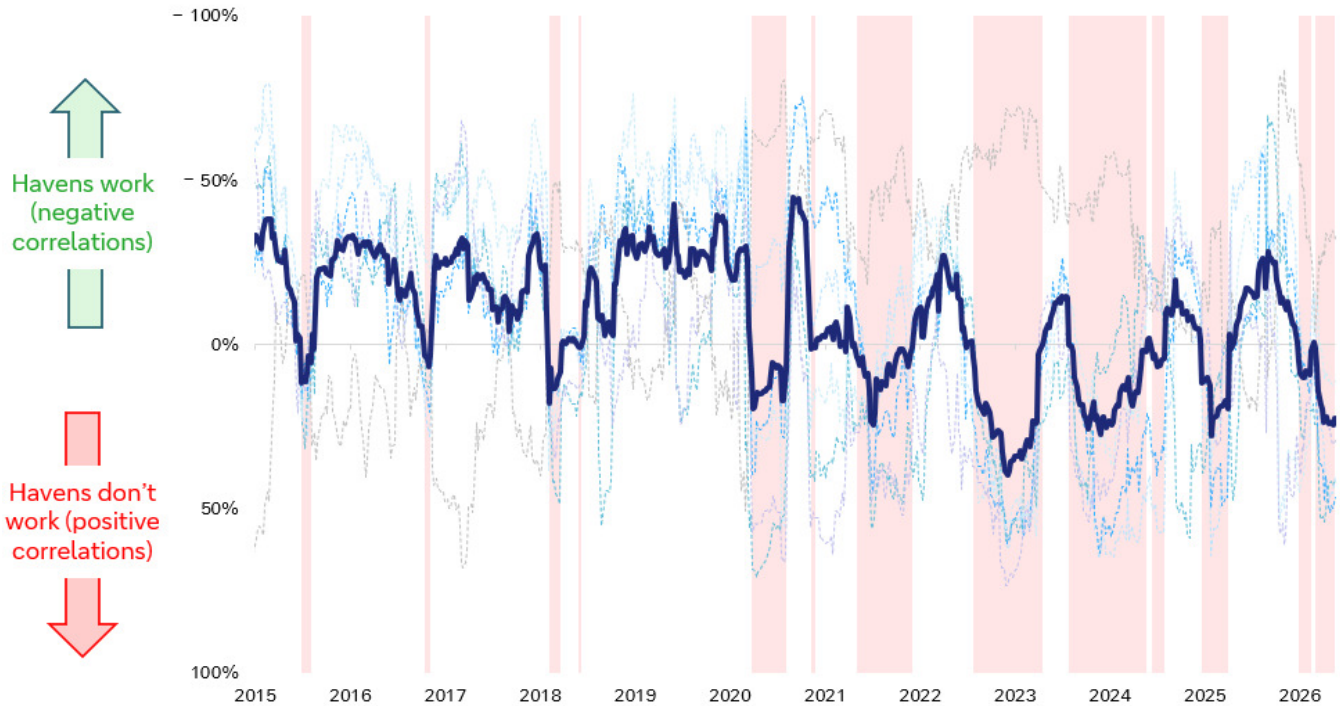
Source : Deutsche Bank.

What about the haven assets?

Understanding megatrends is important because normal hedges may not work anymore.

As we think about the potential market downsides of megatrends the obvious question arises: “What haven assets should I buy?”. The worrying thing is that, in recent times, the answer is increasingly difficult to ascertain. The reason is that many of the traditional haven assets simply do not work well anymore. To examine this, we looked at six haven assets: Gold, the US dollar, the Swiss Franc, the Japanese yen, 10-year treasuries and 10-year bunds. We then aggregated their performance when is shown on the following chart:

Figure 21: Our haven indicator (20-wk rolling, avg across pairwise correlations with S&P 500)



Source : Bloomberg Finance LP, Deutsche Bank.

The most important takeaway from this is that haven assets have not worked during the most important times, ie. Major risk-off moments, including Covid in 2020, the interest rate hikes of 2022, the US tariff announcements in 2025, and the Iran war in 2026.

This breakdown in havens was highlighted by the IMF earlier this year. It warned that this effect may "amplify systemic vulnerabilities" due to forced deleveraging. That is a particular concern as central banks are reducing their bond holdings and the greater supply is being absorbed by price-sensitive private investors.

8. Appendix: Each megatrend in brief

Technology

Overall view

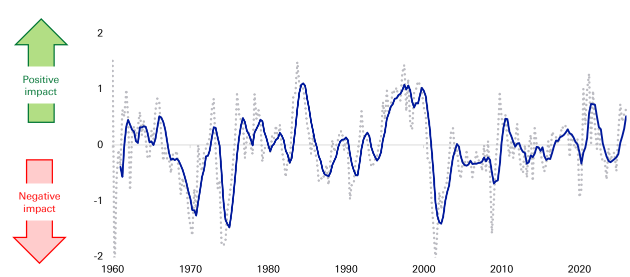
AI has been distributed and adopted far more rapidly than any other technology in human history. It has very low barriers to adoption and its applications will directly drive productivity for corporates. In the next phase of AI, institutions like firms, schools, health care, public administration, are adapting their organisation and processes to better accommodate AI. In some ways, this is analogous to the building of better roads to accommodate more cars in the early part of the 20th century.

The third phase will likely start in the early 2030s. Here, genAI systems will likely interconnect, share knowledge, self-improve exponentially, work across multiple domains. They will likely become partially self-aware and autonomous.

Investor perceptions

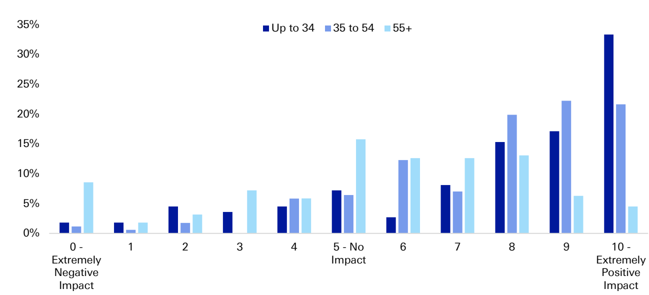
The technology megatrend is strongly correlated with GDP and equity prices. It also tends to push down inflation a little. The impact of technology on markets and economies has recently been driven by the huge investment in information processing equipment, along with its contribution to GDP. This is coupled with the continued investment in software (in its broadest definition).

Figure 22: The impact of the technology megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 23: Investors' anticipated impact of technology on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

The above right chart shows investors are very excited about the potential for technology to positively impact their investments.

Three key technology variables to watch

1. Investment in information processing equipment: Growth has been extremely strong over the last few years as the AI boom has catalysed capex spending. A plateau (or drop) in spending would signal some near-term cooling.
2. Capacity utilisation for Hi-tech industries: This is the greatest level of output a plant can maintain within the framework of a realistic work

schedule. This has fallen from 88% just before the release of ChatGPT to 71% now. We look for increased capacity to signal greater AI adoption.

3. Industrial production of computers, comms, & semis: Growth here was poor several years ago but over the last 18 months has been trending up. Sustained growth will indicate greater domestic diffusion of technology capabilities.

Sovereign deficits

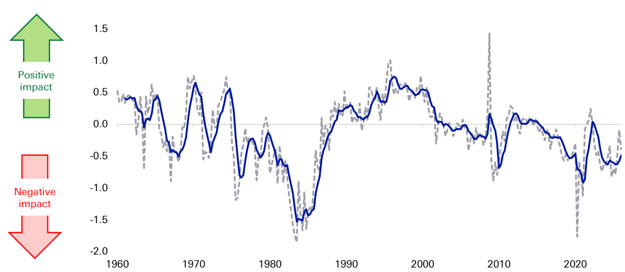
Overall view

In the era of normalised interest rates, we expect fiscal deficits, debt, and fiscal austerity to become contentious political topics. Should deficits continue on their current trajectory, we expect sustained levels of inflation above target and frequent, sovereign debt sell-offs. Associated fear would ripple through equity and credit markets. We do not expect a major sovereign default by 2030.

Investor perceptions

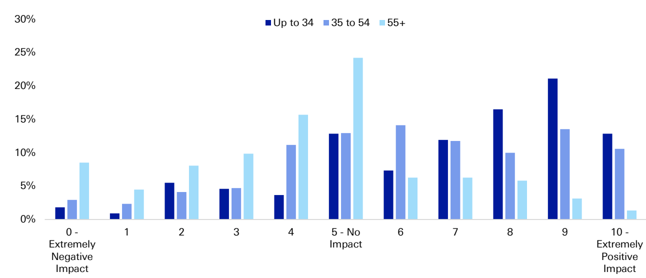
The impact of sovereign deficits on markets and economies has varied significantly over time. The below-right chart shows that investors are tending somewhat positive about the potential for sovereign deficits to impact their investments over the next three years. This may be because expansionary fiscal policy can boost markets over that timeframe.

Figure 24: The impact of the sovereign deficits megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 25: Investors' anticipated impact of sovereign deficits on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Looking out further, it becomes more serious. Indeed, the megatrend indicator (above left chart) has been trending down for the last three decades and the ageing population will apply further negative pressure. Indeed, US debt is expected to rise from about 100% of GDP to 120% by 2036 (CBO).

Three key sovereign deficit variables to watch

1. The contribution of government spending on GDP: This is a potential bright spot as this indicator has recently shot up. Although it is early days, if the continued trend here combines with greater AI adoption, there is potential for sovereign deficits to become less of a drag on markets and economies.

2. US 10yr yield vs. the G7 average: Treasury yields have been higher than the G7 average for most of this century. However, recently that gap has begun to shrink. Should this trend continue, it will signal that global demand for treasuries remains strong which, in turn, will boost the resilience of the US' twin deficits.

3. Fiscal policy uncertainty index: This turned deeply negative in 2024 and 2025, pushing below the Covid lows. While this is a somewhat subjective index, it is an important trend indicator because it can impact the short-term market view, particularly during an exogenous shock which can lead to fear-based sell-offs.

Geopolitics & globalisation

Overall view

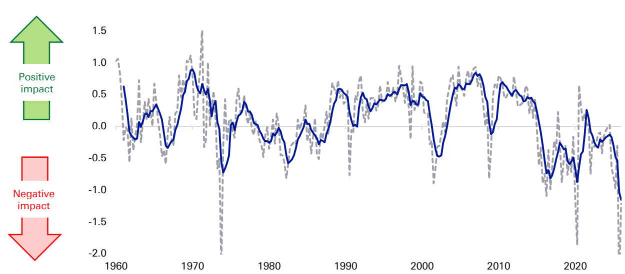
There are many myths associated with this megatrend. We believe globalisation and international trade will, in aggregate, continue to be strong, but it will reorganise itself around the US and Chinese spheres of influence. The US and China will likely continue to decouple their dependence on each other but the shocks of the 2020s show that corporates are very good at reorganising supply chains.

Many commentators have discussed how geopolitical shocks tend to only have a short-term impact on markets. Our megatrend indicator shows this is not true if you consider the broader picture and wider economy. That is because G&G impacts many things that are two, three, or more steps removed from the initial impact – and these may be incorporated into economies and markets slowly and over time.

Investor perceptions

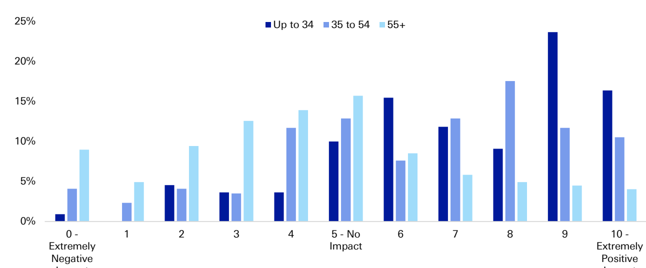
Investors are relatively nonplussed with G&G in terms of the potential impact on their investment portfolio. The below right chart shows that most regard it as neutral, although there is a leaning, particularly among young people, to be positive about the potential benefits of G&G.

Figure 26: The impact of geopolitics and globalisation megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Figure 27: Investors' anticipated impact of geopolitics & globalisation on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Our G&G indicator is somewhat positively correlated with the S&P 500 and GDP growth, and there is a small negative correlation with inflation.

Three key geopolitics & globalisation variables to watch

1. Shortage index: This covers US shortages of industrial goods, energy, food and labour. The combined shocks of this decade have caused the index to average its highest level since the oil shocks of the 1970s and then, before that, WWII. To the extent this remains high, it will indicate a worsening G&G impact

2. Trade openness: This is essentially exports + imports as a proportion of GDP. This has not changed much over the last decade despite much geopolitical upheaval. Thus, in itself, we see this as a positive indicator.

3. Kilian index of global activity: This is derived from a panel of dollar-denominated global bulk dry cargo shipping rates and may be viewed as a proxy for the volume of shipping in global industrial commodity markets

Domestic politics & social discontent

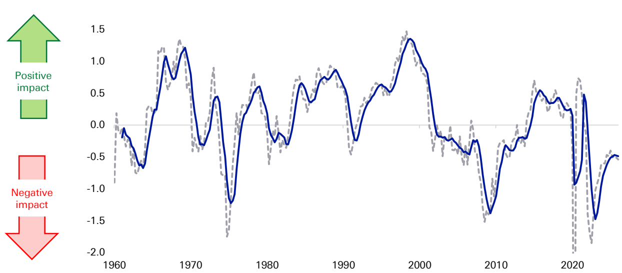
Overall view

We are recently past the 'tipping point' for populism (of all political colours). As such, it will likely become more widespread as the populist left grows to compete with the already-strong populist right. These politicians tend to encourage and amplify anger in voters. This would increasingly impact markets and economies as some countries potentially enact policies with negative long-term implications.

Investor perceptions

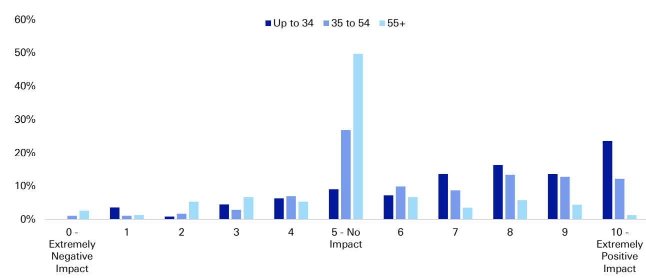
The way investors see domestic politics is similar to the way they view its sister trend, geopolitics & globalisation. Most investors tend to be neutral on the prospects for it to influence their portfolio, with a small proportion leaning positive.

Figure 28: The impact of domestic politics & social discontent megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 29: Investors' anticipated impact of domestic politics & social discontent on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Our megatrend indicator has a very strong correlation with GDP growth. Positive politics and lower social discontent also coincide with lower inflation and bond yields. Yet, we find that many people ignore the impact of domestic politics on markets as it is less impactful on equity markets.

Three key domestic politics variables to watch

1. Consumer sentiment: This gauge has done poorly at predicting what people usually assume – retail sales. But it has been good at estimating the level of social discontent. That directly translates into support for populist parties (on both the left and right) and can increase the likelihood that governments will embark on fiscal and institutional policies that have negative medium-term effects

2. Housing affordability ratio: For almost this entire decade, this indicator has been negative, indicating that affordability is growing worse. Most developed countries are stuck with the problem of incentivising more house building at the same time that regulation and local objections hold up construction.

3. Labour share of income: While growing real wages is positive, people care about their relative position in society. That is why the labour share is so critical. Labour share has been steadily trending down for decades and AI risks worsening this. This is one of those issues which typically lies dormant until there is an exogenous catalyst. When the frustration surfaces, various populist politicians may channel that anger in an unproductive direction.

Demography

Overall view

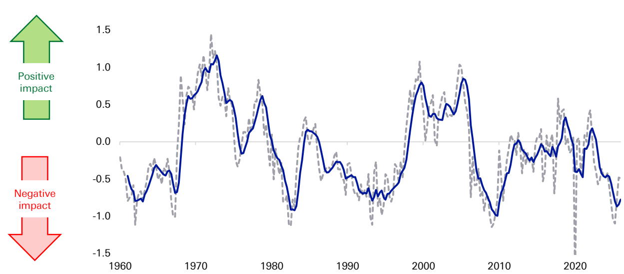
Demography is about far more than just the ageing population. People's 'health span' is rising, they are working longer and spending more. Still, pension savings are generally poor. That means people will take greater investment risk for longer, a positive for equities. Corporates will be forced to use technology to adapt to a tighter workforce.

Investor perception

The below left chart shows that the demography megatrend has deteriorated markedly since Covid. This is not only due to the declining working population but also falling growth in household numbers, lower net migration, and rising objection to migration.

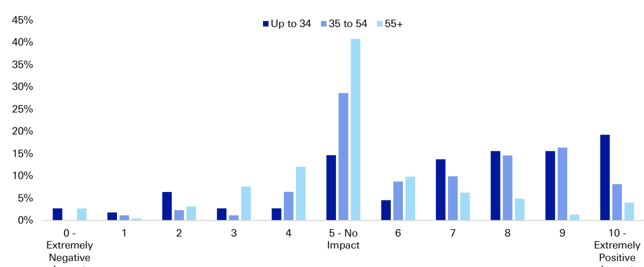
Younger investors err on the positive. They see demography as a positive for their investments over the coming three years. Partly, that could reflect optimism at how AI transformation is being accelerated by the need to address the declining workforce.

Figure 30: The impact of demography & immigration megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Figure 31: Investors' anticipated impact of demography on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Three key demography variables to watch

1. Net migration rate: A big one for almost all developed countries where birth rates are below replacement levels. Politics will be a key driver here. That will be made even more pertinent if populist parties gain more traction in Europe. Some believe in closing borders, others believe in far more open borders.

2. One-family house sales: These are key as people are likely to have more children if they can buy a family home. The good news is that growth here has trended up over the last two years after a strong downswing in 2022 and 2023

3. Number of households: Household formation is generally a critical precursor to building a family (regardless of family home sales). The decade-long upswing in household growth abruptly ended with Covid. Since then, growth has been trending down. The overlapping social dynamics that caused this shift are complicated and poorly understood but this remains a key indicator for demography moving forward.

Energy disruption & transition

Overall view

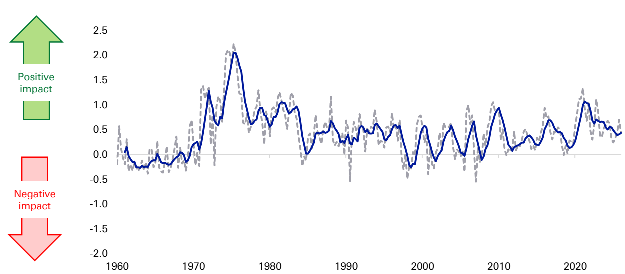
This megatrend is nuanced. Because we assess the medium term, we regard it as a positive when events happen that encourage greater energy security, diversification of supply, and adaptation that reduces the energy intensity of GDP.

The current energy crisis, therefore, can be a positive for economies and markets in the medium term as it is encouraging all the above points – even though the short-term effects are causing pain for consumers. In addition, the ongoing geopolitical tensions between the US and China is leading to a period of 'strategic stability' where countries are working to become more self-sufficient (or 'friend-sufficient') with critical minerals, commodities and other 'chokepoint' inputs.

Investor perception

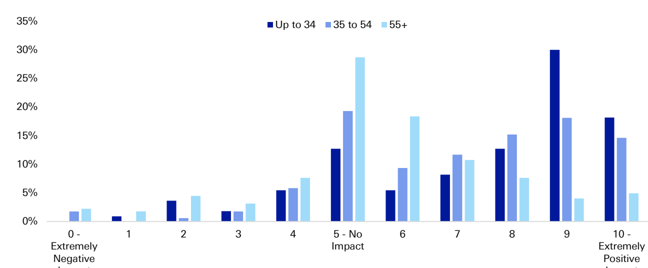
The good news is this megatrend has been positive for markets and economies for most of the last 50 years. That is because the world is gradually decreasing its dependence on traditional 'chokepoint' energy sources. Yet, there is a long way to go and the below right-hand chart shows that investors are generally positive about the prospect.

Figure 32: The impact of energy disruption & transition megatrend on economies and markets (trendline is 5qtr mov avg)



Source : Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Figure 33: Investors' anticipated impact of the energy transition on their main investment portfolio over the next 3 years (0-10) by age group



Source : dbDataInsights, Deutsche Bank.

Three key energy disruption & transition variables to watch

1. Metals price volatility: Big spikes in various metal prices tend to cause panic in industry but, on the whole, they tend to revert to mean over time as either alternative supplies are sources, or alternative production methods are adopted. We expect commodities will be used as geopolitical weapons with increased frequency in the coming years, and we expect markets to respond in turn.
2. Rig count: There are several reasons why these may rise or fall over time. However, a falling rig count over the medium term is one indicator of the level of reliance on fossil fuels.
3. Energy inflation: In any one quarter or year, a rise or fall here can be noise, however, when energy inflation is falling over the medium term, it can indicate that supplies are becoming more diversified, stable and reliable.

Appendix 1

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Deutsche Bank
Research Institute

How defence spending will shrink global imbalances and weaken US

May 12, 2026

In this note we argue that US withdrawal from security alliances established after the Second World War will result in a fall in global savings. This is as allies seek greater strategic autonomy in three key areas: defence, energy and technology. These areas are highly investment intensive and history suggests that when spending on these sectors rise, private consumption and investment do not fall sufficiently to offset the effect on savings. The result should be lower global savings and higher interest rates.

Because the regions most affected by a changing US relationship are also the world's largest net savers (in other words, they run the lion's share of global current account surpluses), the result should be a fall in global imbalances. There will be less recycling of savings out of Europe, Asia and the Middle East into the US and this will act as a material headwind for the dollar. For the sake of coherence, we focus in this note specifically on the defence angle but aim to build on this framework in the future when it comes to both energy infrastructure investment and technology.

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The rapid global rise in defence expenditure

The recent conflict in Iran will accelerate an existing global trend: rising expenditure on defence. According to data from the SIPRI database, 2024 saw the largest rise in defence spending in real terms since data is available (the late 1980s - figure 1). Defence spending is set to rise over coming years, both in real terms and as a percentage of GDP. At the 2025 NATO summit in The Hague, member states committed to a 5% GDP spending target on defence and security-related spending by 2035.[1] Outside of NATO, Japan has committed to doubling its defence spending to 2% GDP by 2027, two years ahead of its original schedule. The Korean government has pledged to increase defence spending by 8.2% this year.[2] China has also been significantly increasing defence spending, by 7% in 2026 and Russian defence expenditure has grown dramatically since the start of its invasion of Ukraine.[3]

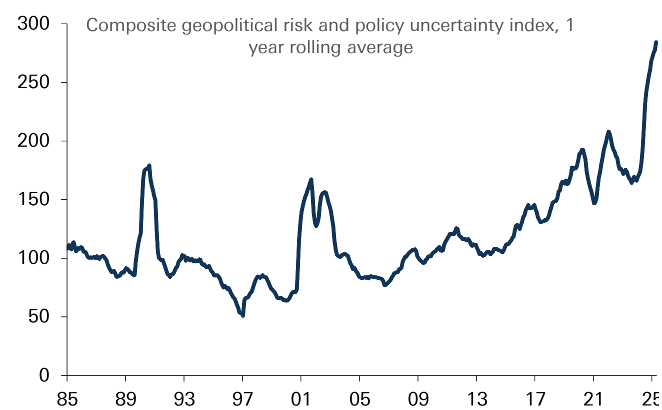
[1] NATO, April 2026
[2] Yonhap news agency, October 2025
[3] Reuters March 5th 2026

While the drivers of increasing defence spending are various (including rising nationalism, increased competition for raw materials), recent shifts in US policy have undoubtedly played an important role. The security umbrella established after the Second World War has come under strain from recent US policy decisions, including the withdrawal of military and financial support for Ukraine, tensions with EU allies over Greenland and most recently criticism of NATO allies over their decision not to back the recent US war against Iran. We have written in detail about increasing US reluctance to continue to provide 'global public goods' such as upholding an open international economic system and underpin western security, with the US pursuing a more self-interested policy agenda and exercising hard power and its economic networks more forcefully.[4] This in turn is an important motivating factor for US allies to pursue strategic autonomy in defence, as well as other areas such as energy and technology.

Figure 1: Fastest increase in global defence spending in decades in 2024

Source : Deutsche Bank, SIPRI Defence Database

Figure 2: Reflecting extremely elevated levels of geopolitical risk and policy uncertainty



Source : Deutsche Bank, Caldara, Dario and Matteo Iacoviello (2022), "Measuring Geopolitical Risk," American Economic Review, April, 112(4), pp.1194-1225. Baker, Bloom and Davis.

Rising defence expenditure: falling global savings

To understand better what impact rising defence expenditure could have on the global economy, we turn to the historical record. We make three important observations.

First, never underestimate how quickly defence expenditure can increase, especially if threat perceptions change. Back in September last year, we highlighted the period of rearmament during the 1930s as a useful historical parallel. Defence expenditure rose dramatically across future belligerents in only a few years. For example, defence spending as a share of GDP between 1935 and 1938 rose 2.5% for Italy, 3.7% for the UK, 8% for Germany and 17% for Japan (figure 3).

Second, fiscal constraints can be overstated. Although Britain's debt to GDP stood at well above 160% in the mid 1930s, the country was still able to significantly ramp up defence expenditure. Arguably, the most basic function of a state is to protect its citizens and guard its sovereignty. As such, if threat perceptions increase, defence spending tends to trump other spending priorities.

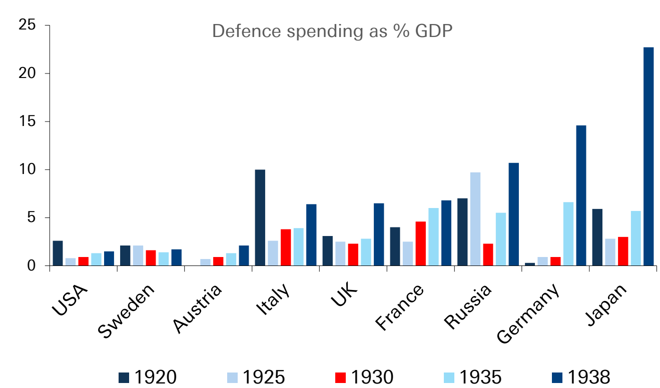
4 See, DB Special Report, "The Unruly hegemon; what a decline in US leadership means for markets," March 2025, EM Blog: Venezuela, hard power, geoeconomics and hemispheric dominance, January 2026 and "Geoeconomics 101," January 2026

Third, rising defence expenditure tends to be associated with falling savings. From a national income perspective, defence expenditure can largely be accounted for through increased investment and government consumption. An important if not crucial question is whether this comes at the expense of private consumption (leaving the savings rate unchanged) or whether consumption does not fully compensate (reducing overall saving).

There is evidence to suggest that household consumption does fall during periods of re-armament. For example, in 1930s Nazi Germany, economic historians have described forms of moral suasion such as national savings days to increase available funding for the military, before the 1936 Four Year Plan saw rationing of raw materials and consumption goods which limited household spending opportunities. It is also possible that the uncertainty caused by a more volatile world prompts households to save for precautionary purposes.

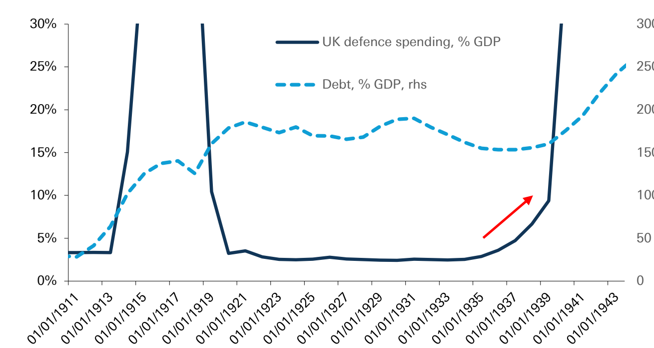
Nevertheless, efforts to increase national saving were never powerful enough to offset the demands of military investment at an aggregate level, reflected Germany's chronic current account deficit. And at the less extreme end of the spectrum, evidence also suggests that savings rates do fall. This was true in the UK during the late 1930s and the US following the start of the Vietnam War (figures 5 & 6).[5] There is also strong theoretical evidence to suggest that defence spending is not subject to the same Ricardian constraints as other government spending.Government spending on defence does not crowd out other forms of spending because without it, the government is no longer able to protect the ability of households to save for the future or bass on bequests to future generations. [6]

Figure 3: Defence expenditure rose dramatically among belligerents in the run up to WW2



Source : Deutsche Bank, Eloranta, J. "Pro Bono Publico? Demand for Military Spending Between the World Wars," Economic and Business History Society, 2017

Figure 4: British defence spending rose in the 1930s in spite of very high debt levels

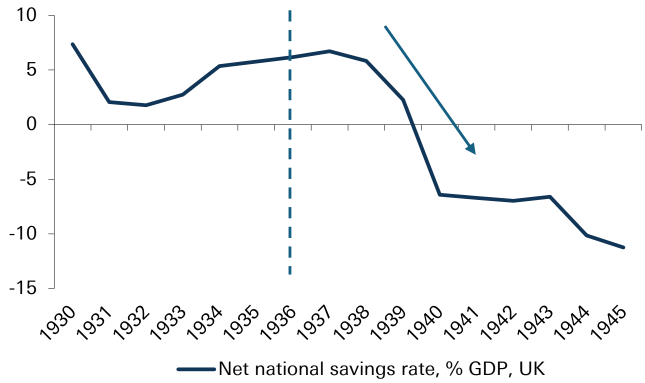


Source : Deutsche Bank, Bank of England Millennium of Macroeconomic Data

5 The UK savings rate does initially rise in the immediate aftermath of rearmament. However, this continued a trend established since the 1932 recession and rearmament did not truly start until 1937 after which national savings fell

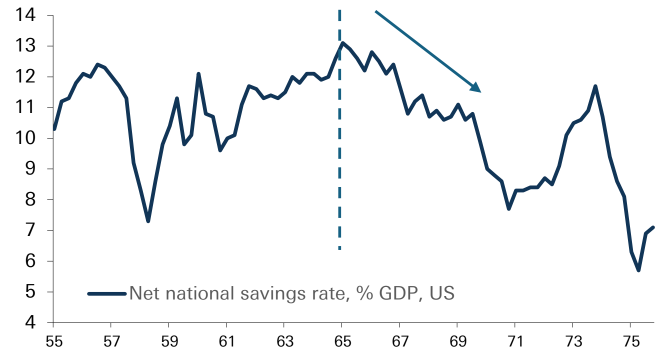
6 See Seiglie, "Defence Spending in a Neo-Ricardian World." 1998

Figure 5: Savings rates initially rose then fell rapidly following re-armament in the 1930s



*Source : Deutsche Bank, Bank of England Millennium of Macroeconomic Data, * dotted lines represent start of rearmament*

Figure 6: They fell almost immediately in the Vietnam War



Source : Deutsche Bank, St Louis Fed, dotted lines represent start of rearmament

As we shall see, these observations have potentially highly important implications for global asset prices. We would note that while this report focuses specifically on defence, there are other sectors in which investment is likely to increase due to changing global political economy, especially energy infrastructure and technology. This should act to magnify the effect on global savings, even if the theoretical effect is clearest in the defence sector.

Falling global savings: shrinking global imbalances

Should increased defence expenditure globally act to increase investment without a corresponding fall in consumption, as we expect, the effect should be to raise interest rates, all being equal. In the long run, economic theory posits that the neutral rate of interest should be determined by the supply of and demand for savings.[7]

In the long run is an important caveat. In the short term, fiscal and monetary policy can also act to dampen the transmission channel. For example, during the Second World War, when British savings rates fell to very low levels, market interest rates remained extremely low. In part this was due to widespread financial repression policies, including capital controls, mandatory holdings of government debt and borrowing and consumer restrictions.[8]

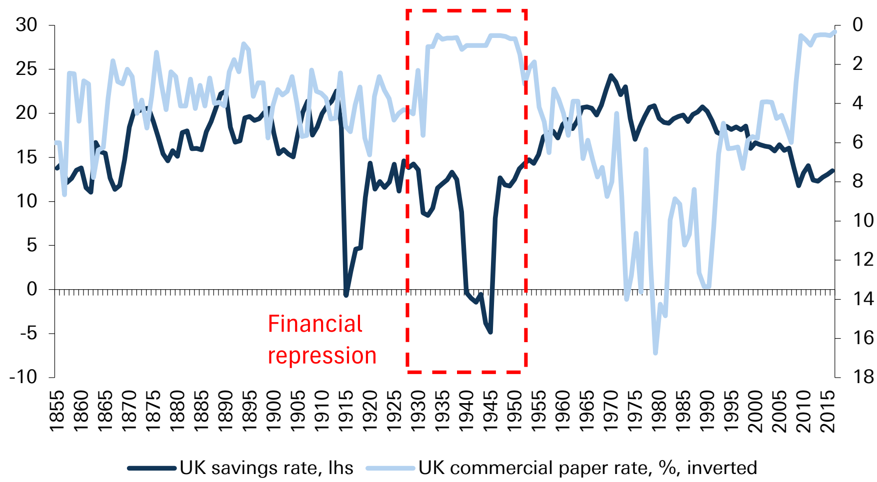
We should be careful in taking the 1930s comparison too far when it comes to the current geopolitical environment. But governments will be confronted with hard choices when it comes to financing additional defence expenditure, especially where public debt to GDP is already at high levels, and given that welfare spending is considerably higher as a share of GDP than 90 years ago. The interplay between defence expenditure, interest rates and government policy is especially relevant when it comes to international capital flows, which we focus on next.

This is because in an era of capital mobility, another important consequence of rising defence expenditure will be its effect on global imbalances. The world's largest net savers (those running the largest current account surpluses) are

7 For example, Federal Reserve Bank of San Francisco Economic Letter, April 2025
8 Mark Harrison, "Paying the price for war." University of Warwick

concentrated in three regions: Europe, East Asia and the Middle East (figure 9). These regions are also those seeing the largest increases in defence expenditure (figure 8), and where defence expenditure needs are likely to be the most pronounced over coming years.

Figure 7: In the Second World War, the government used financial repression to keep rates low despite collapsing savings



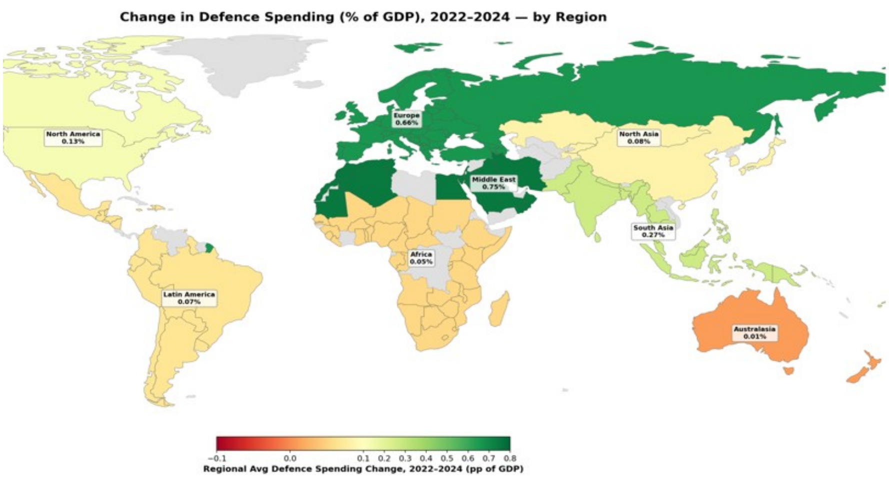
Source : Deutsche Bank, Bank of England Millennium of Macroeconomic Data

Increased defence expenditure will act to reduce global imbalances by reducing net savings in these three regions, in other words falling current account surpluses. But how will defence expenditure be financed? The quid pro quo of smaller current account surpluses should be reduced recycling of savings abroad via the financial account. From a political perspective, there are strong incentives to meet domestic financing needs from private savings rather than increased taxation or central bank financing. For one, it would be less politically unpopular than raising tax or inflation. For another, it reduces economic interdependencies which can make sovereign states vulnerable.

Indeed, as we noted in a recent note, tapping private savings was was a key channel of financing defence expenditure in World Wars 1 & 2. An even more extreme version of this could be outright repatriation of foreign assets to fund defence spending. Leading into the First World War, the economic historian Niall Ferguson has highlighted how international bond markets increasingly tilted towards 'home bias,' and during World War 2 Britain coerced private savers to swap dollar holdings in order to boost the Bank of England's foreign exchange reserves.

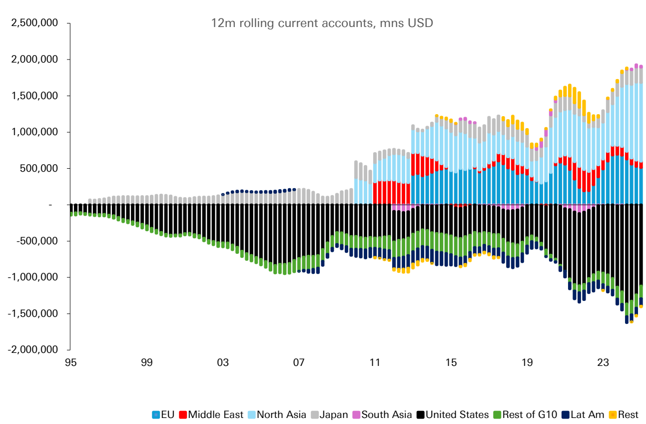
On the deficit side of global imbalances, as figure 9 shows, one country contributes more than any other combined, namely the United States. And it is also the case, when it comes to stocks, that the largest holders of US assets (when excluding the Caribbean holdings which are largely reflective of tax benefits rather than ultimate beneficial ownership) are once again: Europe, East Asia and the Middle East. In other words, if global imbalances fall due to increased defence spending, it is US deficits that may end up being the most vulnerable.

Figure 8: Europe, North Asia and the Middle East are seeing the biggest increases in defence expenditure



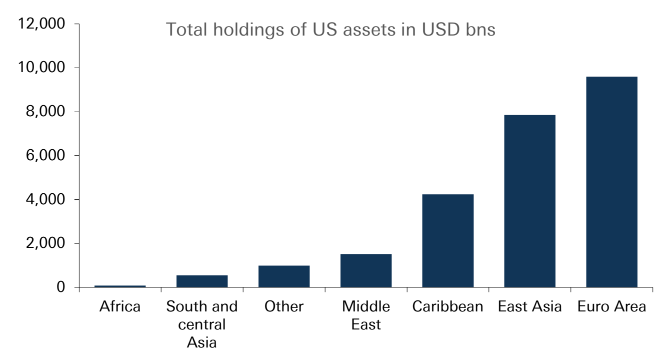
Source : Deutsche Bank, SIPRI Defence Database

Figure 9: Global surpluses driven by similar regions to where defence spending is going to rise the most, global deficits driven by the US



*Source : Deutsche Bank, Haver Analytics, *note time series not continuous due to data availability issues*

Figure 10: Similarly, holdings of US assets are concentrated in regions that need to re-arm



Source : Deutsche Bank, US Treasury TIC Data

This, however, is logical. With a reduced US security commitment to traditional allies, it is inconsistent for the private savings of those allies to be channeled to the US to the same extent as domestic financing needs increase.

The current account channel: can US defence exports compensate?

Reduced recycling of private savings into the US on account of growing defence spending may not necessarily be negative for the dollar. If US deficits are vulnerable to less recycling of private savings in the rest of the world, the US current account should adjust. But this could happen for dollar-positive reasons if the US is able to provide defence equipment and services through its exports.

The push for defence spending is already evident in the data. For example, although volatile, defence-related machinery orders in Japan and capital goods orders in

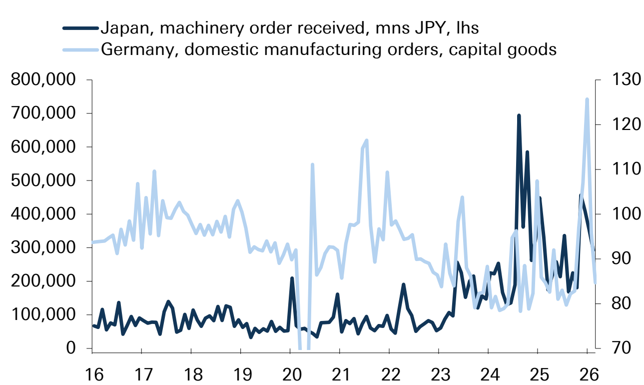
Germany have risen to record high levels in the last four years (figure 11). At the same time, US defence exports have also risen to record levels (figure 12). Nevertheless, while we think that increased US defence exports may compensate the effects on the financing of US deficits in the short term, over the medium term we think this is less likely.

First, although there is a large investment and expertise gap in the defence industry in regions such as Europe, East Asia and the Middle East, which will require imports to fill, over the medium term policymakers are attempting to build up domestic industrial bases and increase strategic autonomy in defence.[9] For example, the EU has identified seven capability gaps, from air and missile defence to AI, Quantum, Cyber and Electronic Warfare with the aim of building a defence base that will enable the EU to conduct the entire spectrum of military tasks. The European Defence Industrial Strategy has put a target of at least 50% defence procurement being sourced from the European industrial base by 2030 and 60% by 2035. To be sure, the starting point presents a challenge - our European economists estimated that 78% of defence procurement went to non-EU providers of which 63% went to the US at the start of last year.[10] But the European Commission also note that European defence industry turnover has grown from EUR 124bn in 2021 to EUR 183bn in 2024.

Second, procurement is only one part of the defence budget. For example, although in the EU it makes up the majority of the investment budget, the EUR 88bn spent in 2024 on procurement is less than a quarter of the total (EUR 381bn) spent overall.

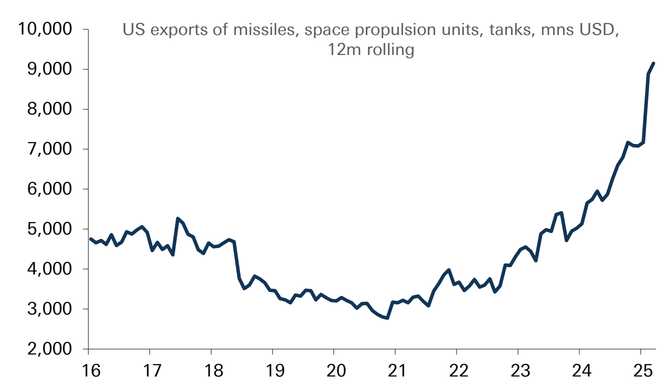
Another reason to doubt that US exports will fully compensate for the reduced ability of allies to finance US deficits is that the Trump administration is itself proposing a major increase in defence expenditure as part of the 2027 Budget, amounting to up to USD 1.5 trillion, nearly a USD 500bn increase from the previous year. It seems unlikely that US industry could simultaneously fully meet global demand for defence equipment and its own domestic demand under government spending plans.

Figure 11: Defence orders in Europe and Japan have already surged in the last three years



Source : Deutsche Bank, Haver Analytics

Figure 12: US defence exports are also climbing rapidly but are unlikely to fully compensate



Source : Deutsche Bank, Haver Analytics

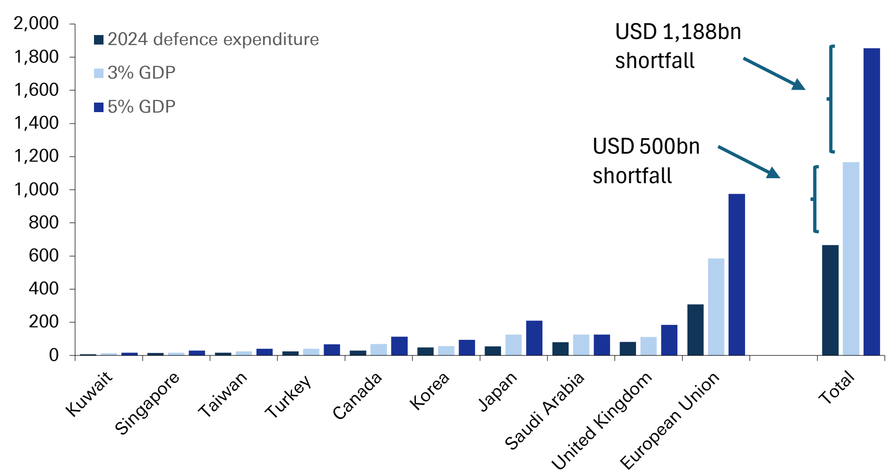
9 See the ReArm Europe Plan, Readiness 2030
10 See Mark Wall and team, "Focus Europe: Defending Europe Part 3, January 2025

Quantifying the impact

This analysis has suggested that rising defence spending will reduce global savings and compress global imbalances. Because the US deficit is the counter-point to savings in Europe, East Asia and the Middle East, recycling of private savings into the US is likely to diminish, putting pressure on the dollar. In the short term, increased US defence exports may compensate somewhat, but these are unlikely in the medium term to fill the gap.

Can we quantify the impact on USD? Future defence needs are subject to a huge amount of uncertainty but for the sake of simplicity we identify 'defence gaps' among a sub-set of economies where pressure to raise defence spending may be most acute. The first defence gap is based on a hypothetical 3% GDP defence spending target, the second a 5% target. We calculate this based on 2024 GDP in current prices, and compare it to updated defence spending from the SIPRI database for the same year. The results are striking. Under the 3% 'gap', the shortfall amounts to USD 500bn and under the 5% 'gap', the shortfall amounts to USD 1.2 trillion. By way of comparison, the 2024 US current account deficit amounted to USD 1.185 trillion. Of course, this is a highly partial equilibrium analysis, ignoring other aspects of supply and demand for global savings. But at a minimum, it suggests that the effect of increased defence spending on savings rates and current accounts will not be trivial.

Figure 13: Quantifying the defence gap - the additional spending is not trivial



Source : Deutsche Bank, IMF , SIPRI database

Conclusions

This report has focused on providing a framework for how increased global defence spending - a trend likely to be accelerated by the current conflict in the Middle East - will act upon global macro, focusing on savings rates and global imbalances.

We have concentrated on only one sector where investment is likely to rise among US allies over coming years. As noted at the start of this piece, other sectors where investment is likely to rise is energy infrastructure and technology. Our colleagues have focused on other areas in which the furling of the US security umbrella could have major ramifications, for example by reducing incentives to spend dollars in energy markets or accelerating the transition towards new energy sources.[11] When dovetailing with defence spending, the investment in strategic autonomy in these areas will act as another powerful force depressing global imbalances. And even within the defence sector, there are other implications we have not explored, such as the currency invoicing of trade and the multipliers of defence spending. This will provide fruitful ground for further research.

11 Mallika Sachdeva, "What Iran means for the dollar: a perfect storm for the petrodollar."

Appendix 1

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